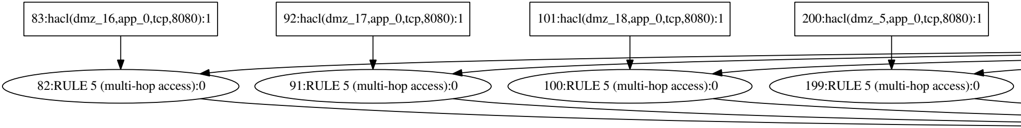
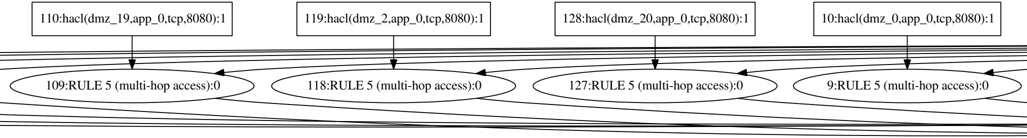
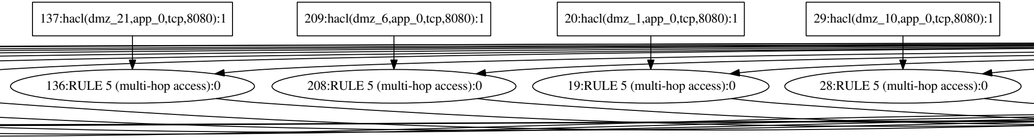
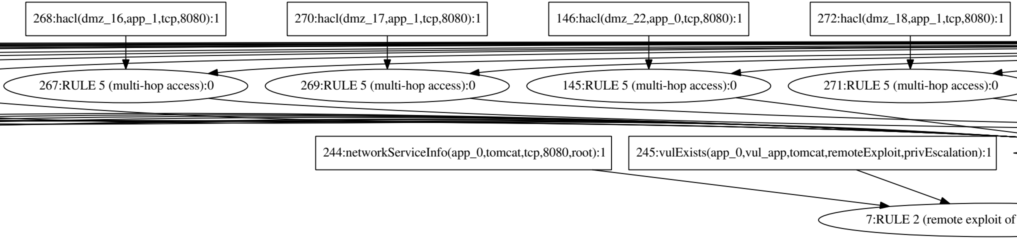
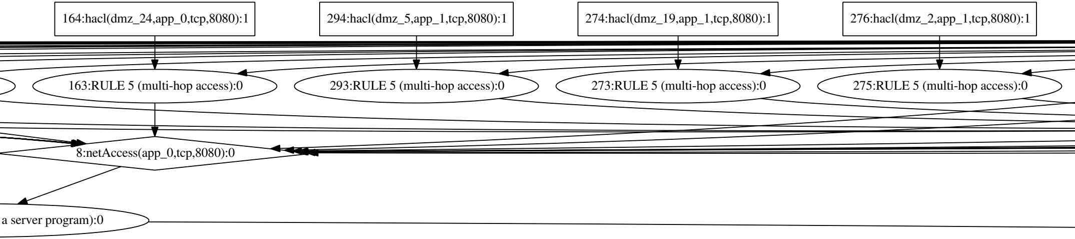
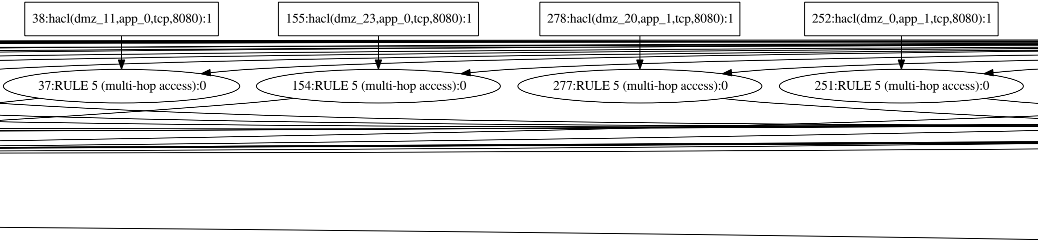

83:hacl(dmz_16,app_0,tcp,8080):1
92:hacl(dmz_17,app_0,tcp,8080):1
101:hacl(dmz_18,app_0,tcp,8080):1
200:hacl(dmz_5,app_0,tcp,8080):1
82:RULE 5 (multi-hop access):0
91:RULE 5 (multi-hop access):0
100:RULE 5 (multi-hop access):0
199:RULE 5 (multi-hop access):0

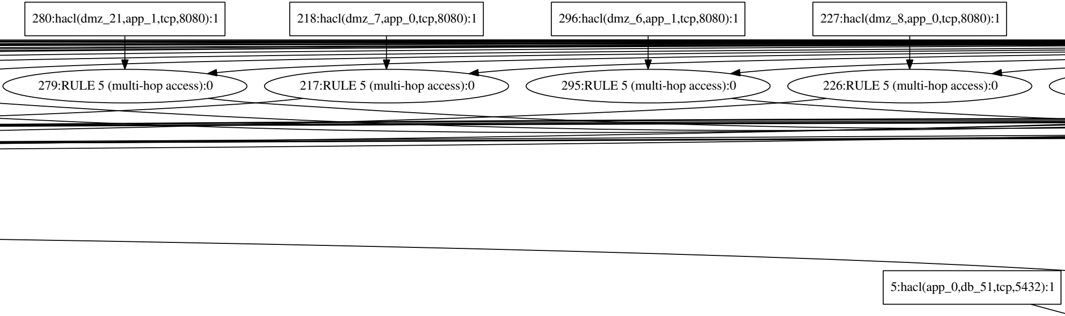

110:hacl(dmz_19,app_0,tcp,8080):1
119:hacl(dmz_2,app_0,tcp,8080):1
128:hacl(dmz_20,app_0,tcp,8080):1
10:hacl(dmz_0,app_0,tcp,8080):1
109:RULE 5 (multi-hop access):0
118:RULE 5 (multi-hop access):0
127:RULE 5 (multi-hop access):0
9:RULE 5 (multi-hop access):0

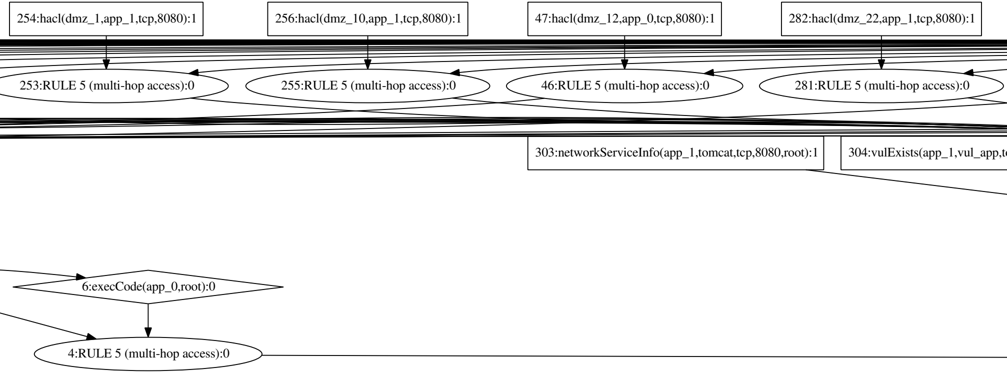

137:hacl(dmz_21,app_0,tcp,8080):1
209:hacl(dmz_6,app_0,tcp,8080):1
20:hacl(dmz_1,app_0,tcp,8080):1
29:hacl(dmz_10,app_0,tcp,8080):1
136:RULE 5 (multi-hop access):0
208:RULE 5 (multi-hop access):0
19:RULE 5 (multi-hop access):0
28:RULE 5 (multi-hop access):0

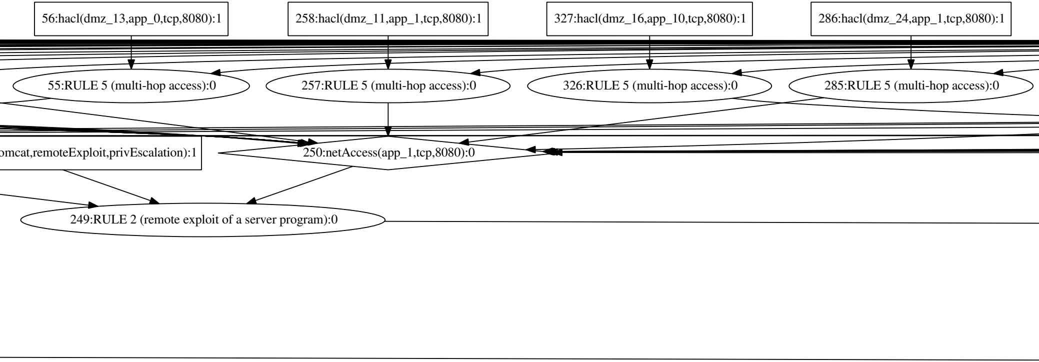

268:hacl(dmz_16,app_1,tcp,8080):1
270:hacl(dmz_17,app_1,tcp,8080):1
146:hacl(dmz_22,app_0,tcp,8080):1
272:hacl(dmz_18,app_1,tcp,8080):1
267:RULE 5 (multi-hop access):0
269:RULE 5 (multi-hop access):0
145:RULE 5 (multi-hop access):0
271:RULE 5 (multi-hop access):0
244:networkServiceInfo(app_0,tomcat,tcp,8080,root):1
245:vulExists(app_0,vul_app,tomcat,remoteExploit,privEscalation):1
7:RULE 2 (remote exploit of

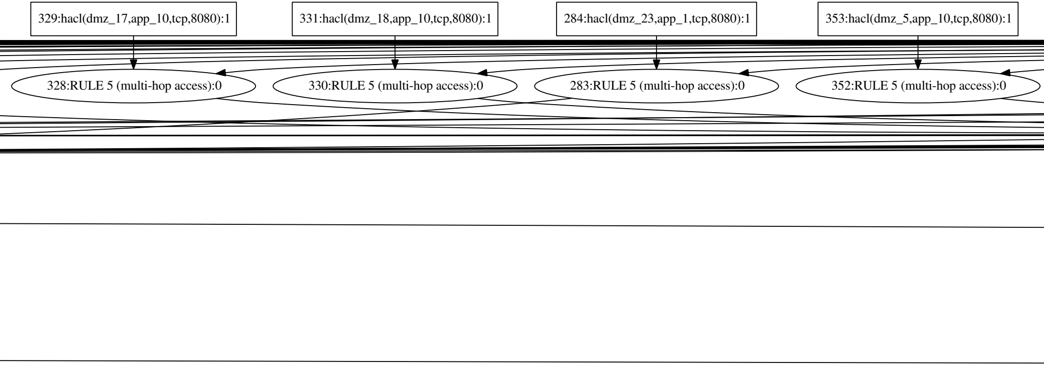

164:hacl(dmz_24,app_0,tcp,8080):1
294:hacl(dmz_5,app_1,tcp,8080):1
274:hacl(dmz_19,app_1,tcp,8080):1
276:hacl(dmz_2,app_1,tcp,8080):1
163:RULE 5 (multi-hop access):0
293:RULE 5 (multi-hop access):0
273:RULE 5 (multi-hop access):0
275:RULE 5 (multi-hop access):0
8:netAccess(app_0,tcp,8080):0
a server program):0

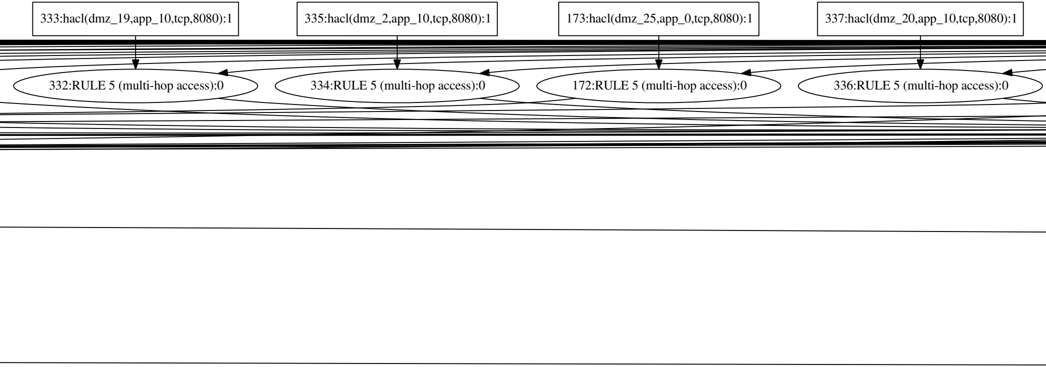

38:hacl(dmz_11,app_0,tcp,8080):1
155:hacl(dmz_23,app_0,tcp,8080):1
278:hacl(dmz_20,app_1,tcp,8080):1
252:hacl(dmz_0,app_1,tcp,8080):1
37:RULE 5 (multi-hop access):0
154:RULE 5 (multi-hop access):0
277:RULE 5 (multi-hop access):0
251:RULE 5 (multi-hop access):0

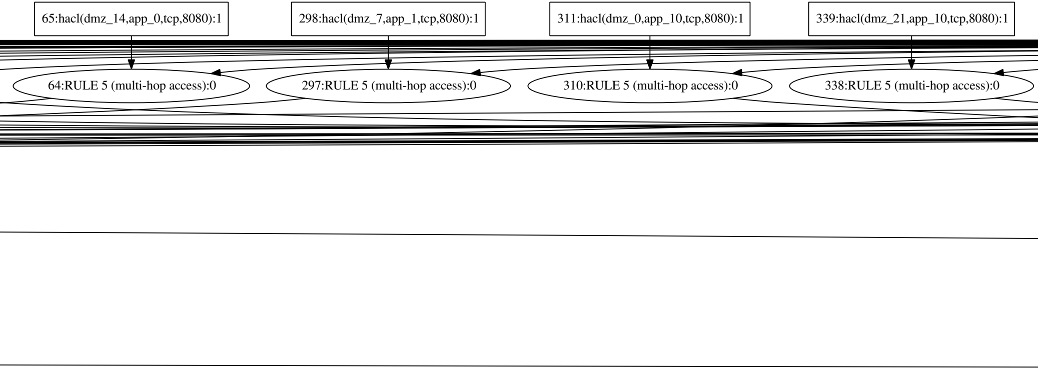

280:hacl(dmz_21,app_1,tcp,8080):1
218:hacl(dmz_7,app_0,tcp,8080):1
296:hacl(dmz_6,app_1,tcp,8080):1
227:hacl(dmz_8,app_0,tcp,8080):1
279:RULE 5 (multi-hop access):0
217:RULE 5 (multi-hop access):0
295:RULE 5 (multi-hop access):0
226:RULE 5 (multi-hop access):0
5:hacl(app_0,db_51,tcp,5432):1

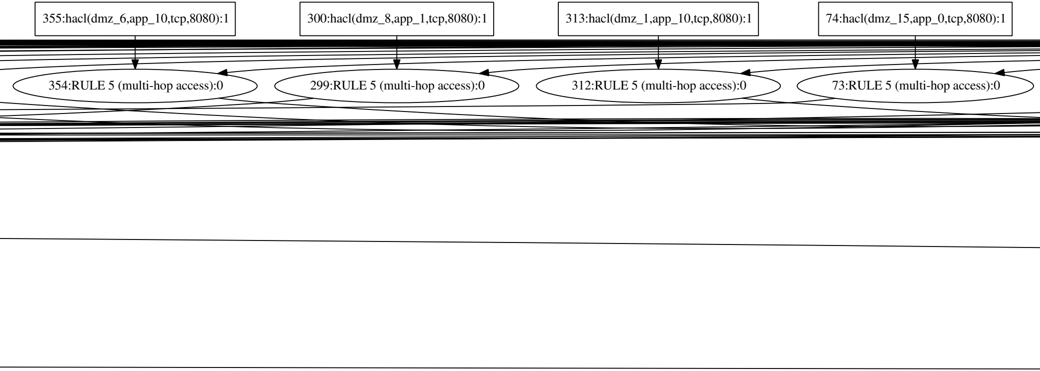

254:hacl(dmz_1,app_1,tcp,8080):1
256:hacl(dmz_10,app_1,tcp,8080):1
47:hacl(dmz_12,app_0,tcp,8080):1
282:hacl(dmz_22,app_1,tcp,8080):1
253:RULE 5 (multi-hop access):0
255:RULE 5 (multi-hop access):0
46:RULE 5 (multi-hop access):0
281:RULE 5 (multi-hop access):0
303:networkServiceInfo(app_1,tomcat,tcp,8080,root):1
304:vulExists(app_1,vul_app,t
6:execCode(app_0,root):0
4:RULE 5 (multi-hop access):0

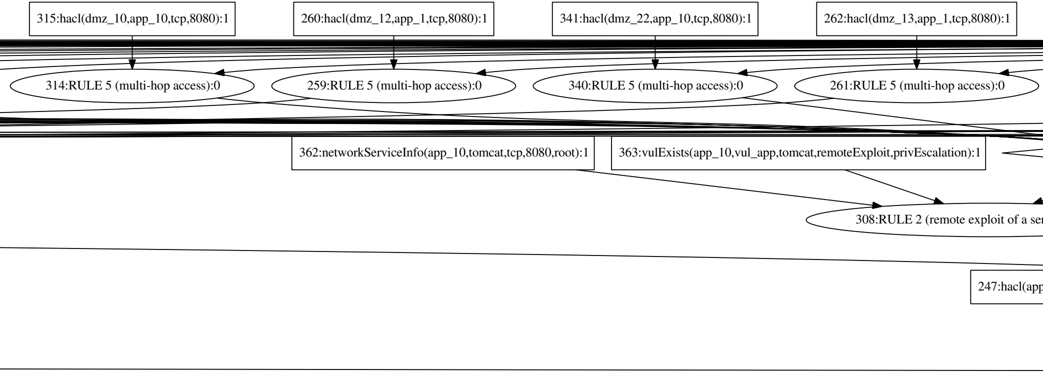

56:hacl(dmz_13,app_0,tcp,8080):1
258:hacl(dmz_11,app_1,tcp,8080):1
327:hacl(dmz_16,app_10,tcp,8080):1
286:hacl(dmz_24,app_1,tcp,8080):1
55:RULE 5 (multi-hop access):0
257:RULE 5 (multi-hop access):0
326:RULE 5 (multi-hop access):0
285:RULE 5 (multi-hop access):0
omcat,remoteExploit,privEscalation):1
250:netAccess(app_1,tcp,8080):0
249:RULE 2 (remote exploit of a server program):0

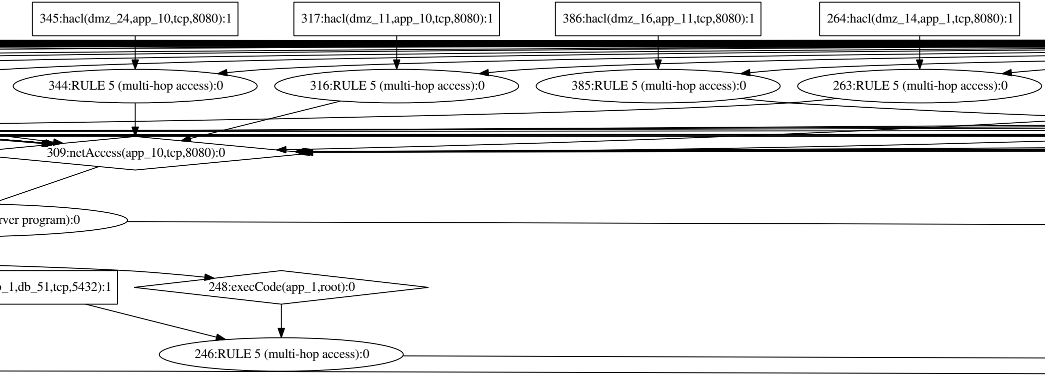

329:hacl(dmz_17,app_10,tcp,8080):1
331:hacl(dmz_18,app_10,tcp,8080):1
284:hacl(dmz_23,app_1,tcp,8080):1
353:hacl(dmz_5,app_10,tcp,8080):1
328:RULE 5 (multi-hop access):0
330:RULE 5 (multi-hop access):0
283:RULE 5 (multi-hop access):0
352:RULE 5 (multi-hop access):0

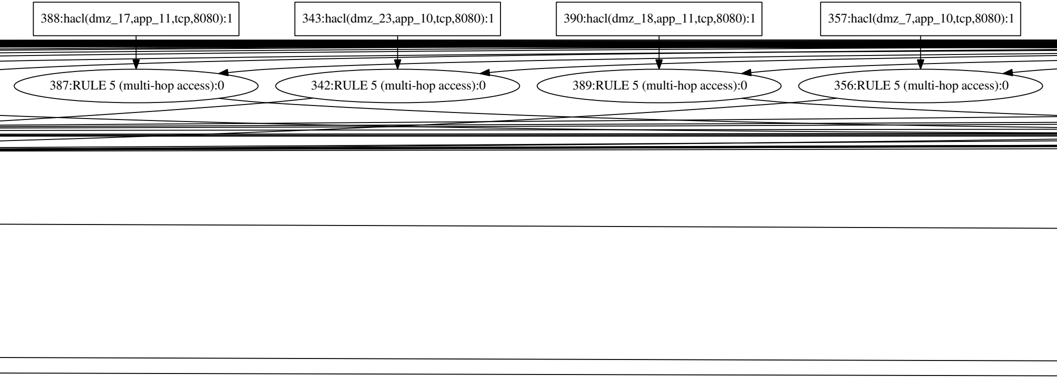

333:hacl(dmz_19,app_10,tcp,8080):1
335:hacl(dmz_2,app_10,tcp,8080):1
173:hacl(dmz_25,app_0,tcp,8080):1
337:hacl(dmz_20,app_10,tcp,8080):1
332:RULE 5 (multi-hop access):0
334:RULE 5 (multi-hop access):0
172:RULE 5 (multi-hop access):0
336:RULE 5 (multi-hop access):0

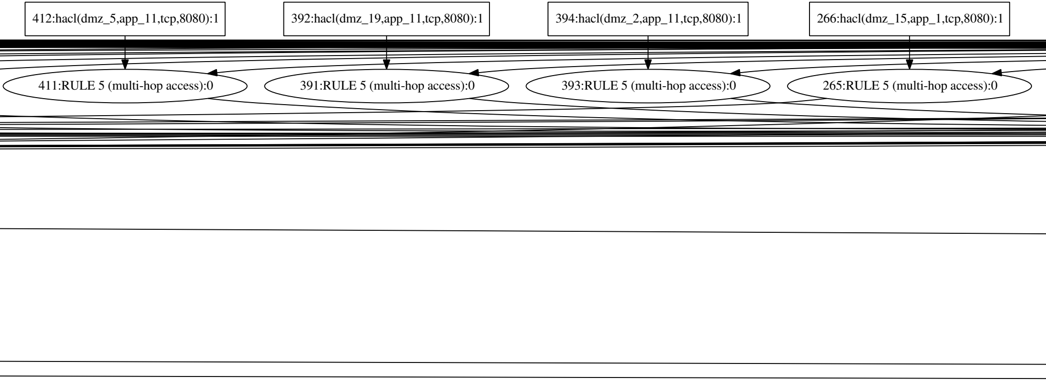

65:hacl(dmz_14,app_0,tcp,8080):1
298:hacl(dmz_7,app_1,tcp,8080):1
311:hacl(dmz_0,app_10,tcp,8080):1
339:hacl(dmz_21,app_10,tcp,8080):1
64:RULE 5 (multi-hop access):0
297:RULE 5 (multi-hop access):0
310:RULE 5 (multi-hop access):0
338:RULE 5 (multi-hop access):0

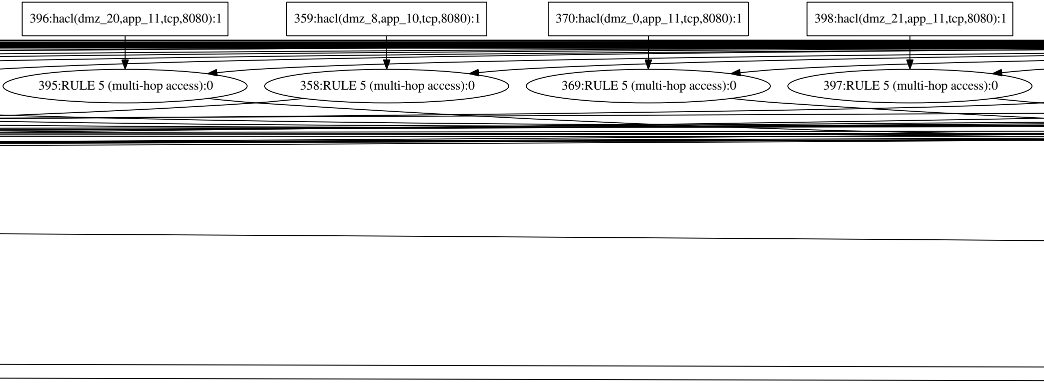

355:hacl(dmz_6,app_10,tcp,8080):1
300:hacl(dmz_8,app_1,tcp,8080):1
313:hacl(dmz_1,app_10,tcp,8080):1
74:hacl(dmz_15,app_0,tcp,8080):1
354:RULE 5 (multi-hop access):0
299:RULE 5 (multi-hop access):0
312:RULE 5 (multi-hop access):0
73:RULE 5 (multi-hop access):0

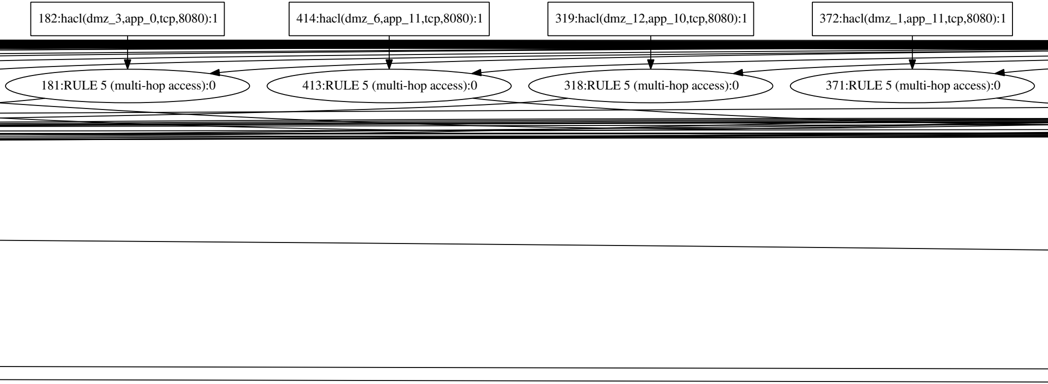

315:hacl(dmz_10,app_10,tcp,8080):1
260:hacl(dmz_12,app_1,tcp,8080):1
341:hacl(dmz_22,app_10,tcp,8080):1
262:hacl(dmz_13,app_1,tcp,8080):1
314:RULE 5 (multi-hop access):0
259:RULE 5 (multi-hop access):0
340:RULE 5 (multi-hop access):0
261:RULE 5 (multi-hop access):0
362:networkServiceInfo(app_10,tomcat,tcp,8080,root):1
363:vulExists(app_10,vul_app,tomcat,remoteExploit,privEscalation):1
308:RULE 2 (remote exploit of a ser
247:hacl(app

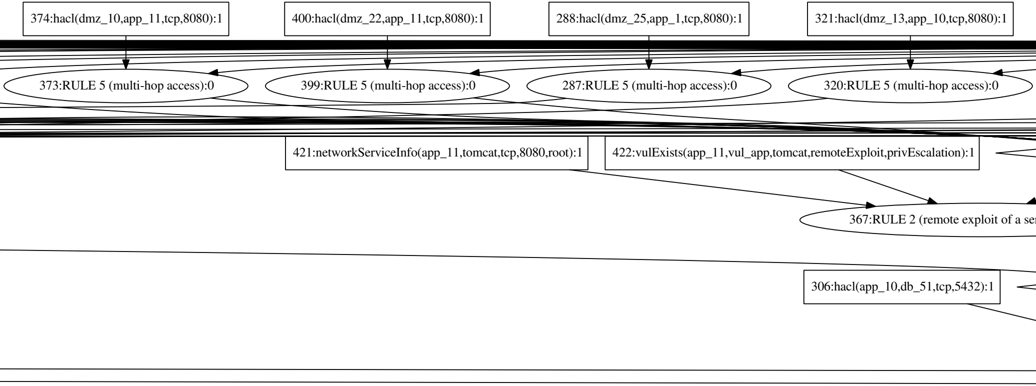

345:hacl(dmz_24,app_10,tcp,8080):1
317:hacl(dmz_11,app_10,tcp,8080):1
386:hacl(dmz_16,app_11,tcp,8080):1
264:hacl(dmz_14,app_1,tcp,8080):1
344:RULE 5 (multi-hop access):0
316:RULE 5 (multi-hop access):0
385:RULE 5 (multi-hop access):0
263:RULE 5 (multi-hop access):0
309:netAccess(app_10,tcp,8080):0
rver program):0
_1,db_51,tcp,5432):1
248:execCode(app_1,root):0
246:RULE 5 (multi-hop access):0

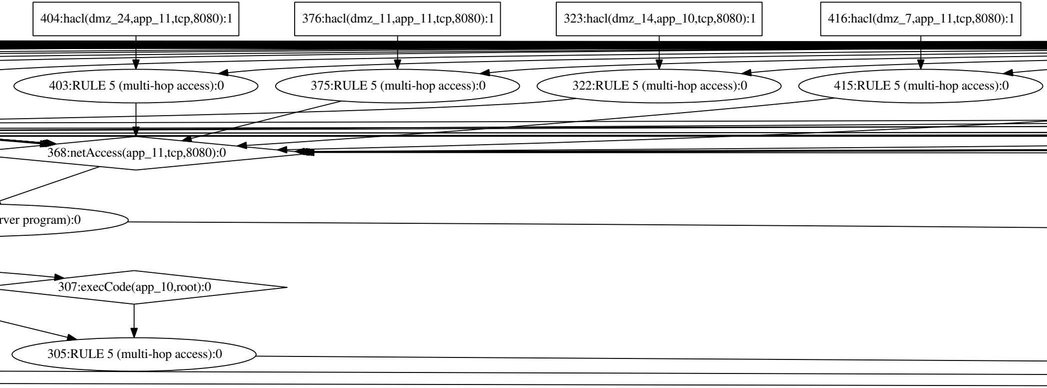

388:hacl(dmz_17,app_11,tcp,8080):1
343:hacl(dmz_23,app_10,tcp,8080):1
390:hacl(dmz_18,app_11,tcp,8080):1
357:hacl(dmz_7,app_10,tcp,8080):1
387:RULE 5 (multi-hop access):0
342:RULE 5 (multi-hop access):0
389:RULE 5 (multi-hop access):0
356:RULE 5 (multi-hop access):0

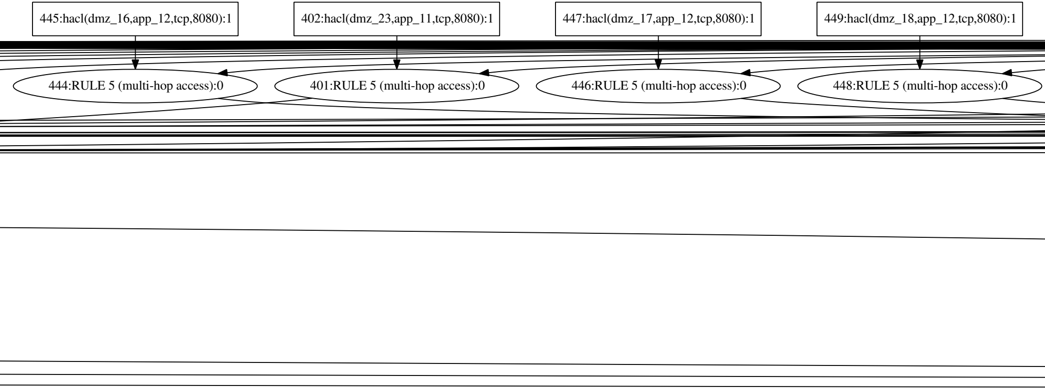

412:hacl(dmz_5,app_11,tcp,8080):1
392:hacl(dmz_19,app_11,tcp,8080):1
394:hacl(dmz_2,app_11,tcp,8080):1
266:hacl(dmz_15,app_1,tcp,8080):1
411:RULE 5 (multi-hop access):0
391:RULE 5 (multi-hop access):0
393:RULE 5 (multi-hop access):0
265:RULE 5 (multi-hop access):0

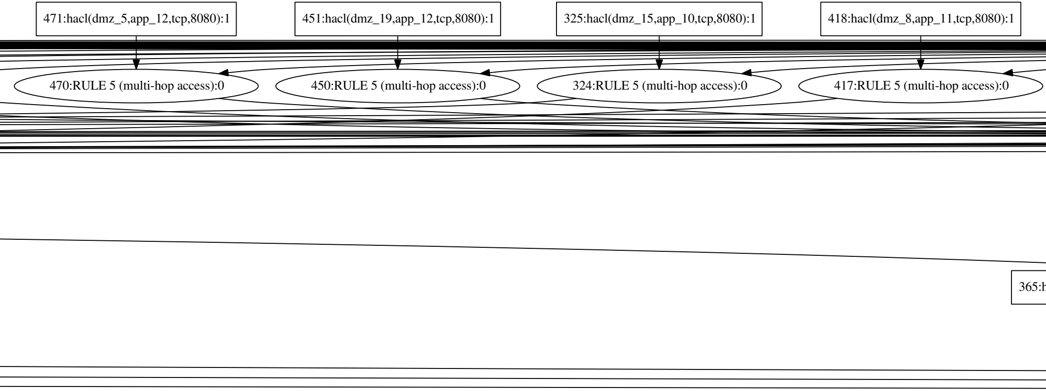

396:hacl(dmz_20,app_11,tcp,8080):1
359:hacl(dmz_8,app_10,tcp,8080):1
370:hacl(dmz_0,app_11,tcp,8080):1
398:hacl(dmz_21,app_11,tcp,8080):1
395:RULE 5 (multi-hop access):0
358:RULE 5 (multi-hop access):0
369:RULE 5 (multi-hop access):0
397:RULE 5 (multi-hop access):0

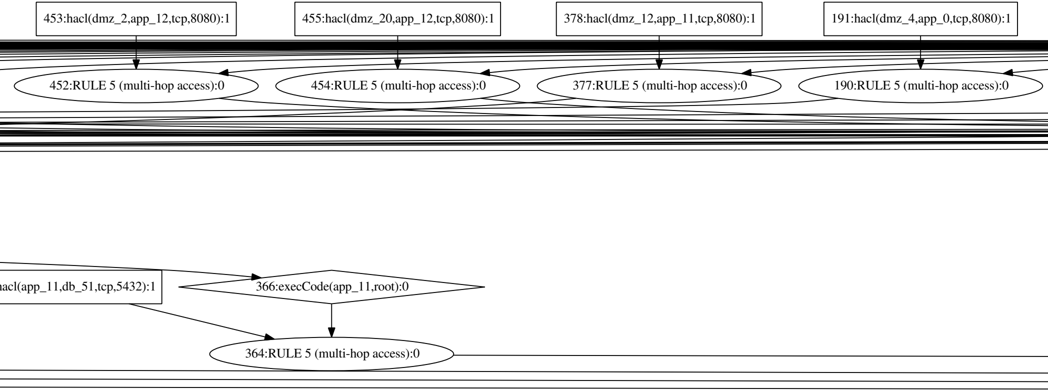

182:hacl(dmz_3,app_0,tcp,8080):1
414:hacl(dmz_6,app_11,tcp,8080):1
319:hacl(dmz_12,app_10,tcp,8080):1
372:hacl(dmz_1,app_11,tcp,8080):1
181:RULE 5 (multi-hop access):0
413:RULE 5 (multi-hop access):0
318:RULE 5 (multi-hop access):0
371:RULE 5 (multi-hop access):0

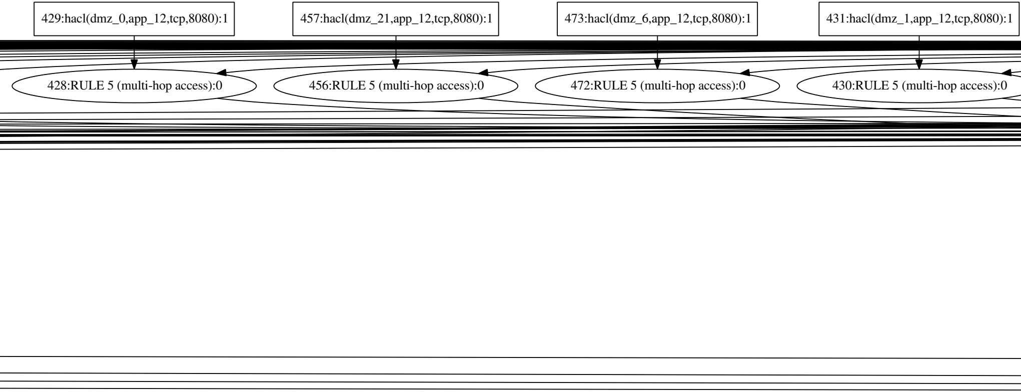

374:hacl(dmz_10,app_11,tcp,8080):1
400:hacl(dmz_22,app_11,tcp,8080):1
288:hacl(dmz_25,app_1,tcp,8080):1
321:hacl(dmz_13,app_10,tcp,8080):1
373:RULE 5 (multi-hop access):0
399:RULE 5 (multi-hop access):0
287:RULE 5 (multi-hop access):0
320:RULE 5 (multi-hop access):0
421:networkServiceInfo(app_11,tomcat,tcp,8080,root):1
422:vulExists(app_11,vul_app,tomcat,remoteExploit,privEscalation):1
367:RULE 2 (remote exploit of a ser
306:hacl(app_10,db_51,tcp,5432):1

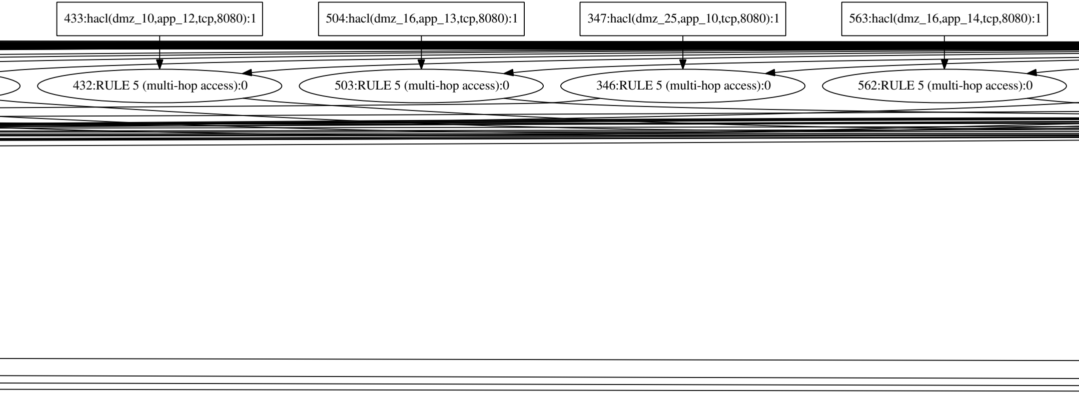

404:hacl(dmz_24,app_11,tcp,8080):1
376:hacl(dmz_11,app_11,tcp,8080):1
323:hacl(dmz_14,app_10,tcp,8080):1
416:hacl(dmz_7,app_11,tcp,8080):1
403:RULE 5 (multi-hop access):0
375:RULE 5 (multi-hop access):0
322:RULE 5 (multi-hop access):0
415:RULE 5 (multi-hop access):0
368:netAccess(app_11,tcp,8080):0
rver program):0
307:execCode(app_10,root):0
305:RULE 5 (multi-hop access):0

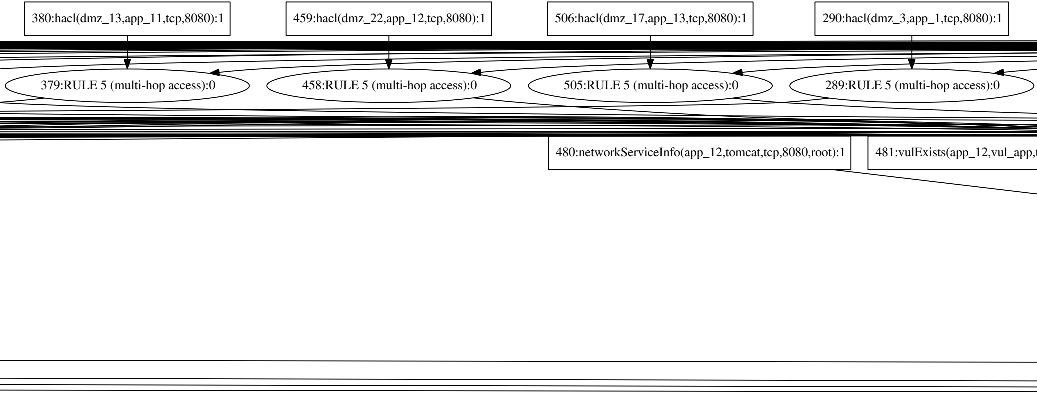

445:hacl(dmz_16,app_12,tcp,8080):1
402:hacl(dmz_23,app_11,tcp,8080):1
447:hacl(dmz_17,app_12,tcp,8080):1
449:hacl(dmz_18,app_12,tcp,8080):1
444:RULE 5 (multi-hop access):0
401:RULE 5 (multi-hop access):0
446:RULE 5 (multi-hop access):0
448:RULE 5 (multi-hop access):0

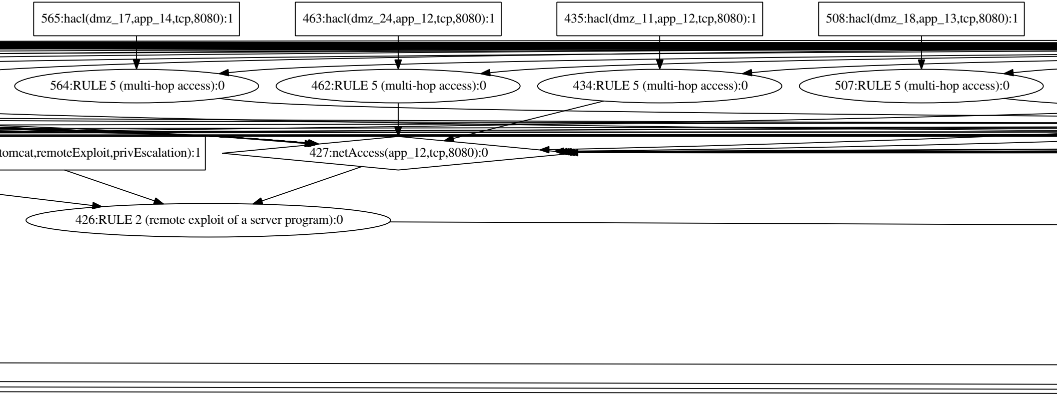

471:hacl(dmz_5,app_12,tcp,8080):1
451:hacl(dmz_19,app_12,tcp,8080):1
325:hacl(dmz_15,app_10,tcp,8080):1
418:hacl(dmz_8,app_11,tcp,8080):1
470:RULE 5 (multi-hop access):0
450:RULE 5 (multi-hop access):0
324:RULE 5 (multi-hop access):0
417:RULE 5 (multi-hop access):0
365:h

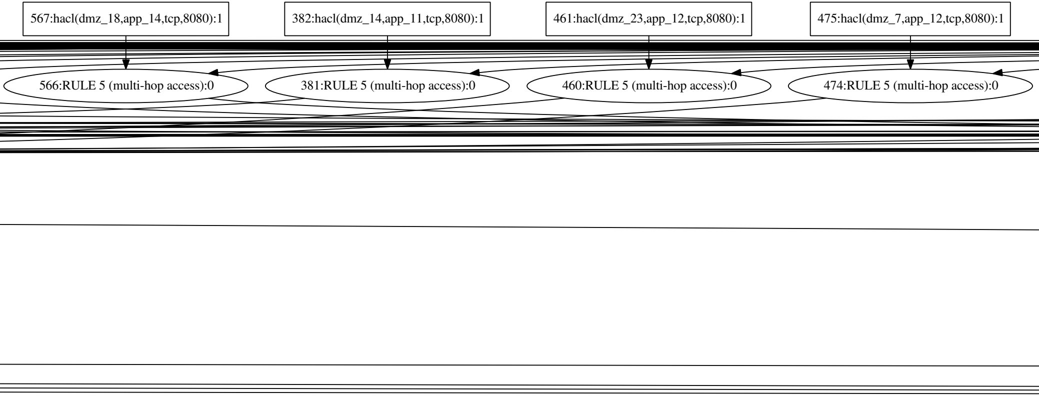

453:hacl(dmz_2,app_12,tcp,8080):1
455:hacl(dmz_20,app_12,tcp,8080):1
378:hacl(dmz_12,app_11,tcp,8080):1
191:hacl(dmz_4,app_0,tcp,8080):1
452:RULE 5 (multi-hop access):0
454:RULE 5 (multi-hop access):0
377:RULE 5 (multi-hop access):0
190:RULE 5 (multi-hop access):0
hacl(app_11,db_51,tcp,5432):1
366:execCode(app_11,root):0
364:RULE 5 (multi-hop access):0

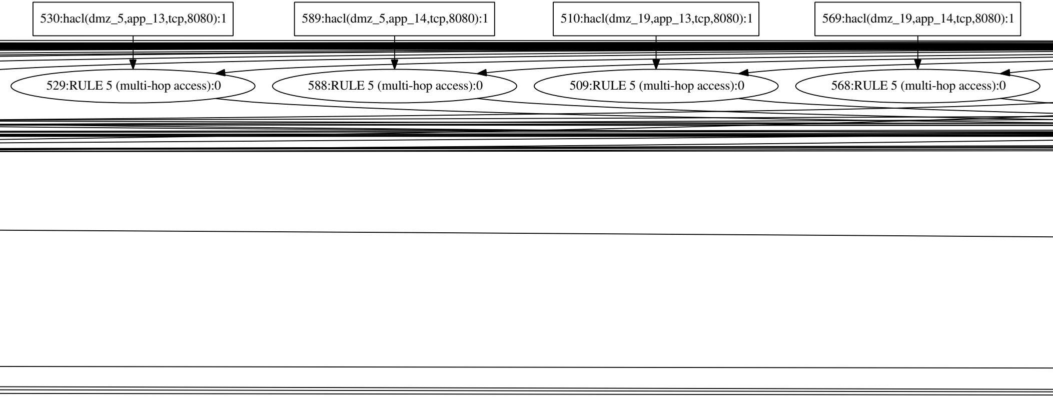

429:hacl(dmz_0,app_12,tcp,8080):1
457:hacl(dmz_21,app_12,tcp,8080):1
473:hacl(dmz_6,app_12,tcp,8080):1
431:hacl(dmz_1,app_12,tcp,8080):1
428:RULE 5 (multi-hop access):0
456:RULE 5 (multi-hop access):0
472:RULE 5 (multi-hop access):0
430:RULE 5 (multi-hop access):0

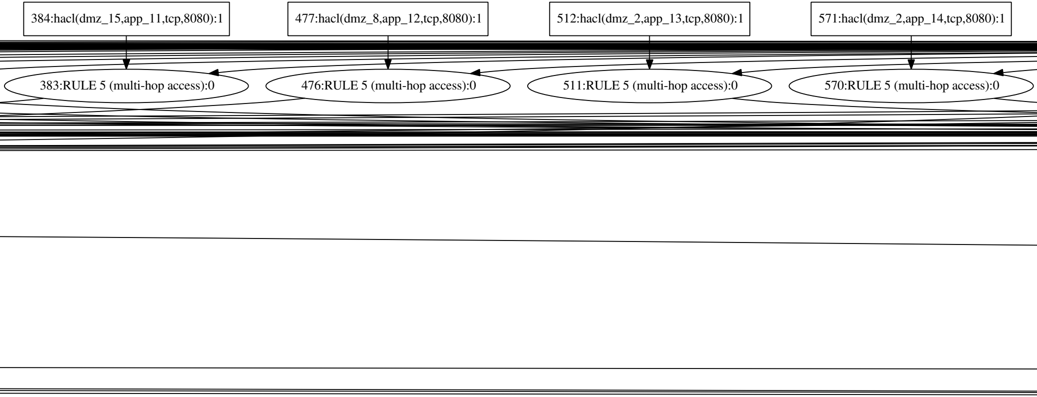

433:hacl(dmz_10,app_12,tcp,8080):1
504:hacl(dmz_16,app_13,tcp,8080):1
347:hacl(dmz_25,app_10,tcp,8080):1
563:hacl(dmz_16,app_14,tcp,8080):1
432:RULE 5 (multi-hop access):0
503:RULE 5 (multi-hop access):0
346:RULE 5 (multi-hop access):0
562:RULE 5 (multi-hop access):0

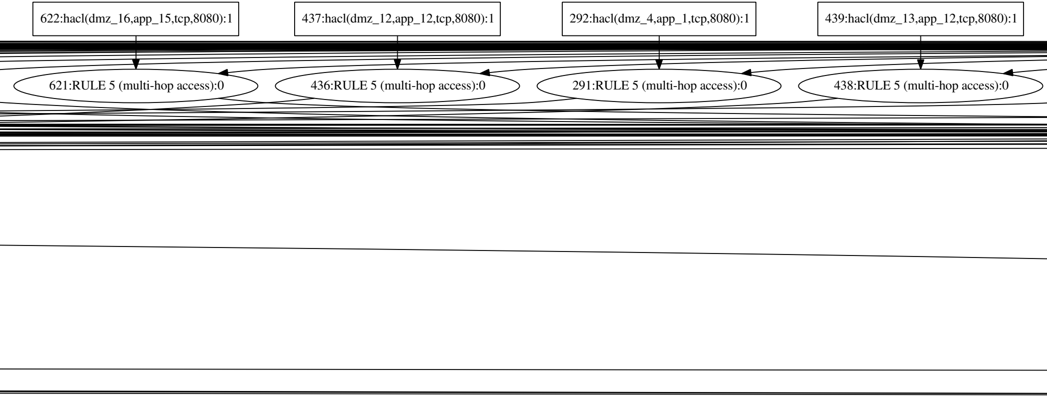

380:hacl(dmz_13,app_11,tcp,8080):1
459:hacl(dmz_22,app_12,tcp,8080):1
506:hacl(dmz_17,app_13,tcp,8080):1
290:hacl(dmz_3,app_1,tcp,8080):1
379:RULE 5 (multi-hop access):0
458:RULE 5 (multi-hop access):0
505:RULE 5 (multi-hop access):0
289:RULE 5 (multi-hop access):0
480:networkServiceInfo(app_12,tomcat,tcp,8080,root):1
481:vulExists(app_12,vul_app,t

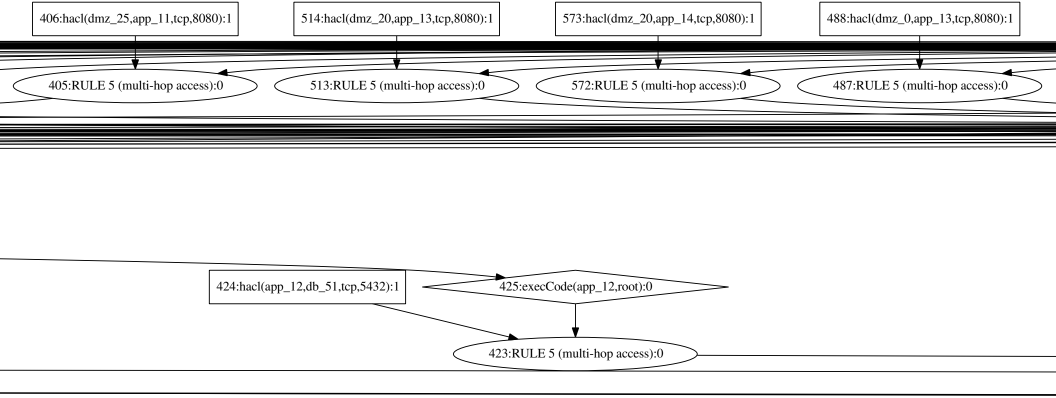

565:hacl(dmz_17,app_14,tcp,8080):1
463:hacl(dmz_24,app_12,tcp,8080):1
435:hacl(dmz_11,app_12,tcp,8080):1
508:hacl(dmz_18,app_13,tcp,8080):1
564:RULE 5 (multi-hop access):0
462:RULE 5 (multi-hop access):0
434:RULE 5 (multi-hop access):0
507:RULE 5 (multi-hop access):0
tomcat,remoteExploit,privEscalation):1
427:netAccess(app_12,tcp,8080):0
426:RULE 2 (remote exploit of a server program):0

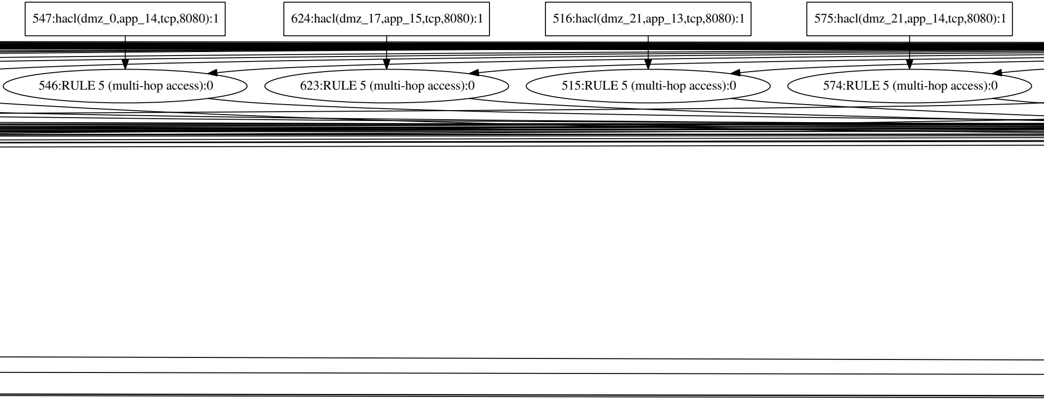

567:hacl(dmz_18,app_14,tcp,8080):1
382:hacl(dmz_14,app_11,tcp,8080):1
461:hacl(dmz_23,app_12,tcp,8080):1
475:hacl(dmz_7,app_12,tcp,8080):1
566:RULE 5 (multi-hop access):0
381:RULE 5 (multi-hop access):0
460:RULE 5 (multi-hop access):0
474:RULE 5 (multi-hop access):0

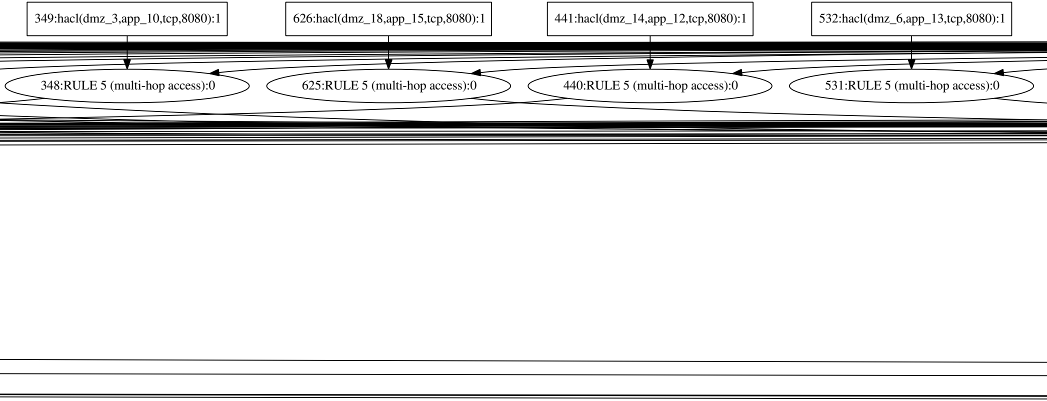

530:hacl(dmz_5,app_13,tcp,8080):1
589:hacl(dmz_5,app_14,tcp,8080):1
510:hacl(dmz_19,app_13,tcp,8080):1
569:hacl(dmz_19,app_14,tcp,8080):1
529:RULE 5 (multi-hop access):0
588:RULE 5 (multi-hop access):0
509:RULE 5 (multi-hop access):0
568:RULE 5 (multi-hop access):0

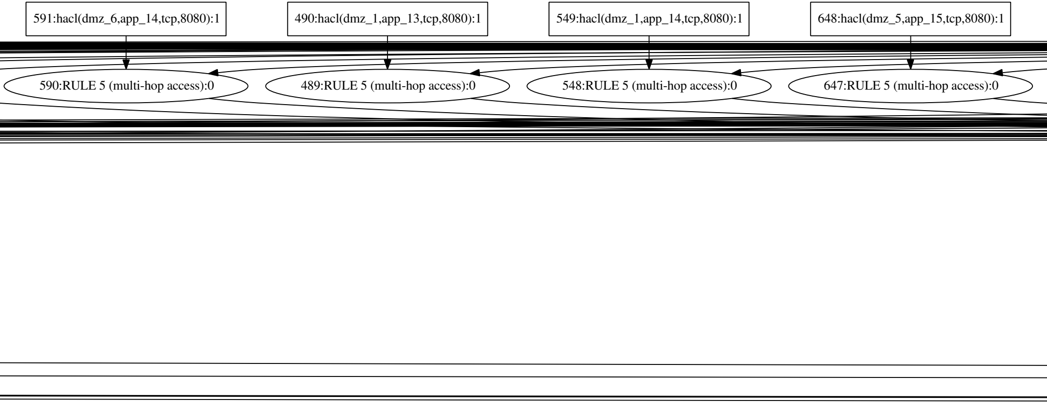

384:hacl(dmz_15,app_11,tcp,8080):1
477:hacl(dmz_8,app_12,tcp,8080):1
512:hacl(dmz_2,app_13,tcp,8080):1
571:hacl(dmz_2,app_14,tcp,8080):1
383:RULE 5 (multi-hop access):0
476:RULE 5 (multi-hop access):0
511:RULE 5 (multi-hop access):0
570:RULE 5 (multi-hop access):0

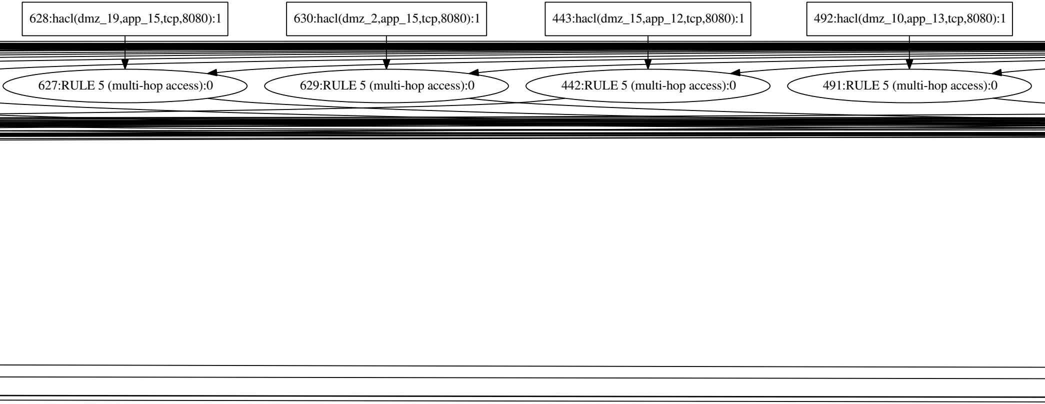

622:hacl(dmz_16,app_15,tcp,8080):1
437:hacl(dmz_12,app_12,tcp,8080):1
292:hacl(dmz_4,app_1,tcp,8080):1
439:hacl(dmz_13,app_12,tcp,8080):1
621:RULE 5 (multi-hop access):0
436:RULE 5 (multi-hop access):0
291:RULE 5 (multi-hop access):0
438:RULE 5 (multi-hop access):0

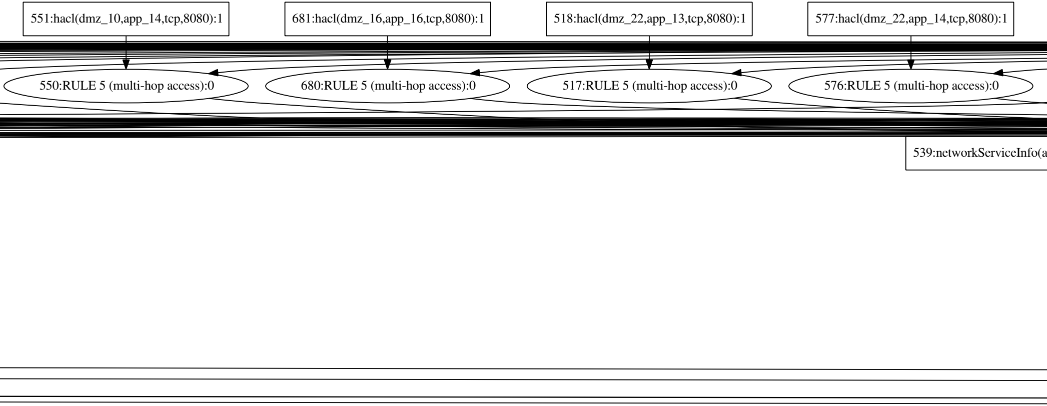

406:hacl(dmz_25,app_11,tcp,8080):1
514:hacl(dmz_20,app_13,tcp,8080):1
573:hacl(dmz_20,app_14,tcp,8080):1
488:hacl(dmz_0,app_13,tcp,8080):1
405:RULE 5 (multi-hop access):0
513:RULE 5 (multi-hop access):0
572:RULE 5 (multi-hop access):0
487:RULE 5 (multi-hop access):0
424:hacl(app_12,db_51,tcp,5432):1
425:execCode(app_12,root):0
423:RULE 5 (multi-hop access):0

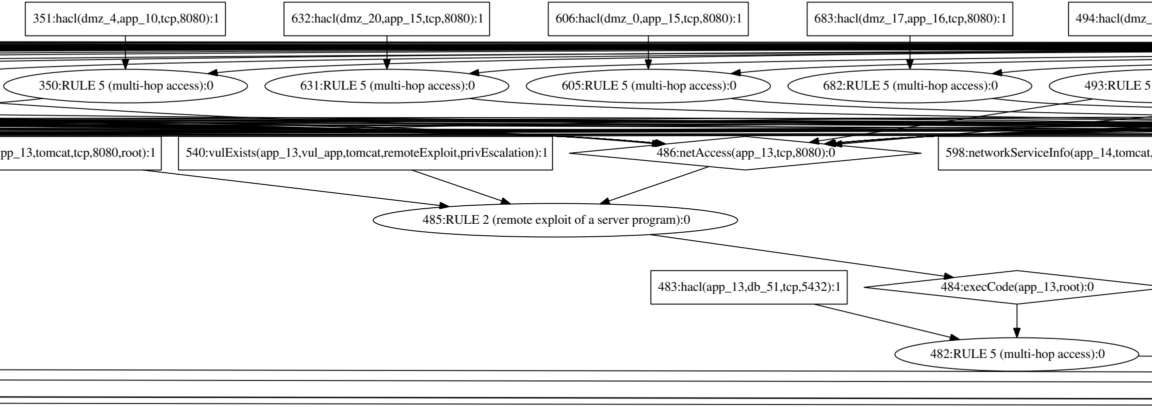

547:hacl(dmz_0,app_14,tcp,8080):1
624:hacl(dmz_17,app_15,tcp,8080):1
516:hacl(dmz_21,app_13,tcp,8080):1
575:hacl(dmz_21,app_14,tcp,8080):1
546:RULE 5 (multi-hop access):0
623:RULE 5 (multi-hop access):0
515:RULE 5 (multi-hop access):0
574:RULE 5 (multi-hop access):0

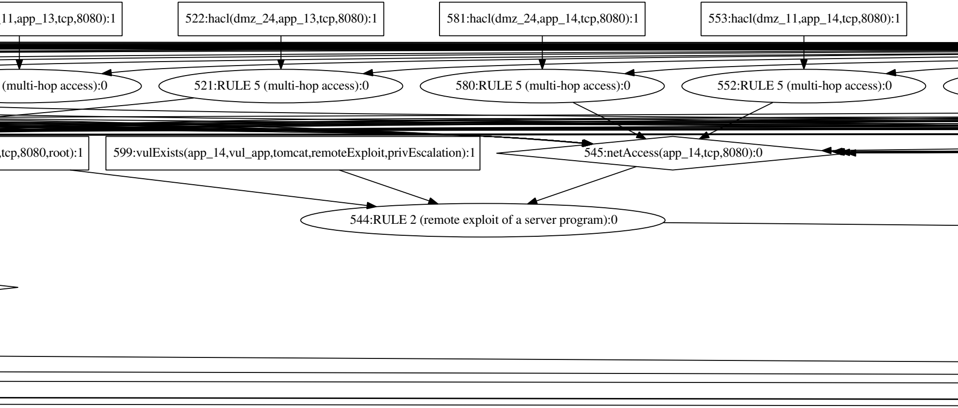

349:hacl(dmz_3,app_10,tcp,8080):1
626:hacl(dmz_18,app_15,tcp,8080):1
441:hacl(dmz_14,app_12,tcp,8080):1
532:hacl(dmz_6,app_13,tcp,8080):1
348:RULE 5 (multi-hop access):0
625:RULE 5 (multi-hop access):0
440:RULE 5 (multi-hop access):0
531:RULE 5 (multi-hop access):0

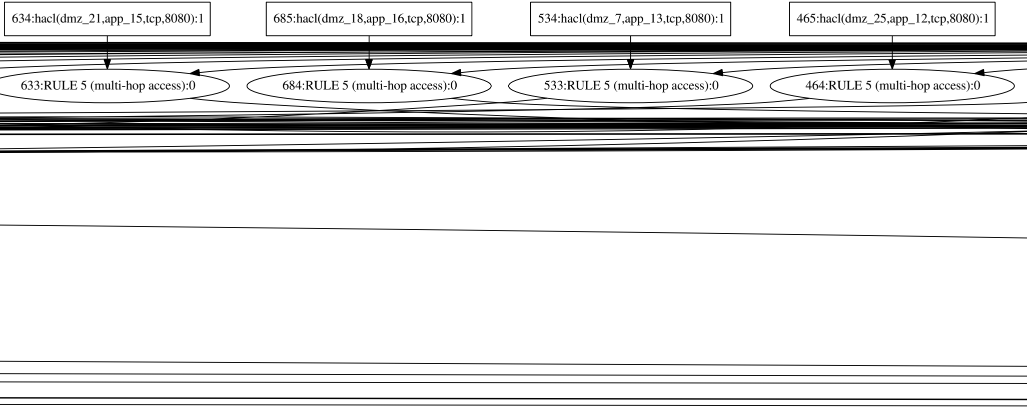

591:hacl(dmz_6,app_14,tcp,8080):1
490:hacl(dmz_1,app_13,tcp,8080):1
549:hacl(dmz_1,app_14,tcp,8080):1
648:hacl(dmz_5,app_15,tcp,8080):1
590:RULE 5 (multi-hop access):0
489:RULE 5 (multi-hop access):0
548:RULE 5 (multi-hop access):0
647:RULE 5 (multi-hop access):0

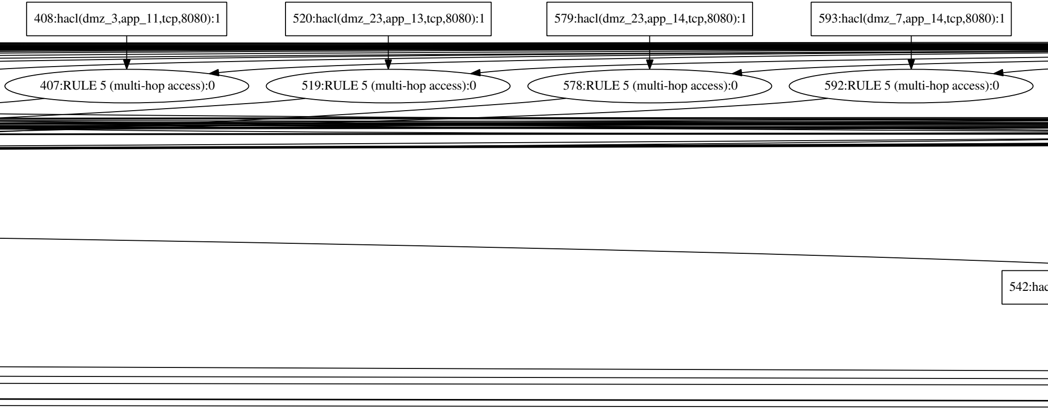

628:hacl(dmz_19,app_15,tcp,8080):1
630:hacl(dmz_2,app_15,tcp,8080):1
443:hacl(dmz_15,app_12,tcp,8080):1
492:hacl(dmz_10,app_13,tcp,8080):1
627:RULE 5 (multi-hop access):0
629:RULE 5 (multi-hop access):0
442:RULE 5 (multi-hop access):0
491:RULE 5 (multi-hop access):0

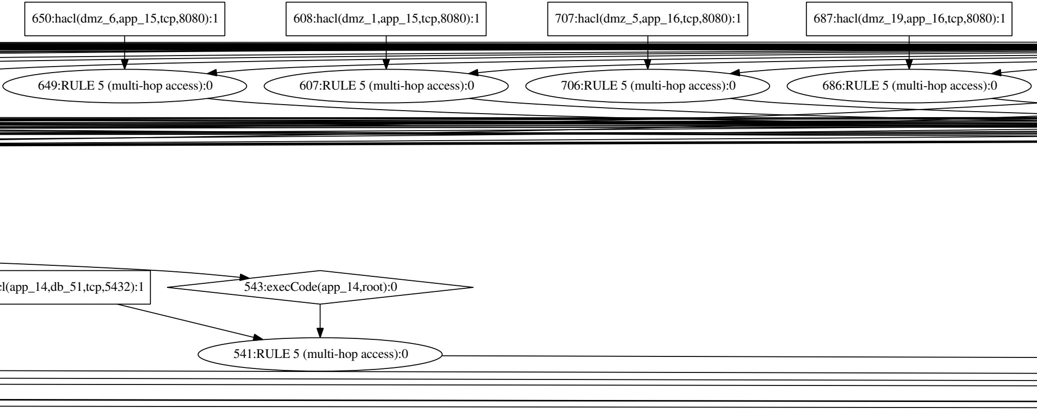

551:hacl(dmz_10,app_14,tcp,8080):1
681:hacl(dmz_16,app_16,tcp,8080):1
518:hacl(dmz_22,app_13,tcp,8080):1
577:hacl(dmz_22,app_14,tcp,8080):1
550:RULE 5 (multi-hop access):0
680:RULE 5 (multi-hop access):0
517:RULE 5 (multi-hop access):0
576:RULE 5 (multi-hop access):0
539:networkServiceInfo(a

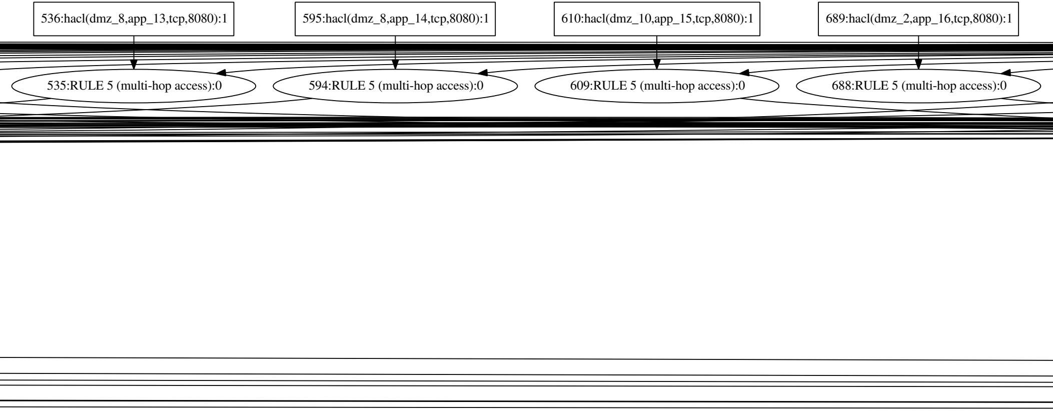

351:hacl(dmz_4,app_10,tcp,8080):1
632:hacl(dmz_20,app_15,tcp,8080):1
606:hacl(dmz_0,app_15,tcp,8080):1
683:hacl(dmz_17,app_16,tcp,8080):1
494:hacl(dmz_
350:RULE 5 (multi-hop access):0
631:RULE 5 (multi-hop access):0
605:RULE 5 (multi-hop access):0
682:RULE 5 (multi-hop access):0
493:RULE 5
pp_13,tomcat,tcp,8080,root):1
540:vulExists(app_13,vul_app,tomcat,remoteExploit,privEscalation):1
486:netAccess(app_13,tcp,8080):0
598:networkServiceInfo(app_14,tomcat,
485:RULE 2 (remote exploit of a server program):0
483:hacl(app_13,db_51,tcp,5432):1
484:execCode(app_13,root):0
482:RULE 5 (multi-hop access):0

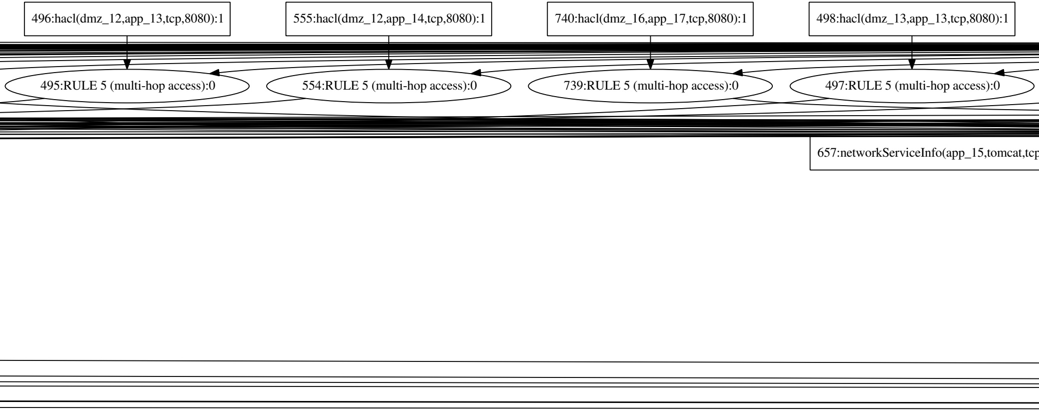

11,app_13,tcp,8080):1
522:hacl(dmz_24,app_13,tcp,8080):1
581:hacl(dmz_24,app_14,tcp,8080):1
553:hacl(dmz_11,app_14,tcp,8080):1
(multi-hop access):0
521:RULE 5 (multi-hop access):0
580:RULE 5 (multi-hop access):0
552:RULE 5 (multi-hop access):0
,tcp,8080,root):1
599:vulExists(app_14,vul_app,tomcat,remoteExploit,privEscalation):1
545:netAccess(app_14,tcp,8080):0
544:RULE 2 (remote exploit of a server program):0

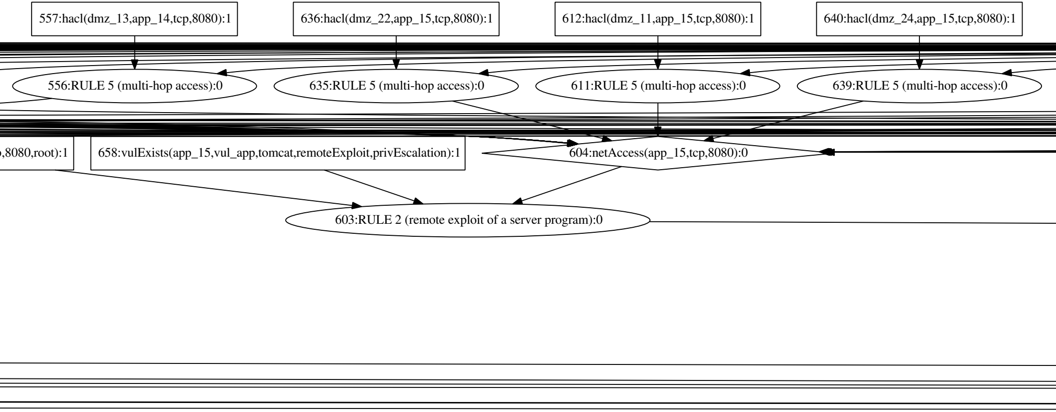

634:hacl(dmz_21,app_15,tcp,8080):1
685:hacl(dmz_18,app_16,tcp,8080):1
534:hacl(dmz_7,app_13,tcp,8080):1
465:hacl(dmz_25,app_12,tcp,8080):1
633:RULE 5 (multi-hop access):0
684:RULE 5 (multi-hop access):0
533:RULE 5 (multi-hop access):0
464:RULE 5 (multi-hop access):0

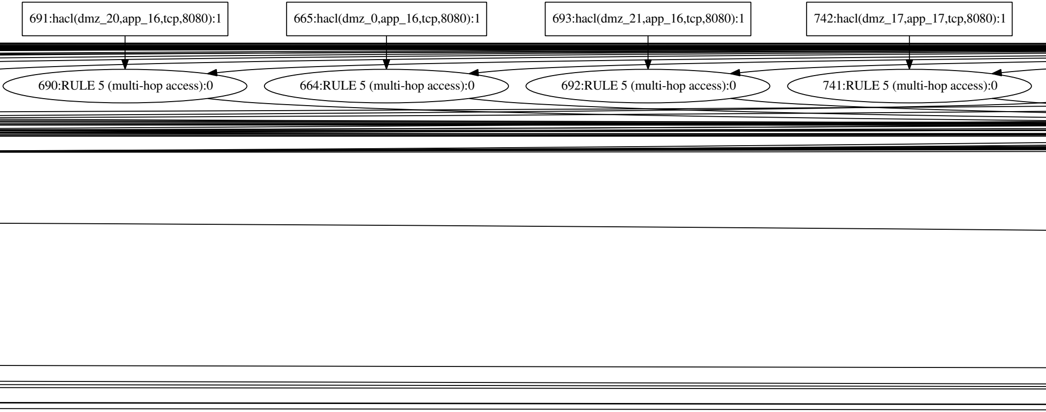

408:hacl(dmz_3,app_11,tcp,8080):1
520:hacl(dmz_23,app_13,tcp,8080):1
579:hacl(dmz_23,app_14,tcp,8080):1
593:hacl(dmz_7,app_14,tcp,8080):1
407:RULE 5 (multi-hop access):0
519:RULE 5 (multi-hop access):0
578:RULE 5 (multi-hop access):0
592:RULE 5 (multi-hop access):0
542:hac

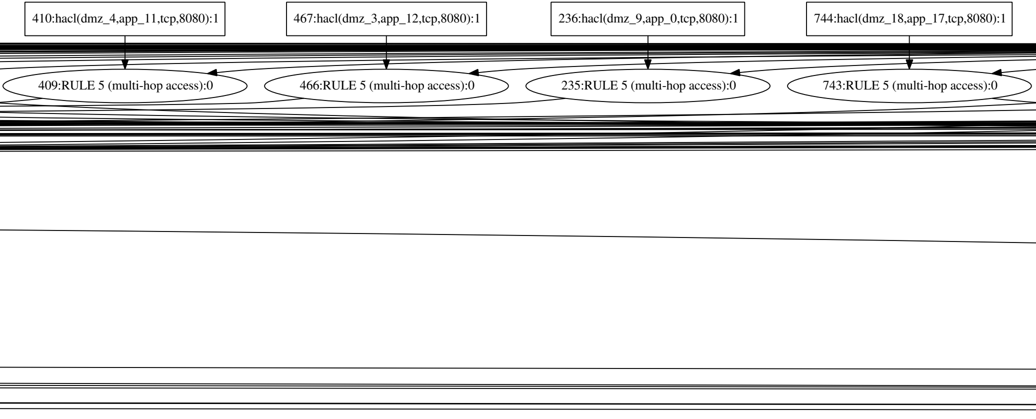

650:hacl(dmz_6,app_15,tcp,8080):1
608:hacl(dmz_1,app_15,tcp,8080):1
707:hacl(dmz_5,app_16,tcp,8080):1
687:hacl(dmz_19,app_16,tcp,8080):1
649:RULE 5 (multi-hop access):0
607:RULE 5 (multi-hop access):0
706:RULE 5 (multi-hop access):0
686:RULE 5 (multi-hop access):0
cl(app_14,db_51,tcp,5432):1
543:execCode(app_14,root):0
541:RULE 5 (multi-hop access):0

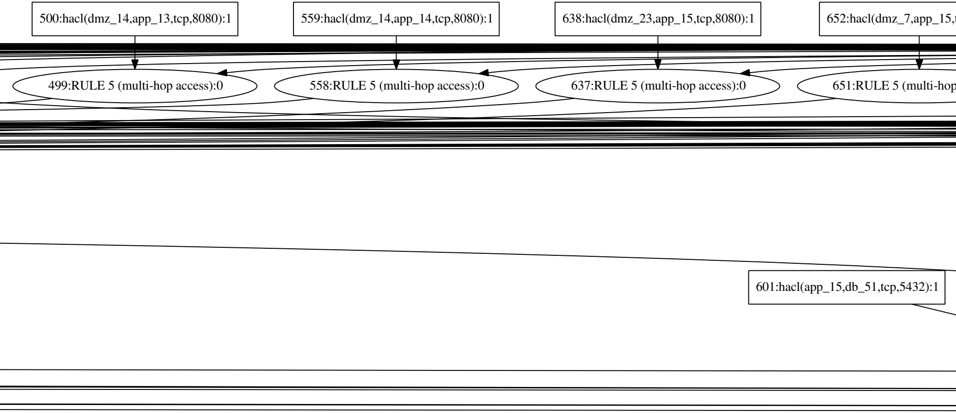

536:hacl(dmz_8,app_13,tcp,8080):1
595:hacl(dmz_8,app_14,tcp,8080):1
610:hacl(dmz_10,app_15,tcp,8080):1
689:hacl(dmz_2,app_16,tcp,8080):1
535:RULE 5 (multi-hop access):0
594:RULE 5 (multi-hop access):0
609:RULE 5 (multi-hop access):0
688:RULE 5 (multi-hop access):0

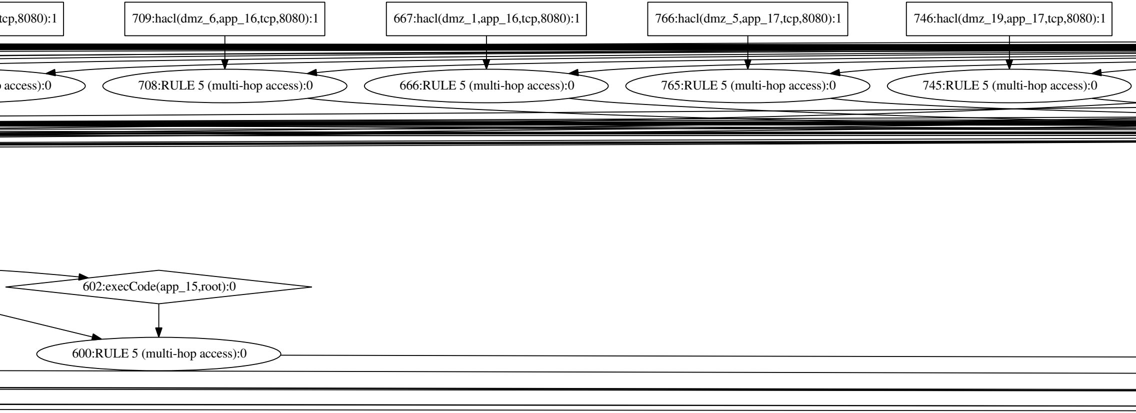

496:hacl(dmz_12,app_13,tcp,8080):1
555:hacl(dmz_12,app_14,tcp,8080):1
740:hacl(dmz_16,app_17,tcp,8080):1
498:hacl(dmz_13,app_13,tcp,8080):1
495:RULE 5 (multi-hop access):0
554:RULE 5 (multi-hop access):0
739:RULE 5 (multi-hop access):0
497:RULE 5 (multi-hop access):0
657:networkServiceInfo(app_15,tomcat,tcp

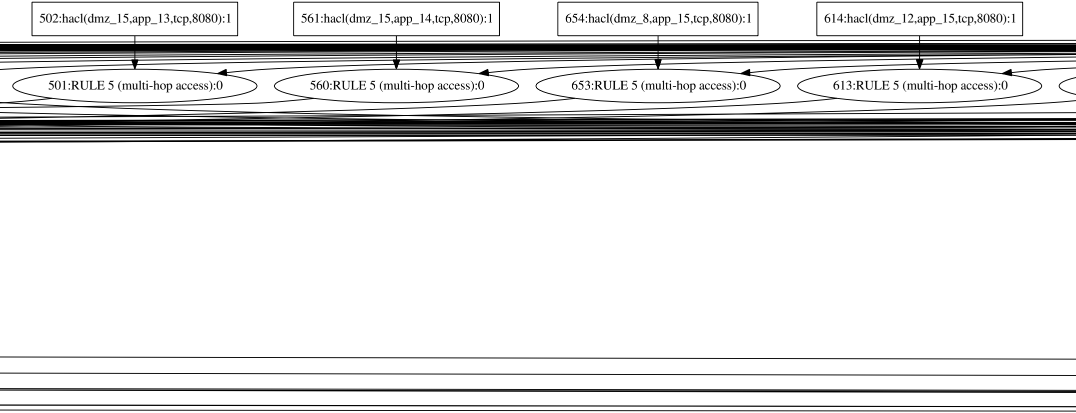

557:hacl(dmz_13,app_14,tcp,8080):1
636:hacl(dmz_22,app_15,tcp,8080):1
612:hacl(dmz_11,app_15,tcp,8080):1
640:hacl(dmz_24,app_15,tcp,8080):1
556:RULE 5 (multi-hop access):0
635:RULE 5 (multi-hop access):0
611:RULE 5 (multi-hop access):0
639:RULE 5 (multi-hop access):0
,8080,root):1
658:vulExists(app_15,vul_app,tomcat,remoteExploit,privEscalation):1
604:netAccess(app_15,tcp,8080):0
603:RULE 2 (remote exploit of a server program):0

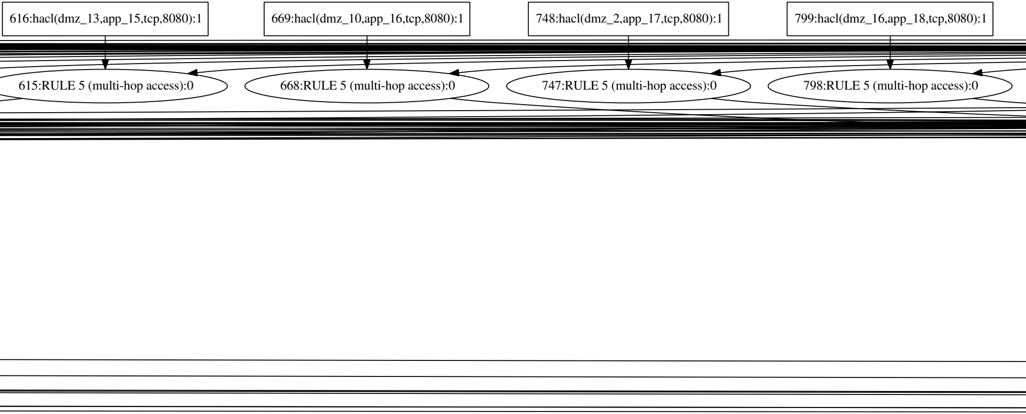

691:hacl(dmz_20,app_16,tcp,8080):1
665:hacl(dmz_0,app_16,tcp,8080):1
693:hacl(dmz_21,app_16,tcp,8080):1
742:hacl(dmz_17,app_17,tcp,8080):1
690:RULE 5 (multi-hop access):0
664:RULE 5 (multi-hop access):0
692:RULE 5 (multi-hop access):0
741:RULE 5 (multi-hop access):0

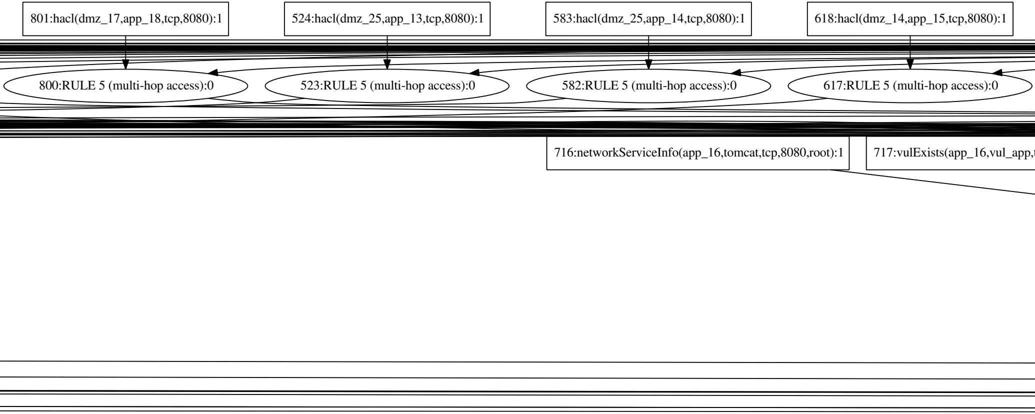

410:hacl(dmz_4,app_11,tcp,8080):1
467:hacl(dmz_3,app_12,tcp,8080):1
236:hacl(dmz_9,app_0,tcp,8080):1
744:hacl(dmz_18,app_17,tcp,8080):1
409:RULE 5 (multi-hop access):0
466:RULE 5 (multi-hop access):0
235:RULE 5 (multi-hop access):0
743:RULE 5 (multi-hop access):0

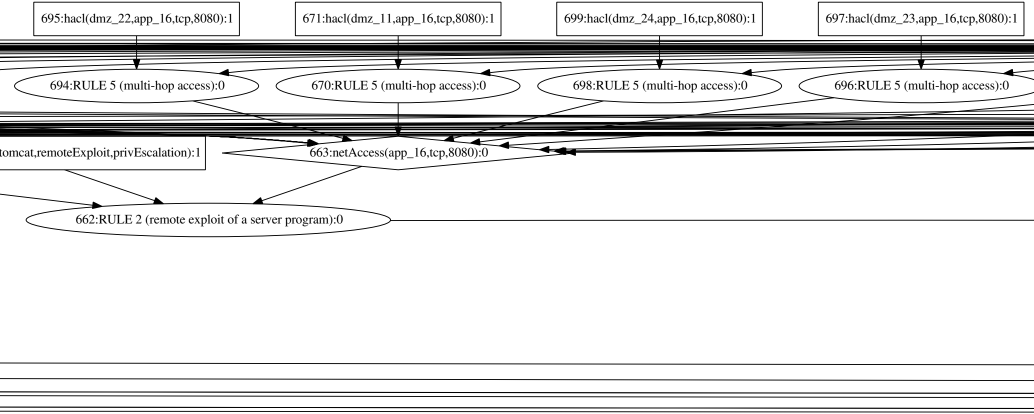

500:hacl(dmz_14,app_13,tcp,8080):1
559:hacl(dmz_14,app_14,tcp,8080):1
638:hacl(dmz_23,app_15,tcp,8080):1
652:hacl(dmz_7,app_15,t
499:RULE 5 (multi-hop access):0
558:RULE 5 (multi-hop access):0
637:RULE 5 (multi-hop access):0
651:RULE 5 (multi-hop
601:hacl(app_15,db_51,tcp,5432):1

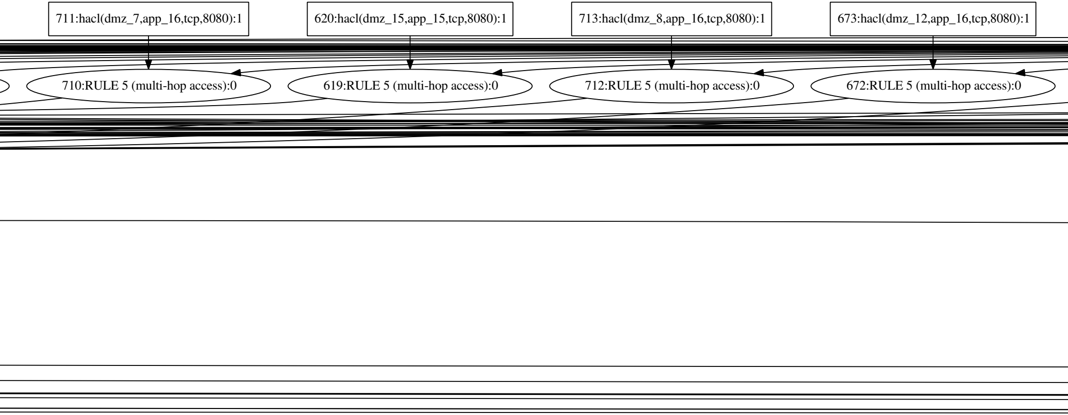

tcp,8080):1
709:hacl(dmz_6,app_16,tcp,8080):1
667:hacl(dmz_1,app_16,tcp,8080):1
766:hacl(dmz_5,app_17,tcp,8080):1
746:hacl(dmz_19,app_17,tcp,8080):1
access):0
708:RULE 5 (multi-hop access):0
666:RULE 5 (multi-hop access):0
765:RULE 5 (multi-hop access):0
745:RULE 5 (multi-hop access):0
602:execCode(app_15,root):0
600:RULE 5 (multi-hop access):0

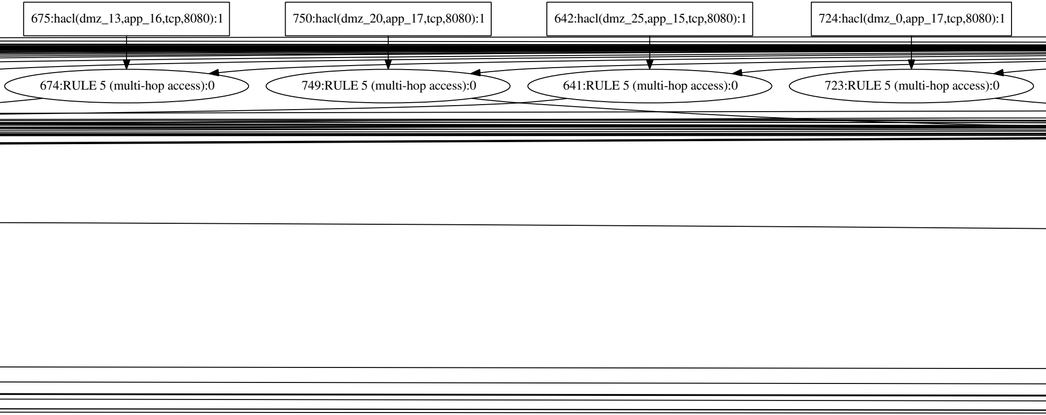

502:hacl(dmz_15,app_13,tcp,8080):1
561:hacl(dmz_15,app_14,tcp,8080):1
654:hacl(dmz_8,app_15,tcp,8080):1
614:hacl(dmz_12,app_15,tcp,8080):1
501:RULE 5 (multi-hop access):0
560:RULE 5 (multi-hop access):0
653:RULE 5 (multi-hop access):0
613:RULE 5 (multi-hop access):0

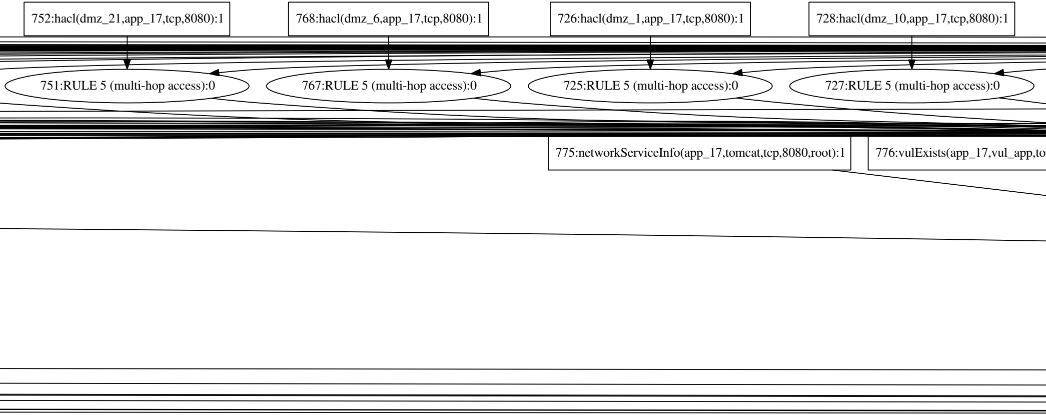

616:hacl(dmz_13,app_15,tcp,8080):1
669:hacl(dmz_10,app_16,tcp,8080):1
748:hacl(dmz_2,app_17,tcp,8080):1
799:hacl(dmz_16,app_18,tcp,8080):1
615:RULE 5 (multi-hop access):0
668:RULE 5 (multi-hop access):0
747:RULE 5 (multi-hop access):0
798:RULE 5 (multi-hop access):0

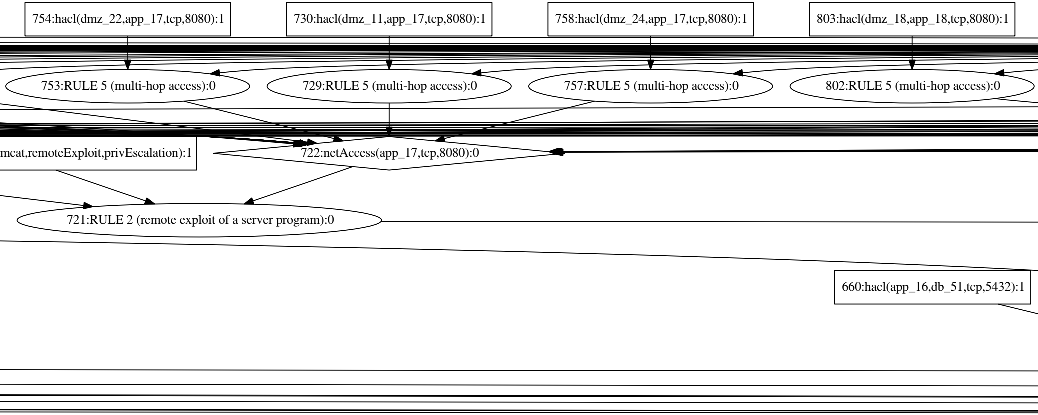

801:hacl(dmz_17,app_18,tcp,8080):1
524:hacl(dmz_25,app_13,tcp,8080):1
583:hacl(dmz_25,app_14,tcp,8080):1
618:hacl(dmz_14,app_15,tcp,8080):1
800:RULE 5 (multi-hop access):0
523:RULE 5 (multi-hop access):0
582:RULE 5 (multi-hop access):0
617:RULE 5 (multi-hop access):0
716:networkServiceInfo(app_16,tomcat,tcp,8080,root):1
717:vulExists(app_16,vul_app,

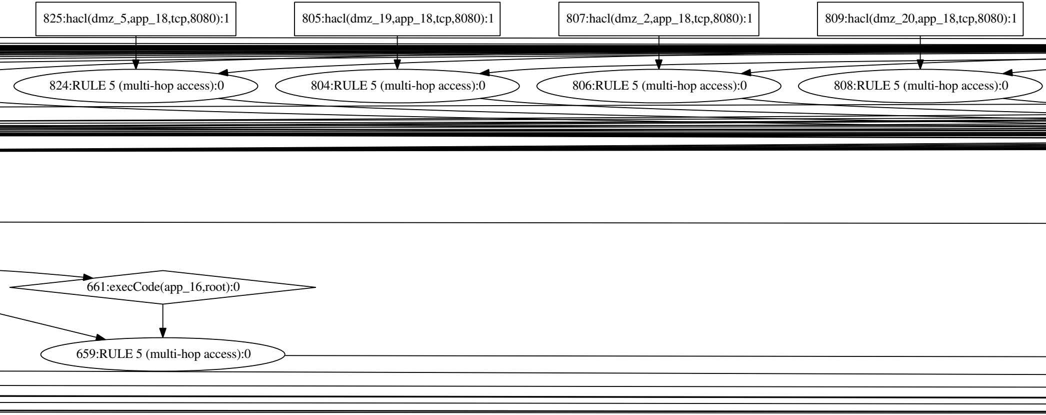

695:hacl(dmz_22,app_16,tcp,8080):1
671:hacl(dmz_11,app_16,tcp,8080):1
699:hacl(dmz_24,app_16,tcp,8080):1
697:hacl(dmz_23,app_16,tcp,8080):1
694:RULE 5 (multi-hop access):0
670:RULE 5 (multi-hop access):0
698:RULE 5 (multi-hop access):0
696:RULE 5 (multi-hop access):0
tomcat,remoteExploit,privEscalation):1
663:netAccess(app_16,tcp,8080):0
662:RULE 2 (remote exploit of a server program):0

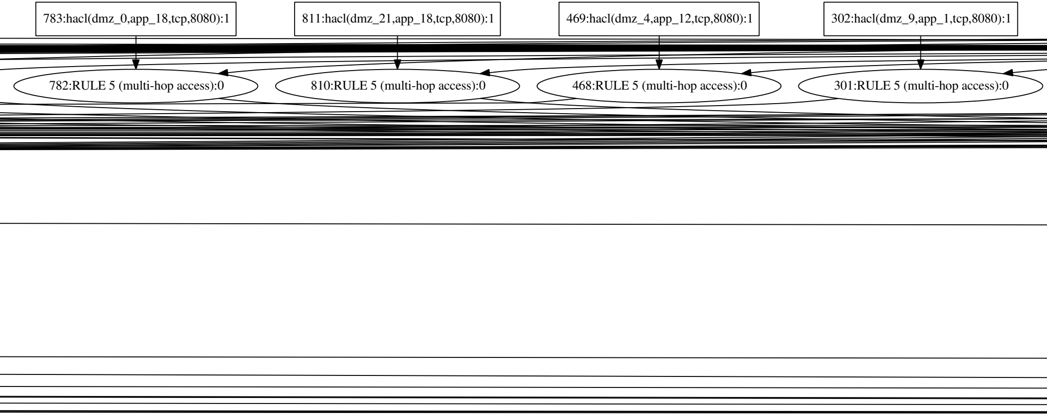

711:hacl(dmz_7,app_16,tcp,8080):1
620:hacl(dmz_15,app_15,tcp,8080):1
713:hacl(dmz_8,app_16,tcp,8080):1
673:hacl(dmz_12,app_16,tcp,8080):1
710:RULE 5 (multi-hop access):0
619:RULE 5 (multi-hop access):0
712:RULE 5 (multi-hop access):0
672:RULE 5 (multi-hop access):0

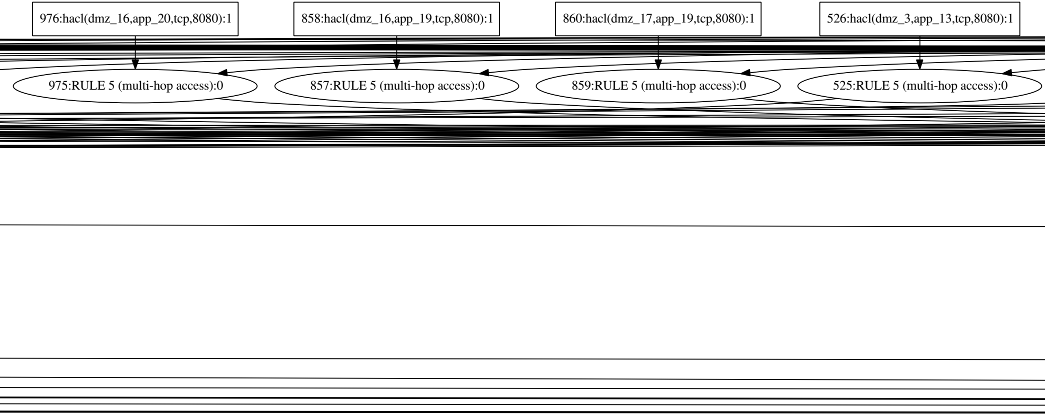

675:hacl(dmz_13,app_16,tcp,8080):1
750:hacl(dmz_20,app_17,tcp,8080):1
642:hacl(dmz_25,app_15,tcp,8080):1
724:hacl(dmz_0,app_17,tcp,8080):1
674:RULE 5 (multi-hop access):0
749:RULE 5 (multi-hop access):0
641:RULE 5 (multi-hop access):0
723:RULE 5 (multi-hop access):0

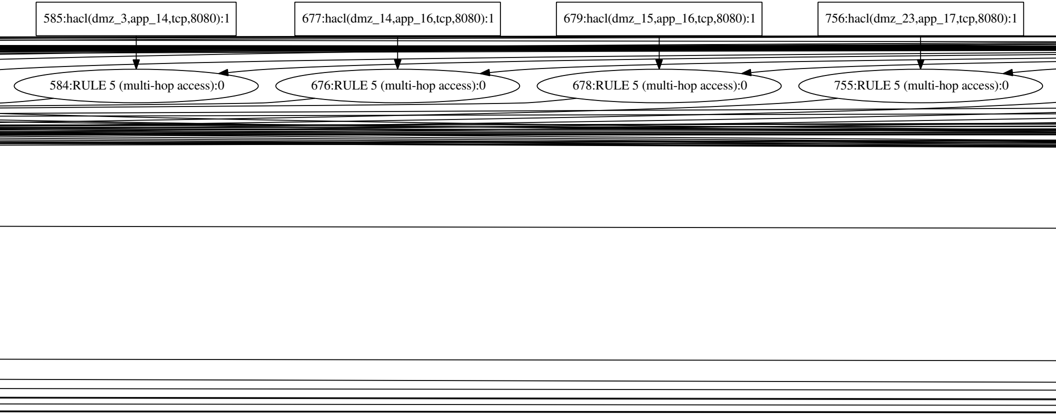

752:hacl(dmz_21,app_17,tcp,8080):1
768:hacl(dmz_6,app_17,tcp,8080):1
726:hacl(dmz_1,app_17,tcp,8080):1
728:hacl(dmz_10,app_17,tcp,8080):1
751:RULE 5 (multi-hop access):0
767:RULE 5 (multi-hop access):0
725:RULE 5 (multi-hop access):0
727:RULE 5 (multi-hop access):0
775:networkServiceInfo(app_17,tomcat,tcp,8080,root):1
776:vulExists(app_17,vul_app,to

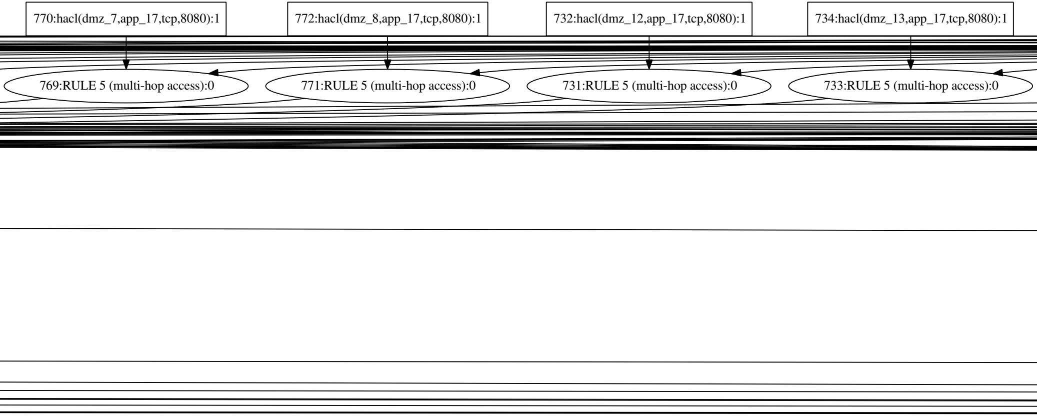

754:hacl(dmz_22,app_17,tcp,8080):1
730:hacl(dmz_11,app_17,tcp,8080):1
758:hacl(dmz_24,app_17,tcp,8080):1
803:hacl(dmz_18,app_18,tcp,8080):1
753:RULE 5 (multi-hop access):0
729:RULE 5 (multi-hop access):0
757:RULE 5 (multi-hop access):0
802:RULE 5 (multi-hop access):0
mcat,remoteExploit,privEscalation):1
722:netAccess(app_17,tcp,8080):0
721:RULE 2 (remote exploit of a server program):0
660:hacl(app_16,db_51,tcp,5432):1

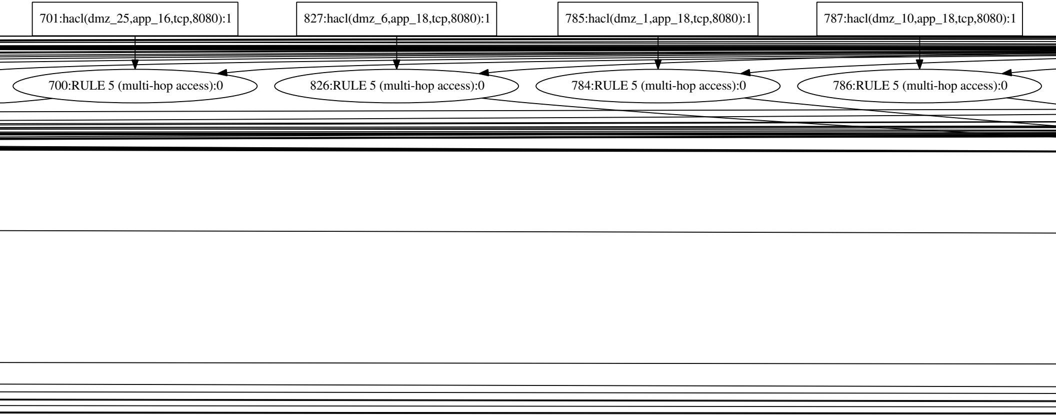

825:hacl(dmz_5,app_18,tcp,8080):1
805:hacl(dmz_19,app_18,tcp,8080):1
807:hacl(dmz_2,app_18,tcp,8080):1
809:hacl(dmz_20,app_18,tcp,8080):1
824:RULE 5 (multi-hop access):0
804:RULE 5 (multi-hop access):0
806:RULE 5 (multi-hop access):0
808:RULE 5 (multi-hop access):0
661:execCode(app_16,root):0
659:RULE 5 (multi-hop access):0

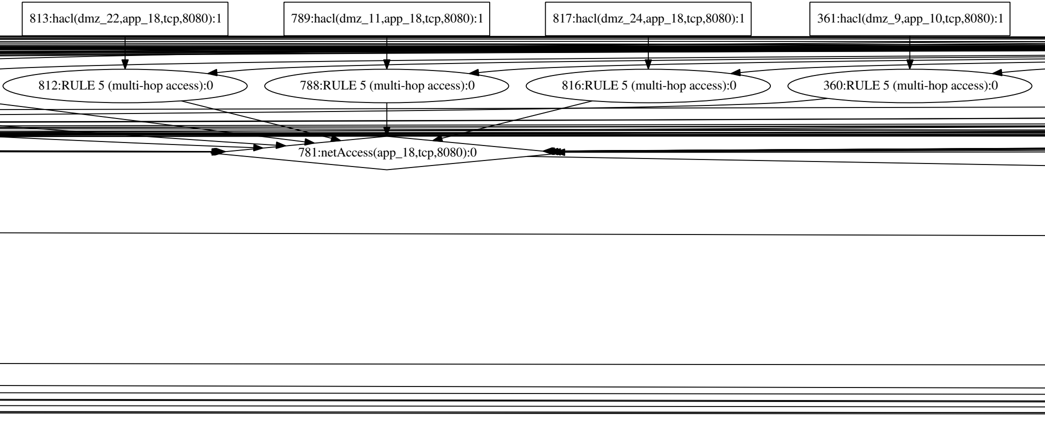

783:hacl(dmz_0,app_18,tcp,8080):1
811:hacl(dmz_21,app_18,tcp,8080):1
469:hacl(dmz_4,app_12,tcp,8080):1
302:hacl(dmz_9,app_1,tcp,8080):1
782:RULE 5 (multi-hop access):0
810:RULE 5 (multi-hop access):0
468:RULE 5 (multi-hop access):0
301:RULE 5 (multi-hop access):0

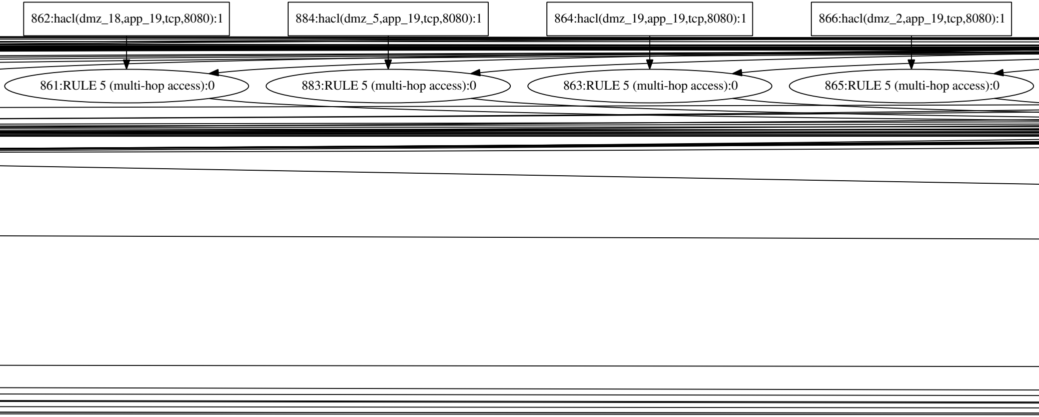

976:hacl(dmz_16,app_20,tcp,8080):1
858:hacl(dmz_16,app_19,tcp,8080):1
860:hacl(dmz_17,app_19,tcp,8080):1
526:hacl(dmz_3,app_13,tcp,8080):1
975:RULE 5 (multi-hop access):0
857:RULE 5 (multi-hop access):0
859:RULE 5 (multi-hop access):0
525:RULE 5 (multi-hop access):0

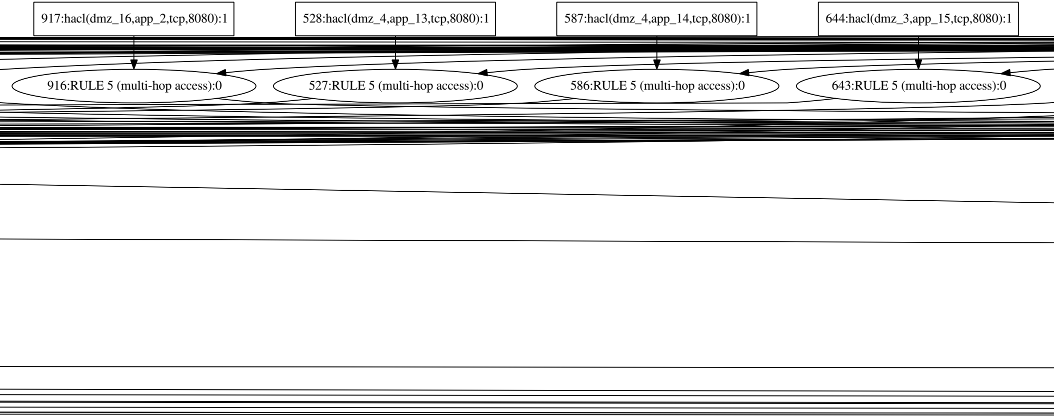

585:hacl(dmz_3,app_14,tcp,8080):1
677:hacl(dmz_14,app_16,tcp,8080):1
679:hacl(dmz_15,app_16,tcp,8080):1
756:hacl(dmz_23,app_17,tcp,8080):1
584:RULE 5 (multi-hop access):0
676:RULE 5 (multi-hop access):0
678:RULE 5 (multi-hop access):0
755:RULE 5 (multi-hop access):0

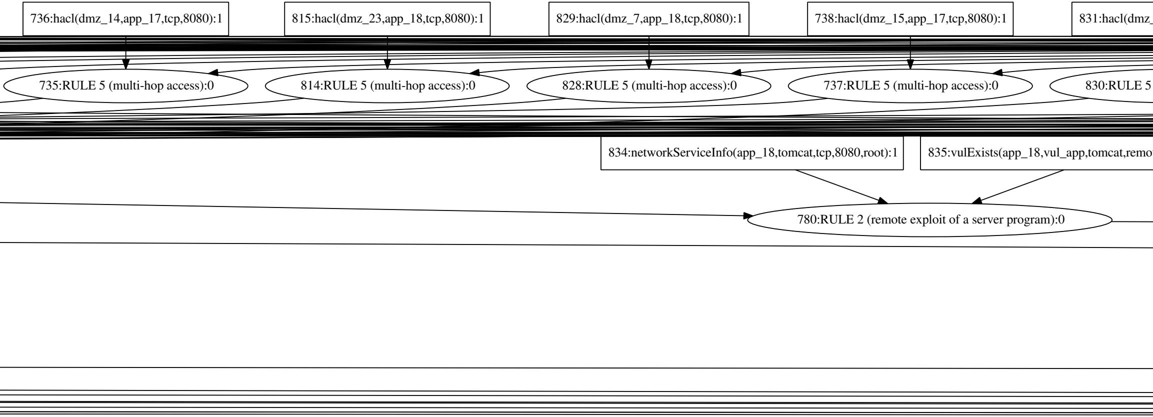

770:hacl(dmz_7,app_17,tcp,8080):1
772:hacl(dmz_8,app_17,tcp,8080):1
732:hacl(dmz_12,app_17,tcp,8080):1
734:hacl(dmz_13,app_17,tcp,8080):1
769:RULE 5 (multi-hop access):0
771:RULE 5 (multi-hop access):0
731:RULE 5 (multi-hop access):0
733:RULE 5 (multi-hop access):0

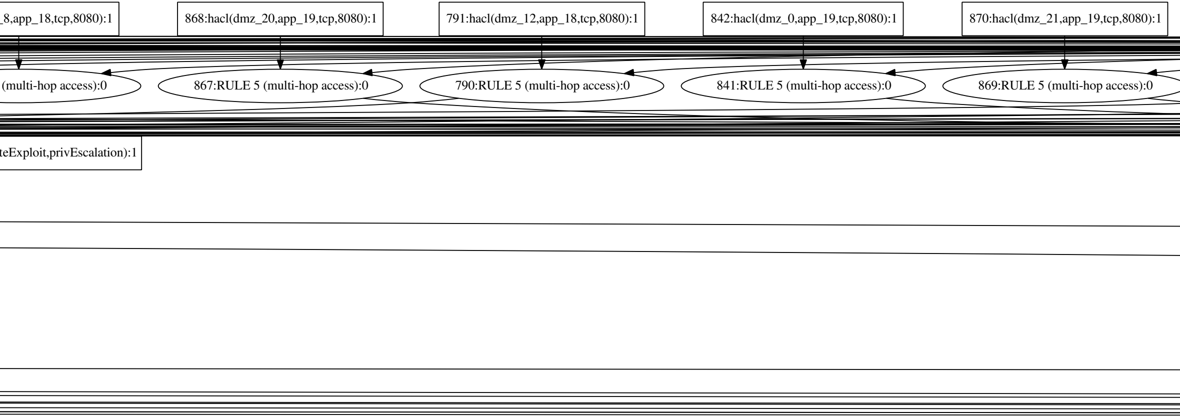

701:hacl(dmz_25,app_16,tcp,8080):1
827:hacl(dmz_6,app_18,tcp,8080):1
785:hacl(dmz_1,app_18,tcp,8080):1
787:hacl(dmz_10,app_18,tcp,8080):1
700:RULE 5 (multi-hop access):0
826:RULE 5 (multi-hop access):0
784:RULE 5 (multi-hop access):0
786:RULE 5 (multi-hop access):0

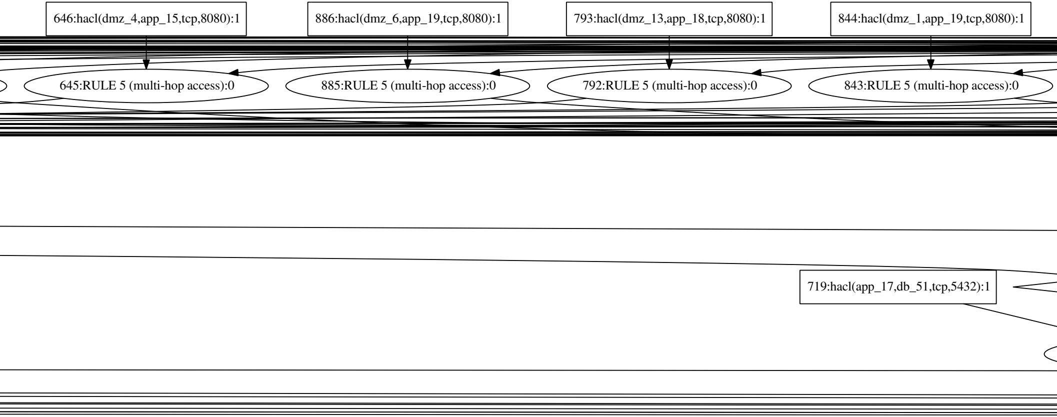

813:hacl(dmz_22,app_18,tcp,8080):1
789:hacl(dmz_11,app_18,tcp,8080):1
817:hacl(dmz_24,app_18,tcp,8080):1
361:hacl(dmz_9,app_10,tcp,8080):1
812:RULE 5 (multi-hop access):0
788:RULE 5 (multi-hop access):0
816:RULE 5 (multi-hop access):0
360:RULE 5 (multi-hop access):0
781:netAccess(app_18,tcp,8080):0

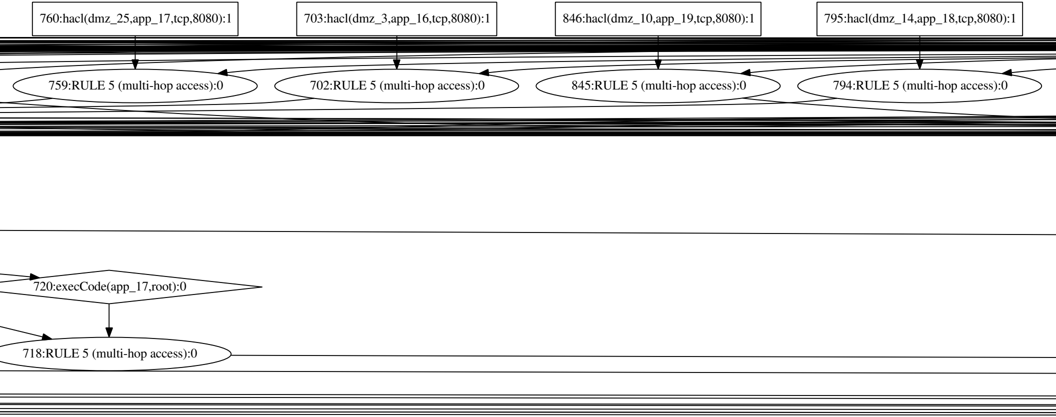

862:hacl(dmz_18,app_19,tcp,8080):1
884:hacl(dmz_5,app_19,tcp,8080):1
864:hacl(dmz_19,app_19,tcp,8080):1
866:hacl(dmz_2,app_19,tcp,8080):1
861:RULE 5 (multi-hop access):0
883:RULE 5 (multi-hop access):0
863:RULE 5 (multi-hop access):0
865:RULE 5 (multi-hop access):0

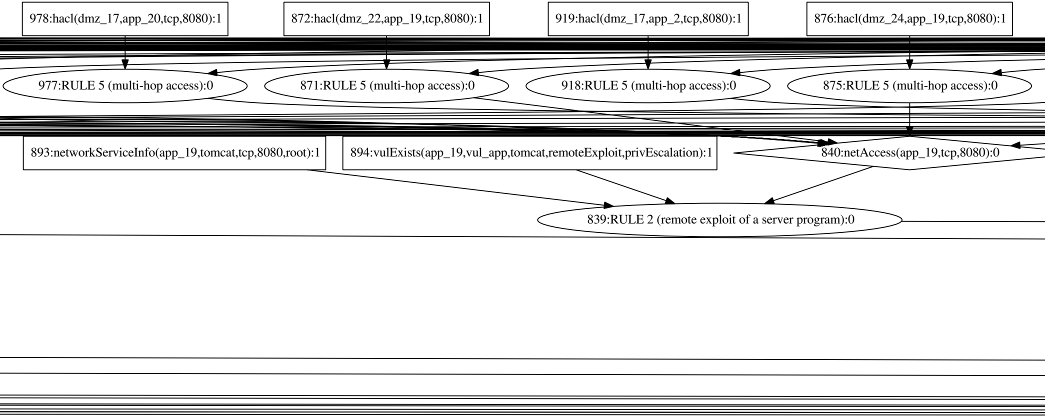

917:hacl(dmz_16,app_2,tcp,8080):1
528:hacl(dmz_4,app_13,tcp,8080):1
587:hacl(dmz_4,app_14,tcp,8080):1
644:hacl(dmz_3,app_15,tcp,8080):1
916:RULE 5 (multi-hop access):0
527:RULE 5 (multi-hop access):0
586:RULE 5 (multi-hop access):0
643:RULE 5 (multi-hop access):0

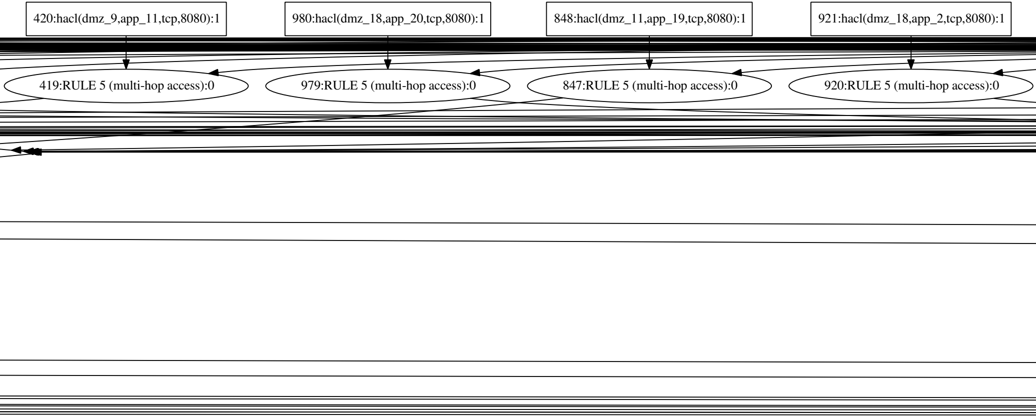

736:hacl(dmz_14,app_17,tcp,8080):1
815:hacl(dmz_23,app_18,tcp,8080):1
829:hacl(dmz_7,app_18,tcp,8080):1
738:hacl(dmz_15,app_17,tcp,8080):1
831:hacl(dmz_
735:RULE 5 (multi-hop access):0
814:RULE 5 (multi-hop access):0
828:RULE 5 (multi-hop access):0
737:RULE 5 (multi-hop access):0
830:RULE 5
834:networkServiceInfo(app_18,tomcat,tcp,8080,root):1
835:vulExists(app_18,vul_app,tomcat,remo
780:RULE 2 (remote exploit of a server program):0

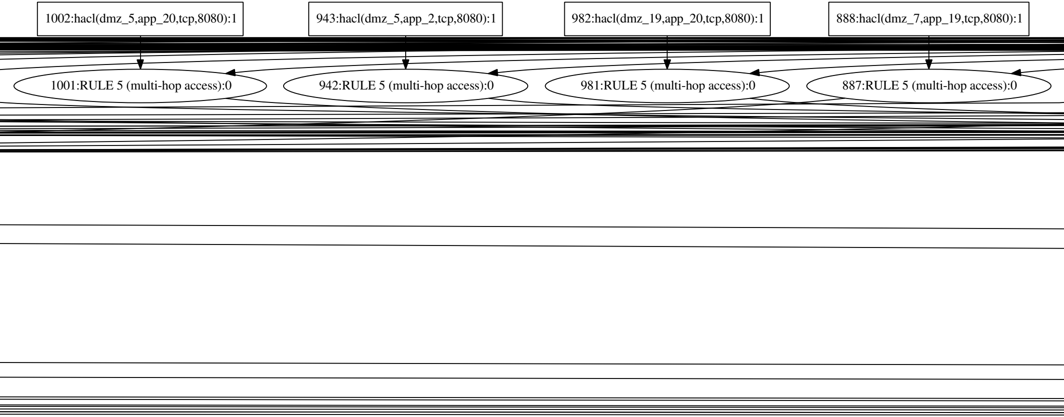

_8,app_18,tcp,8080):1
868:hacl(dmz_20,app_19,tcp,8080):1
791:hacl(dmz_12,app_18,tcp,8080):1
842:hacl(dmz_0,app_19,tcp,8080):1
870:hacl(dmz_21,app_19,tcp,8080):1
(multi-hop access):0
867:RULE 5 (multi-hop access):0
790:RULE 5 (multi-hop access):0
841:RULE 5 (multi-hop access):0
869:RULE 5 (multi-hop access):0
teExploit,privEscalation):1

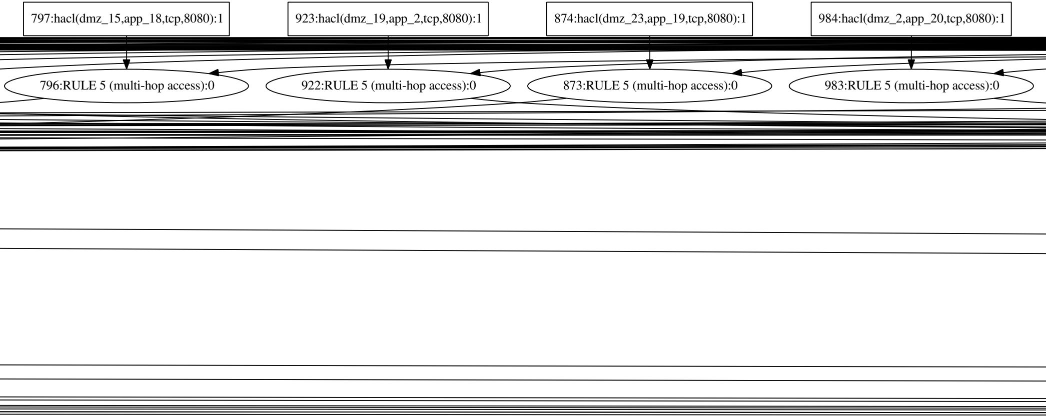

646:hacl(dmz_4,app_15,tcp,8080):1
886:hacl(dmz_6,app_19,tcp,8080):1
793:hacl(dmz_13,app_18,tcp,8080):1
844:hacl(dmz_1,app_19,tcp,8080):1
645:RULE 5 (multi-hop access):0
885:RULE 5 (multi-hop access):0
792:RULE 5 (multi-hop access):0
843:RULE 5 (multi-hop access):0
719:hacl(app_17,db_51,tcp,5432):1

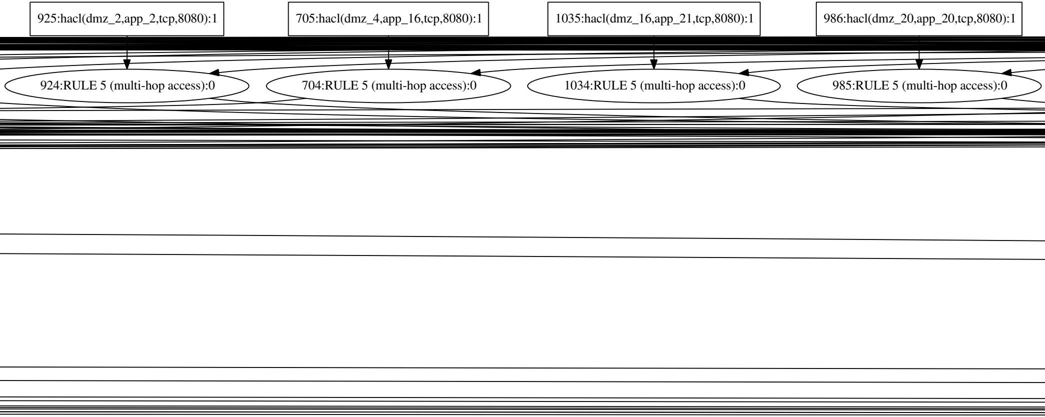

760:hacl(dmz_25,app_17,tcp,8080):1
703:hacl(dmz_3,app_16,tcp,8080):1
846:hacl(dmz_10,app_19,tcp,8080):1
795:hacl(dmz_14,app_18,tcp,8080):1
759:RULE 5 (multi-hop access):0
702:RULE 5 (multi-hop access):0
845:RULE 5 (multi-hop access):0
794:RULE 5 (multi-hop access):0
720:execCode(app_17,root):0
718:RULE 5 (multi-hop access):0

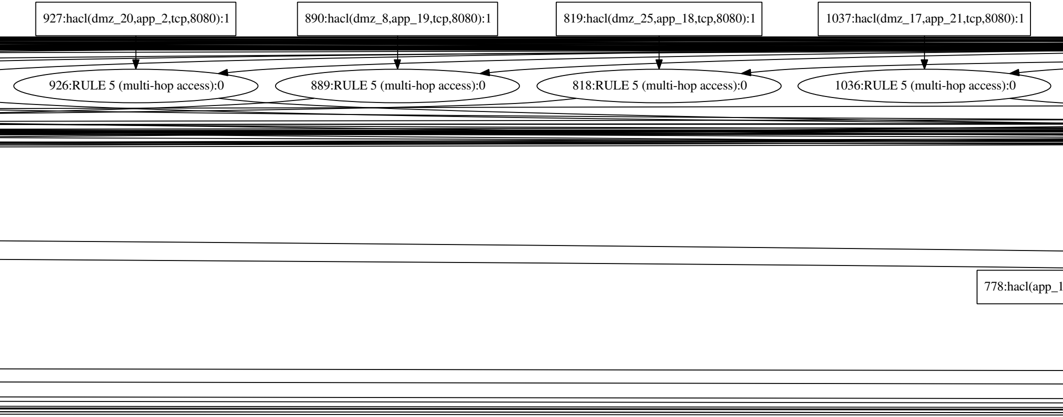

978:hacl(dmz_17,app_20,tcp,8080):1
872:hacl(dmz_22,app_19,tcp,8080):1
919:hacl(dmz_17,app_2,tcp,8080):1
876:hacl(dmz_24,app_19,tcp,8080):1
977:RULE 5 (multi-hop access):0
871:RULE 5 (multi-hop access):0
918:RULE 5 (multi-hop access):0
875:RULE 5 (multi-hop access):0
893:networkServiceInfo(app_19,tomcat,tcp,8080,root):1
894:vulExists(app_19,vul_app,tomcat,remoteExploit,privEscalation):1
840:netAccess(app_19,tcp,8080):0
839:RULE 2 (remote exploit of a server program):0

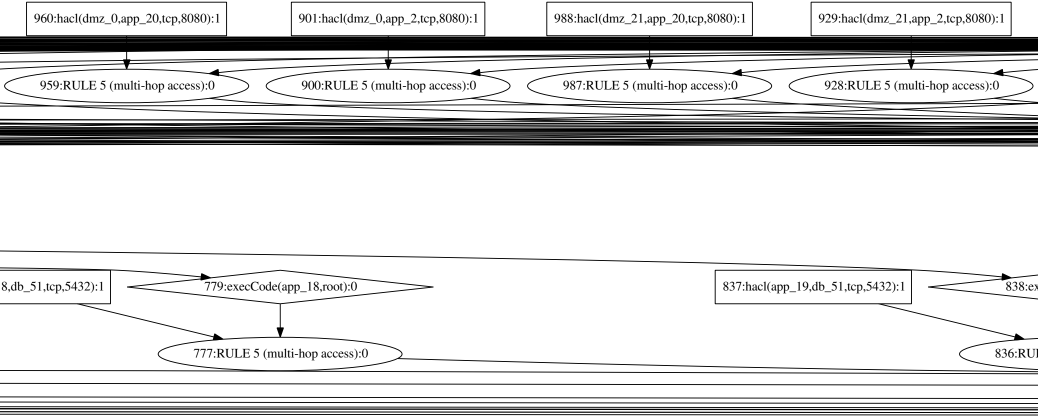

420:hacl(dmz_9,app_11,tcp,8080):1
980:hacl(dmz_18,app_20,tcp,8080):1
848:hacl(dmz_11,app_19,tcp,8080):1
921:hacl(dmz_18,app_2,tcp,8080):1
419:RULE 5 (multi-hop access):0
979:RULE 5 (multi-hop access):0
847:RULE 5 (multi-hop access):0
920:RULE 5 (multi-hop access):0

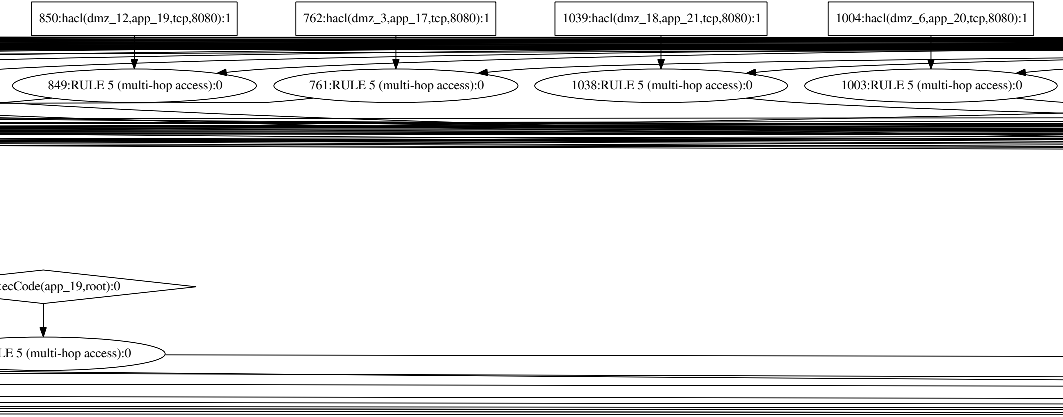

1002:hacl(dmz_5,app_20,tcp,8080):1
943:hacl(dmz_5,app_2,tcp,8080):1
982:hacl(dmz_19,app_20,tcp,8080):1
888:hacl(dmz_7,app_19,tcp,8080):1
1001:RULE 5 (multi-hop access):0
942:RULE 5 (multi-hop access):0
981:RULE 5 (multi-hop access):0
887:RULE 5 (multi-hop access):0

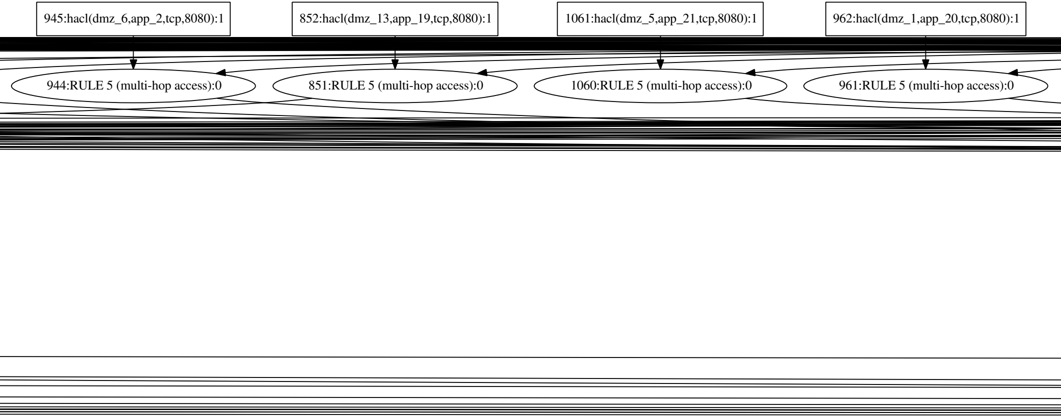

797:hacl(dmz_15,app_18,tcp,8080):1
923:hacl(dmz_19,app_2,tcp,8080):1
874:hacl(dmz_23,app_19,tcp,8080):1
984:hacl(dmz_2,app_20,tcp,8080):1
796:RULE 5 (multi-hop access):0
922:RULE 5 (multi-hop access):0
873:RULE 5 (multi-hop access):0
983:RULE 5 (multi-hop access):0

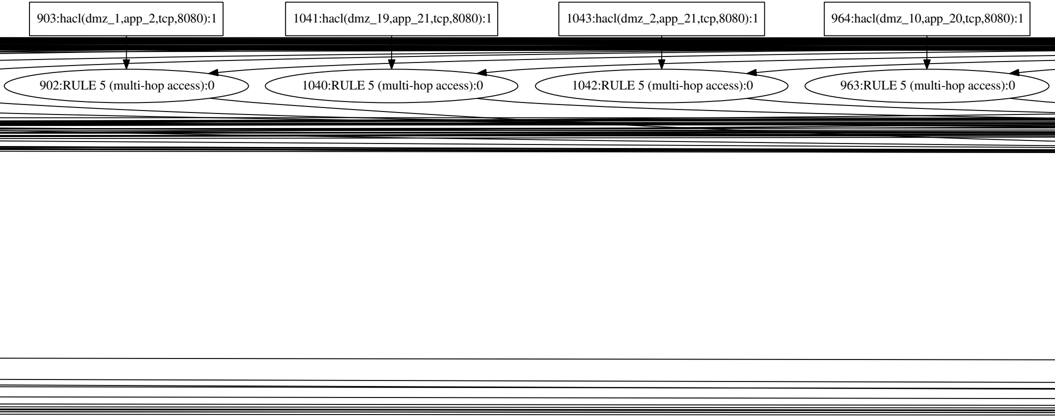

925:hacl(dmz_2,app_2,tcp,8080):1
705:hacl(dmz_4,app_16,tcp,8080):1
1035:hacl(dmz_16,app_21,tcp,8080):1
986:hacl(dmz_20,app_20,tcp,8080):1
924:RULE 5 (multi-hop access):0
704:RULE 5 (multi-hop access):0
1034:RULE 5 (multi-hop access):0
985:RULE 5 (multi-hop access):0

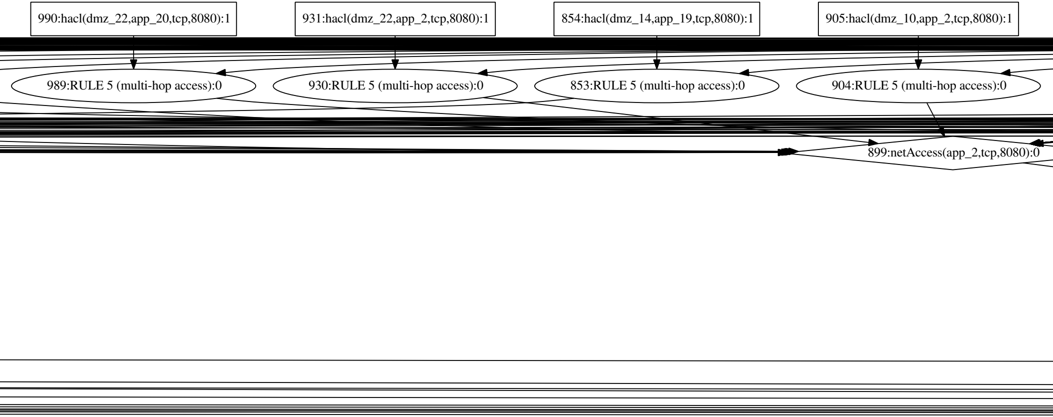

927:hacl(dmz_20,app_2,tcp,8080):1
890:hacl(dmz_8,app_19,tcp,8080):1
819:hacl(dmz_25,app_18,tcp,8080):1
1037:hacl(dmz_17,app_21,tcp,8080):1
926:RULE 5 (multi-hop access):0
889:RULE 5 (multi-hop access):0
818:RULE 5 (multi-hop access):0
1036:RULE 5 (multi-hop access):0
778:hacl(app_1

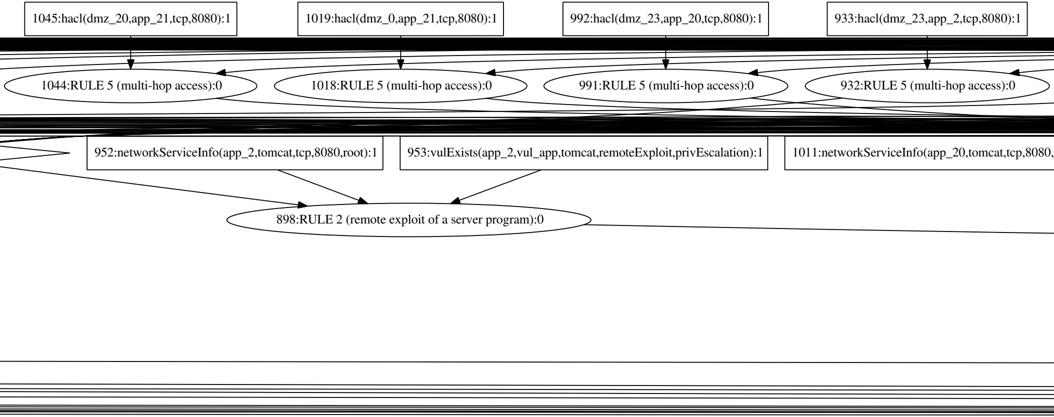

960:hacl(dmz_0,app_20,tcp,8080):1
901:hacl(dmz_0,app_2,tcp,8080):1
988:hacl(dmz_21,app_20,tcp,8080):1
929:hacl(dmz_21,app_2,tcp,8080):1
959:RULE 5 (multi-hop access):0
900:RULE 5 (multi-hop access):0
987:RULE 5 (multi-hop access):0
928:RULE 5 (multi-hop access):0
8,db_51,tcp,5432):1
779:execCode(app_18,root):0
837:hacl(app_19,db_51,tcp,5432):1
838:ex
777:RULE 5 (multi-hop access):0
836:RUI

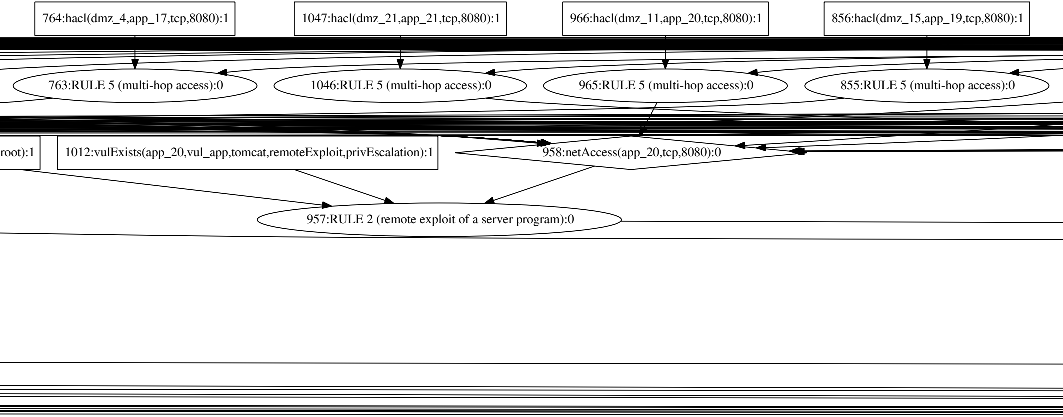

850:hacl(dmz_12,app_19,tcp,8080):1
762:hacl(dmz_3,app_17,tcp,8080):1
1039:hacl(dmz_18,app_21,tcp,8080):1
1004:hacl(dmz_6,app_20,tcp,8080):1
849:RULE 5 (multi-hop access):0
761:RULE 5 (multi-hop access):0
1038:RULE 5 (multi-hop access):0
1003:RULE 5 (multi-hop access):0
ecCode(app_19,root):0
LE 5 (multi-hop access):0

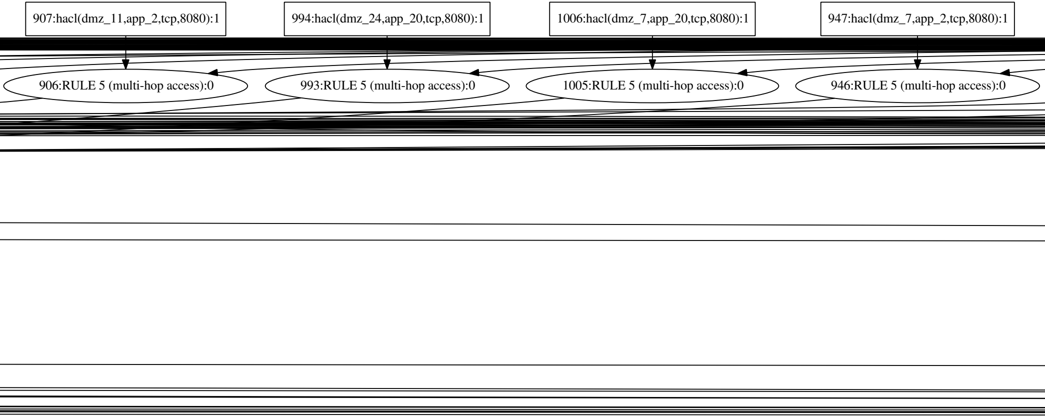

945:hacl(dmz_6,app_2,tcp,8080):1
852:hacl(dmz_13,app_19,tcp,8080):1
1061:hacl(dmz_5,app_21,tcp,8080):1
962:hacl(dmz_1,app_20,tcp,8080):1
944:RULE 5 (multi-hop access):0
851:RULE 5 (multi-hop access):0
1060:RULE 5 (multi-hop access):0
961:RULE 5 (multi-hop access):0

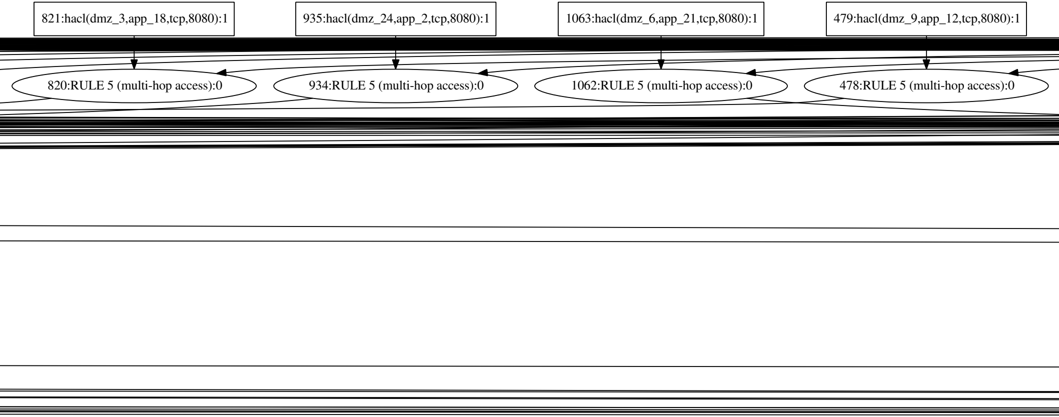

903:hacl(dmz_1,app_2,tcp,8080):1
1041:hacl(dmz_19,app_21,tcp,8080):1
1043:hacl(dmz_2,app_21,tcp,8080):1
964:hacl(dmz_10,app_20,tcp,8080):1
902:RULE 5 (multi-hop access):0
1040:RULE 5 (multi-hop access):0
1042:RULE 5 (multi-hop access):0
963:RULE 5 (multi-hop access):0

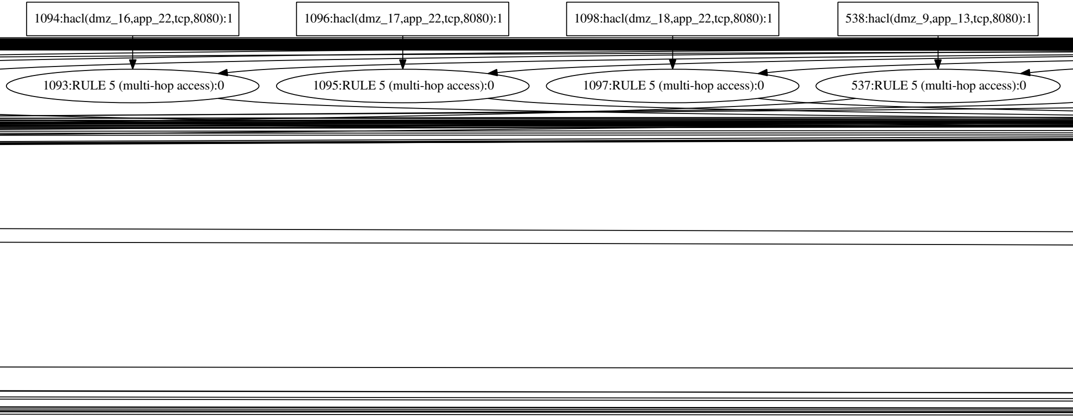

990:hacl(dmz_22,app_20,tcp,8080):1
931:hacl(dmz_22,app_2,tcp,8080):1
854:hacl(dmz_14,app_19,tcp,8080):1
905:hacl(dmz_10,app_2,tcp,8080):1
989:RULE 5 (multi-hop access):0
930:RULE 5 (multi-hop access):0
853:RULE 5 (multi-hop access):0
904:RULE 5 (multi-hop access):0
899:netAccess(app_2,tcp,8080):0

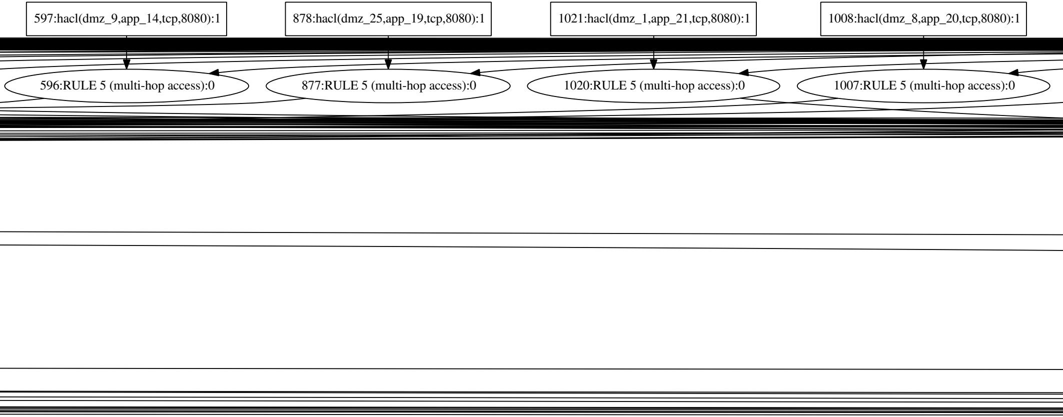

1045:hacl(dmz_20,app_21,tcp,8080):1
1019:hacl(dmz_0,app_21,tcp,8080):1
992:hacl(dmz_23,app_20,tcp,8080):1
933:hacl(dmz_23,app_2,tcp,8080):1
1044:RULE 5 (multi-hop access):0
1018:RULE 5 (multi-hop access):0
991:RULE 5 (multi-hop access):0
932:RULE 5 (multi-hop access):0
952:networkServiceInfo(app_2,tomcat,tcp,8080,root):1
953:vulExists(app_2,vul_app,tomcat,remoteExploit,privEscalation):1
1011:networkServiceInfo(app_20,tomcat,tcp,8080,
898:RULE 2 (remote exploit of a server program):0

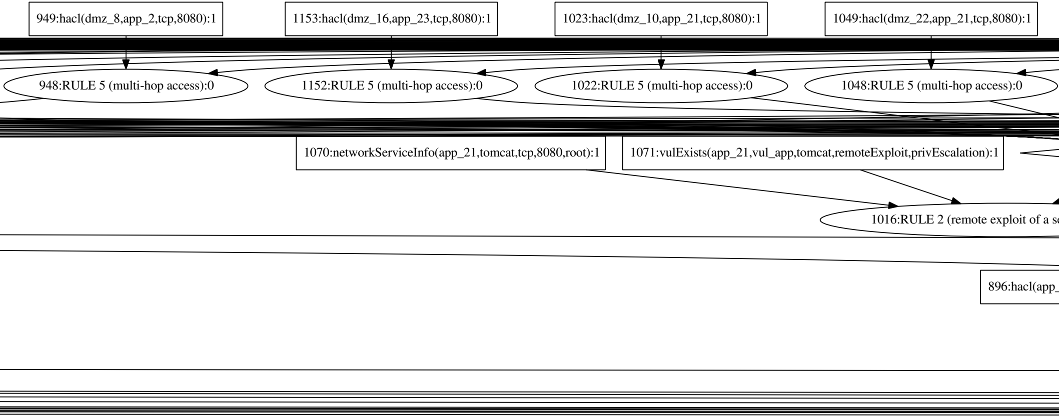

764:hacl(dmz_4,app_17,tcp,8080):1
1047:hacl(dmz_21,app_21,tcp,8080):1
966:hacl(dmz_11,app_20,tcp,8080):1
856:hacl(dmz_15,app_19,tcp,8080):1
763:RULE 5 (multi-hop access):0
1046:RULE 5 (multi-hop access):0
965:RULE 5 (multi-hop access):0
855:RULE 5 (multi-hop access):0
root):1
1012:vulExists(app_20,vul_app,tomcat,remoteExploit,privEscalation):1
958:netAccess(app_20,tcp,8080):0
957:RULE 2 (remote exploit of a server program):0

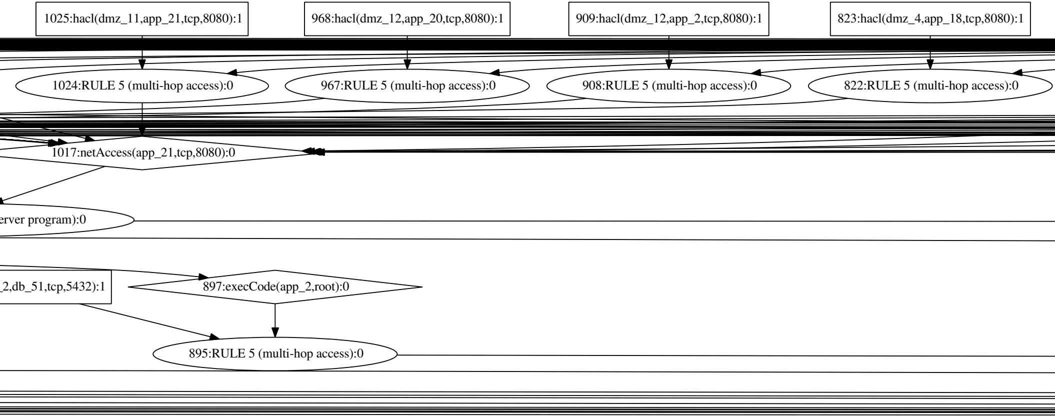

907:hacl(dmz_11,app_2,tcp,8080):1
994:hacl(dmz_24,app_20,tcp,8080):1
1006:hacl(dmz_7,app_20,tcp,8080):1
947:hacl(dmz_7,app_2,tcp,8080):1
906:RULE 5 (multi-hop access):0
993:RULE 5 (multi-hop access):0
1005:RULE 5 (multi-hop access):0
946:RULE 5 (multi-hop access):0

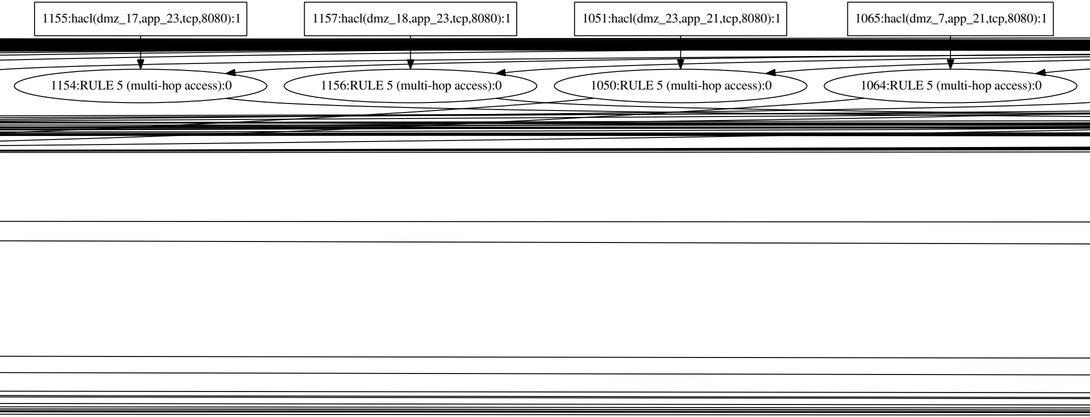

821:hacl(dmz_3,app_18,tcp,8080):1
935:hacl(dmz_24,app_2,tcp,8080):1
1063:hacl(dmz_6,app_21,tcp,8080):1
479:hacl(dmz_9,app_12,tcp,8080):1
820:RULE 5 (multi-hop access):0
934:RULE 5 (multi-hop access):0
1062:RULE 5 (multi-hop access):0
478:RULE 5 (multi-hop access):0

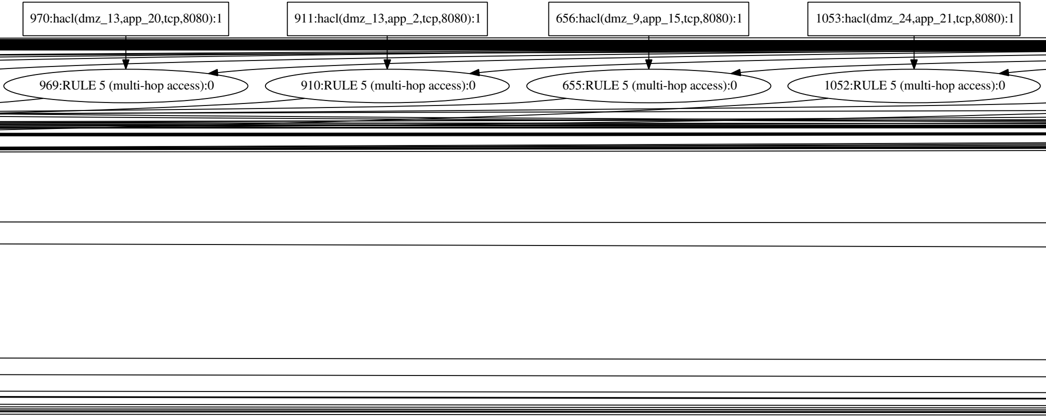

1094:hacl(dmz_16,app_22,tcp,8080):1
1096:hacl(dmz_17,app_22,tcp,8080):1
1098:hacl(dmz_18,app_22,tcp,8080):1
538:hacl(dmz_9,app_13,tcp,8080):1
1093:RULE 5 (multi-hop access):0
1095:RULE 5 (multi-hop access):0
1097:RULE 5 (multi-hop access):0
537:RULE 5 (multi-hop access):0

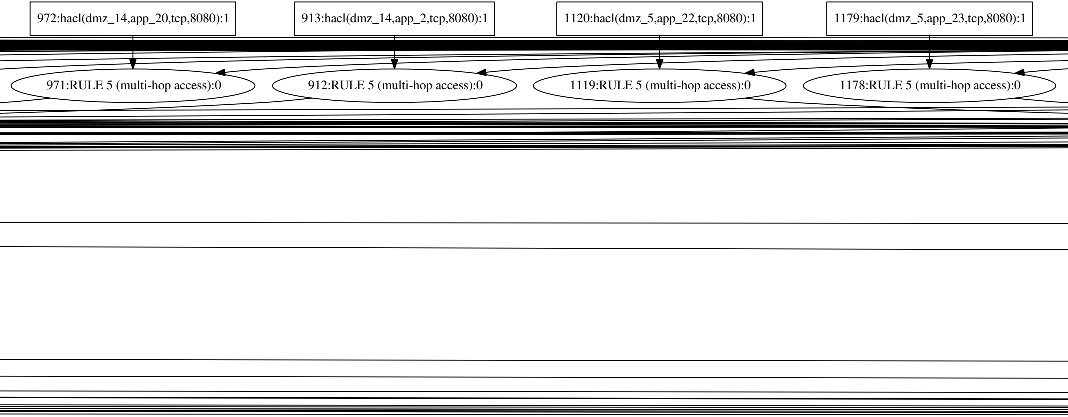

597:hacl(dmz_9,app_14,tcp,8080):1
878:hacl(dmz_25,app_19,tcp,8080):1
1021:hacl(dmz_1,app_21,tcp,8080):1
1008:hacl(dmz_8,app_20,tcp,8080):1
596:RULE 5 (multi-hop access):0
877:RULE 5 (multi-hop access):0
1020:RULE 5 (multi-hop access):0
1007:RULE 5 (multi-hop access):0

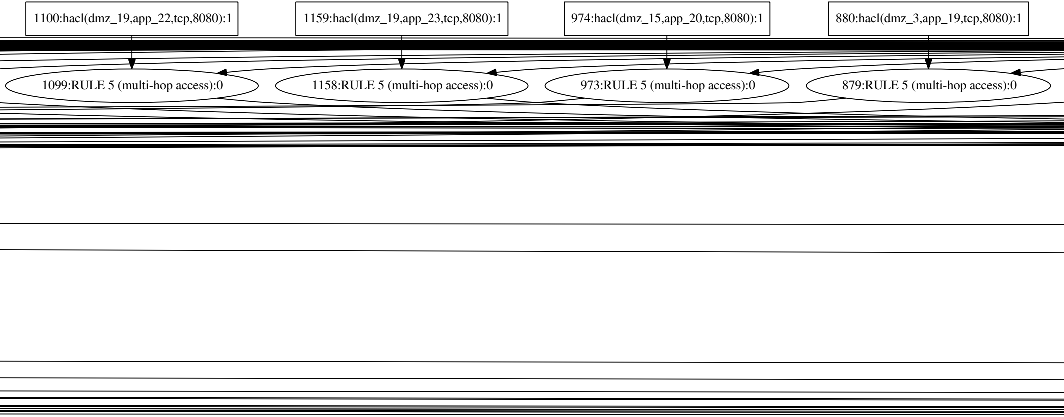

949:hacl(dmz_8,app_2,tcp,8080):1
1153:hacl(dmz_16,app_23,tcp,8080):1
1023:hacl(dmz_10,app_21,tcp,8080):1
1049:hacl(dmz_22,app_21,tcp,8080):1
948:RULE 5 (multi-hop access):0
1152:RULE 5 (multi-hop access):0
1022:RULE 5 (multi-hop access):0
1048:RULE 5 (multi-hop access):0
1070:networkServiceInfo(app_21,tomcat,tcp,8080,root):1
1071:vulExists(app_21,vul_app,tomcat,remoteExploit,privEscalation):1
1016:RULE 2 (remote exploit of a se
896:hacl(app_

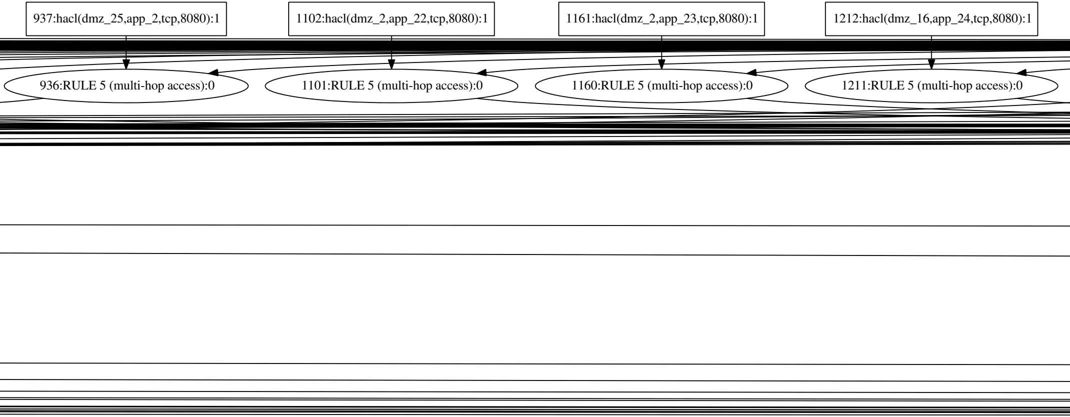

1025:hacl(dmz_11,app_21,tcp,8080):1
968:hacl(dmz_12,app_20,tcp,8080):1
909:hacl(dmz_12,app_2,tcp,8080):1
823:hacl(dmz_4,app_18,tcp,8080):1
1024:RULE 5 (multi-hop access):0
967:RULE 5 (multi-hop access):0
908:RULE 5 (multi-hop access):0
822:RULE 5 (multi-hop access):0
1017:netAccess(app_21,tcp,8080):0
erver program):0
_2,db_51,tcp,5432):1
897:execCode(app_2,root):0
895:RULE 5 (multi-hop access):0

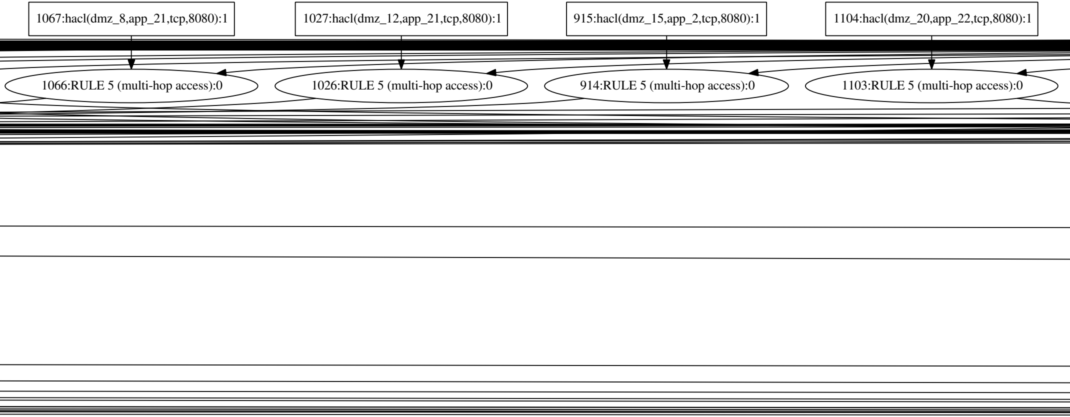

1155:hacl(dmz_17,app_23,tcp,8080):1
1157:hacl(dmz_18,app_23,tcp,8080):1
1051:hacl(dmz_23,app_21,tcp,8080):1
1065:hacl(dmz_7,app_21,tcp,8080):1
1154:RULE 5 (multi-hop access):0
1156:RULE 5 (multi-hop access):0
1050:RULE 5 (multi-hop access):0
1064:RULE 5 (multi-hop access):0

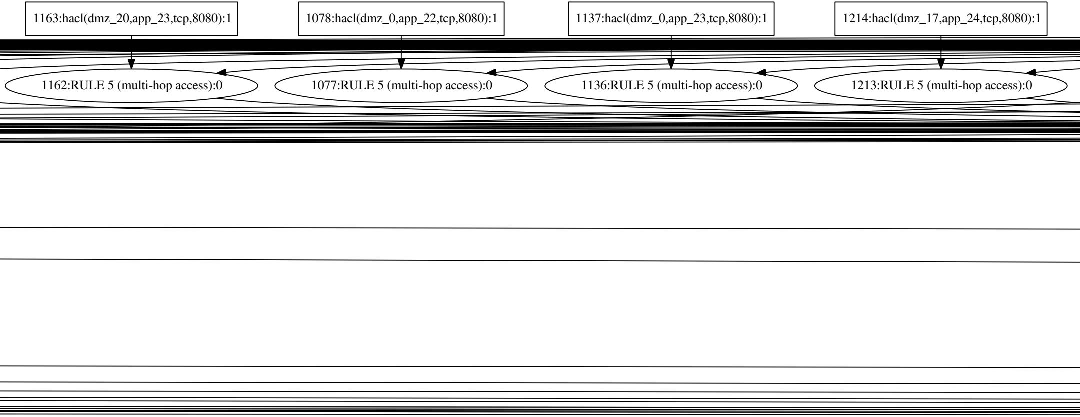

970:hacl(dmz_13,app_20,tcp,8080):1
911:hacl(dmz_13,app_2,tcp,8080):1
656:hacl(dmz_9,app_15,tcp,8080):1
1053:hacl(dmz_24,app_21,tcp,8080):1
969:RULE 5 (multi-hop access):0
910:RULE 5 (multi-hop access):0
655:RULE 5 (multi-hop access):0
1052:RULE 5 (multi-hop access):0

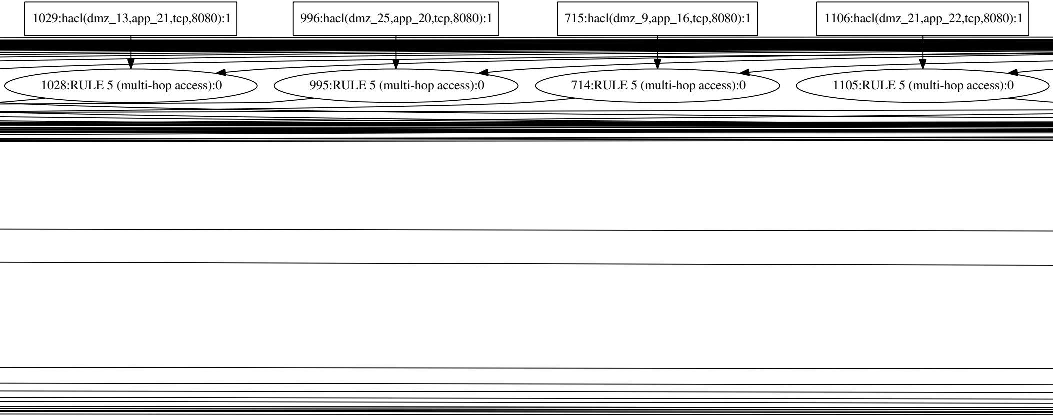

972:hacl(dmz_14,app_20,tcp,8080):1
913:hacl(dmz_14,app_2,tcp,8080):1
1120:hacl(dmz_5,app_22,tcp,8080):1
1179:hacl(dmz_5,app_23,tcp,8080):1
971:RULE 5 (multi-hop access):0
912:RULE 5 (multi-hop access):0
1119:RULE 5 (multi-hop access):0
1178:RULE 5 (multi-hop access):0

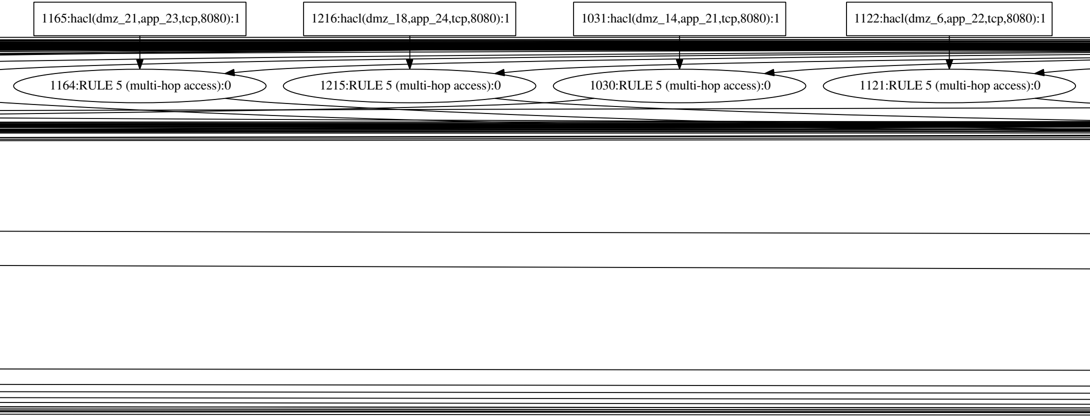

1100:hacl(dmz_19,app_22,tcp,8080):1
1159:hacl(dmz_19,app_23,tcp,8080):1
974:hacl(dmz_15,app_20,tcp,8080):1
880:hacl(dmz_3,app_19,tcp,8080):1
1099:RULE 5 (multi-hop access):0
1158:RULE 5 (multi-hop access):0
973:RULE 5 (multi-hop access):0
879:RULE 5 (multi-hop access):0

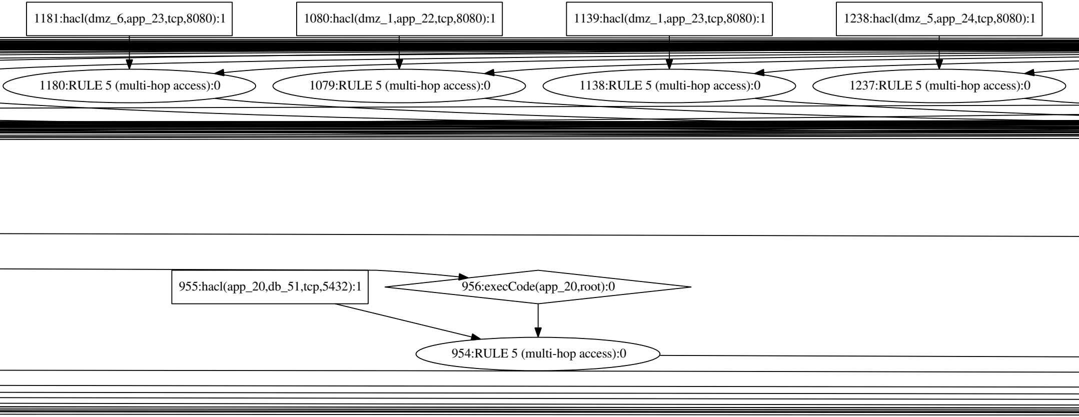

937:hacl(dmz_25,app_2,tcp,8080):1
1102:hacl(dmz_2,app_22,tcp,8080):1
1161:hacl(dmz_2,app_23,tcp,8080):1
1212:hacl(dmz_16,app_24,tcp,8080):1
936:RULE 5 (multi-hop access):0
1101:RULE 5 (multi-hop access):0
1160:RULE 5 (multi-hop access):0
1211:RULE 5 (multi-hop access):0

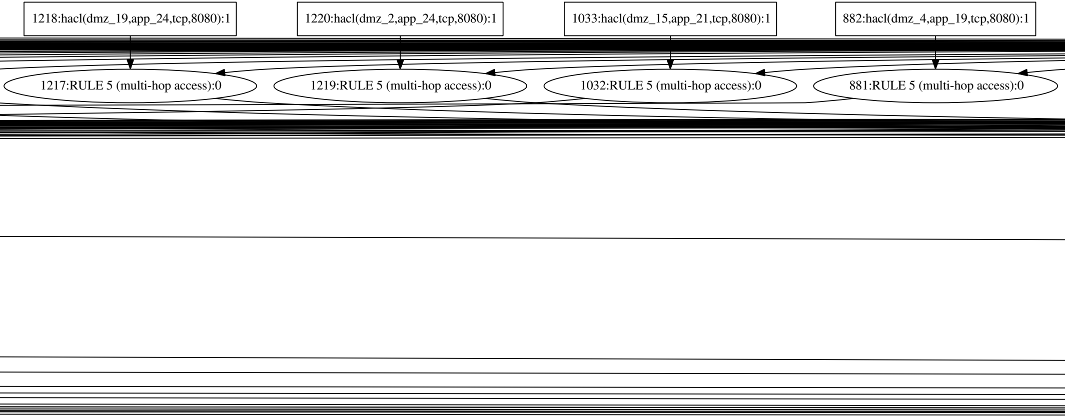

1067:hacl(dmz_8,app_21,tcp,8080):1
1027:hacl(dmz_12,app_21,tcp,8080):1
915:hacl(dmz_15,app_2,tcp,8080):1
1104:hacl(dmz_20,app_22,tcp,8080):1
1066:RULE 5 (multi-hop access):0
1026:RULE 5 (multi-hop access):0
914:RULE 5 (multi-hop access):0
1103:RULE 5 (multi-hop access):0

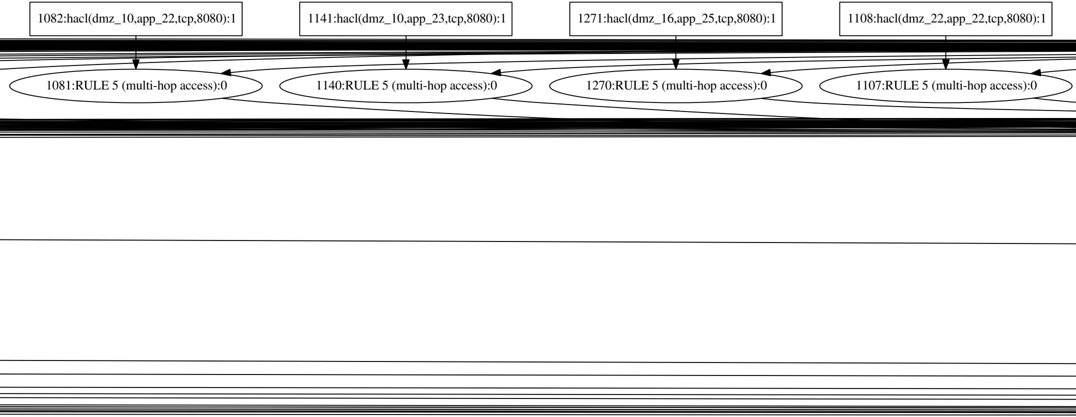

1163:hacl(dmz_20,app_23,tcp,8080):1
1078:hacl(dmz_0,app_22,tcp,8080):1
1137:hacl(dmz_0,app_23,tcp,8080):1
1214:hacl(dmz_17,app_24,tcp,8080):1
1162:RULE 5 (multi-hop access):0
1077:RULE 5 (multi-hop access):0
1136:RULE 5 (multi-hop access):0
1213:RULE 5 (multi-hop access):0

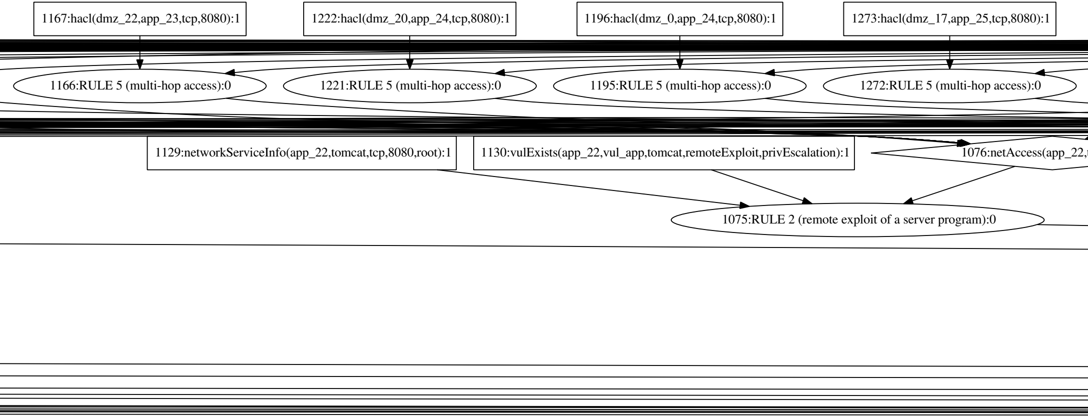

1029:hacl(dmz_13,app_21,tcp,8080):1
996:hacl(dmz_25,app_20,tcp,8080):1
715:hacl(dmz_9,app_16,tcp,8080):1
1106:hacl(dmz_21,app_22,tcp,8080):1
1028:RULE 5 (multi-hop access):0
995:RULE 5 (multi-hop access):0
714:RULE 5 (multi-hop access):0
1105:RULE 5 (multi-hop access):0

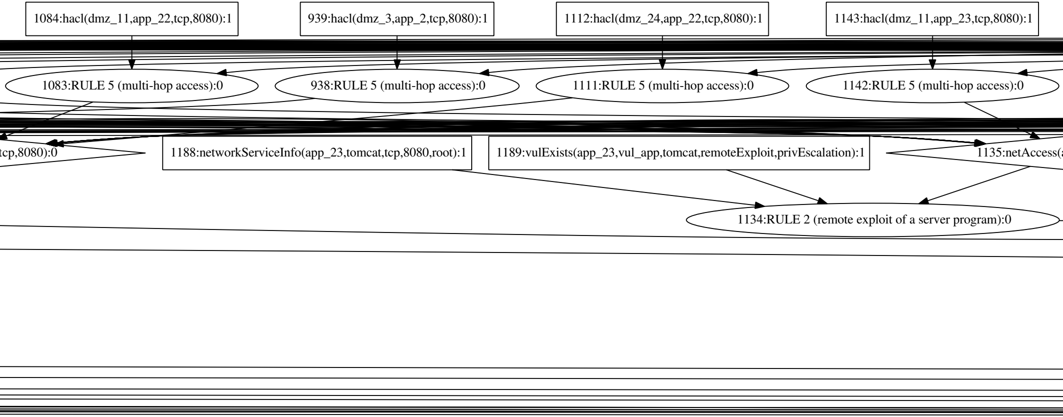

1165:hacl(dmz_21,app_23,tcp,8080):1
1216:hacl(dmz_18,app_24,tcp,8080):1
1031:hacl(dmz_14,app_21,tcp,8080):1
1122:hacl(dmz_6,app_22,tcp,8080):1
1164:RULE 5 (multi-hop access):0
1215:RULE 5 (multi-hop access):0
1030:RULE 5 (multi-hop access):0
1121:RULE 5 (multi-hop access):0

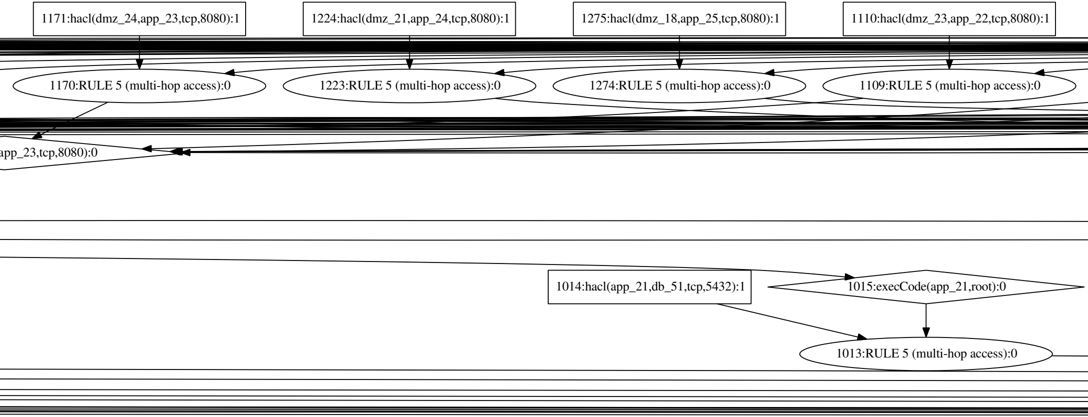

1181:hacl(dmz_6,app_23,tcp,8080):1
1080:hacl(dmz_1,app_22,tcp,8080):1
1139:hacl(dmz_1,app_23,tcp,8080):1
1238:hacl(dmz_5,app_24,tcp,8080):1
1180:RULE 5 (multi-hop access):0
1079:RULE 5 (multi-hop access):0
1138:RULE 5 (multi-hop access):0
1237:RULE 5 (multi-hop access):0
955:hacl(app_20,db_51,tcp,5432):1
956:execCode(app_20,root):0
954:RULE 5 (multi-hop access):0

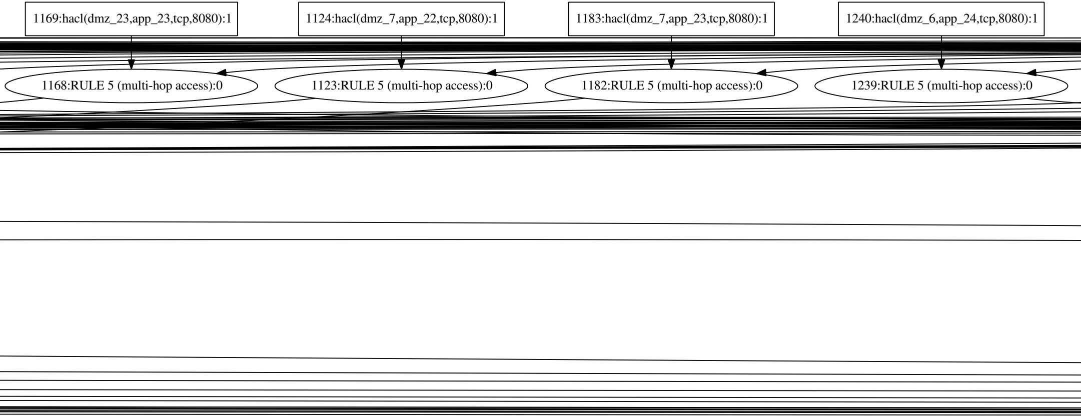

1218:hacl(dmz_19,app_24,tcp,8080):1
1220:hacl(dmz_2,app_24,tcp,8080):1
1033:hacl(dmz_15,app_21,tcp,8080):1
882:hacl(dmz_4,app_19,tcp,8080):1
1217:RULE 5 (multi-hop access):0
1219:RULE 5 (multi-hop access):0
1032:RULE 5 (multi-hop access):0
881:RULE 5 (multi-hop access):0

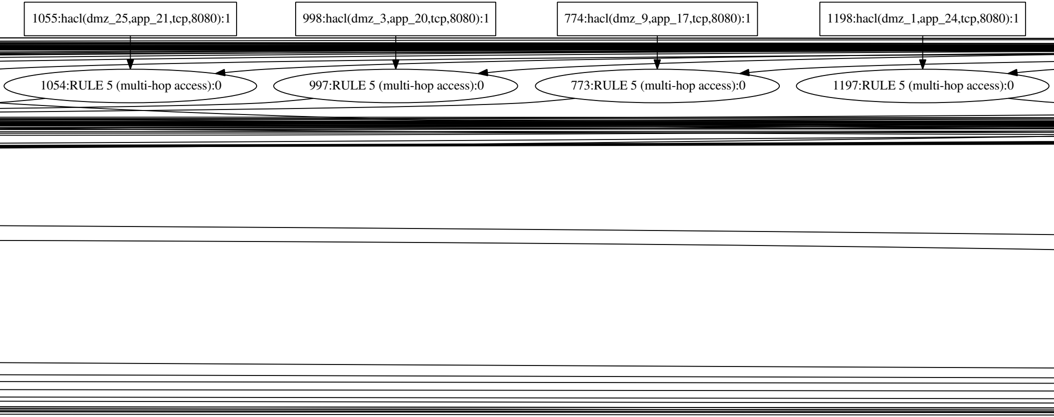

1082:hacl(dmz_10,app_22,tcp,8080):1
1141:hacl(dmz_10,app_23,tcp,8080):1
1271:hacl(dmz_16,app_25,tcp,8080):1
1108:hacl(dmz_22,app_22,tcp,8080):1
1081:RULE 5 (multi-hop access):0
1140:RULE 5 (multi-hop access):0
1270:RULE 5 (multi-hop access):0
1107:RULE 5 (multi-hop access):0

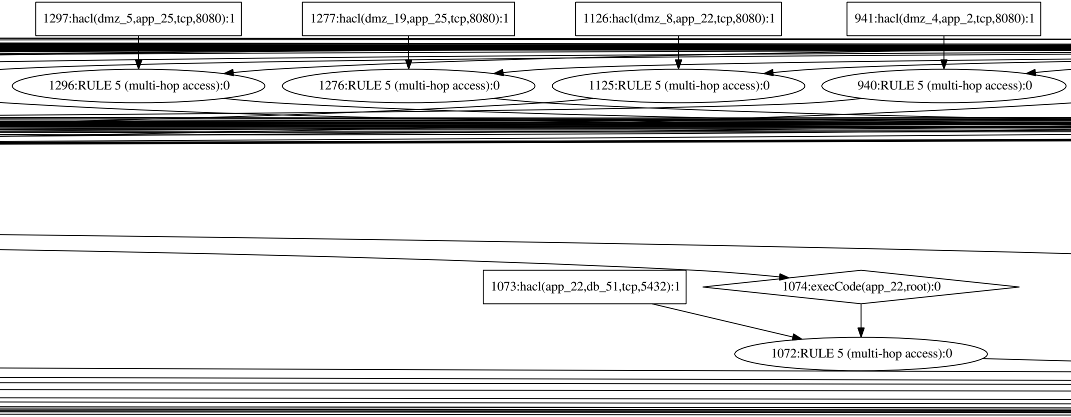

1167:hacl(dmz_22,app_23,tcp,8080):1
1222:hacl(dmz_20,app_24,tcp,8080):1
1196:hacl(dmz_0,app_24,tcp,8080):1
1273:hacl(dmz_17,app_25,tcp,8080):1
1166:RULE 5 (multi-hop access):0
1221:RULE 5 (multi-hop access):0
1195:RULE 5 (multi-hop access):0
1272:RULE 5 (multi-hop access):0
1129:networkServiceInfo(app_22,tomcat,tcp,8080,root):1
1130:vulExists(app_22,vul_app,tomcat,remoteExploit,privEscalation):1
1076:netAccess(app_22,
1075:RULE 2 (remote exploit of a server program):0

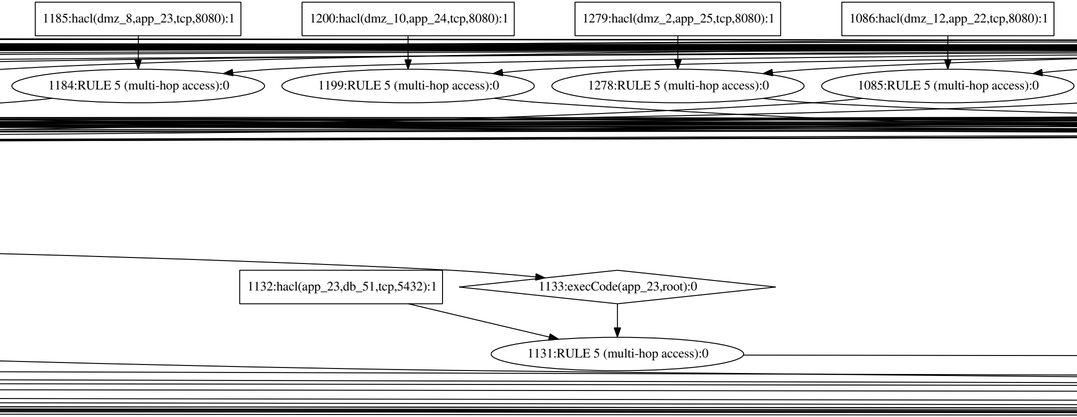

1084:hacl(dmz_11,app_22,tcp,8080):1
939:hacl(dmz_3,app_2,tcp,8080):1
1112:hacl(dmz_24,app_22,tcp,8080):1
1143:hacl(dmz_11,app_23,tcp,8080):1
1083:RULE 5 (multi-hop access):0
938:RULE 5 (multi-hop access):0
1111:RULE 5 (multi-hop access):0
1142:RULE 5 (multi-hop access):0
tcp,8080):0
1188:networkServiceInfo(app_23,tomcat,tcp,8080,root):1
1189:vulExists(app_23,vul_app,tomcat,remoteExploit,privEscalation):1
1135:netAccess(a
1134:RULE 2 (remote exploit of a server program):0

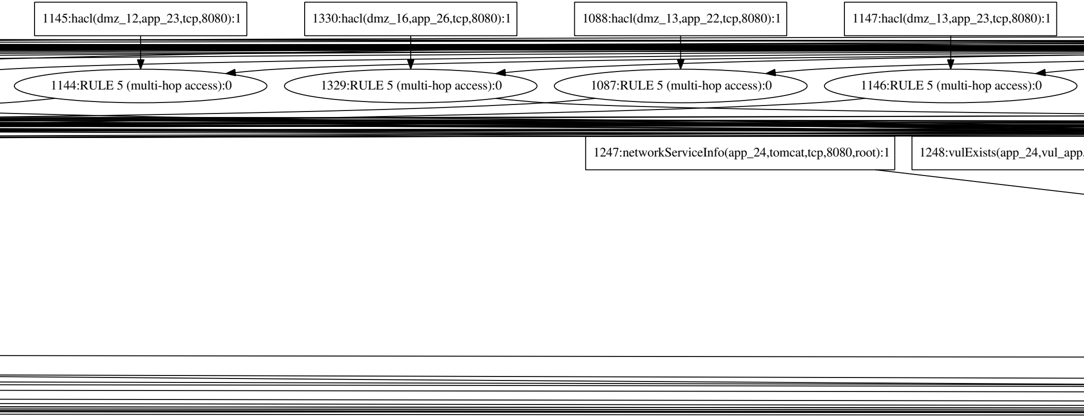

1171:hacl(dmz_24,app_23,tcp,8080):1
1224:hacl(dmz_21,app_24,tcp,8080):1
1275:hacl(dmz_18,app_25,tcp,8080):1
1110:hacl(dmz_23,app_22,tcp,8080):1
1170:RULE 5 (multi-hop access):0
1223:RULE 5 (multi-hop access):0
1274:RULE 5 (multi-hop access):0
1109:RULE 5 (multi-hop access):0
app_23,tcp,8080):0
1014:hacl(app_21,db_51,tcp,5432):1
1015:execCode(app_21,root):0
1013:RULE 5 (multi-hop access):0

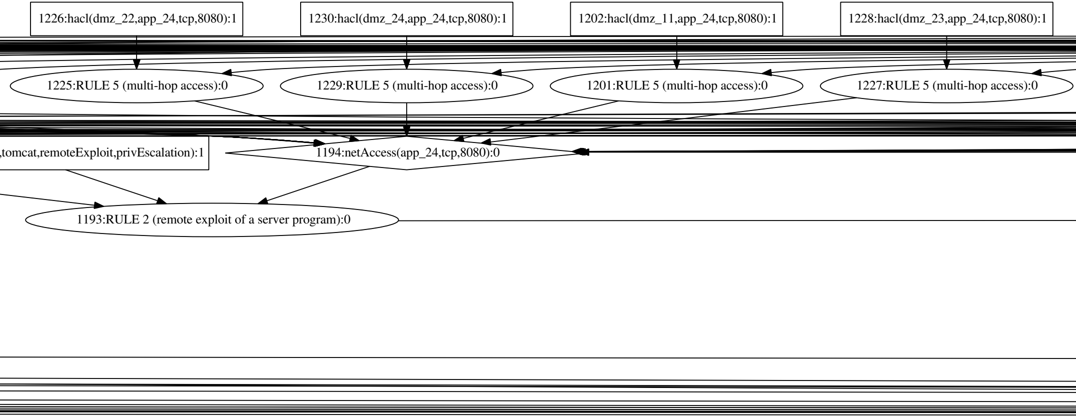

1169:hacl(dmz_23,app_23,tcp,8080):1
1124:hacl(dmz_7,app_22,tcp,8080):1
1183:hacl(dmz_7,app_23,tcp,8080):1
1240:hacl(dmz_6,app_24,tcp,8080):1
1168:RULE 5 (multi-hop access):0
1123:RULE 5 (multi-hop access):0
1182:RULE 5 (multi-hop access):0
1239:RULE 5 (multi-hop access):0

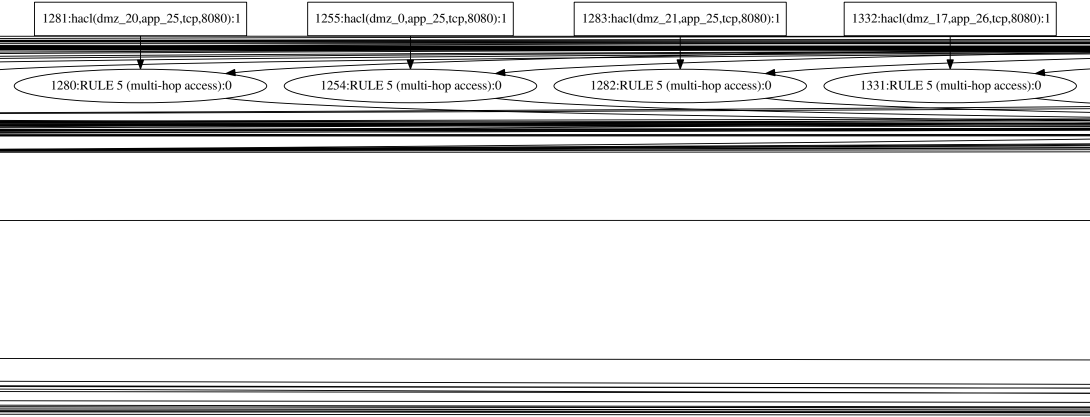

1055:hacl(dmz_25,app_21,tcp,8080):1
998:hacl(dmz_3,app_20,tcp,8080):1
774:hacl(dmz_9,app_17,tcp,8080):1
1198:hacl(dmz_1,app_24,tcp,8080):1
1054:RULE 5 (multi-hop access):0
997:RULE 5 (multi-hop access):0
773:RULE 5 (multi-hop access):0
1197:RULE 5 (multi-hop access):0

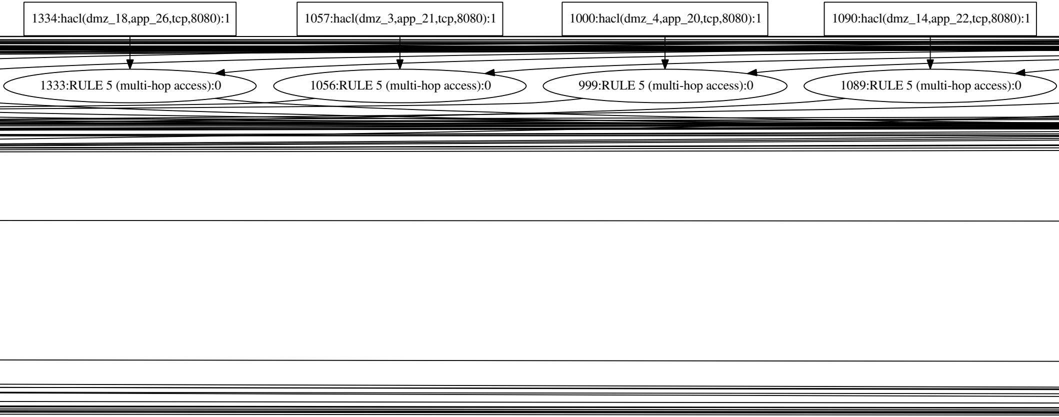

1297:hacl(dmz_5,app_25,tcp,8080):1
1277:hacl(dmz_19,app_25,tcp,8080):1
1126:hacl(dmz_8,app_22,tcp,8080):1
941:hacl(dmz_4,app_2,tcp,8080):1
1296:RULE 5 (multi-hop access):0
1276:RULE 5 (multi-hop access):0
1125:RULE 5 (multi-hop access):0
940:RULE 5 (multi-hop access):0
1073:hacl(app_22,db_51,tcp,5432):1
1074:execCode(app_22,root):0
1072:RULE 5 (multi-hop access):0

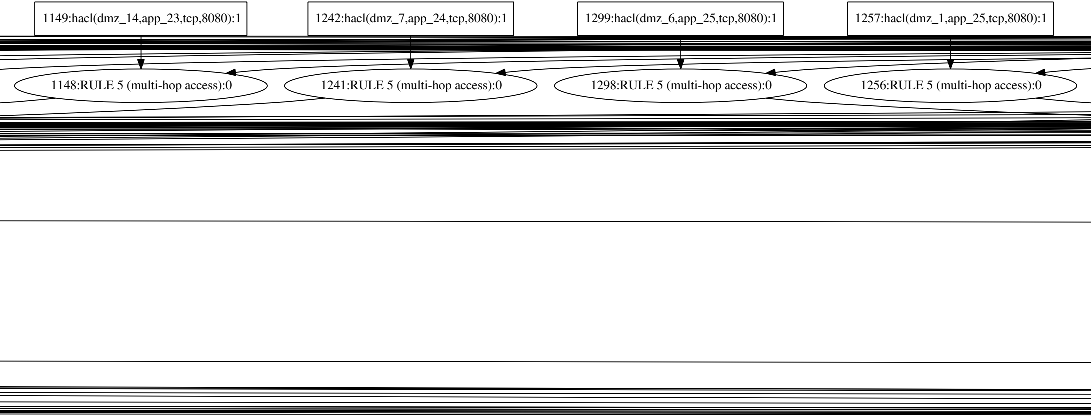

1185:hacl(dmz_8,app_23,tcp,8080):1
1200:hacl(dmz_10,app_24,tcp,8080):1
1279:hacl(dmz_2,app_25,tcp,8080):1
1086:hacl(dmz_12,app_22,tcp,8080):1
1184:RULE 5 (multi-hop access):0
1199:RULE 5 (multi-hop access):0
1278:RULE 5 (multi-hop access):0
1085:RULE 5 (multi-hop access):0
1132:hacl(app_23,db_51,tcp,5432):1
1133:execCode(app_23,root):0
1131:RULE 5 (multi-hop access):0

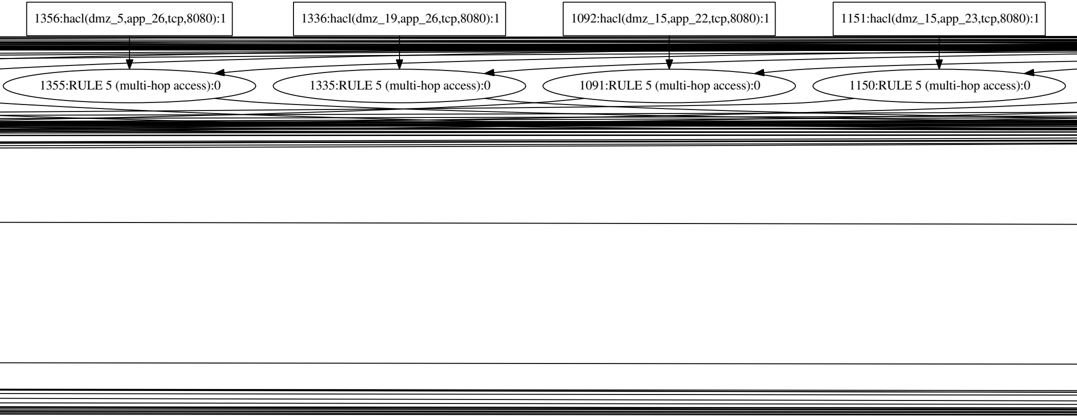

1145:hacl(dmz_12,app_23,tcp,8080):1
1330:hacl(dmz_16,app_26,tcp,8080):1
1088:hacl(dmz_13,app_22,tcp,8080):1
1147:hacl(dmz_13,app_23,tcp,8080):1
1144:RULE 5 (multi-hop access):0
1329:RULE 5 (multi-hop access):0
1087:RULE 5 (multi-hop access):0
1146:RULE 5 (multi-hop access):0
1247:networkServiceInfo(app_24,tomcat,tcp,8080,root):1
1248:vulExists(app_24,vul_app

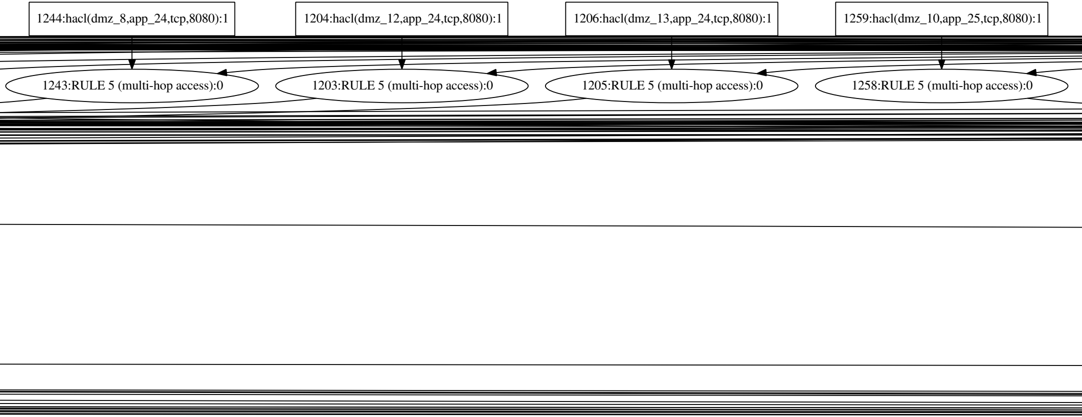

1226:hacl(dmz_22,app_24,tcp,8080):1
1230:hacl(dmz_24,app_24,tcp,8080):1
1202:hacl(dmz_11,app_24,tcp,8080):1
1228:hacl(dmz_23,app_24,tcp,8080):1
1225:RULE 5 (multi-hop access):0
1229:RULE 5 (multi-hop access):0
1201:RULE 5 (multi-hop access):0
1227:RULE 5 (multi-hop access):0
,tomcat,remoteExploit,privEscalation):1
1194:netAccess(app_24,tcp,8080):0
1193:RULE 2 (remote exploit of a server program):0

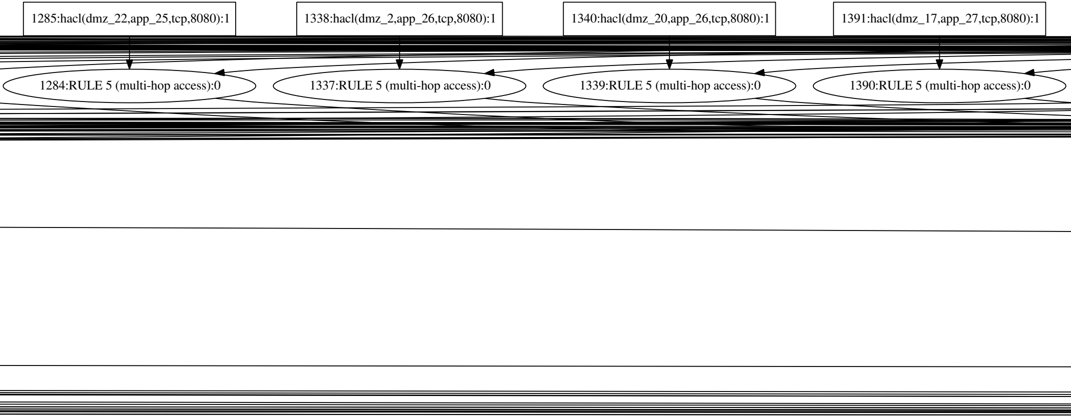

1281:hacl(dmz_20,app_25,tcp,8080):1
1255:hacl(dmz_0,app_25,tcp,8080):1
1283:hacl(dmz_21,app_25,tcp,8080):1
1332:hacl(dmz_17,app_26,tcp,8080):1
1280:RULE 5 (multi-hop access):0
1254:RULE 5 (multi-hop access):0
1282:RULE 5 (multi-hop access):0
1331:RULE 5 (multi-hop access):0

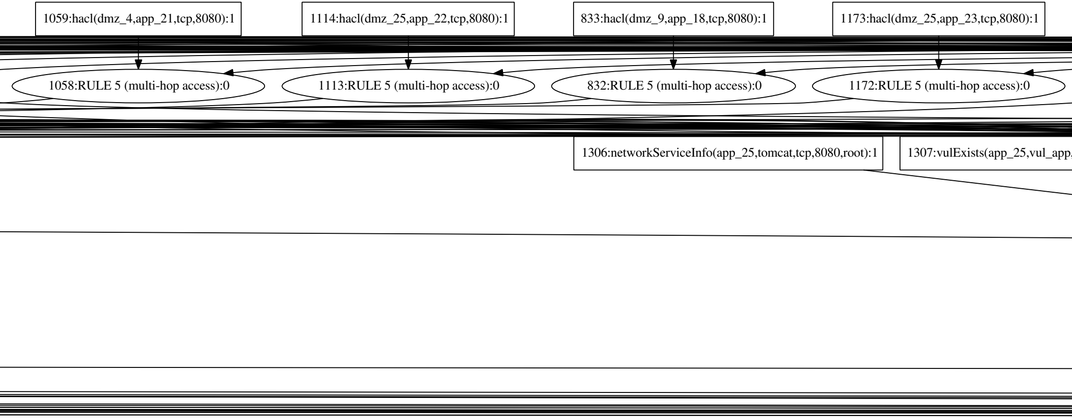

1334:hacl(dmz_18,app_26,tcp,8080):1
1057:hacl(dmz_3,app_21,tcp,8080):1
1000:hacl(dmz_4,app_20,tcp,8080):1
1090:hacl(dmz_14,app_22,tcp,8080):1
1333:RULE 5 (multi-hop access):0
1056:RULE 5 (multi-hop access):0
999:RULE 5 (multi-hop access):0
1089:RULE 5 (multi-hop access):0

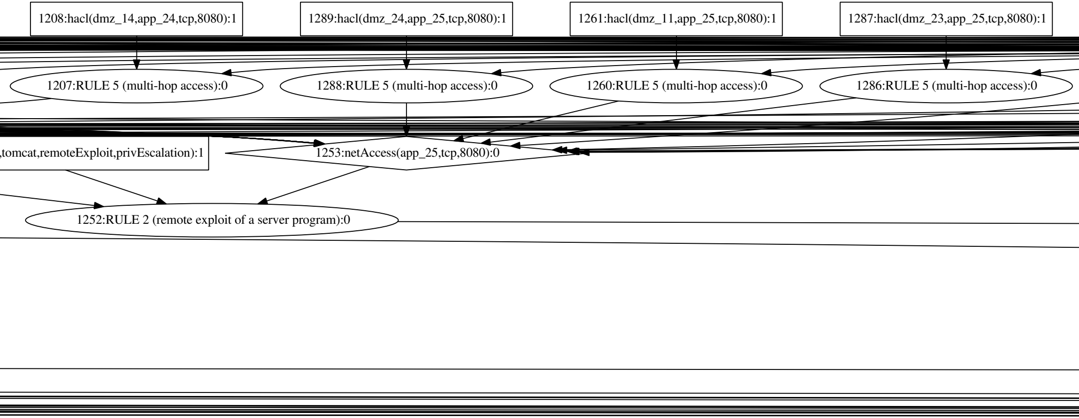

1149:hacl(dmz_14,app_23,tcp,8080):1
1242:hacl(dmz_7,app_24,tcp,8080):1
1299:hacl(dmz_6,app_25,tcp,8080):1
1257:hacl(dmz_1,app_25,tcp,8080):1
1148:RULE 5 (multi-hop access):0
1241:RULE 5 (multi-hop access):0
1298:RULE 5 (multi-hop access):0
1256:RULE 5 (multi-hop access):0

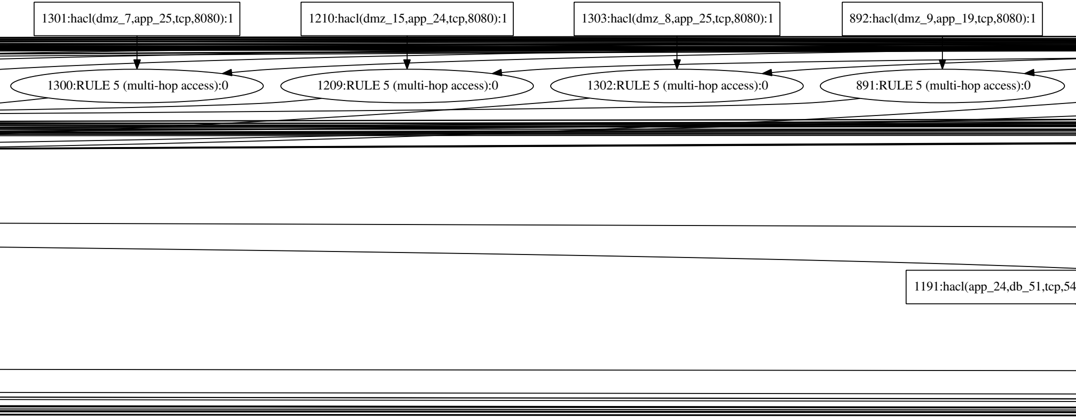

1356:hacl(dmz_5,app_26,tcp,8080):1
1336:hacl(dmz_19,app_26,tcp,8080):1
1092:hacl(dmz_15,app_22,tcp,8080):1
1151:hacl(dmz_15,app_23,tcp,8080):1
1355:RULE 5 (multi-hop access):0
1335:RULE 5 (multi-hop access):0
1091:RULE 5 (multi-hop access):0
1150:RULE 5 (multi-hop access):0

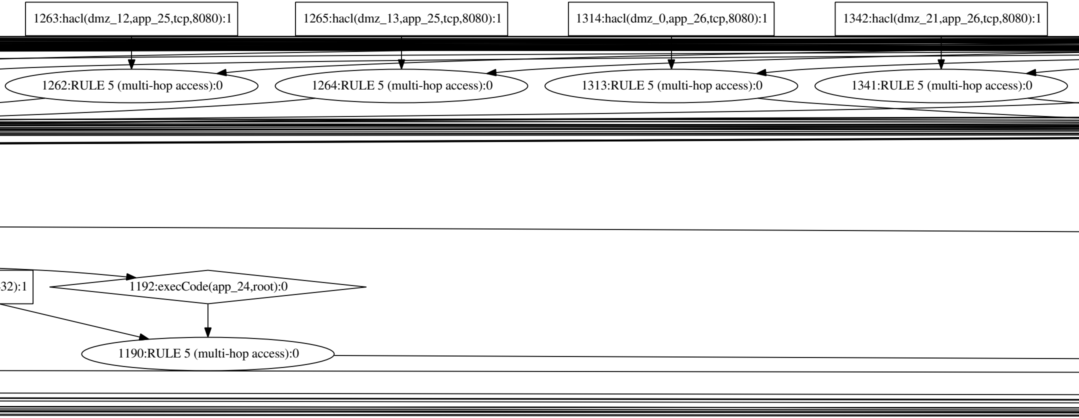

1244:hacl(dmz_8,app_24,tcp,8080):1
1204:hacl(dmz_12,app_24,tcp,8080):1
1206:hacl(dmz_13,app_24,tcp,8080):1
1259:hacl(dmz_10,app_25,tcp,8080):1
1243:RULE 5 (multi-hop access):0
1203:RULE 5 (multi-hop access):0
1205:RULE 5 (multi-hop access):0
1258:RULE 5 (multi-hop access):0

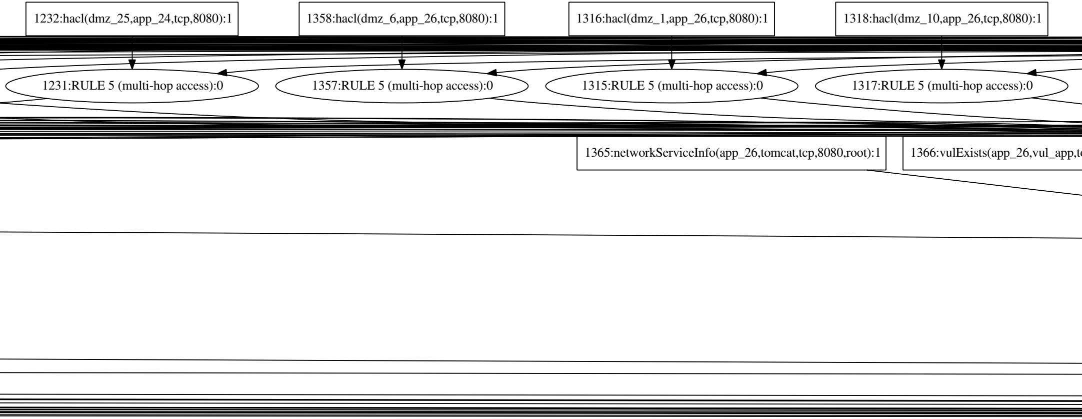

1285:hacl(dmz_22,app_25,tcp,8080):1
1338:hacl(dmz_2,app_26,tcp,8080):1
1340:hacl(dmz_20,app_26,tcp,8080):1
1391:hacl(dmz_17,app_27,tcp,8080):1
1284:RULE 5 (multi-hop access):0
1337:RULE 5 (multi-hop access):0
1339:RULE 5 (multi-hop access):0
1390:RULE 5 (multi-hop access):0

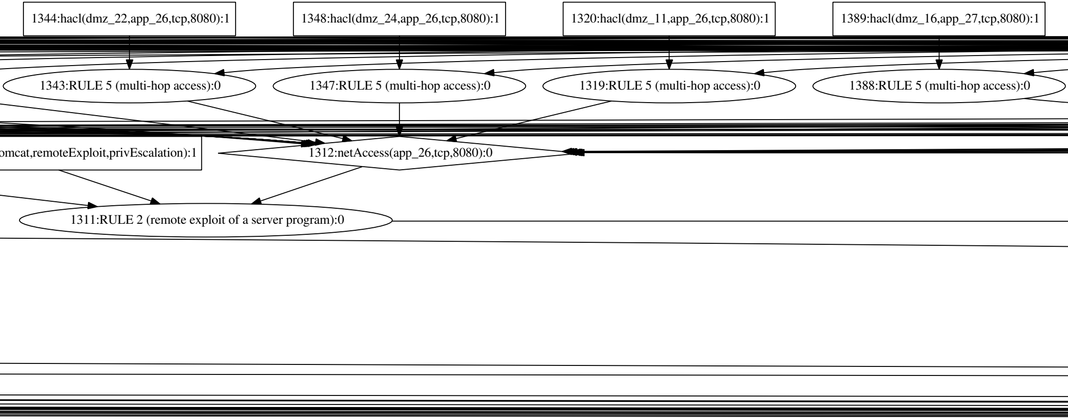

1059:hacl(dmz_4,app_21,tcp,8080):1
1114:hacl(dmz_25,app_22,tcp,8080):1
833:hacl(dmz_9,app_18,tcp,8080):1
1173:hacl(dmz_25,app_23,tcp,8080):1
1058:RULE 5 (multi-hop access):0
1113:RULE 5 (multi-hop access):0
832:RULE 5 (multi-hop access):0
1172:RULE 5 (multi-hop access):0
1306:networkServiceInfo(app_25,tomcat,tcp,8080,root):1
1307:vulExists(app_25,vul_app,

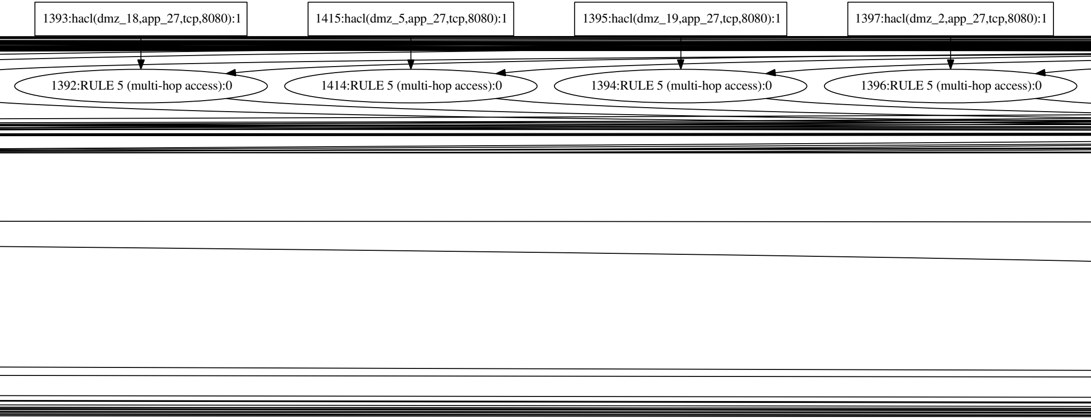

1208:hacl(dmz_14,app_24,tcp,8080):1
1289:hacl(dmz_24,app_25,tcp,8080):1
1261:hacl(dmz_11,app_25,tcp,8080):1
1287:hacl(dmz_23,app_25,tcp,8080):1
1207:RULE 5 (multi-hop access):0
1288:RULE 5 (multi-hop access):0
1260:RULE 5 (multi-hop access):0
1286:RULE 5 (multi-hop access):0
,tomcat,remoteExploit,privEscalation):1
1253:netAccess(app_25,tcp,8080):0
1252:RULE 2 (remote exploit of a server program):0

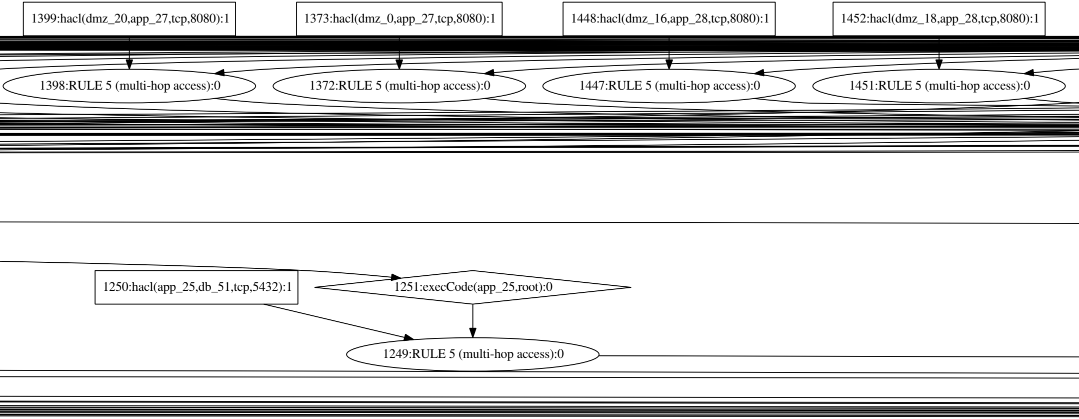

1301:hacl(dmz_7,app_25,tcp,8080):1
1210:hacl(dmz_15,app_24,tcp,8080):1
1303:hacl(dmz_8,app_25,tcp,8080):1
892:hacl(dmz_9,app_19,tcp,8080):1
1300:RULE 5 (multi-hop access):0
1209:RULE 5 (multi-hop access):0
1302:RULE 5 (multi-hop access):0
891:RULE 5 (multi-hop access):0
1191:hacl(app_24,db_51,tcp,54

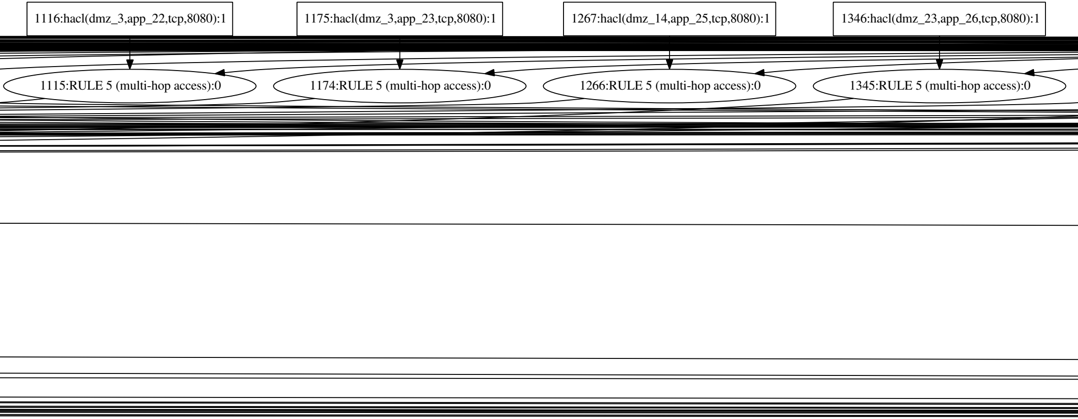

1263:hacl(dmz_12,app_25,tcp,8080):1
1265:hacl(dmz_13,app_25,tcp,8080):1
1314:hacl(dmz_0,app_26,tcp,8080):1
1342:hacl(dmz_21,app_26,tcp,8080):1
1262:RULE 5 (multi-hop access):0
1264:RULE 5 (multi-hop access):0
1313:RULE 5 (multi-hop access):0
1341:RULE 5 (multi-hop access):0
32):1
1192:execCode(app_24,root):0
1190:RULE 5 (multi-hop access):0

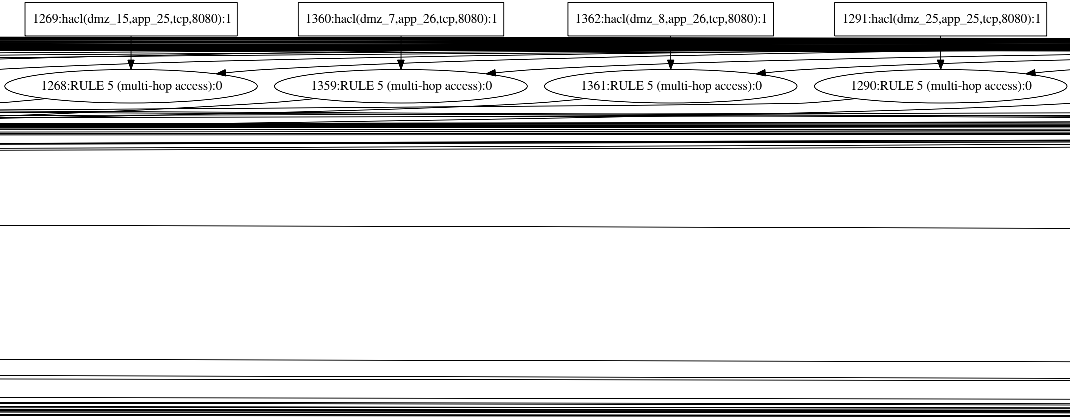

1232:hacl(dmz_25,app_24,tcp,8080):1
1358:hacl(dmz_6,app_26,tcp,8080):1
1316:hacl(dmz_1,app_26,tcp,8080):1
1318:hacl(dmz_10,app_26,tcp,8080):1
1231:RULE 5 (multi-hop access):0
1357:RULE 5 (multi-hop access):0
1315:RULE 5 (multi-hop access):0
1317:RULE 5 (multi-hop access):0
1365:networkServiceInfo(app_26,tomcat,tcp,8080,root):1
1366:vulExists(app_26,vul_app,t

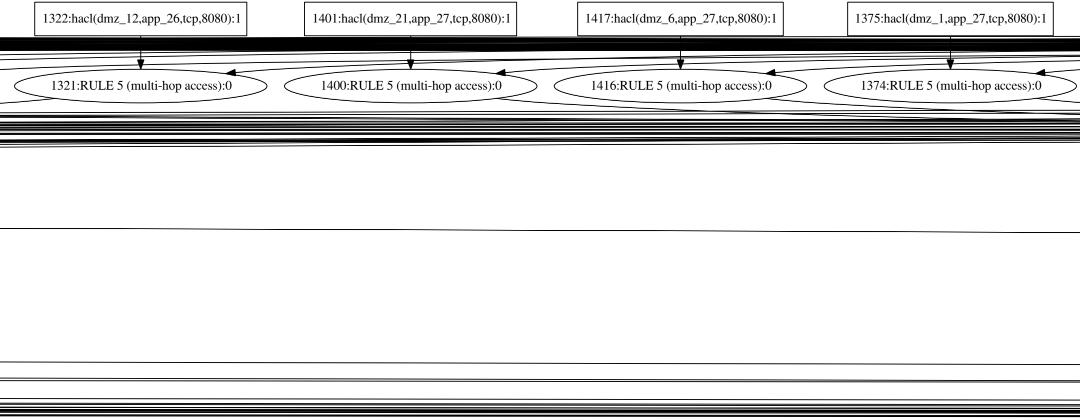

1344:hacl(dmz_22,app_26,tcp,8080):1
1348:hacl(dmz_24,app_26,tcp,8080):1
1320:hacl(dmz_11,app_26,tcp,8080):1
1389:hacl(dmz_16,app_27,tcp,8080):1
1343:RULE 5 (multi-hop access):0
1347:RULE 5 (multi-hop access):0
1319:RULE 5 (multi-hop access):0
1388:RULE 5 (multi-hop access):0
omcat,remoteExploit,privEscalation):1
1312:netAccess(app_26,tcp,8080):0
1311:RULE 2 (remote exploit of a server program):0

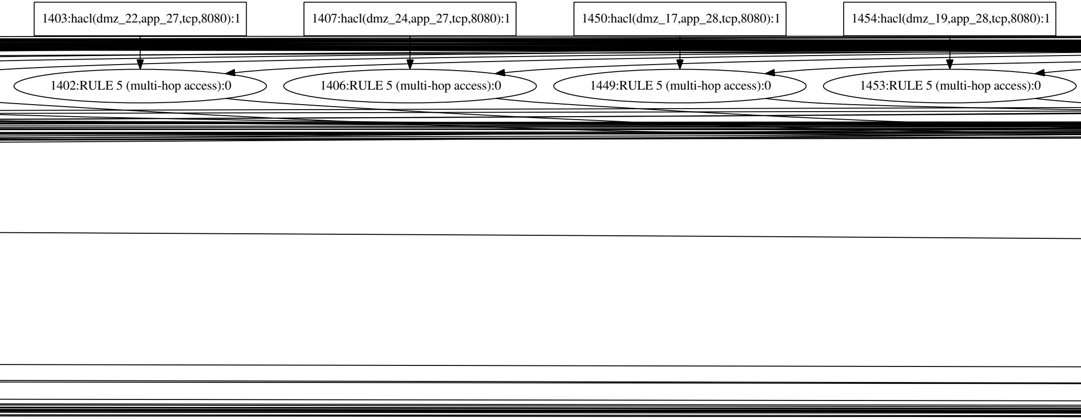

1393:hacl(dmz_18,app_27,tcp,8080):1
1415:hacl(dmz_5,app_27,tcp,8080):1
1395:hacl(dmz_19,app_27,tcp,8080):1
1397:hacl(dmz_2,app_27,tcp,8080):1
1392:RULE 5 (multi-hop access):0
1414:RULE 5 (multi-hop access):0
1394:RULE 5 (multi-hop access):0
1396:RULE 5 (multi-hop access):0

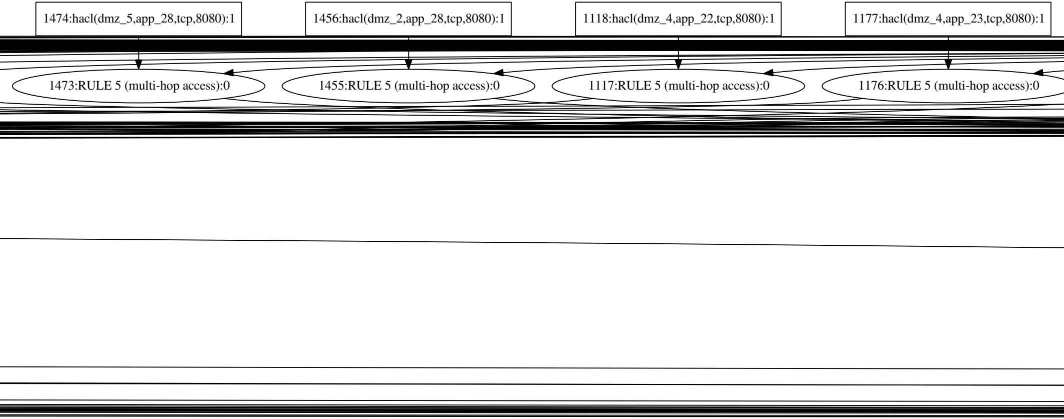

1399:hacl(dmz_20,app_27,tcp,8080):1
1373:hacl(dmz_0,app_27,tcp,8080):1
1448:hacl(dmz_16,app_28,tcp,8080):1
1452:hacl(dmz_18,app_28,tcp,8080):1
1398:RULE 5 (multi-hop access):0
1372:RULE 5 (multi-hop access):0
1447:RULE 5 (multi-hop access):0
1451:RULE 5 (multi-hop access):0
1250:hacl(app_25,db_51,tcp,5432):1
1251:execCode(app_25,root):0
1249:RULE 5 (multi-hop access):0

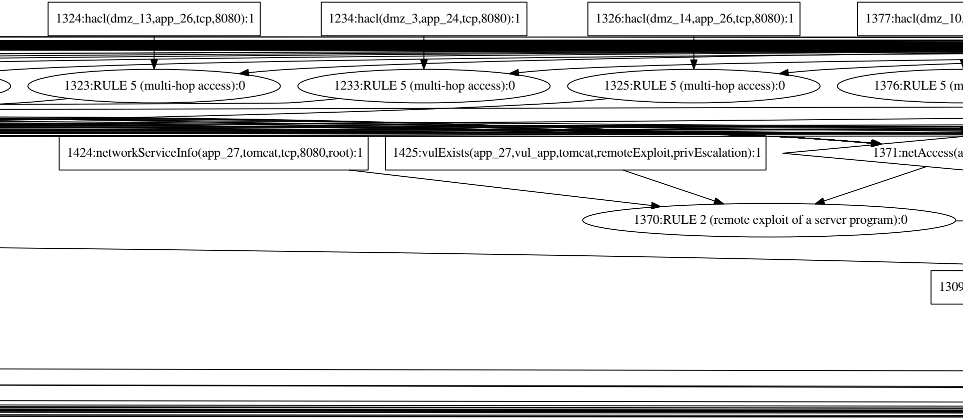

1116:hacl(dmz_3,app_22,tcp,8080):1
1175:hacl(dmz_3,app_23,tcp,8080):1
1267:hacl(dmz_14,app_25,tcp,8080):1
1346:hacl(dmz_23,app_26,tcp,8080):1
1115:RULE 5 (multi-hop access):0
1174:RULE 5 (multi-hop access):0
1266:RULE 5 (multi-hop access):0
1345:RULE 5 (multi-hop access):0

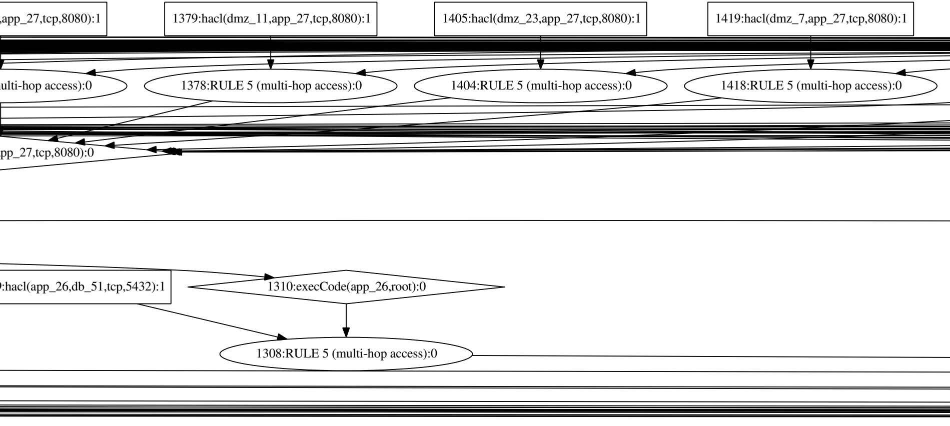

1269:hacl(dmz_15,app_25,tcp,8080):1
1360:hacl(dmz_7,app_26,tcp,8080):1
1362:hacl(dmz_8,app_26,tcp,8080):1
1291:hacl(dmz_25,app_25,tcp,8080):1
1268:RULE 5 (multi-hop access):0
1359:RULE 5 (multi-hop access):0
1361:RULE 5 (multi-hop access):0
1290:RULE 5 (multi-hop access):0

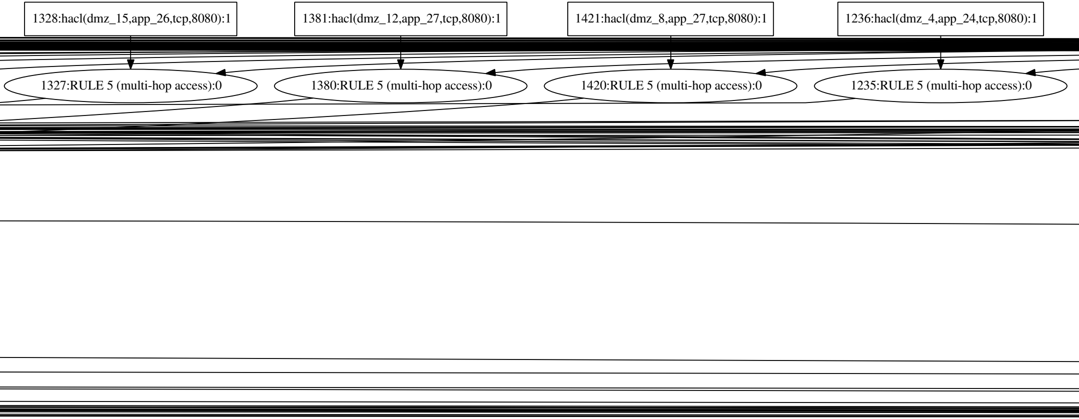

1322:hacl(dmz_12,app_26,tcp,8080):1
1401:hacl(dmz_21,app_27,tcp,8080):1
1417:hacl(dmz_6,app_27,tcp,8080):1
1375:hacl(dmz_1,app_27,tcp,8080):1
1321:RULE 5 (multi-hop access):0
1400:RULE 5 (multi-hop access):0
1416:RULE 5 (multi-hop access):0
1374:RULE 5 (multi-hop access):0

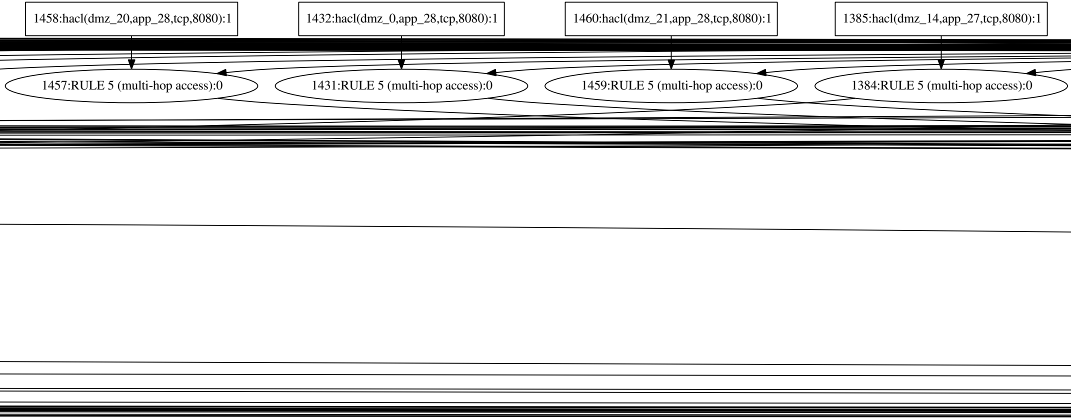

1403:hacl(dmz_22,app_27,tcp,8080):1
1407:hacl(dmz_24,app_27,tcp,8080):1
1450:hacl(dmz_17,app_28,tcp,8080):1
1454:hacl(dmz_19,app_28,tcp,8080):1
1402:RULE 5 (multi-hop access):0
1406:RULE 5 (multi-hop access):0
1449:RULE 5 (multi-hop access):0
1453:RULE 5 (multi-hop access):0

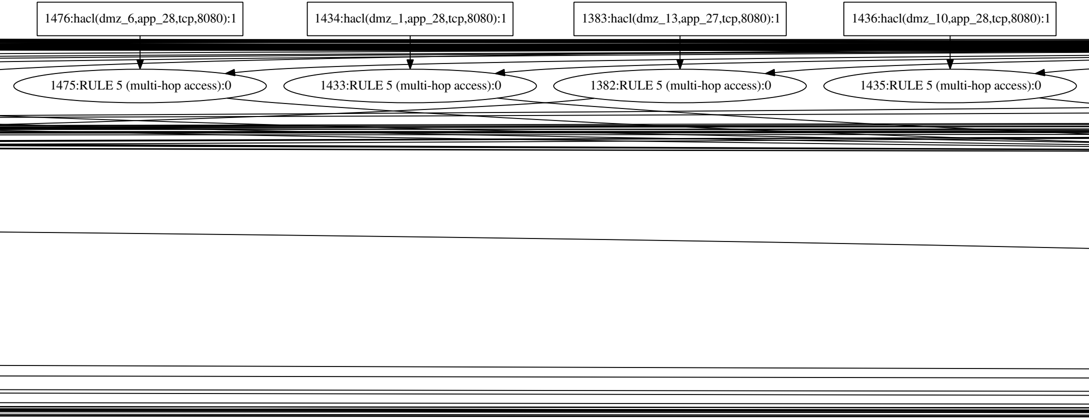

1474:hacl(dmz_5,app_28,tcp,8080):1
1456:hacl(dmz_2,app_28,tcp,8080):1
1118:hacl(dmz_4,app_22,tcp,8080):1
1177:hacl(dmz_4,app_23,tcp,8080):1
1473:RULE 5 (multi-hop access):0
1455:RULE 5 (multi-hop access):0
1117:RULE 5 (multi-hop access):0
1176:RULE 5 (multi-hop access):0

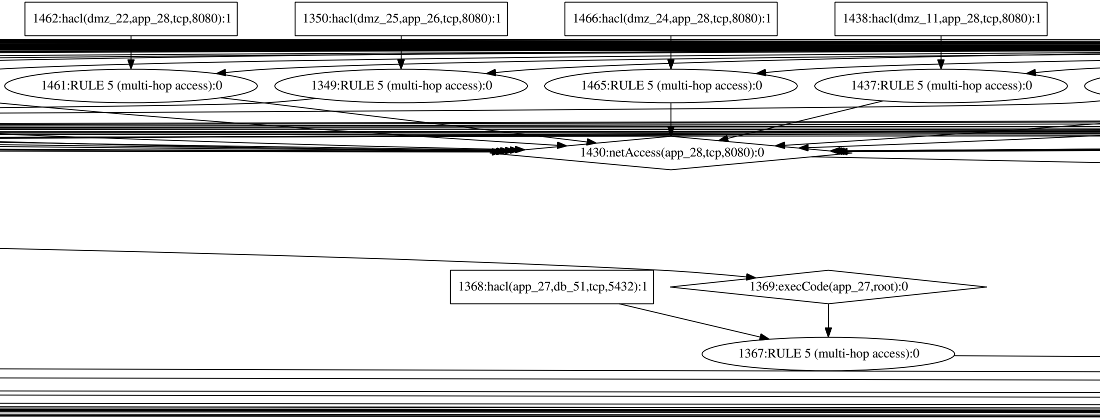

1324:hacl(dmz_13,app_26,tcp,8080):1
1234:hacl(dmz_3,app_24,tcp,8080):1
1326:hacl(dmz_14,app_26,tcp,8080):1
1377:hacl(dmz_10,
1323:RULE 5 (multi-hop access):0
1233:RULE 5 (multi-hop access):0
1325:RULE 5 (multi-hop access):0
1376:RULE 5 (m
1424:networkServiceInfo(app_27,tomcat,tcp,8080,root):1
1425:vulExists(app_27,vul_app,tomcat,remoteExploit,privEscalation):1
1371:netAccess(a
1370:RULE 2 (remote exploit of a server program):0
1309

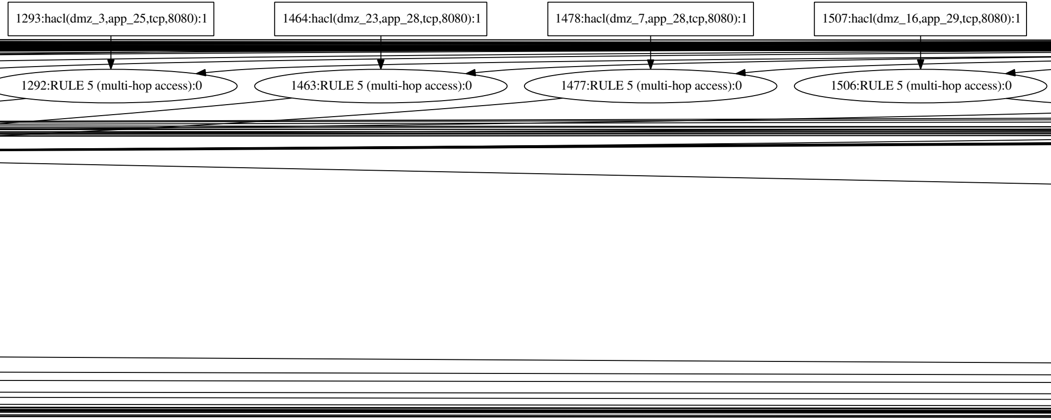

,app_27,tcp,8080):1
1379:hacl(dmz_11,app_27,tcp,8080):1
1405:hacl(dmz_23,app_27,tcp,8080):1
1419:hacl(dmz_7,app_27,tcp,8080):1
ulti-hop access):0
1378:RULE 5 (multi-hop access):0
1404:RULE 5 (multi-hop access):0
1418:RULE 5 (multi-hop access):0
pp_27,tcp,8080):0
9:hacl(app_26,db_51,tcp,5432):1
1310:execCode(app_26,root):0
1308:RULE 5 (multi-hop access):0

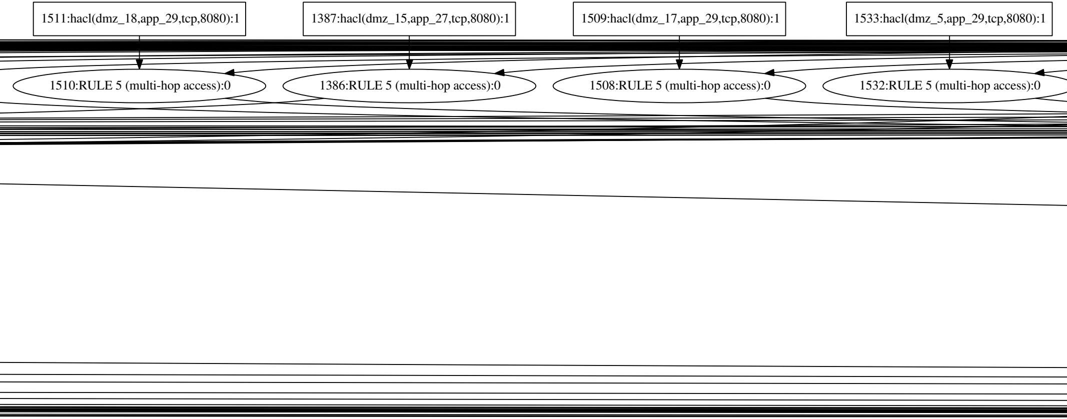

1328:hacl(dmz_15,app_26,tcp,8080):1
1381:hacl(dmz_12,app_27,tcp,8080):1
1421:hacl(dmz_8,app_27,tcp,8080):1
1236:hacl(dmz_4,app_24,tcp,8080):1
1327:RULE 5 (multi-hop access):0
1380:RULE 5 (multi-hop access):0
1420:RULE 5 (multi-hop access):0
1235:RULE 5 (multi-hop access):0

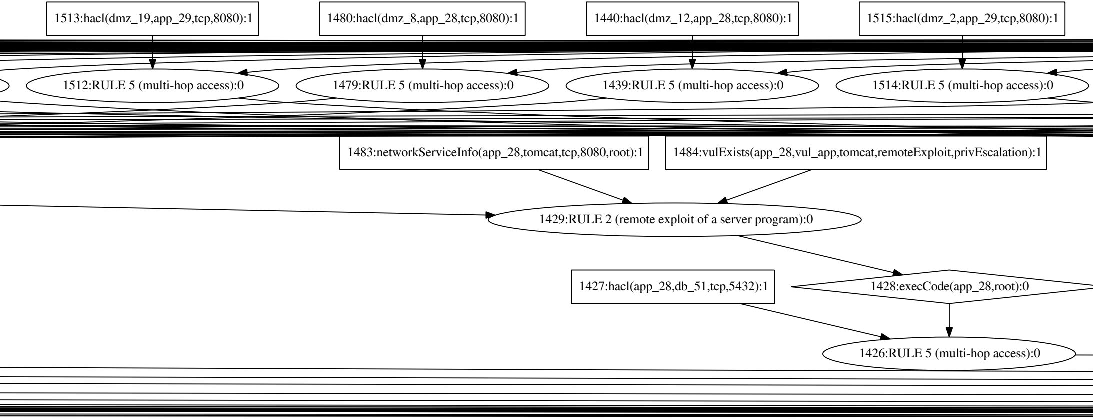

1458:hacl(dmz_20,app_28,tcp,8080):1
1432:hacl(dmz_0,app_28,tcp,8080):1
1460:hacl(dmz_21,app_28,tcp,8080):1
1385:hacl(dmz_14,app_27,tcp,8080):1
1457:RULE 5 (multi-hop access):0
1431:RULE 5 (multi-hop access):0
1459:RULE 5 (multi-hop access):0
1384:RULE 5 (multi-hop access):0

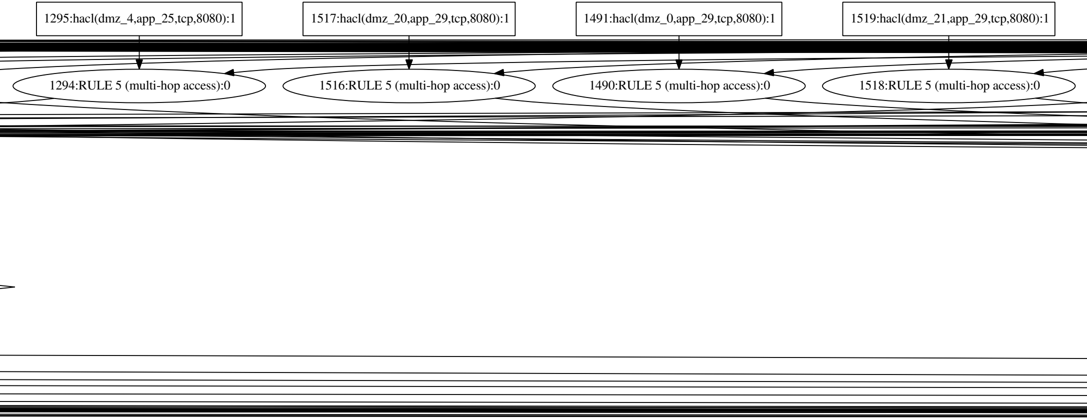

1476:hacl(dmz_6,app_28,tcp,8080):1
1434:hacl(dmz_1,app_28,tcp,8080):1
1383:hacl(dmz_13,app_27,tcp,8080):1
1436:hacl(dmz_10,app_28,tcp,8080):1
1475:RULE 5 (multi-hop access):0
1433:RULE 5 (multi-hop access):0
1382:RULE 5 (multi-hop access):0
1435:RULE 5 (multi-hop access):0

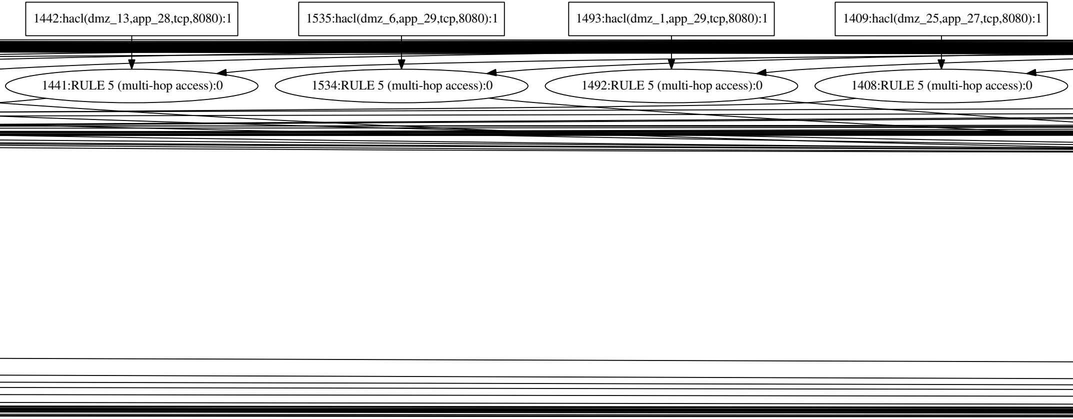

1462:hacl(dmz_22,app_28,tcp,8080):1
1350:hacl(dmz_25,app_26,tcp,8080):1
1466:hacl(dmz_24,app_28,tcp,8080):1
1438:hacl(dmz_11,app_28,tcp,8080):1
1461:RULE 5 (multi-hop access):0
1349:RULE 5 (multi-hop access):0
1465:RULE 5 (multi-hop access):0
1437:RULE 5 (multi-hop access):0
1430:netAccess(app_28,tcp,8080):0
1368:hacl(app_27,db_51,tcp,5432):1
1369:execCode(app_27,root):0
1367:RULE 5 (multi-hop access):0

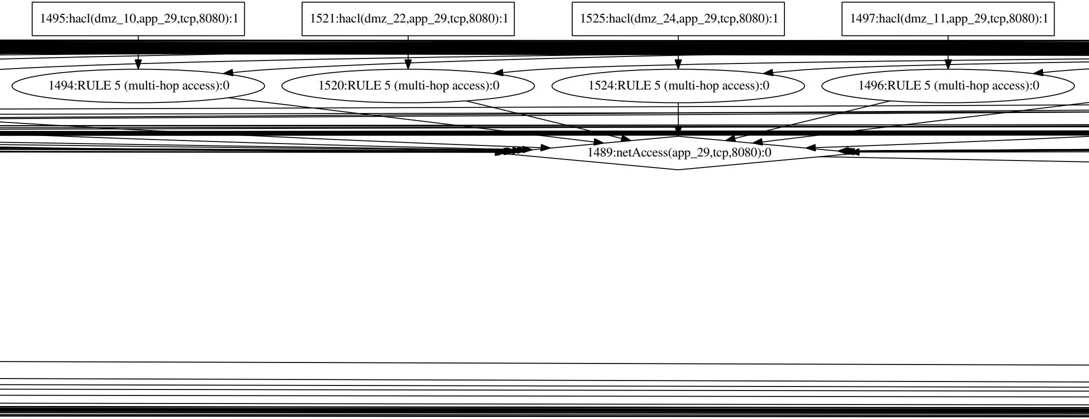

1293:hacl(dmz_3,app_25,tcp,8080):1
1464:hacl(dmz_23,app_28,tcp,8080):1
1478:hacl(dmz_7,app_28,tcp,8080):1
1507:hacl(dmz_16,app_29,tcp,8080):1
1292:RULE 5 (multi-hop access):0
1463:RULE 5 (multi-hop access):0
1477:RULE 5 (multi-hop access):0
1506:RULE 5 (multi-hop access):0

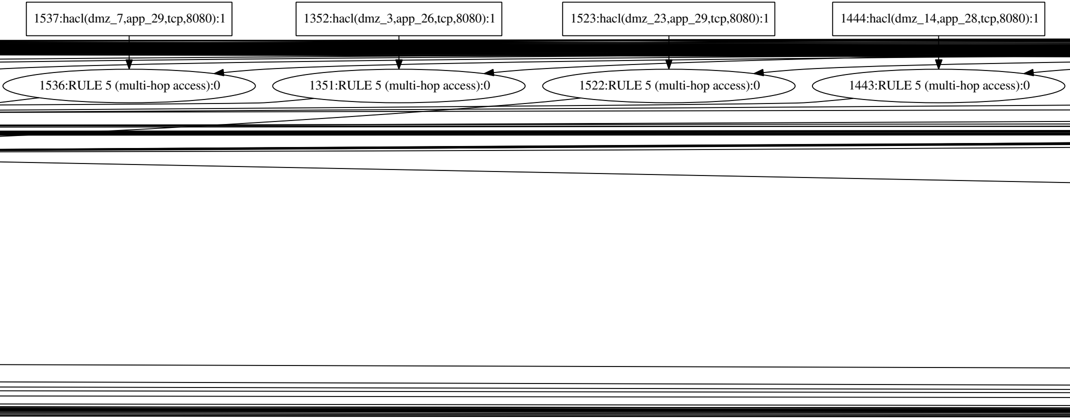

1511:hacl(dmz_18,app_29,tcp,8080):1
1387:hacl(dmz_15,app_27,tcp,8080):1
1509:hacl(dmz_17,app_29,tcp,8080):1
1533:hacl(dmz_5,app_29,tcp,8080):1
1510:RULE 5 (multi-hop access):0
1386:RULE 5 (multi-hop access):0
1508:RULE 5 (multi-hop access):0
1532:RULE 5 (multi-hop access):0

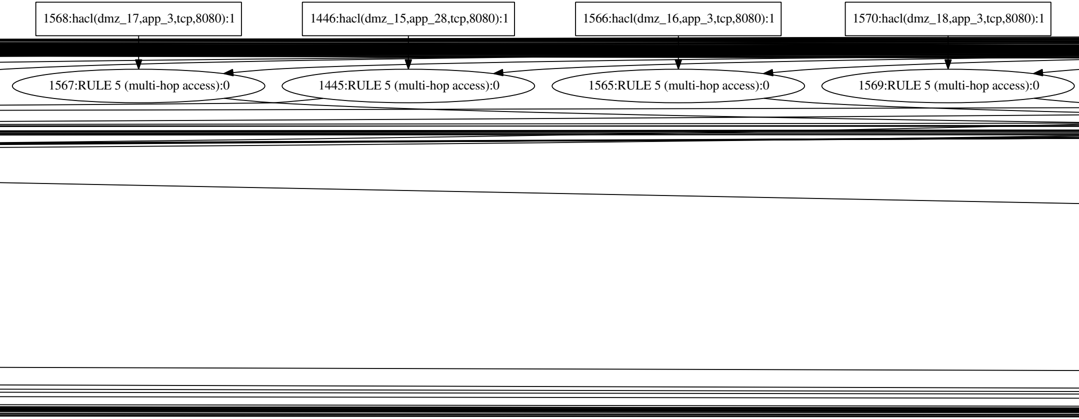

1513:hacl(dmz_19,app_29,tcp,8080):1
1480:hacl(dmz_8,app_28,tcp,8080):1
1440:hacl(dmz_12,app_28,tcp,8080):1
1515:hacl(dmz_2,app_29,tcp,8080):1
1512:RULE 5 (multi-hop access):0
1479:RULE 5 (multi-hop access):0
1439:RULE 5 (multi-hop access):0
1514:RULE 5 (multi-hop access):0
1483:networkServiceInfo(app_28,tomcat,tcp,8080,root):1
1484:vulExists(app_28,vul_app,tomcat,remoteExploit,privEscalation):1
1429:RULE 2 (remote exploit of a server program):0
1427:hacl(app_28,db_51,tcp,5432):1
1428:execCode(app_28,root):0
1426:RULE 5 (multi-hop access):0

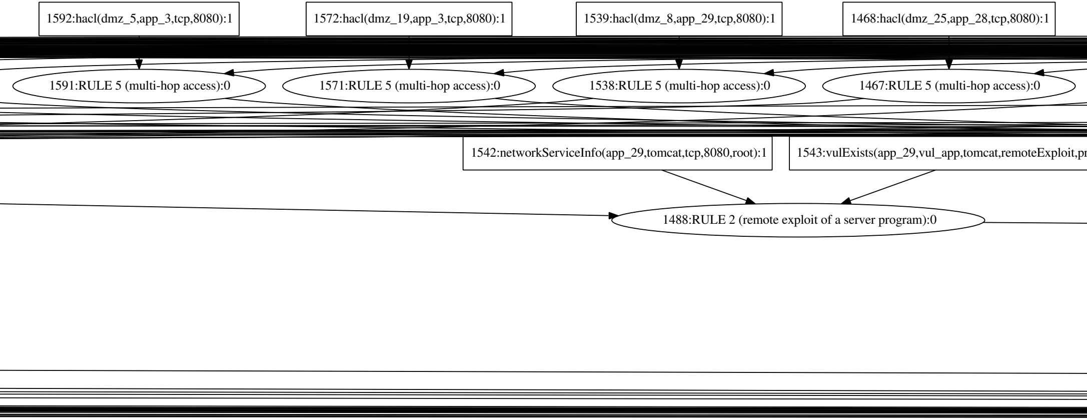

1295:hacl(dmz_4,app_25,tcp,8080):1
1517:hacl(dmz_20,app_29,tcp,8080):1
1491:hacl(dmz_0,app_29,tcp,8080):1
1519:hacl(dmz_21,app_29,tcp,8080):1
1294:RULE 5 (multi-hop access):0
1516:RULE 5 (multi-hop access):0
1490:RULE 5 (multi-hop access):0
1518:RULE 5 (multi-hop access):0

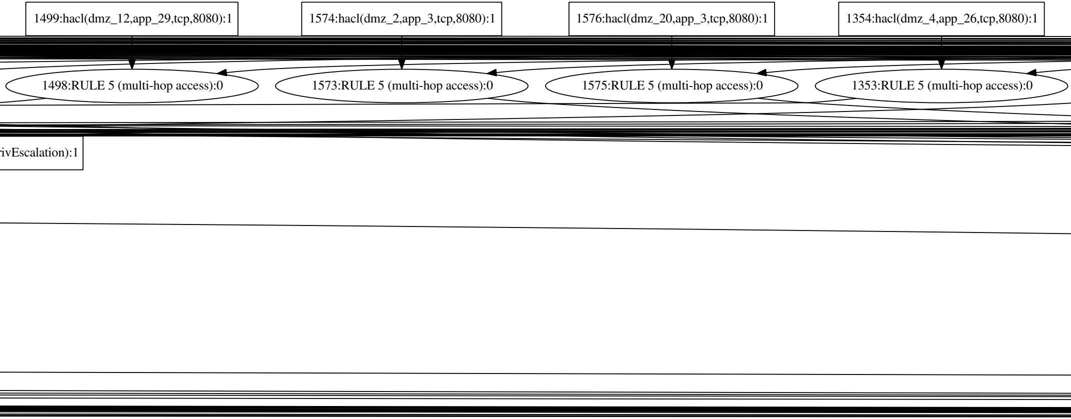

1442:hacl(dmz_13,app_28,tcp,8080):1
1535:hacl(dmz_6,app_29,tcp,8080):1
1493:hacl(dmz_1,app_29,tcp,8080):1
1409:hacl(dmz_25,app_27,tcp,8080):1
1441:RULE 5 (multi-hop access):0
1534:RULE 5 (multi-hop access):0
1492:RULE 5 (multi-hop access):0
1408:RULE 5 (multi-hop access):0

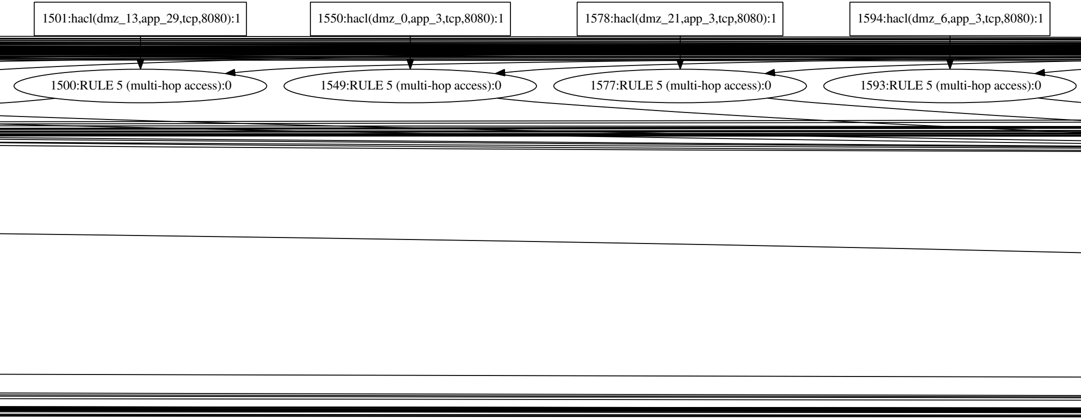

1495:hacl(dmz_10,app_29,tcp,8080):1
1521:hacl(dmz_22,app_29,tcp,8080):1
1525:hacl(dmz_24,app_29,tcp,8080):1
1497:hacl(dmz_11,app_29,tcp,8080):1
1494:RULE 5 (multi-hop access):0
1520:RULE 5 (multi-hop access):0
1524:RULE 5 (multi-hop access):0
1496:RULE 5 (multi-hop access):0
1489:netAccess(app_29,tcp,8080):0

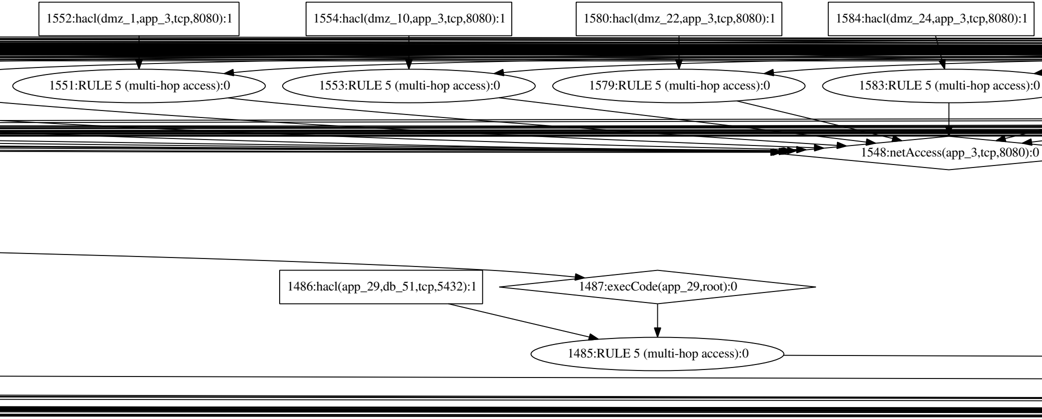

1537:hacl(dmz_7,app_29,tcp,8080):1
1352:hacl(dmz_3,app_26,tcp,8080):1
1523:hacl(dmz_23,app_29,tcp,8080):1
1444:hacl(dmz_14,app_28,tcp,8080):1
1536:RULE 5 (multi-hop access):0
1351:RULE 5 (multi-hop access):0
1522:RULE 5 (multi-hop access):0
1443:RULE 5 (multi-hop access):0

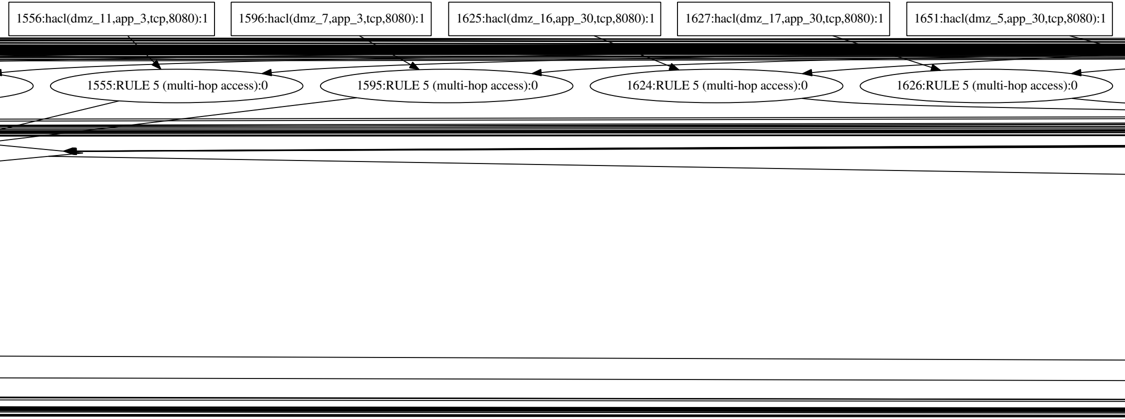

1568:hacl(dmz_17,app_3,tcp,8080):1
1446:hacl(dmz_15,app_28,tcp,8080):1
1566:hacl(dmz_16,app_3,tcp,8080):1
1570:hacl(dmz_18,app_3,tcp,8080):1
1567:RULE 5 (multi-hop access):0
1445:RULE 5 (multi-hop access):0
1565:RULE 5 (multi-hop access):0
1569:RULE 5 (multi-hop access):0

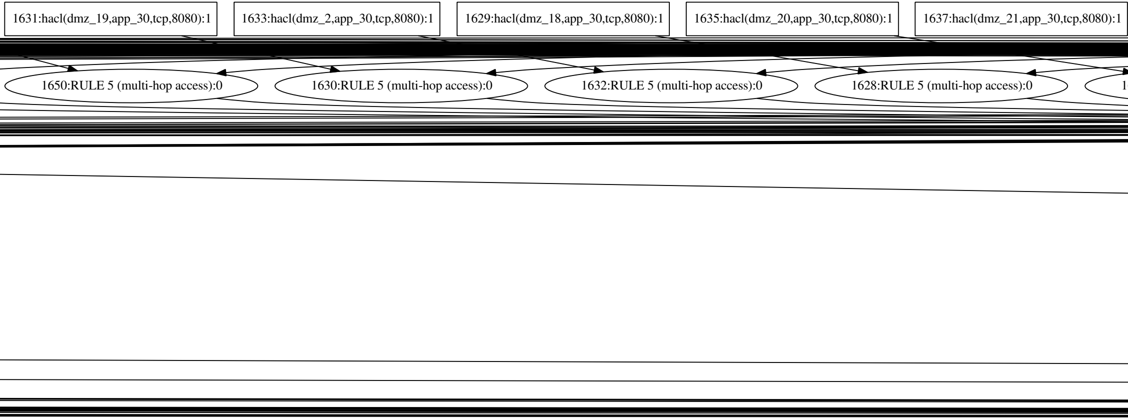

1592:hacl(dmz_5,app_3,tcp,8080):1
1572:hacl(dmz_19,app_3,tcp,8080):1
1539:hacl(dmz_8,app_29,tcp,8080):1
1468:hacl(dmz_25,app_28,tcp,8080):1
1591:RULE 5 (multi-hop access):0
1571:RULE 5 (multi-hop access):0
1538:RULE 5 (multi-hop access):0
1467:RULE 5 (multi-hop access):0
1542:networkServiceInfo(app_29,tomcat,tcp,8080,root):1
1543:vulExists(app_29,vul_app,tomcat,remoteExploit,pr
1488:RULE 2 (remote exploit of a server program):0

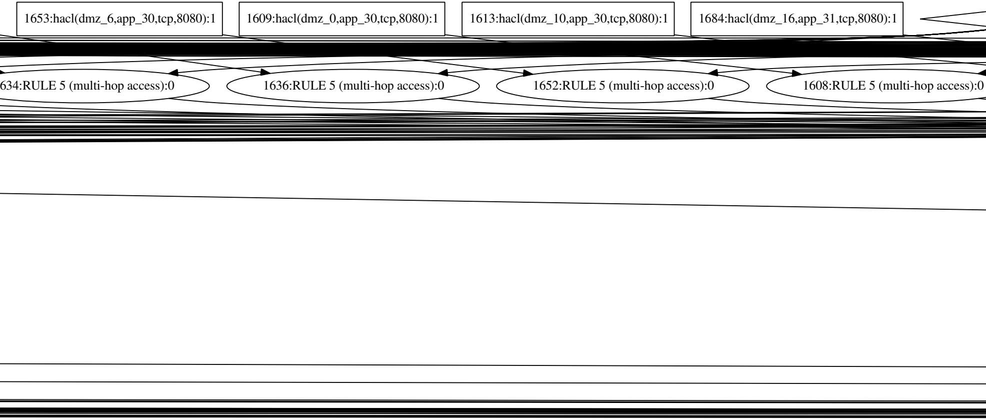

1499:hacl(dmz_12,app_29,tcp,8080):1
1574:hacl(dmz_2,app_3,tcp,8080):1
1576:hacl(dmz_20,app_3,tcp,8080):1
1354:hacl(dmz_4,app_26,tcp,8080):1
1498:RULE 5 (multi-hop access):0
1573:RULE 5 (multi-hop access):0
1575:RULE 5 (multi-hop access):0
1353:RULE 5 (multi-hop access):0
rivEscalation):1

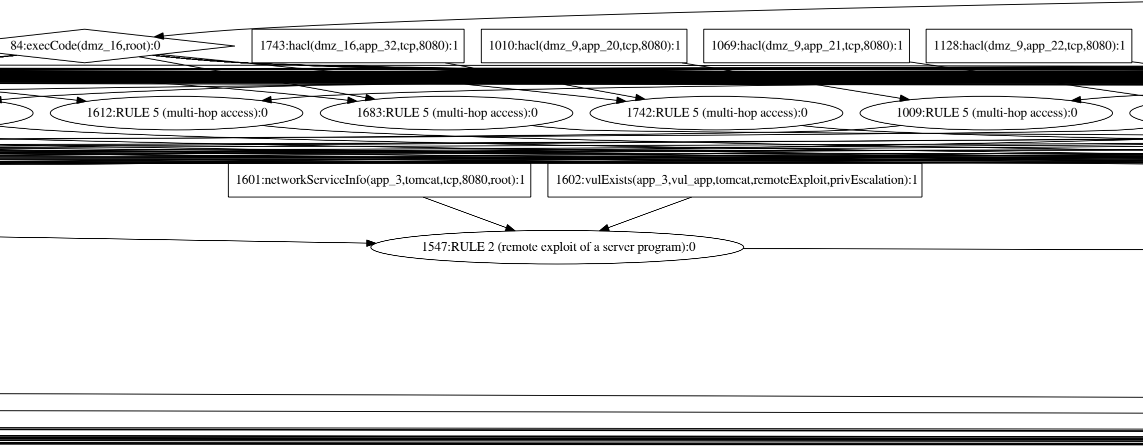

1501:hacl(dmz_13,app_29,tcp,8080):1
1550:hacl(dmz_0,app_3,tcp,8080):1
1578:hacl(dmz_21,app_3,tcp,8080):1
1594:hacl(dmz_6,app_3,tcp,8080):1
1500:RULE 5 (multi-hop access):0
1549:RULE 5 (multi-hop access):0
1577:RULE 5 (multi-hop access):0
1593:RULE 5 (multi-hop access):0

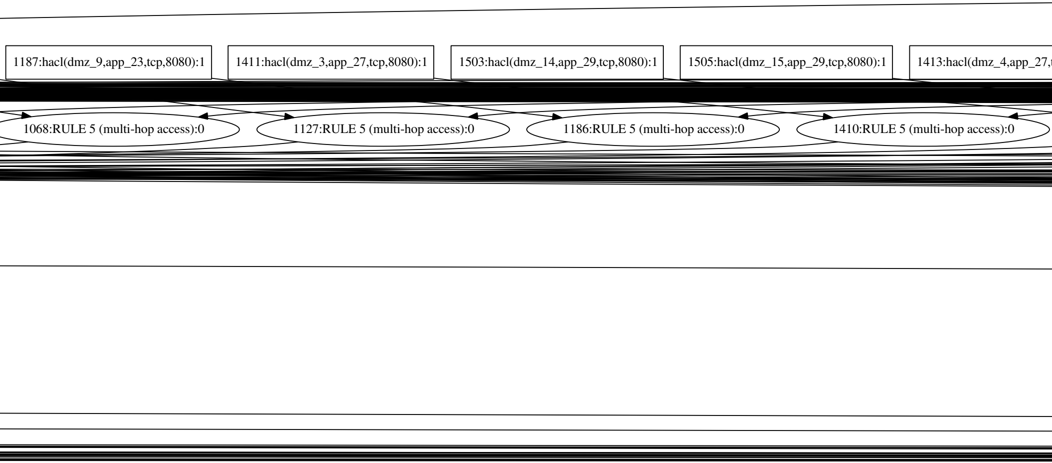

1552:hacl(dmz_1,app_3,tcp,8080):1
1554:hacl(dmz_10,app_3,tcp,8080):1
1580:hacl(dmz_22,app_3,tcp,8080):1
1584:hacl(dmz_24,app_3,tcp,8080):1
1551:RULE 5 (multi-hop access):0
1553:RULE 5 (multi-hop access):0
1579:RULE 5 (multi-hop access):0
1583:RULE 5 (multi-hop access):0
1548:netAccess(app_3,tcp,8080):0
1486:hacl(app_29,db_51,tcp,5432):1
1487:execCode(app_29,root):0
1485:RULE 5 (multi-hop access):0

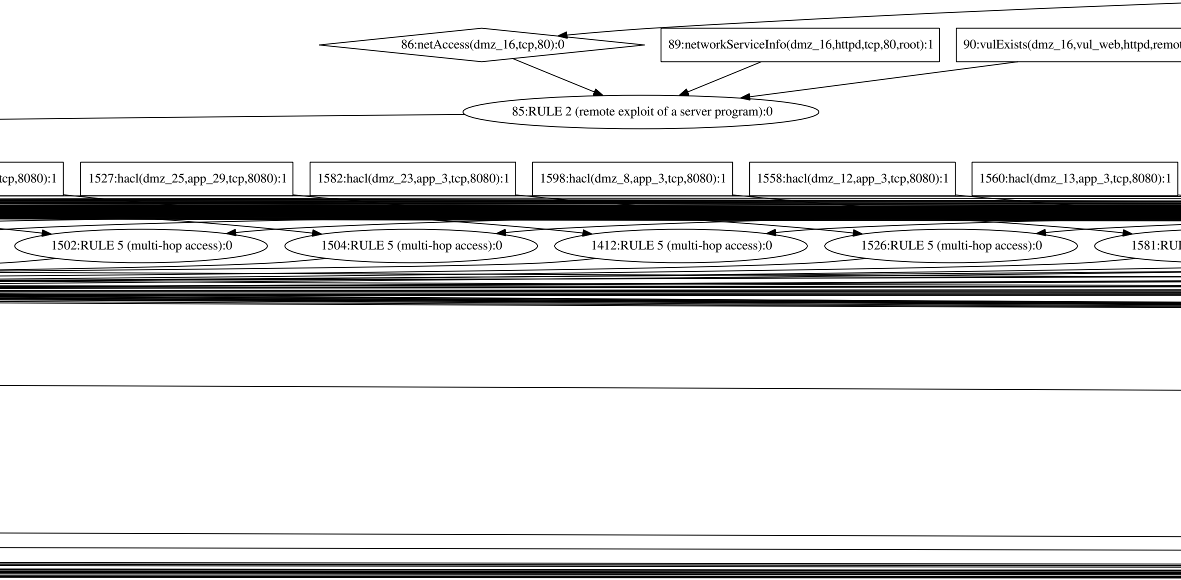

1556:hacl(dmz_11,app_3,tcp,8080):1
1596:hacl(dmz_7,app_3,tcp,8080):1
1625:hacl(dmz_16,app_30,tcp,8080):1
1627:hacl(dmz_17,app_30,tcp,8080):1
1651:hacl(dmz_5,app_30,tcp,8080):1
1555:RULE 5 (multi-hop access):0
1595:RULE 5 (multi-hop access):0
1624:RULE 5 (multi-hop access):0
1626:RULE 5 (multi-hop access):0

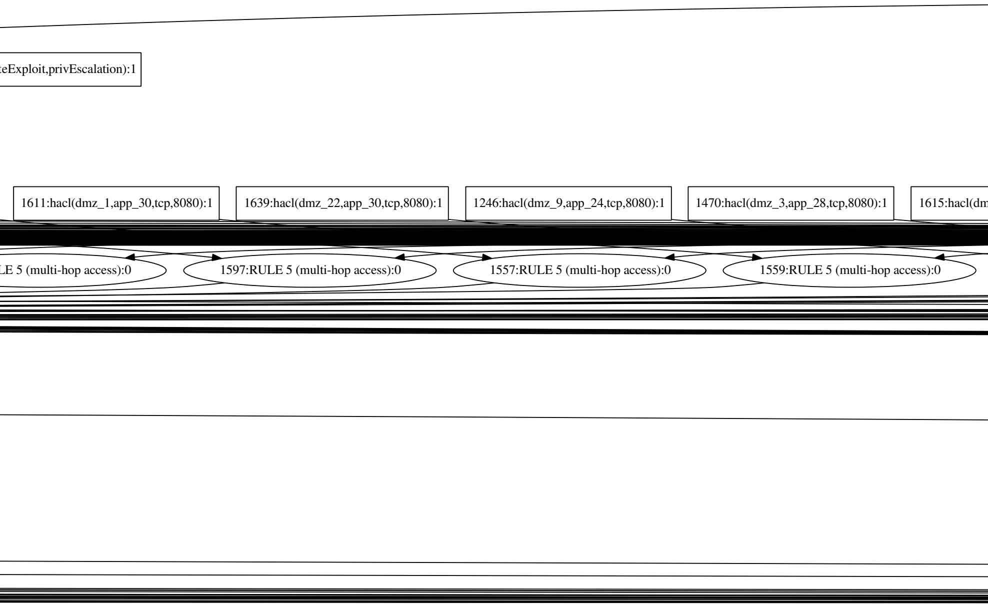

1631:hacl(dmz_19,app_30,tcp,8080):1
1633:hacl(dmz_2,app_30,tcp,8080):1
1629:hacl(dmz_18,app_30,tcp,8080):1
1635:hacl(dmz_20,app_30,tcp,8080):1
1637:hacl(dmz_21,app_30,tcp,8080):1
1650:RULE 5 (multi-hop access):0
1630:RULE 5 (multi-hop access):0
1632:RULE 5 (multi-hop access):0
1628:RULE 5 (multi-hop access):0
1

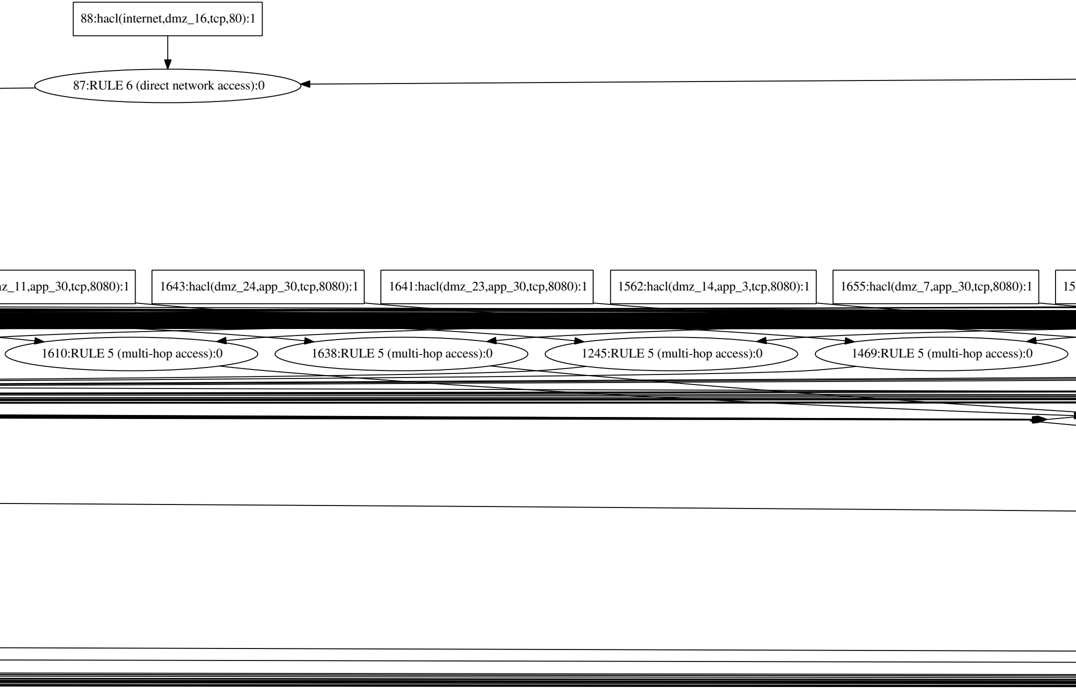

1653:hacl(dmz_6,app_30,tcp,8080):1
1609:hacl(dmz_0,app_30,tcp,8080):1
1613:hacl(dmz_10,app_30,tcp,8080):1
1684:hacl(dmz_16,app_31,tcp,8080):1
634:RULE 5 (multi-hop access):0
1636:RULE 5 (multi-hop access):0
1652:RULE 5 (multi-hop access):0
1608:RULE 5 (multi-hop access):0

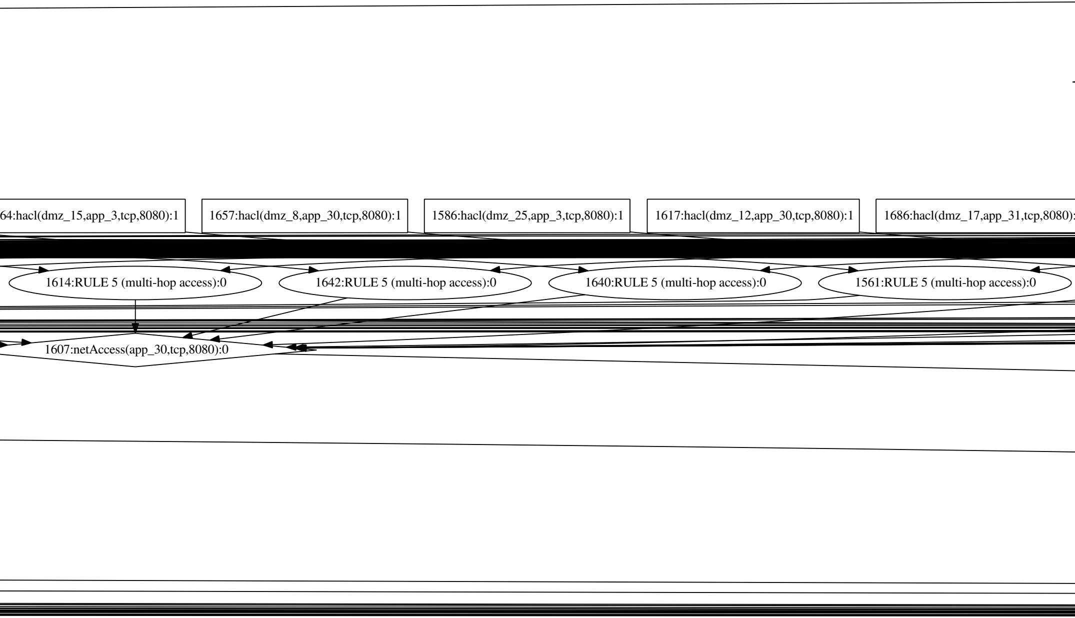

84:execCode(dmz_16,root):0
1743:hacl(dmz_16,app_32,tcp,8080):1
1010:hacl(dmz_9,app_20,tcp,8080):1
1069:hacl(dmz_9,app_21,tcp,8080):1
1128:hacl(dmz_9,app_22,tcp,8080):1
1612:RULE 5 (multi-hop access):0
1683:RULE 5 (multi-hop access):0
1742:RULE 5 (multi-hop access):0
1009:RULE 5 (multi-hop access):0
1601:networkServiceInfo(app_3,tomcat,tcp,8080,root):1
1602:vulExists(app_3,vul_app,tomcat,remoteExploit,privEscalation):1
1547:RULE 2 (remote exploit of a server program):0

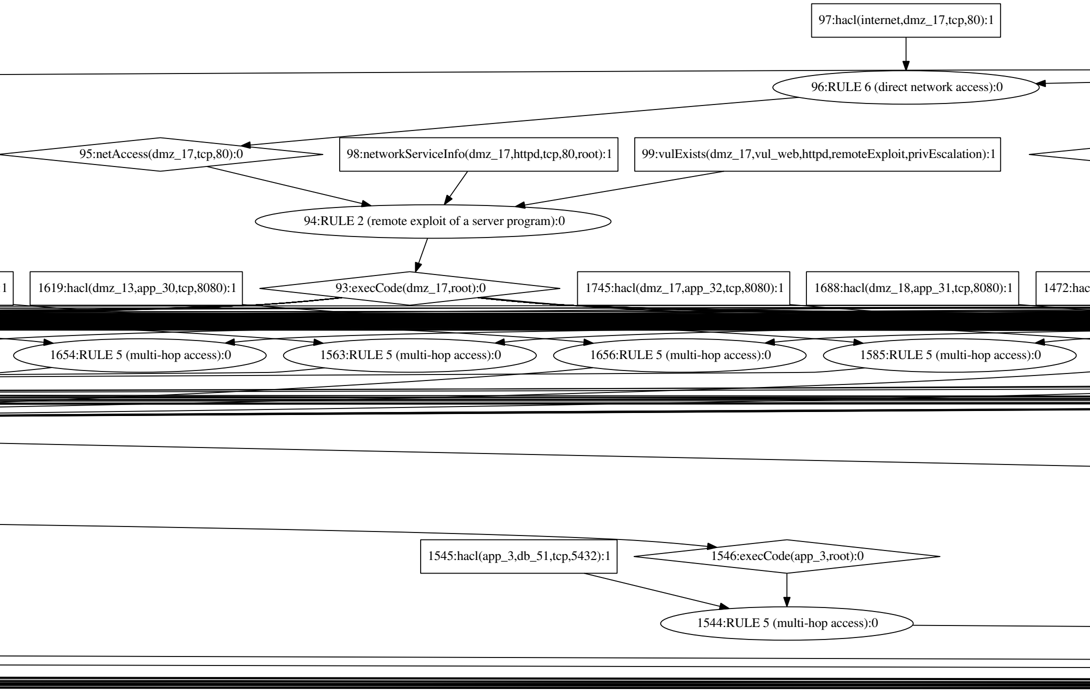

1187:hacl(dmz_9,app_23,tcp,8080):1
1411:hacl(dmz_3,app_27,tcp,8080):1
1503:hacl(dmz_14,app_29,tcp,8080):1
1505:hacl(dmz_15,app_29,tcp,8080):1
1413:hacl(dmz_4,app_27,t
1068:RULE 5 (multi-hop access):0
1127:RULE 5 (multi-hop access):0
1186:RULE 5 (multi-hop access):0
1410:RULE 5 (multi-hop access):0

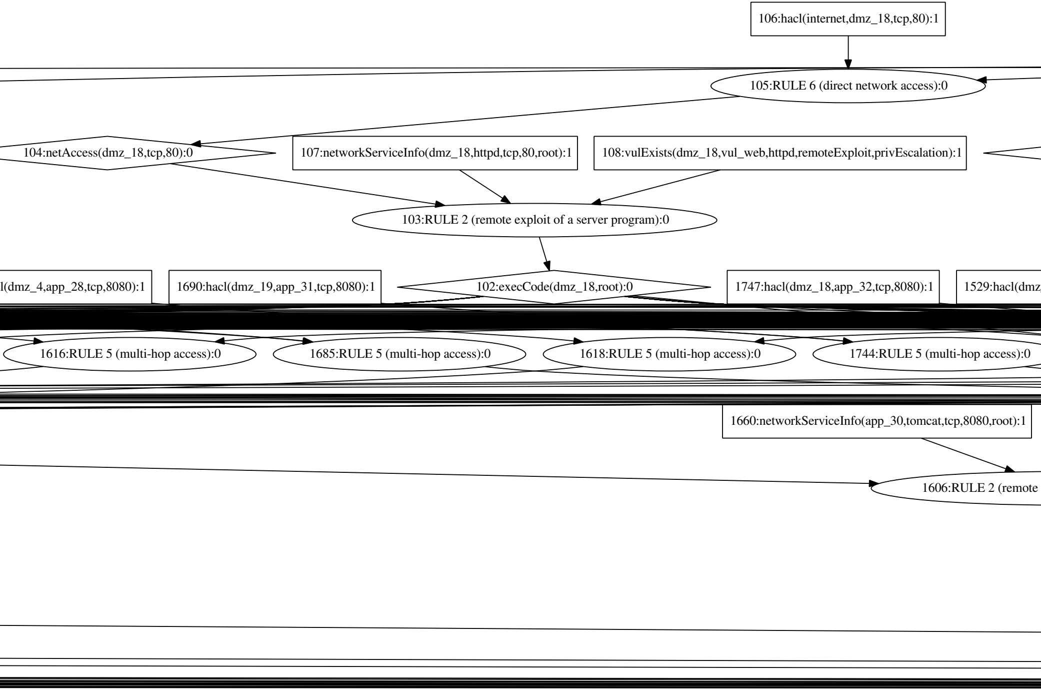

86:netAccess(dmz_16,tcp,80):0
89:networkServiceInfo(dmz_16,httpd,tcp,80,root):1
90:vulExists(dmz_16,vul_web,httpd,remot
85:RULE 2 (remote exploit of a server program):0
tcp,8080):1
1527:hacl(dmz_25,app_29,tcp,8080):1
1582:hacl(dmz_23,app_3,tcp,8080):1
1598:hacl(dmz_8,app_3,tcp,8080):1
1558:hacl(dmz_12,app_3,tcp,8080):1
1560:hacl(dmz_13,app_3,tcp,8080):1
1502:RULE 5 (multi-hop access):0
1504:RULE 5 (multi-hop access):0
1412:RULE 5 (multi-hop access):0
1526:RULE 5 (multi-hop access):0
1581:RUL

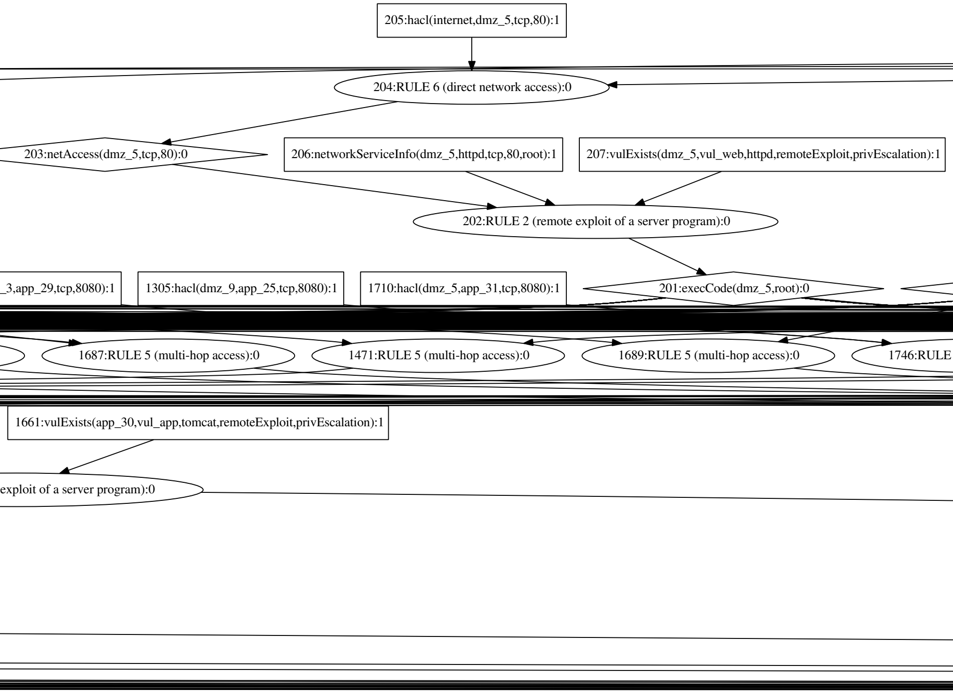

teExploit,privEscalation):1
1611:hacl(dmz_1,app_30,tcp,8080):1
1639:hacl(dmz_22,app_30,tcp,8080):1
1246:hacl(dmz_9,app_24,tcp,8080):1
1470:hacl(dmz_3,app_28,tcp,8080):1
1615:hacl(dm
LE 5 (multi-hop access):0
1597:RULE 5 (multi-hop access):0
1557:RULE 5 (multi-hop access):0
1559:RULE 5 (multi-hop access):0

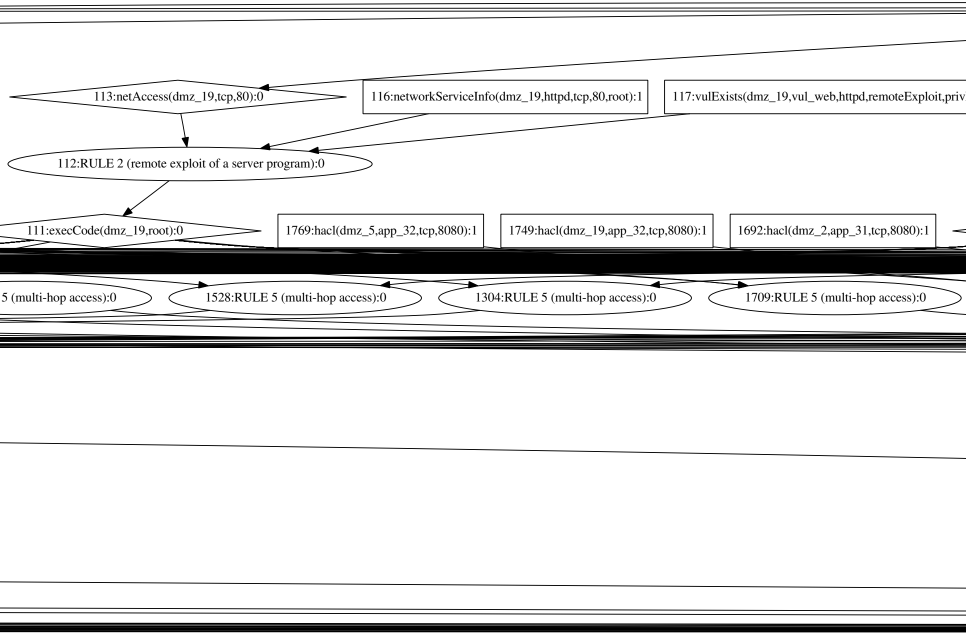

88:hacl(internet,dmz_16,tcp,80):1
87:RULE 6 (direct network access):0
z_11,app_30,tcp,8080):1
1643:hacl(dmz_24,app_30,tcp,8080):1
1641:hacl(dmz_23,app_30,tcp,8080):1
1562:hacl(dmz_14,app_3,tcp,8080):1
1655:hacl(dmz_7,app_30,tcp,8080):1
15
1610:RULE 5 (multi-hop access):0
1638:RULE 5 (multi-hop access):0
1245:RULE 5 (multi-hop access):0
1469:RULE 5 (multi-hop access):0

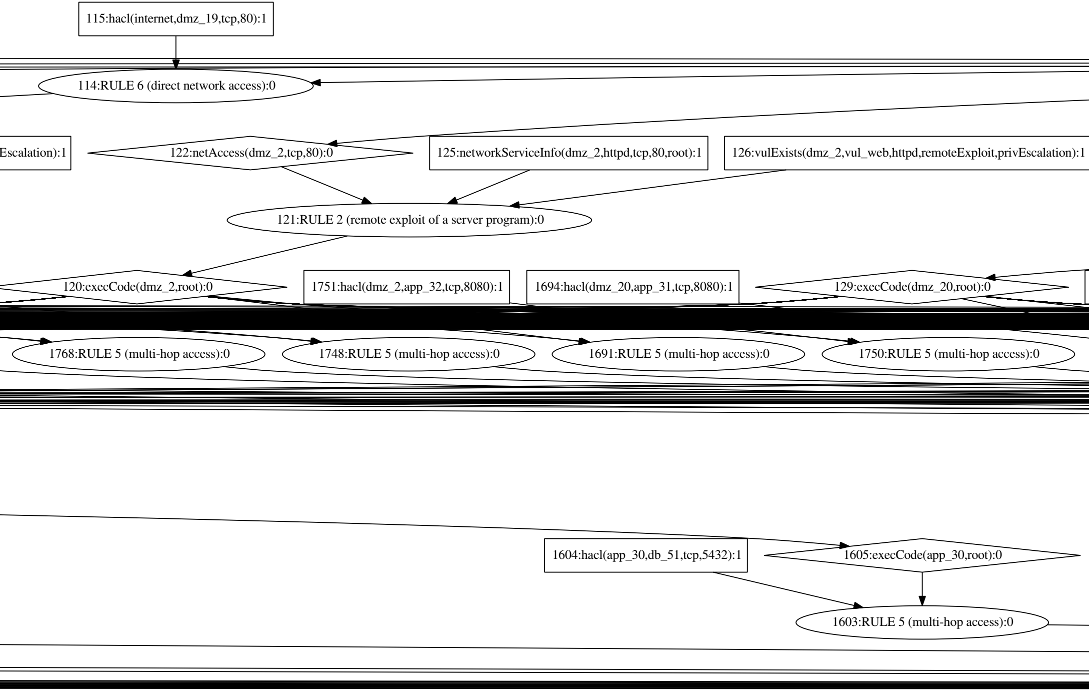

64:hacl(dmz_15,app_3,tcp,8080):1
1657:hacl(dmz_8,app_30,tcp,8080):1
1586:hacl(dmz_25,app_3,tcp,8080):1
1617:hacl(dmz_12,app_30,tcp,8080):1
1686:hacl(dmz_17,app_31,tcp,8080):
1614:RULE 5 (multi-hop access):0
1642:RULE 5 (multi-hop access):0
1640:RULE 5 (multi-hop access):0
1561:RULE 5 (multi-hop access):0
1607:netAccess(app_30,tcp,8080):0

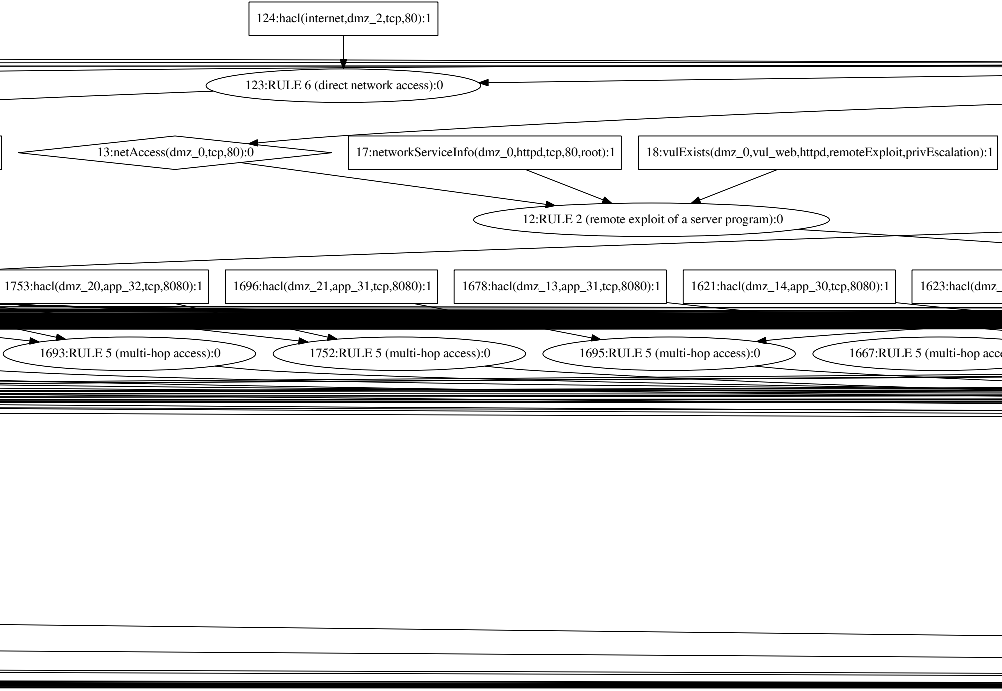

97:hacl(internet,dmz_17,tcp,80):1
96:RULE 6 (direct network access):0
95:netAccess(dmz_17,tcp,80):0
98:networkServiceInfo(dmz_17,httpd,tcp,80,root):1
99:vulExists(dmz_17,vul_web,httpd,remoteExploit,privEscalation):1
94:RULE 2 (remote exploit of a server program):0
1
1619:hacl(dmz_13,app_30,tcp,8080):1
93:execCode(dmz_17,root):0
1745:hacl(dmz_17,app_32,tcp,8080):1
1688:hacl(dmz_18,app_31,tcp,8080):1
1472:hac
1654:RULE 5 (multi-hop access):0
1563:RULE 5 (multi-hop access):0
1656:RULE 5 (multi-hop access):0
1585:RULE 5 (multi-hop access):0
1545:hacl(app_3,db_51,tcp,5432):1
1546:execCode(app_3,root):0
1544:RULE 5 (multi-hop access):0

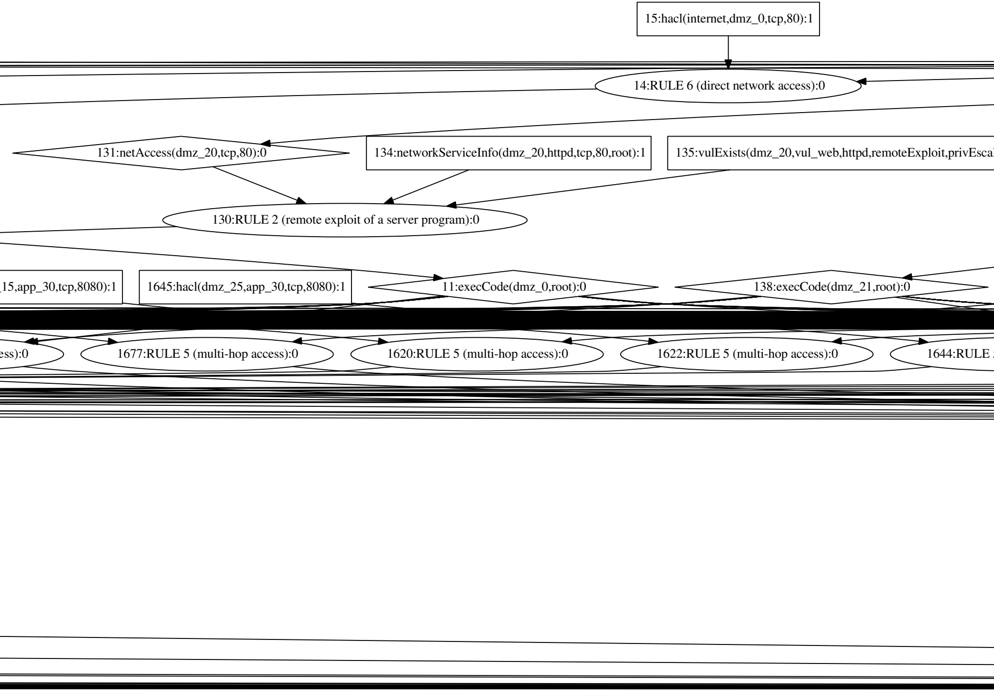

106:hacl(internet,dmz_18,tcp,80):1
105:RULE 6 (direct network access):0
104:netAccess(dmz_18,tcp,80):0
107:networkServiceInfo(dmz_18,httpd,tcp,80,root):1
108:vulExists(dmz_18,vul_web,httpd,remoteExploit,privEscalation):1
103:RULE 2 (remote exploit of a server program):0
l(dmz_4,app_28,tcp,8080):1
1690:hacl(dmz_19,app_31,tcp,8080):1
102:execCode(dmz_18,root):0
1747:hacl(dmz_18,app_32,tcp,8080):1
1529:hacl(dmz_
1616:RULE 5 (multi-hop access):0
1685:RULE 5 (multi-hop access):0
1618:RULE 5 (multi-hop access):0
1744:RULE 5 (multi-hop access):0
1660:networkServiceInfo(app_30,tomcat,tcp,8080,root):1
1606:RULE 2 (remote

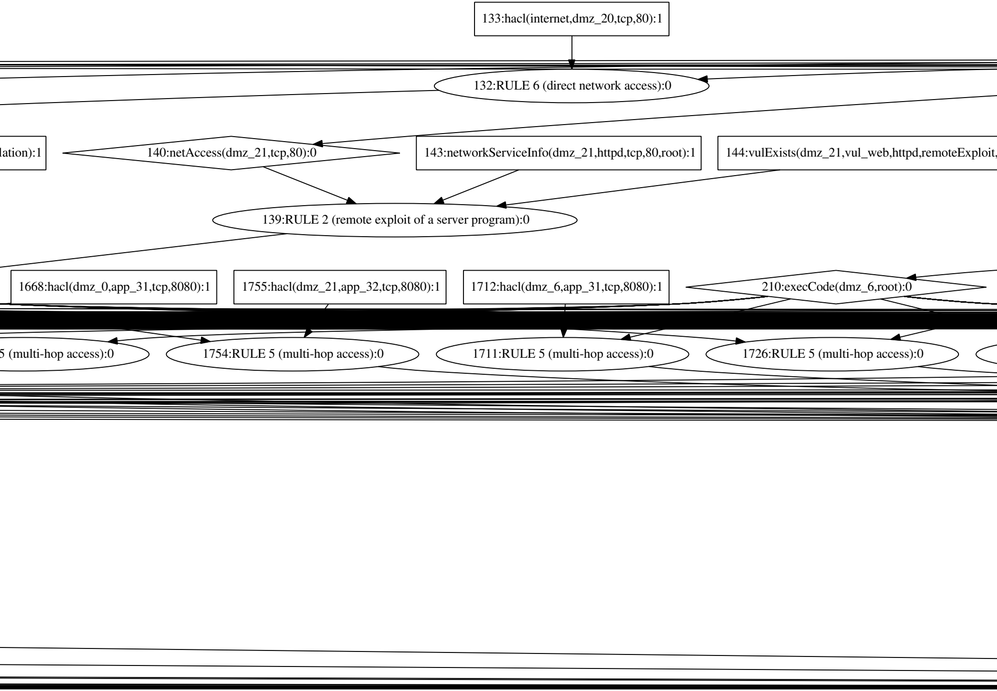

205:hacl(internet,dmz_5,tcp,80):1
204:RULE 6 (direct network access):0
203:netAccess(dmz_5,tcp,80):0
206:networkServiceInfo(dmz_5,httpd,tcp,80,root):1
207:vulExists(dmz_5,vul_web,httpd,remoteExploit,privEscalation):1
202:RULE 2 (remote exploit of a server program):0
_3,app_29,tcp,8080):1
1305:hacl(dmz_9,app_25,tcp,8080):1
1710:hacl(dmz_5,app_31,tcp,8080):1
201:execCode(dmz_5,root):0
1687:RULE 5 (multi-hop access):0
1471:RULE 5 (multi-hop access):0
1689:RULE 5 (multi-hop access):0
1746:RULE
1661:vulExists(app_30,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0

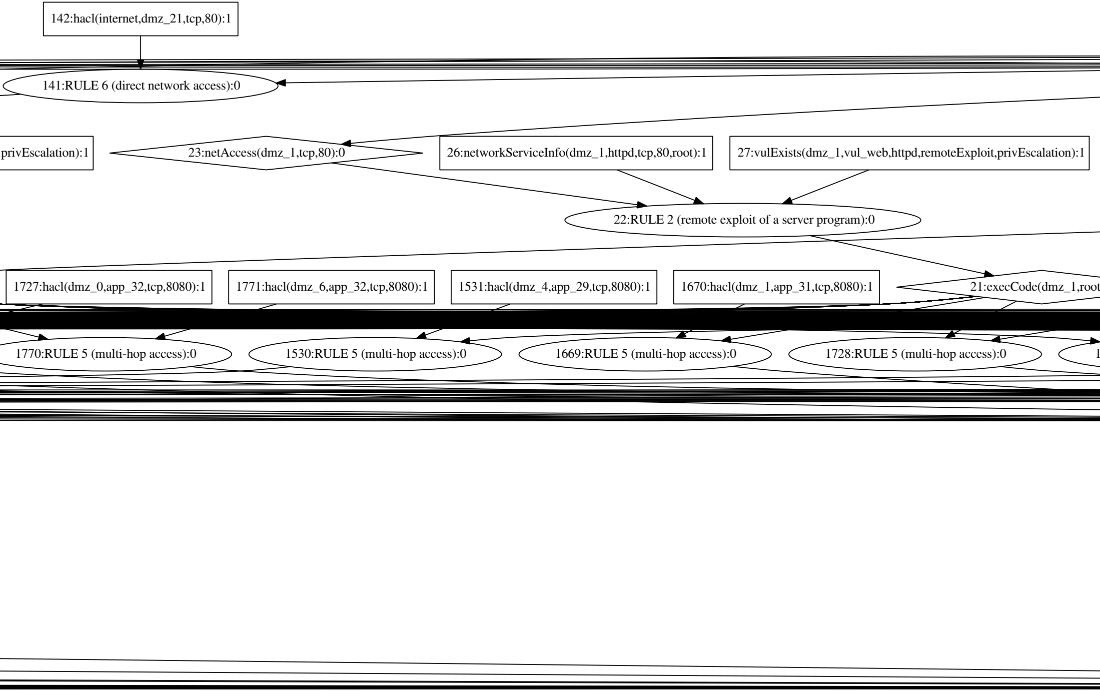

113:netAccess(dmz_19,tcp,80):0
116:networkServiceInfo(dmz_19,httpd,tcp,80,root):1
117:vulExists(dmz_19,vul_web,httpd,remoteExploit,priv
112:RULE 2 (remote exploit of a server program):0
111:execCode(dmz_19,root):0
1769:hacl(dmz_5,app_32,tcp,8080):1
1749:hacl(dmz_19,app_32,tcp,8080):1
1692:hacl(dmz_2,app_31,tcp,8080):1
5 (multi-hop access):0
1528:RULE 5 (multi-hop access):0
1304:RULE 5 (multi-hop access):0
1709:RULE 5 (multi-hop access):0

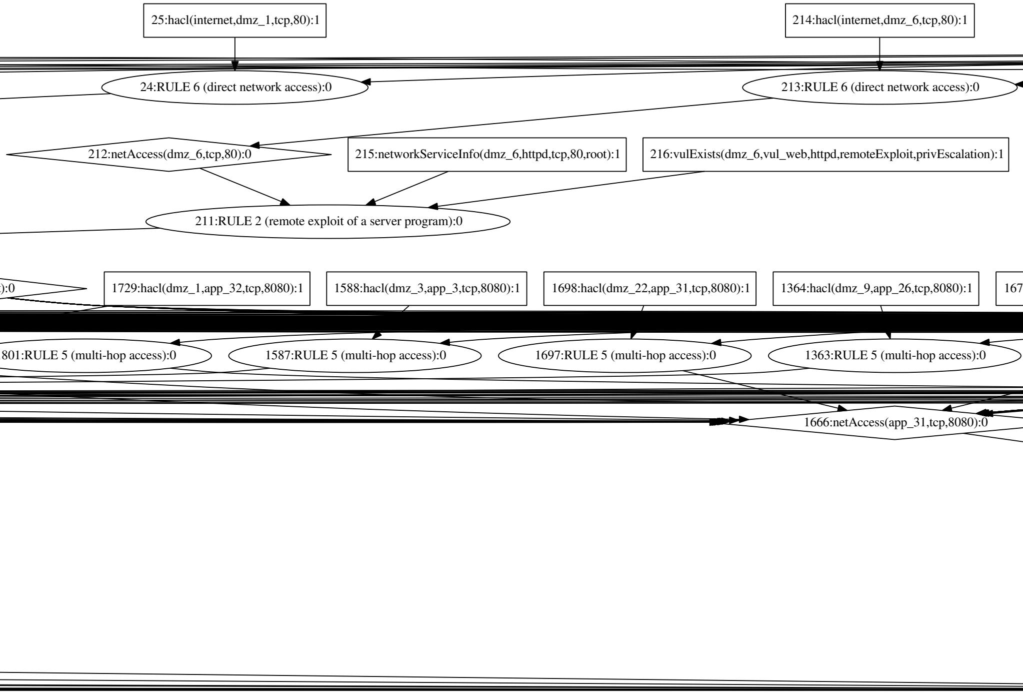

115:hacl(internet,dmz_19,tcp,80):1
114:RULE 6 (direct network access):0
Escalation):1
122:netAccess(dmz_2,tcp,80):0
125:networkServiceInfo(dmz_2,httpd,tcp,80,root):1
126:vulExists(dmz_2,vul_web,httpd,remoteExploit,privEscalation):1
121:RULE 2 (remote exploit of a server program):0
120:execCode(dmz_2,root):0
1751:hacl(dmz_2,app_32,tcp,8080):1
1694:hacl(dmz_20,app_31,tcp,8080):1
129:execCode(dmz_20,root):0
1768:RULE 5 (multi-hop access):0
1748:RULE 5 (multi-hop access):0
1691:RULE 5 (multi-hop access):0
1750:RULE 5 (multi-hop access):0
1604:hacl(app_30,db_51,tcp,5432):1
1605:execCode(app_30,root):0
1603:RULE 5 (multi-hop access):0

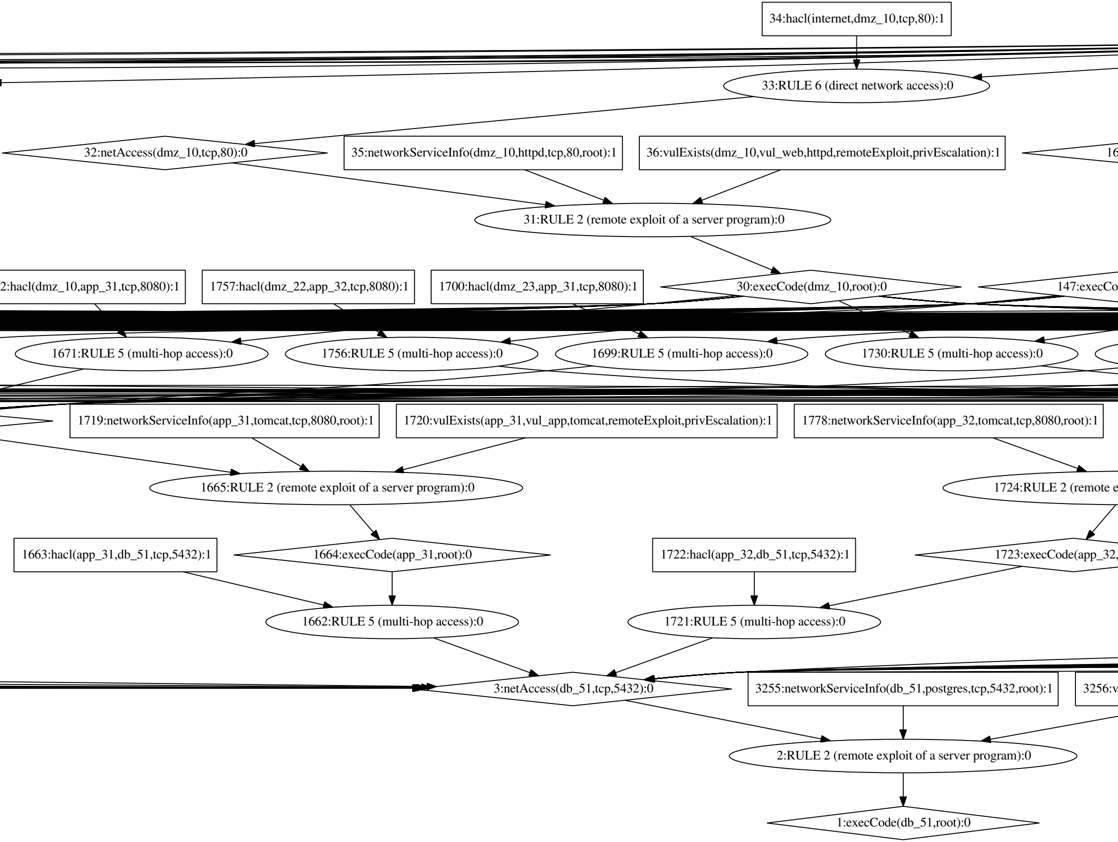

124:hacl(internet,dmz_2,tcp,80):1
123:RULE 6 (direct network access):0
13:netAccess(dmz_0,tcp,80):0
17:networkServiceInfo(dmz_0,httpd,tcp,80,root):1
18:vulExists(dmz_0,vul_web,httpd,remoteExploit,privEscalation):1
12:RULE 2 (remote exploit of a server program):0
1753:hacl(dmz_20,app_32,tcp,8080):1
1696:hacl(dmz_21,app_31,tcp,8080):1
1678:hacl(dmz_13,app_31,tcp,8080):1
1621:hacl(dmz_14,app_30,tcp,8080):1
1623:hacl(dmz_
1693:RULE 5 (multi-hop access):0
1752:RULE 5 (multi-hop access):0
1695:RULE 5 (multi-hop access):0
1667:RULE 5 (multi-hop acc

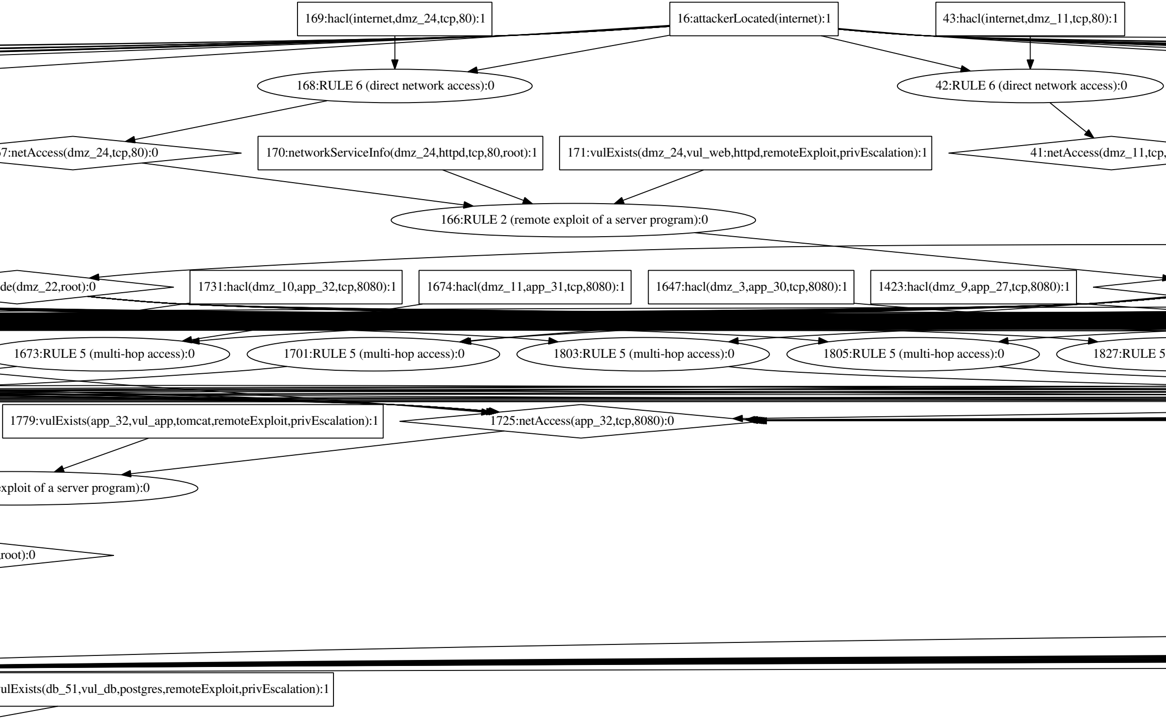

15:hacl(internet,dmz_0,tcp,80):1
14:RULE 6 (direct network access):0
131:netAccess(dmz_20,tcp,80):0
134:networkServiceInfo(dmz_20,httpd,tcp,80,root):1
135:vulExists(dmz_20,vul_web,httpd,remoteExploit,privEsca
130:RULE 2 (remote exploit of a server program):0
_15,app_30,tcp,8080):1
1645:hacl(dmz_25,app_30,tcp,8080):1
11:execCode(dmz_0,root):0
138:execCode(dmz_21,root):0
ss):0
1677:RULE 5 (multi-hop access):0
1620:RULE 5 (multi-hop access):0
1622:RULE 5 (multi-hop access):0
1644:RULE

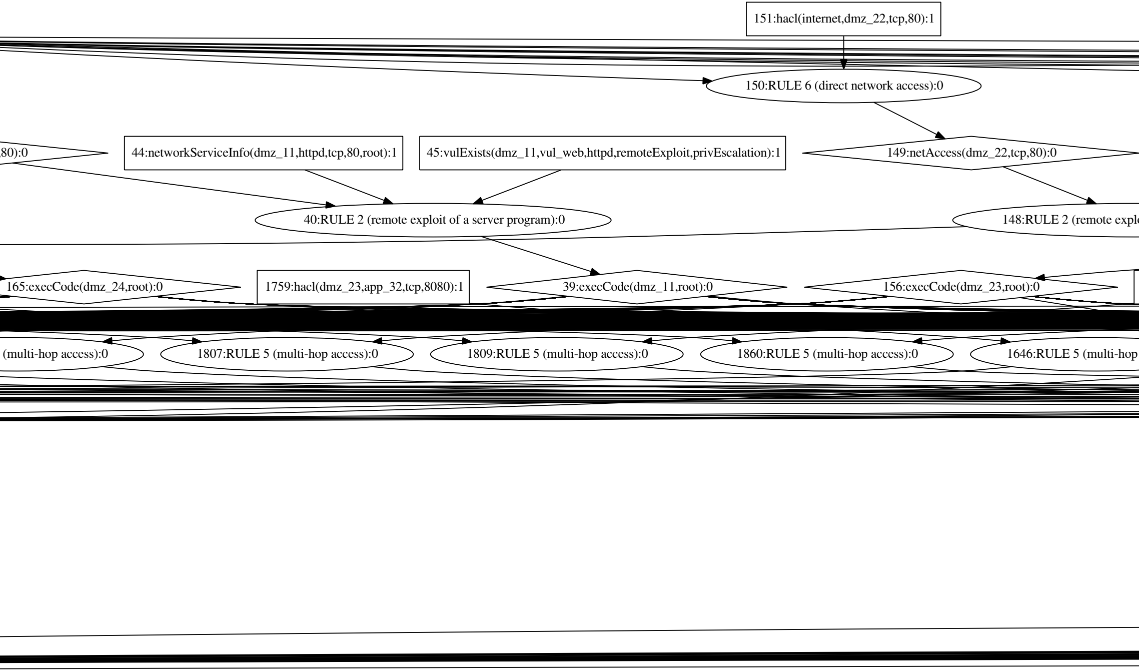

133:hacl(internet,dmz_20,tcp,80):1
132:RULE 6 (direct network access):0
lation):1
140:netAccess(dmz_21,tcp,80):0
143:networkServiceInfo(dmz_21,httpd,tcp,80,root):1
144:vulExists(dmz_21,vul_web,httpd,remoteExploit,
139:RULE 2 (remote exploit of a server program):0
1668:hacl(dmz_0,app_31,tcp,8080):1
1755:hacl(dmz_21,app_32,tcp,8080):1
1712:hacl(dmz_6,app_31,tcp,8080):1
210:execCode(dmz_6,root):0
5 (multi-hop access):0
1754:RULE 5 (multi-hop access):0
1711:RULE 5 (multi-hop access):0
1726:RULE 5 (multi-hop access):0

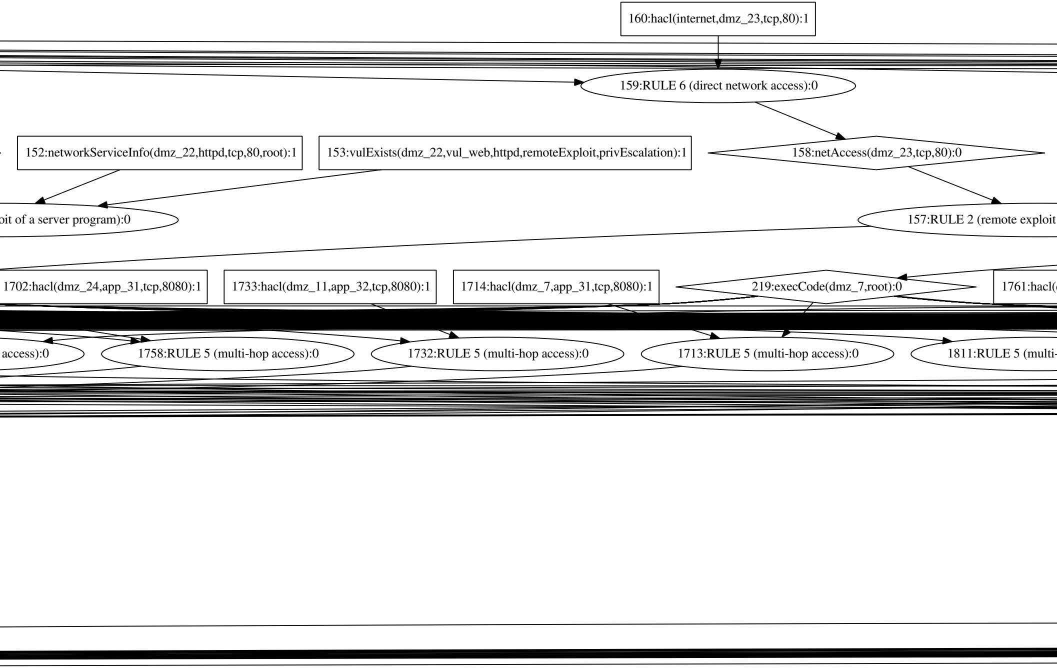

142:hacl(internet,dmz_21,tcp,80):1
141:RULE 6 (direct network access):0
privEscalation):1
23:netAccess(dmz_1,tcp,80):0
26:networkServiceInfo(dmz_1,httpd,tcp,80,root):1
27:vulExists(dmz_1,vul_web,httpd,remoteExploit,privEscalation):1
22:RULE 2 (remote exploit of a server program):0
1727:hacl(dmz_0,app_32,tcp,8080):1
1771:hacl(dmz_6,app_32,tcp,8080):1
1531:hacl(dmz_4,app_29,tcp,8080):1
1670:hacl(dmz_1,app_31,tcp,8080):1
21:execCode(dmz_1,root
1770:RULE 5 (multi-hop access):0
1530:RULE 5 (multi-hop access):0
1669:RULE 5 (multi-hop access):0
1728:RULE 5 (multi-hop access):0
1

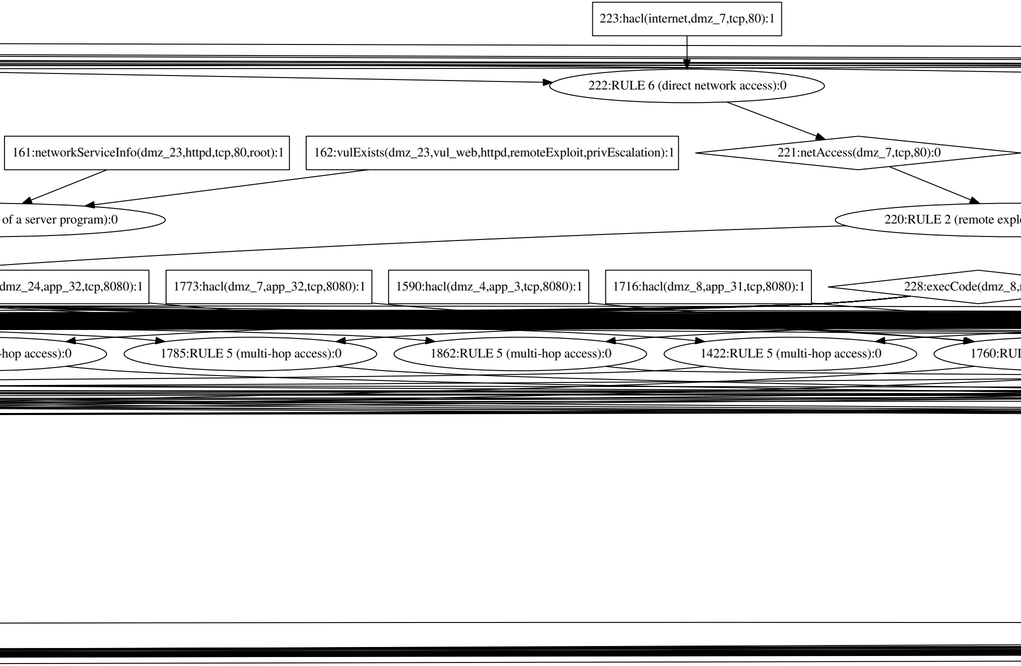

25:hacl(internet,dmz_1,tcp,80):1
214:hacl(internet,dmz_6,tcp,80):1
24:RULE 6 (direct network access):0
213:RULE 6 (direct network access):0
212:netAccess(dmz_6,tcp,80):0
215:networkServiceInfo(dmz_6,httpd,tcp,80,root):1
216:vulExists(dmz_6,vul_web,httpd,remoteExploit,privEscalation):1
211:RULE 2 (remote exploit of a server program):0
):0
1729:hacl(dmz_1,app_32,tcp,8080):1
1588:hacl(dmz_3,app_3,tcp,8080):1
1698:hacl(dmz_22,app_31,tcp,8080):1
1364:hacl(dmz_9,app_26,tcp,8080):1
167
801:RULE 5 (multi-hop access):0
1587:RULE 5 (multi-hop access):0
1697:RULE 5 (multi-hop access):0
1363:RULE 5 (multi-hop access):0
1666:netAccess(app_31,tcp,8080):0

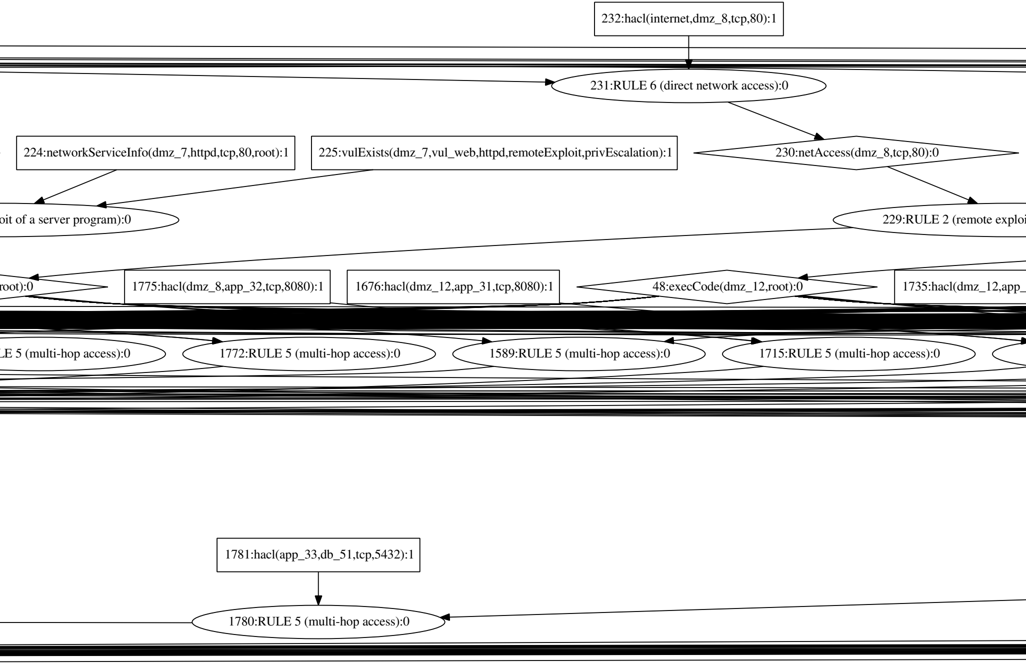

34:hacl(internet,dmz_10,tcp,80):1
33:RULE 6 (direct network access):0
32:netAccess(dmz_10,tcp,80):0
35:networkServiceInfo(dmz_10,httpd,tcp,80,root):1
36:vulExists(dmz_10,vul_web,httpd,remoteExploit,privEscalation):1
16
31:RULE 2 (remote exploit of a server program):0
2:hacl(dmz_10,app_31,tcp,8080):1
1757:hacl(dmz_22,app_32,tcp,8080):1
1700:hacl(dmz_23,app_31,tcp,8080):1
30:execCode(dmz_10,root):0
147:execCo
1671:RULE 5 (multi-hop access):0
1756:RULE 5 (multi-hop access):0
1699:RULE 5 (multi-hop access):0
1730:RULE 5 (multi-hop access):0
1719:networkServiceInfo(app_31,tomcat,tcp,8080,root):1
1720:vulExists(app_31,vul_app,tomcat,remoteExploit,privEscalation):1
1778:networkServiceInfo(app_32,tomcat,tcp,8080,root):1
1665:RULE 2 (remote exploit of a server program):0
1724:RULE 2 (remote e
1663:hacl(app_31,db_51,tcp,5432):1
1664:execCode(app_31,root):0
1722:hacl(app_32,db_51,tcp,5432):1
1723:execCode(app_32,
1662:RULE 5 (multi-hop access):0
1721:RULE 5 (multi-hop access):0
3:netAccess(db_51,tcp,5432):0
3255:networkServiceInfo(db_51,postgres,tcp,5432,root):1
3256:v
2:RULE 2 (remote exploit of a server program):0
1:execCode(db_51,root):0

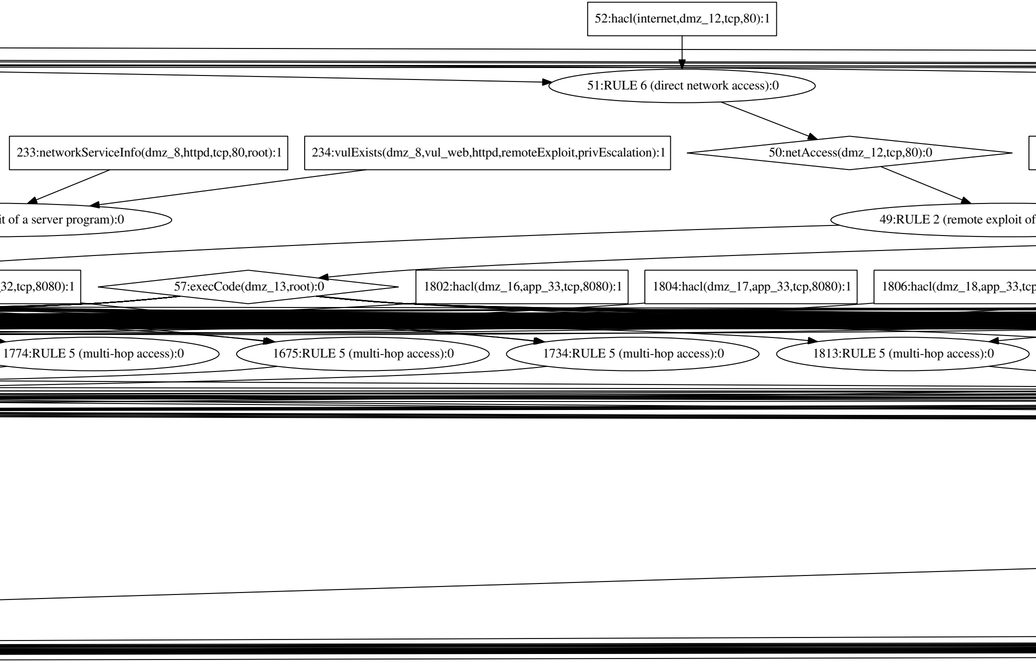

169:hacl(internet,dmz_24,tcp,80):1
16:attackerLocated(internet):1
43:hacl(internet,dmz_11,tcp,80):1
168:RULE 6 (direct network access):0
42:RULE 6 (direct network access):0
7:netAccess(dmz_24,tcp,80):0
170:networkServiceInfo(dmz_24,httpd,tcp,80,root):1
171:vulExists(dmz_24,vul_web,httpd,remoteExploit,privEscalation):1
41:netAccess(dmz_11,tcp,
166:RULE 2 (remote exploit of a server program):0
de(dmz_22,root):0
1731:hacl(dmz_10,app_32,tcp,8080):1
1674:hacl(dmz_11,app_31,tcp,8080):1
1647:hacl(dmz_3,app_30,tcp,8080):1
1423:hacl(dmz_9,app_27,tcp,8080):1
1673:RULE 5 (multi-hop access):0
1701:RULE 5 (multi-hop access):0
1803:RULE 5 (multi-hop access):0
1805:RULE 5 (multi-hop access):0
1827:RULE 5
1779:vulExists(app_32,vul_app,tomcat,remoteExploit,privEscalation):1
1725:netAccess(app_32,tcp,8080):0
xploit of a server program):0
root):0
ulExists(db_51,vul_db,postgres,remoteExploit,privEscalation):1

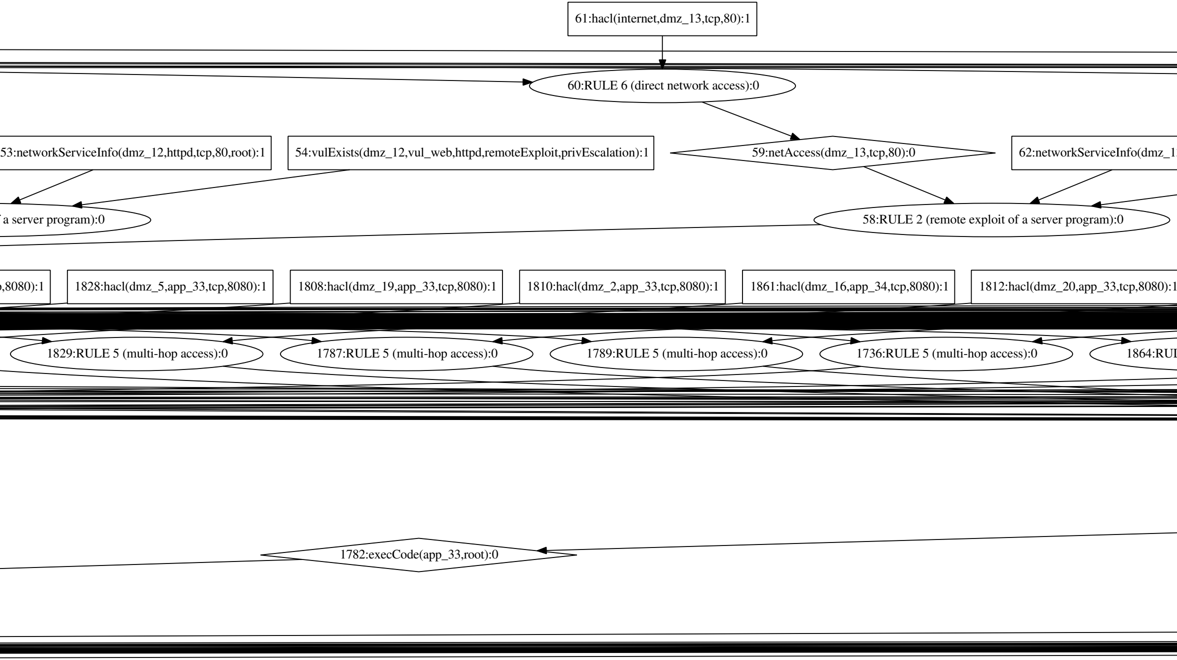

151:hacl(internet,dmz_22,tcp,80):1
150:RULE 6 (direct network access):0
80):0
44:networkServiceInfo(dmz_11,httpd,tcp,80,root):1
45:vulExists(dmz_11,vul_web,httpd,remoteExploit,privEscalation):1
149:netAccess(dmz_22,tcp,80):0
40:RULE 2 (remote exploit of a server program):0
148:RULE 2 (remote explo
165:execCode(dmz_24,root):0
1759:hacl(dmz_23,app_32,tcp,8080):1
39:execCode(dmz_11,root):0
156:execCode(dmz_23,root):0
(multi-hop access):0
1807:RULE 5 (multi-hop access):0
1809:RULE 5 (multi-hop access):0
1860:RULE 5 (multi-hop access):0
1646:RULE 5 (multi-hop

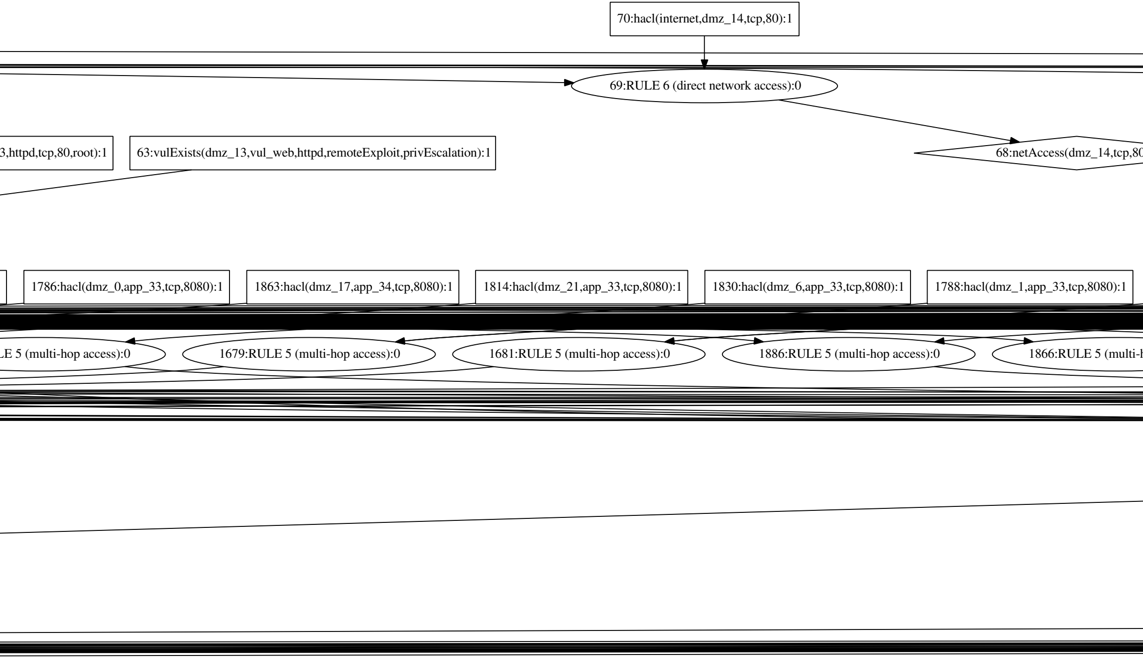

160:hacl(internet,dmz_23,tcp,80):1
159:RULE 6 (direct network access):0
152:networkServiceInfo(dmz_22,httpd,tcp,80,root):1
153:vulExists(dmz_22,vul_web,httpd,remoteExploit,privEscalation):1
158:netAccess(dmz_23,tcp,80):0
oit of a server program):0
157:RULE 2 (remote exploit
1702:hacl(dmz_24,app_31,tcp,8080):1
1733:hacl(dmz_11,app_32,tcp,8080):1
1714:hacl(dmz_7,app_31,tcp,8080):1
219:execCode(dmz_7,root):0
1761:hacl(
access):0
1758:RULE 5 (multi-hop access):0
1732:RULE 5 (multi-hop access):0
1713:RULE 5 (multi-hop access):0
1811:RULE 5 (multi-

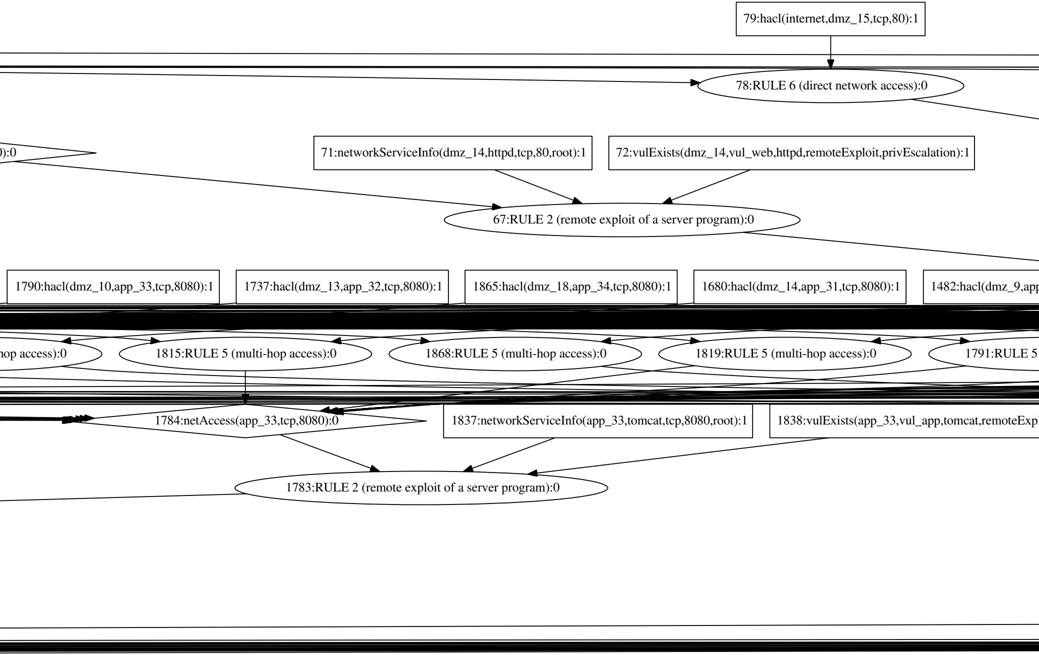

223:hacl(internet,dmz_7,tcp,80):1
222:RULE 6 (direct network access):0
161:networkServiceInfo(dmz_23,httpd,tcp,80,root):1
162:vulExists(dmz_23,vul_web,httpd,remoteExploit,privEscalation):1
221:netAccess(dmz_7,tcp,80):0
of a server program):0
220:RULE 2 (remote expl
dmz_24,app_32,tcp,8080):1
1773:hacl(dmz_7,app_32,tcp,8080):1
1590:hacl(dmz_4,app_3,tcp,8080):1
1716:hacl(dmz_8,app_31,tcp,8080):1
228:execCode(dmz_8,
-hop access):0
1785:RULE 5 (multi-hop access):0
1862:RULE 5 (multi-hop access):0
1422:RULE 5 (multi-hop access):0
1760:RUL

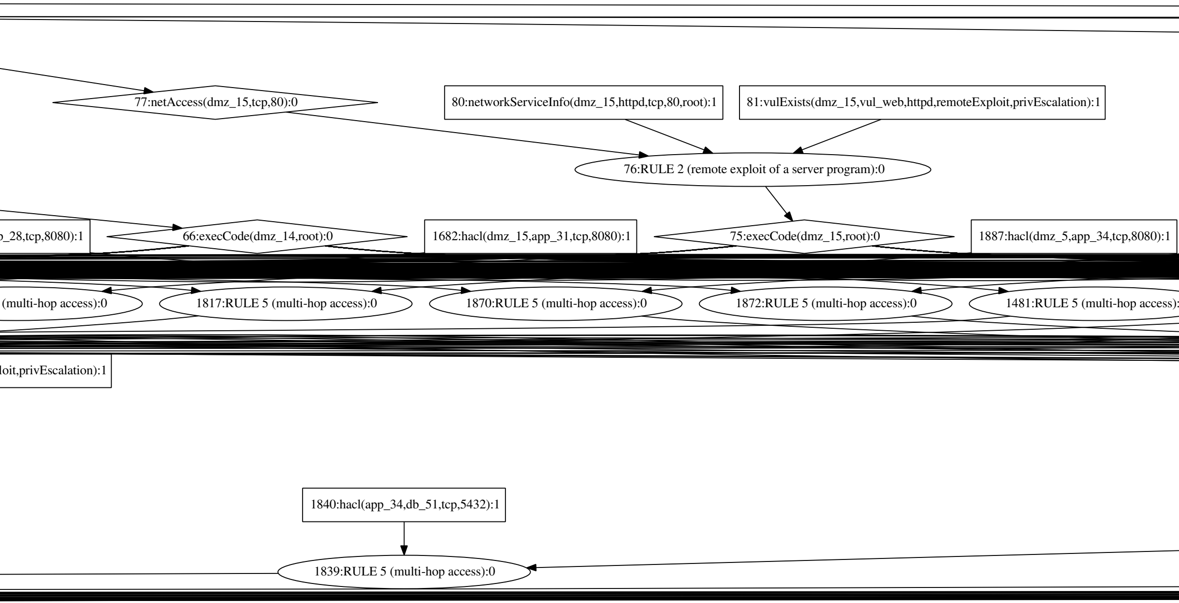

232:hacl(internet,dmz_8,tcp,80):1
231:RULE 6 (direct network access):0
224:networkServiceInfo(dmz_7,httpd,tcp,80,root):1
225:vulExists(dmz_7,vul_web,httpd,remoteExploit,privEscalation):1
230:netAccess(dmz_8,tcp,80):0
oit of a server program):0
229:RULE 2 (remote exploi
root):0
1775:hacl(dmz_8,app_32,tcp,8080):1
1676:hacl(dmz_12,app_31,tcp,8080):1
48:execCode(dmz_12,root):0
1735:hacl(dmz_12,app_
LE 5 (multi-hop access):0
1772:RULE 5 (multi-hop access):0
1589:RULE 5 (multi-hop access):0
1715:RULE 5 (multi-hop access):0
1781:hacl(app_33,db_51,tcp,5432):1
1780:RULE 5 (multi-hop access):0

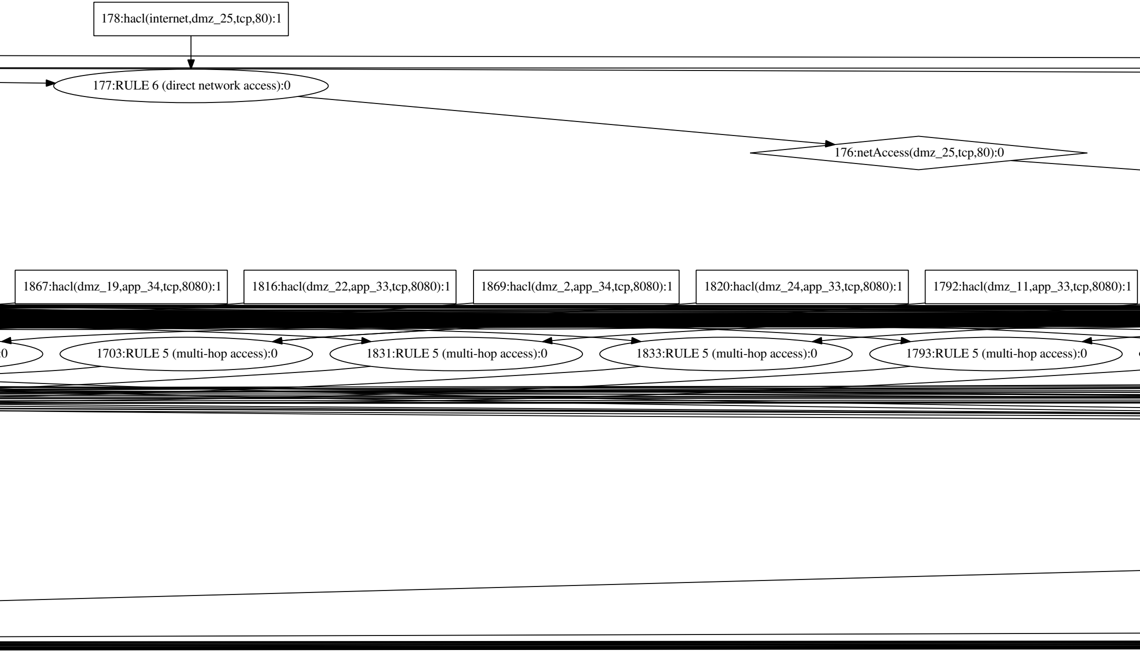

52:hacl(internet,dmz_12,tcp,80):1
51:RULE 6 (direct network access):0
233:networkServiceInfo(dmz_8,httpd,tcp,80,root):1
234:vulExists(dmz_8,vul_web,httpd,remoteExploit,privEscalation):1
50:netAccess(dmz_12,tcp,80):0
it of a server program):0
49:RULE 2 (remote exploit of
32,tcp,8080):1
57:execCode(dmz_13,root):0
1802:hacl(dmz_16,app_33,tcp,8080):1
1804:hacl(dmz_17,app_33,tcp,8080):1
1806:hacl(dmz_18,app_33,tcp
1774:RULE 5 (multi-hop access):0
1675:RULE 5 (multi-hop access):0
1734:RULE 5 (multi-hop access):0
1813:RULE 5 (multi-hop access):0

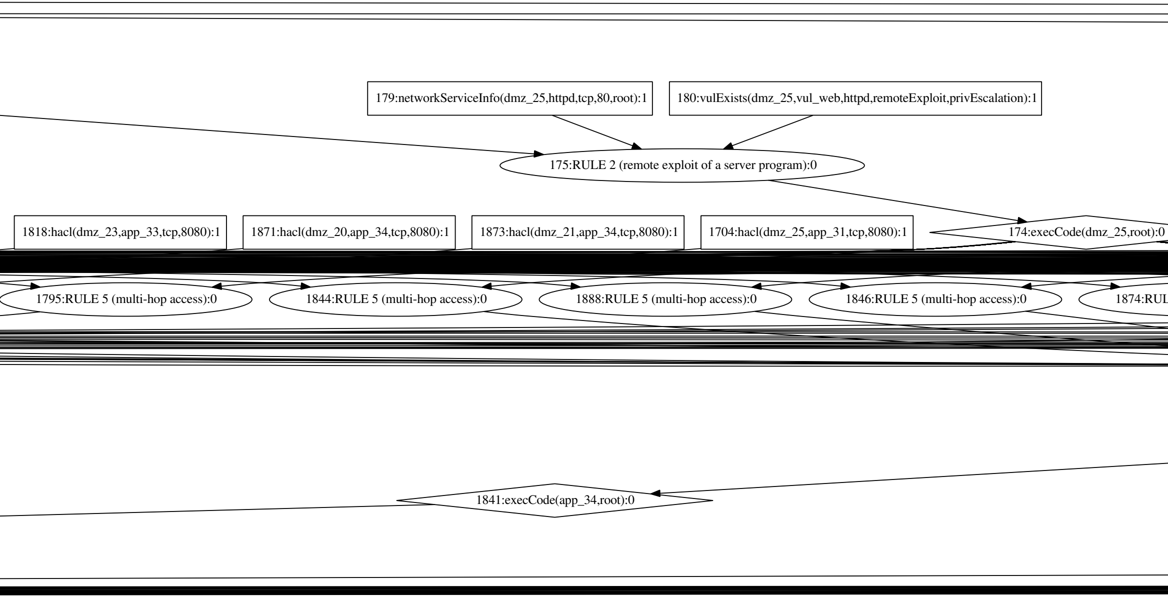

61:hacl(internet,dmz_13,tcp,80):1
60:RULE 6 (direct network access):0
53:networkServiceInfo(dmz_12,httpd,tcp,80,root):1
54:vulExists(dmz_12,vul_web,httpd,remoteExploit,privEscalation):1
59:netAccess(dmz_13,tcp,80):0
62:networkServiceInfo(dmz_1
f a server program):0
58:RULE 2 (remote exploit of a server program):0
,8080):1
1828:hacl(dmz_5,app_33,tcp,8080):1
1808:hacl(dmz_19,app_33,tcp,8080):1
1810:hacl(dmz_2,app_33,tcp,8080):1
1861:hacl(dmz_16,app_34,tcp,8080):1
1812:hacl(dmz_20,app_33,tcp,8080):1
1829:RULE 5 (multi-hop access):0
1787:RULE 5 (multi-hop access):0
1789:RULE 5 (multi-hop access):0
1736:RULE 5 (multi-hop access):0
1864:RUL
1782:execCode(app_33,root):0

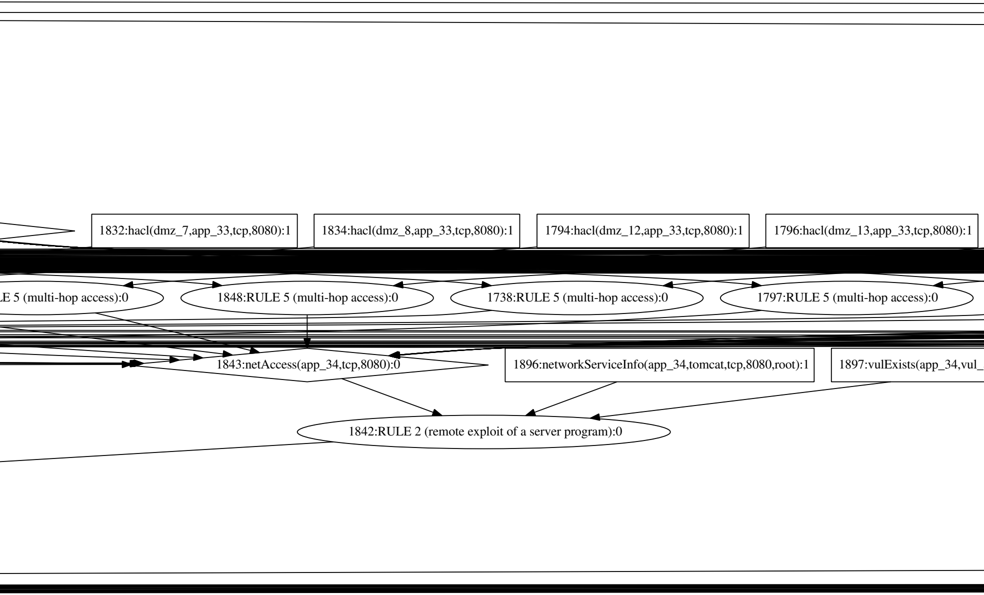

70:hacl(internet,dmz_14,tcp,80):1
69:RULE 6 (direct network access):0
3,httpd,tcp,80,root):1
63:vulExists(dmz_13,vul_web,httpd,remoteExploit,privEscalation):1
68:netAccess(dmz_14,tcp,80
1786:hacl(dmz_0,app_33,tcp,8080):1
1863:hacl(dmz_17,app_34,tcp,8080):1
1814:hacl(dmz_21,app_33,tcp,8080):1
1830:hacl(dmz_6,app_33,tcp,8080):1
1788:hacl(dmz_1,app_33,tcp,8080):1
LE 5 (multi-hop access):0
1679:RULE 5 (multi-hop access):0
1681:RULE 5 (multi-hop access):0
1886:RULE 5 (multi-hop access):0
1866:RULE 5 (multi-h

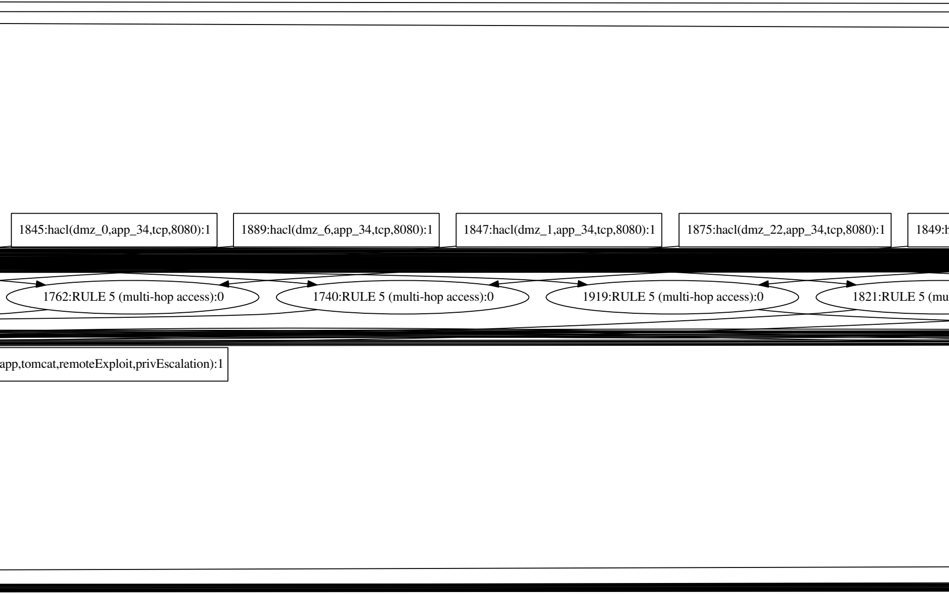

79:hacl(internet,dmz_15,tcp,80):1
78:RULE 6 (direct network access):0
)):0
71:networkServiceInfo(dmz_14,httpd,tcp,80,root):1
72:vulExists(dmz_14,vul_web,httpd,remoteExploit,privEscalation):1
67:RULE 2 (remote exploit of a server program):0
1790:hacl(dmz_10,app_33,tcp,8080):1
1737:hacl(dmz_13,app_32,tcp,8080):1
1865:hacl(dmz_18,app_34,tcp,8080):1
1680:hacl(dmz_14,app_31,tcp,8080):1
1482:hacl(dmz_9,app
hop access):0
1815:RULE 5 (multi-hop access):0
1868:RULE 5 (multi-hop access):0
1819:RULE 5 (multi-hop access):0
1791:RULE 5
1784:netAccess(app_33,tcp,8080):0
1837:networkServiceInfo(app_33,tomcat,tcp,8080,root):1
1838:vulExists(app_33,vul_app,tomcat,remoteExpl
1783:RULE 2 (remote exploit of a server program):0

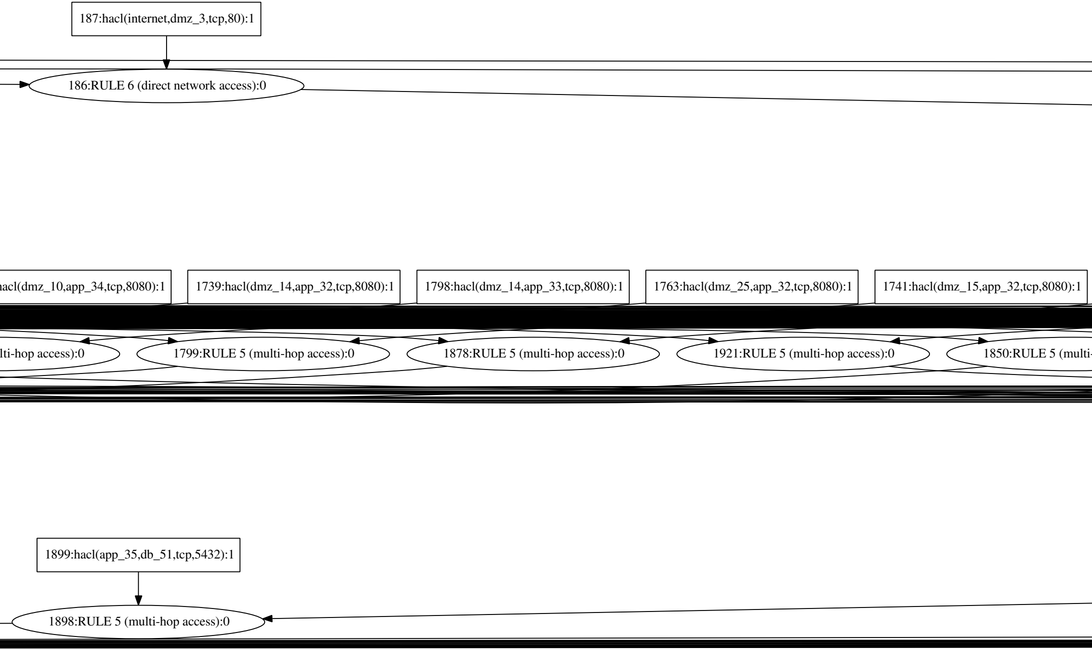

77:netAccess(dmz_15,tcp,80):0
80:networkServiceInfo(dmz_15,httpd,tcp,80,root):1
81:vulExists(dmz_15,vul_web,httpd,remoteExploit,privEscalation):1
76:RULE 2 (remote exploit of a server program):0
_28,tcp,8080):1
66:execCode(dmz_14,root):0
1682:hacl(dmz_15,app_31,tcp,8080):1
75:execCode(dmz_15,root):0
1887:hacl(dmz_5,app_34,tcp,8080):1
(multi-hop access):0
1817:RULE 5 (multi-hop access):0
1870:RULE 5 (multi-hop access):0
1872:RULE 5 (multi-hop access):0
1481:RULE 5 (multi-hop access)
loit,privEscalation):1
1840:hacl(app_34,db_51,tcp,5432):1
1839:RULE 5 (multi-hop access):0

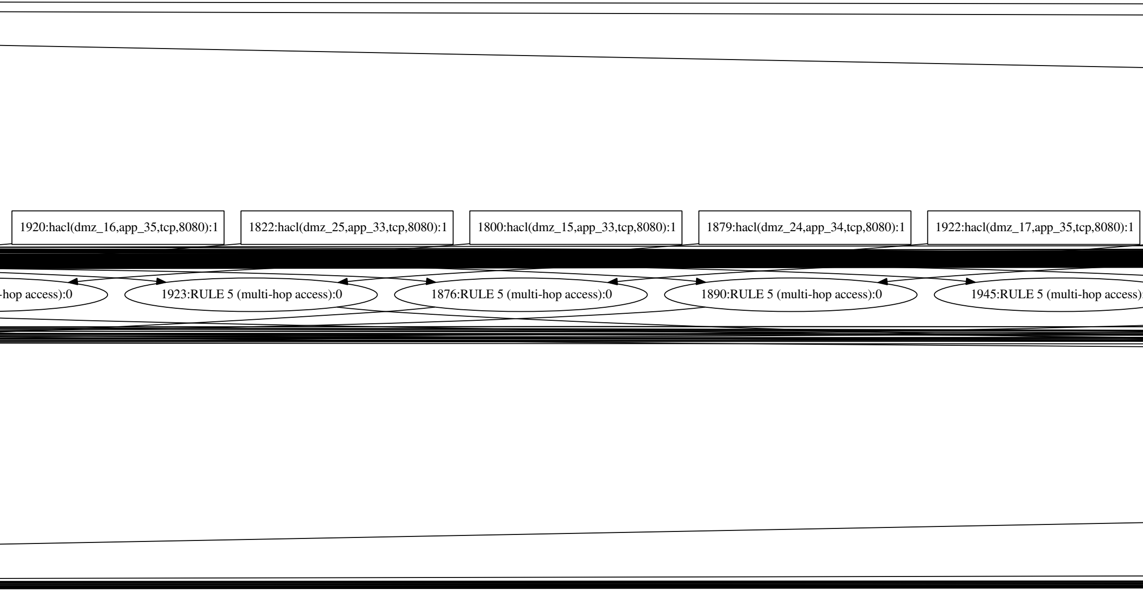

178:hacl(internet,dmz_25,tcp,80):1
177:RULE 6 (direct network access):0
176:netAccess(dmz_25,tcp,80):0
1867:hacl(dmz_19,app_34,tcp,8080):1
1816:hacl(dmz_22,app_33,tcp,8080):1
1869:hacl(dmz_2,app_34,tcp,8080):1
1820:hacl(dmz_24,app_33,tcp,8080):1
1792:hacl(dmz_11,app_33,tcp,8080):1
1703:RULE 5 (multi-hop access):0
1831:RULE 5 (multi-hop access):0
1833:RULE 5 (multi-hop access):0
1793:RULE 5 (multi-hop access):0

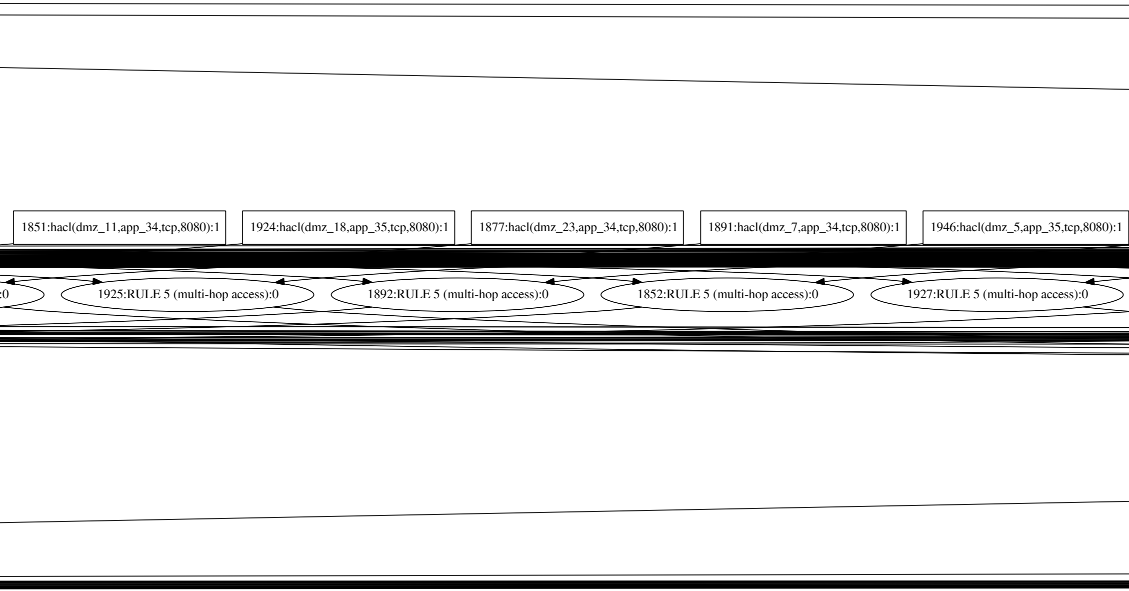

179:networkServiceInfo(dmz_25,httpd,tcp,80,root):1
180:vulExists(dmz_25,vul_web,httpd,remoteExploit,privEscalation):1
175:RULE 2 (remote exploit of a server program):0
1818:hacl(dmz_23,app_33,tcp,8080):1
1871:hacl(dmz_20,app_34,tcp,8080):1
1873:hacl(dmz_21,app_34,tcp,8080):1
1704:hacl(dmz_25,app_31,tcp,8080):1
174:execCode(dmz_25,root):0
1795:RULE 5 (multi-hop access):0
1844:RULE 5 (multi-hop access):0
1888:RULE 5 (multi-hop access):0
1846:RULE 5 (multi-hop access):0
1874:RUL
1841:execCode(app_34,root):0

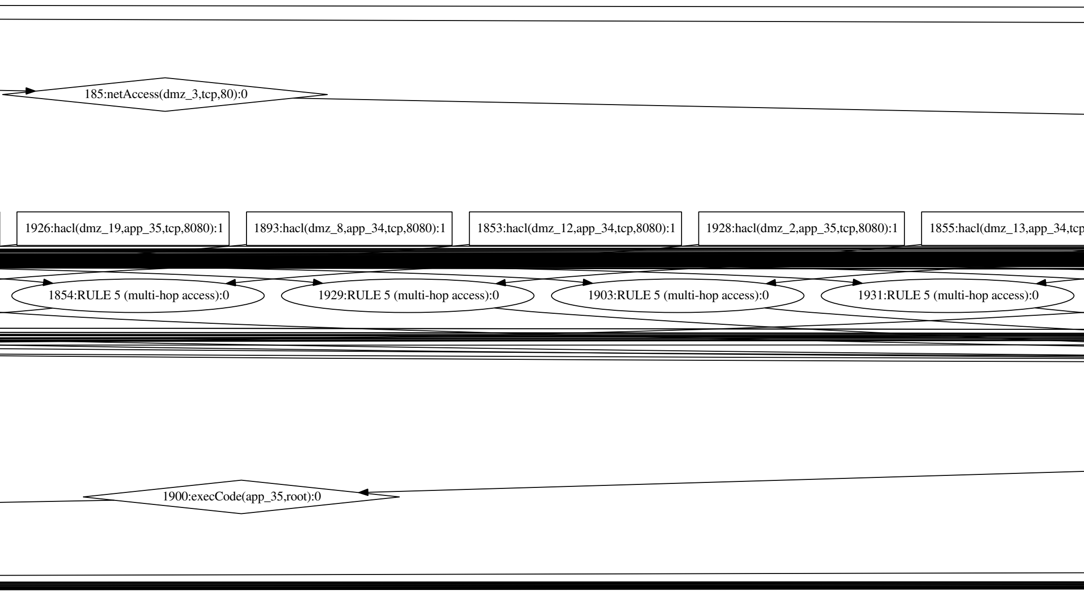

1832:hacl(dmz_7,app_33,tcp,8080):1
1834:hacl(dmz_8,app_33,tcp,8080):1
1794:hacl(dmz_12,app_33,tcp,8080):1
1796:hacl(dmz_13,app_33,tcp,8080):1
LE 5 (multi-hop access):0
1848:RULE 5 (multi-hop access):0
1738:RULE 5 (multi-hop access):0
1797:RULE 5 (multi-hop access):0
1843:netAccess(app_34,tcp,8080):0
1896:networkServiceInfo(app_34,tomcat,tcp,8080,root):1
1897:vulExists(app_34,vul_
1842:RULE 2 (remote exploit of a server program):0

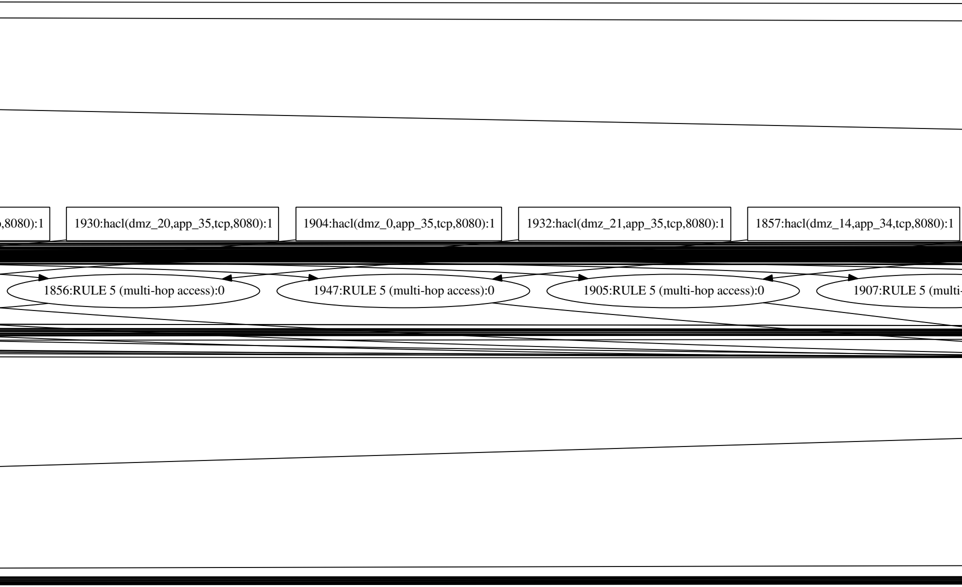

1845:hacl(dmz_0,app_34,tcp,8080):1
1889:hacl(dmz_6,app_34,tcp,8080):1
1847:hacl(dmz_1,app_34,tcp,8080):1
1875:hacl(dmz_22,app_34,tcp,8080):1
1849:h
1762:RULE 5 (multi-hop access):0
1740:RULE 5 (multi-hop access):0
1919:RULE 5 (multi-hop access):0
1821:RULE 5 (mu
app,tomcat,remoteExploit,privEscalation):1

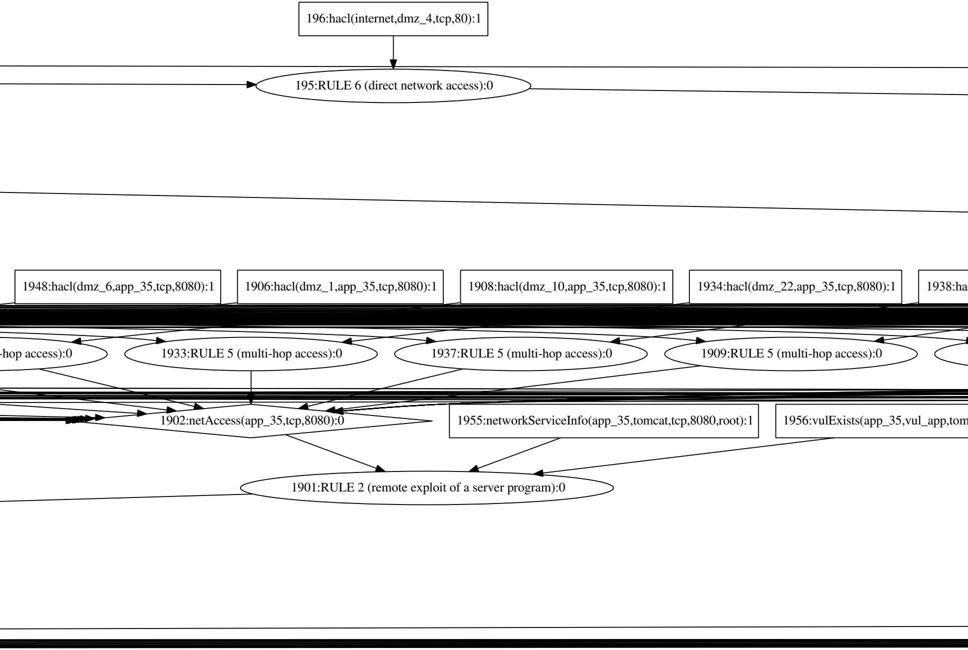

187:hacl(internet,dmz_3,tcp,80):1
186:RULE 6 (direct network access):0
acl(dmz_10,app_34,tcp,8080):1
1739:hacl(dmz_14,app_32,tcp,8080):1
1798:hacl(dmz_14,app_33,tcp,8080):1
1763:hacl(dmz_25,app_32,tcp,8080):1
1741:hacl(dmz_15,app_32,tcp,8080):1
lti-hop access):0
1799:RULE 5 (multi-hop access):0
1878:RULE 5 (multi-hop access):0
1921:RULE 5 (multi-hop access):0
1850:RULE 5 (multi-
1899:hacl(app_35,db_51,tcp,5432):1
1898:RULE 5 (multi-hop access):0

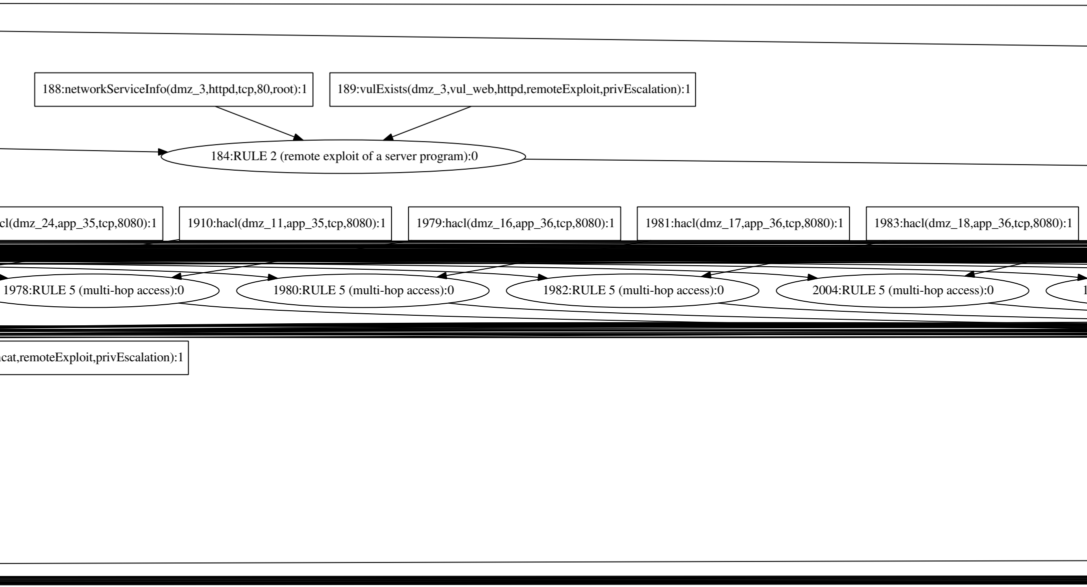

1920:hacl(dmz_16,app_35,tcp,8080):1
1822:hacl(dmz_25,app_33,tcp,8080):1
1800:hacl(dmz_15,app_33,tcp,8080):1
1879:hacl(dmz_24,app_34,tcp,8080):1
1922:hacl(dmz_17,app_35,tcp,8080):1
-hop access):0
1923:RULE 5 (multi-hop access):0
1876:RULE 5 (multi-hop access):0
1890:RULE 5 (multi-hop access):0
1945:RULE 5 (multi-hop access)

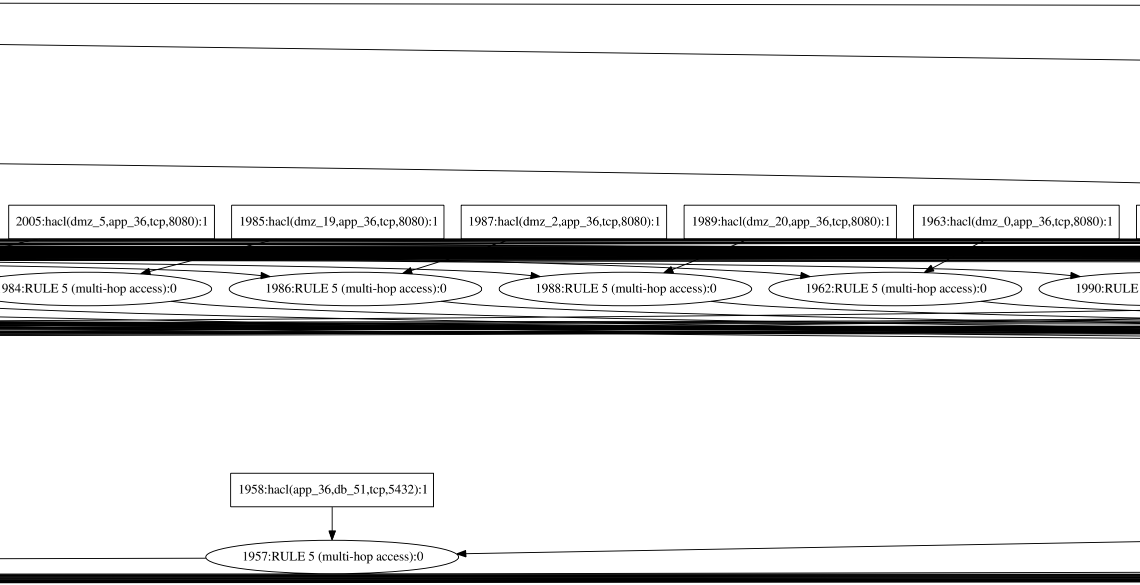

1851:hacl(dmz_11,app_34,tcp,8080):1
1924:hacl(dmz_18,app_35,tcp,8080):1
1877:hacl(dmz_23,app_34,tcp,8080):1
1891:hacl(dmz_7,app_34,tcp,8080):1
1946:hacl(dmz_5,app_35,tcp,8080):1
:0
1925:RULE 5 (multi-hop access):0
1892:RULE 5 (multi-hop access):0
1852:RULE 5 (multi-hop access):0
1927:RULE 5 (multi-hop access):0

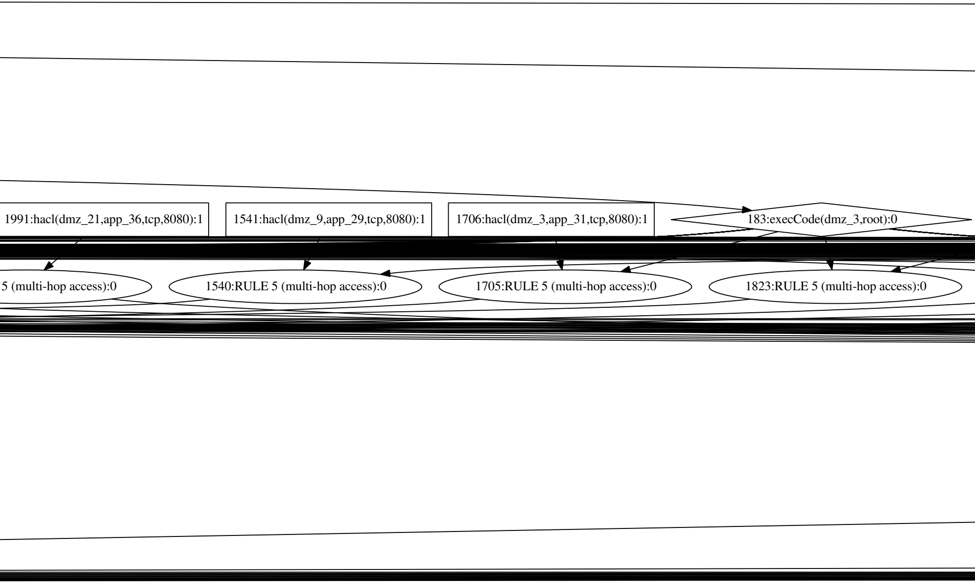

185:netAccess(dmz_3,tcp,80):0
1926:hacl(dmz_19,app_35,tcp,8080):1
1893:hacl(dmz_8,app_34,tcp,8080):1
1853:hacl(dmz_12,app_34,tcp,8080):1
1928:hacl(dmz_2,app_35,tcp,8080):1
1855:hacl(dmz_13,app_34,tcp
1854:RULE 5 (multi-hop access):0
1929:RULE 5 (multi-hop access):0
1903:RULE 5 (multi-hop access):0
1931:RULE 5 (multi-hop access):0
1900:execCode(app_35,root):0

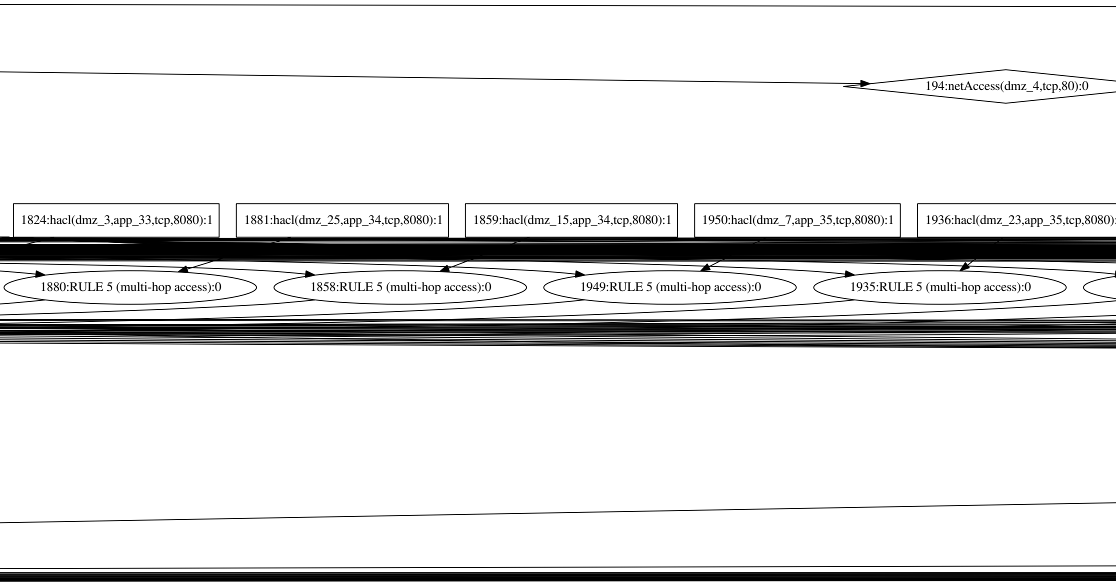

,8080):1
1930:hacl(dmz_20,app_35,tcp,8080):1
1904:hacl(dmz_0,app_35,tcp,8080):1
1932:hacl(dmz_21,app_35,tcp,8080):1
1857:hacl(dmz_14,app_34,tcp,8080):1
1856:RULE 5 (multi-hop access):0
1947:RULE 5 (multi-hop access):0
1905:RULE 5 (multi-hop access):0
1907:RULE 5 (multi-

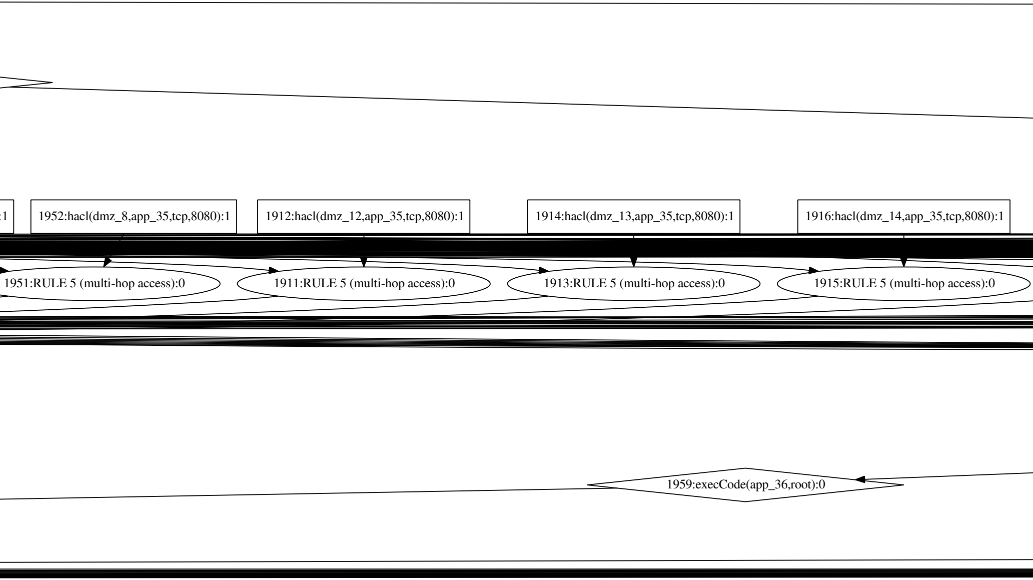

196:hacl(internet,dmz_4,tcp,80):1
195:RULE 6 (direct network access):0
1948:hacl(dmz_6,app_35,tcp,8080):1
1906:hacl(dmz_1,app_35,tcp,8080):1
1908:hacl(dmz_10,app_35,tcp,8080):1
1934:hacl(dmz_22,app_35,tcp,8080):1
1938:ha
-hop access):0
1933:RULE 5 (multi-hop access):0
1937:RULE 5 (multi-hop access):0
1909:RULE 5 (multi-hop access):0
1902:netAccess(app_35,tcp,8080):0
1955:networkServiceInfo(app_35,tomcat,tcp,8080,root):1
1956:vulExists(app_35,vul_app,tom
1901:RULE 2 (remote exploit of a server program):0

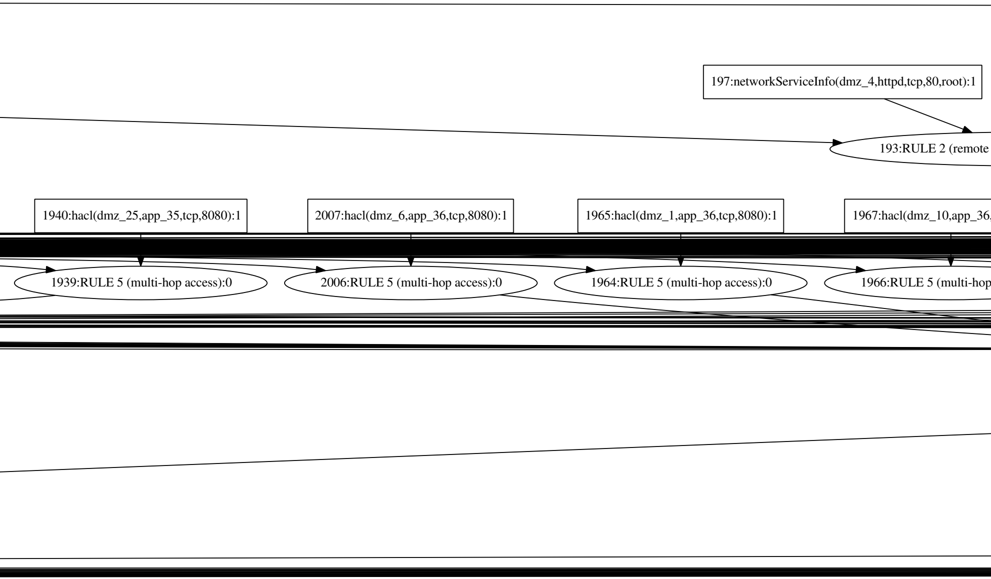

188:networkServiceInfo(dmz_3,httpd,tcp,80,root):1
189:vulExists(dmz_3,vul_web,httpd,remoteExploit,privEscalation):1
184:RULE 2 (remote exploit of a server program):0
cl(dmz_24,app_35,tcp,8080):1
1910:hacl(dmz_11,app_35,tcp,8080):1
1979:hacl(dmz_16,app_36,tcp,8080):1
1981:hacl(dmz_17,app_36,tcp,8080):1
1983:hacl(dmz_18,app_36,tcp,8080):1
1978:RULE 5 (multi-hop access):0
1980:RULE 5 (multi-hop access):0
1982:RULE 5 (multi-hop access):0
2004:RULE 5 (multi-hop access):0
cat,remoteExploit,privEscalation):1

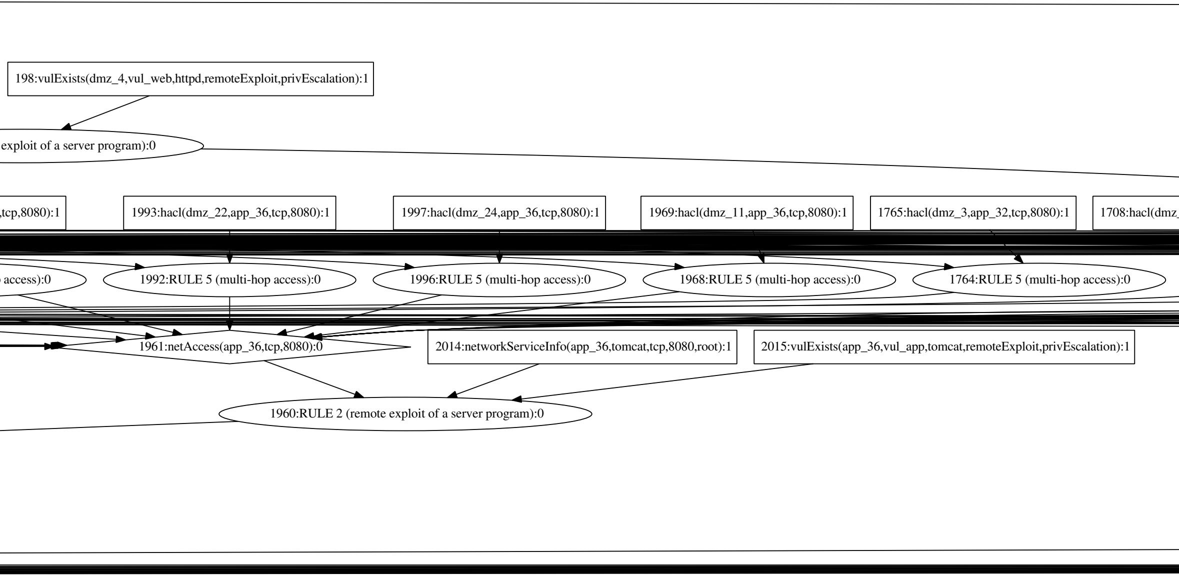

2005:hacl(dmz_5,app_36,tcp,8080):1
1985:hacl(dmz_19,app_36,tcp,8080):1
1987:hacl(dmz_2,app_36,tcp,8080):1
1989:hacl(dmz_20,app_36,tcp,8080):1
1963:hacl(dmz_0,app_36,tcp,8080):1
984:RULE 5 (multi-hop access):0
1986:RULE 5 (multi-hop access):0
1988:RULE 5 (multi-hop access):0
1962:RULE 5 (multi-hop access):0
1990:RULE
1958:hacl(app_36,db_51,tcp,5432):1
1957:RULE 5 (multi-hop access):0

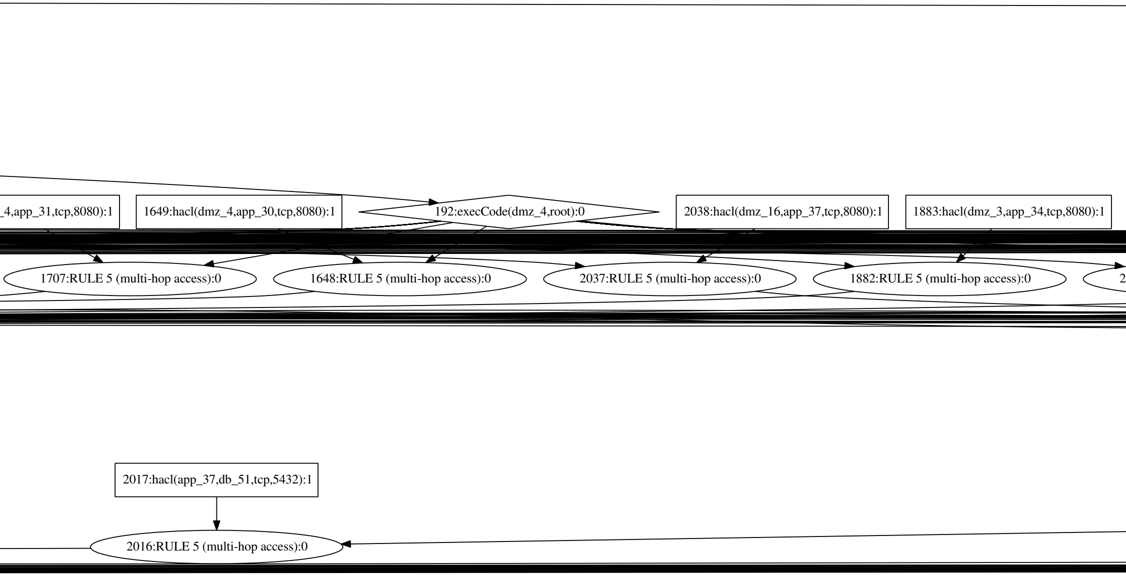

1991:hacl(dmz_21,app_36,tcp,8080):1
1541:hacl(dmz_9,app_29,tcp,8080):1
1706:hacl(dmz_3,app_31,tcp,8080):1
183:execCode(dmz_3,root):0
5 (multi-hop access):0
1540:RULE 5 (multi-hop access):0
1705:RULE 5 (multi-hop access):0
1823:RULE 5 (multi-hop access):0

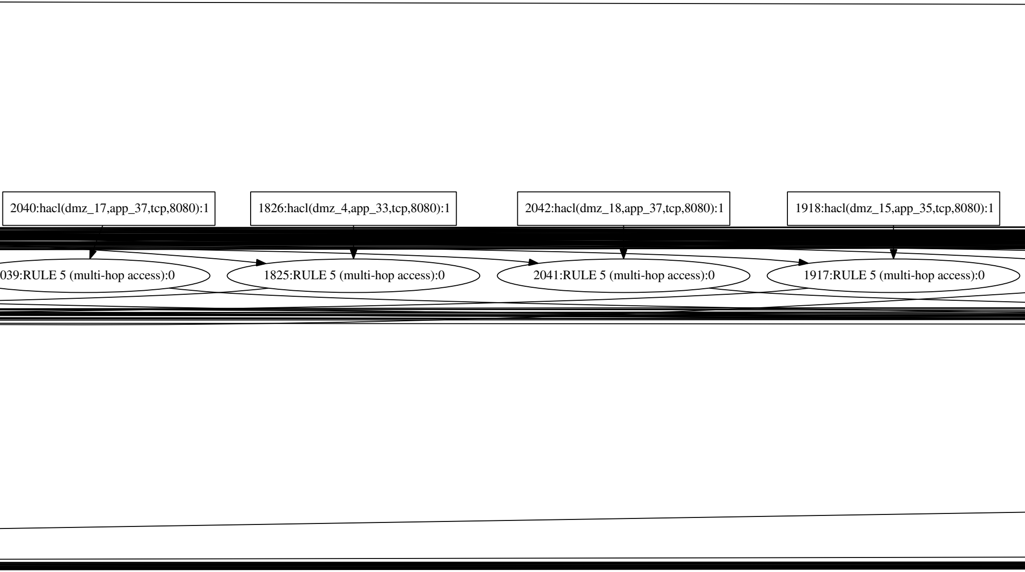

194:netAccess(dmz_4,tcp,80):0
1824:hacl(dmz_3,app_33,tcp,8080):1
1881:hacl(dmz_25,app_34,tcp,8080):1
1859:hacl(dmz_15,app_34,tcp,8080):1
1950:hacl(dmz_7,app_35,tcp,8080):1
1936:hacl(dmz_23,app_35,tcp,8080)
1880:RULE 5 (multi-hop access):0
1858:RULE 5 (multi-hop access):0
1949:RULE 5 (multi-hop access):0
1935:RULE 5 (multi-hop access):0

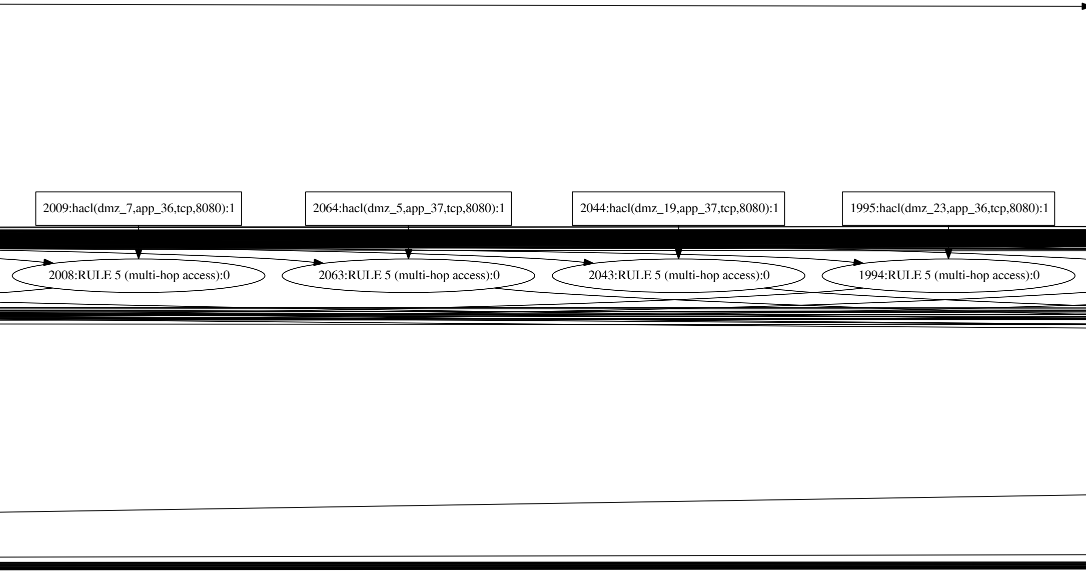

:1
1952:hacl(dmz_8,app_35,tcp,8080):1
1912:hacl(dmz_12,app_35,tcp,8080):1
1914:hacl(dmz_13,app_35,tcp,8080):1
1916:hacl(dmz_14,app_35,tcp,8080):1
1951:RULE 5 (multi-hop access):0
1911:RULE 5 (multi-hop access):0
1913:RULE 5 (multi-hop access):0
1915:RULE 5 (multi-hop access):0
1959:execCode(app_36,root):0

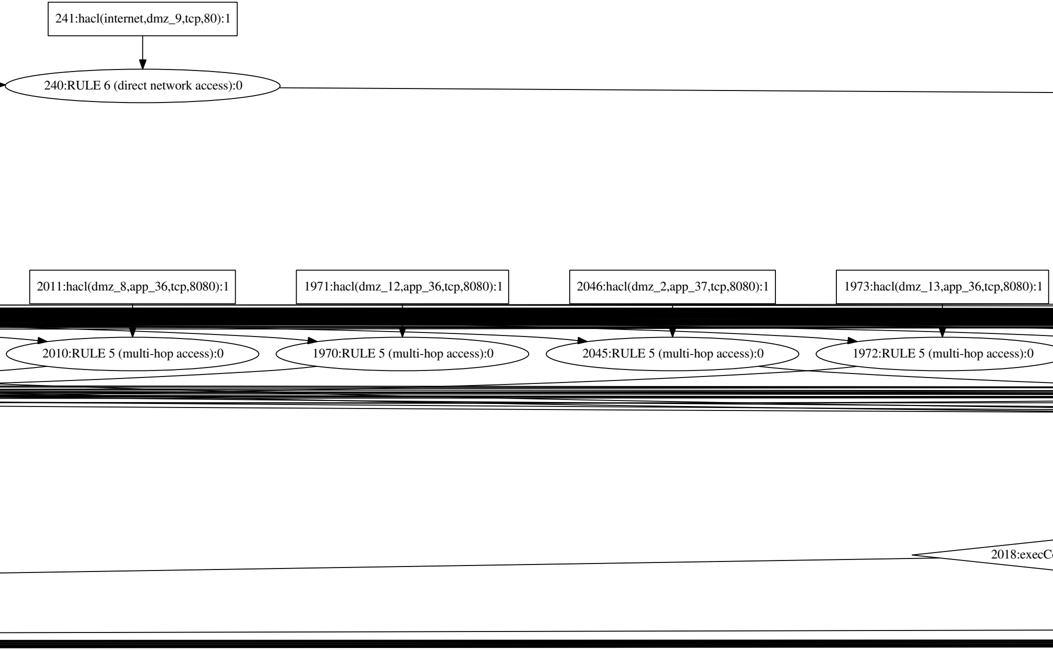

197:networkServiceInfo(dmz_4,httpd,tcp,80,root):1
193:RULE 2 (remote
1940:hacl(dmz_25,app_35,tcp,8080):1
2007:hacl(dmz_6,app_36,tcp,8080):1
1965:hacl(dmz_1,app_36,tcp,8080):1
1967:hacl(dmz_10,app_36,
1939:RULE 5 (multi-hop access):0
2006:RULE 5 (multi-hop access):0
1964:RULE 5 (multi-hop access):0
1966:RULE 5 (multi-hop

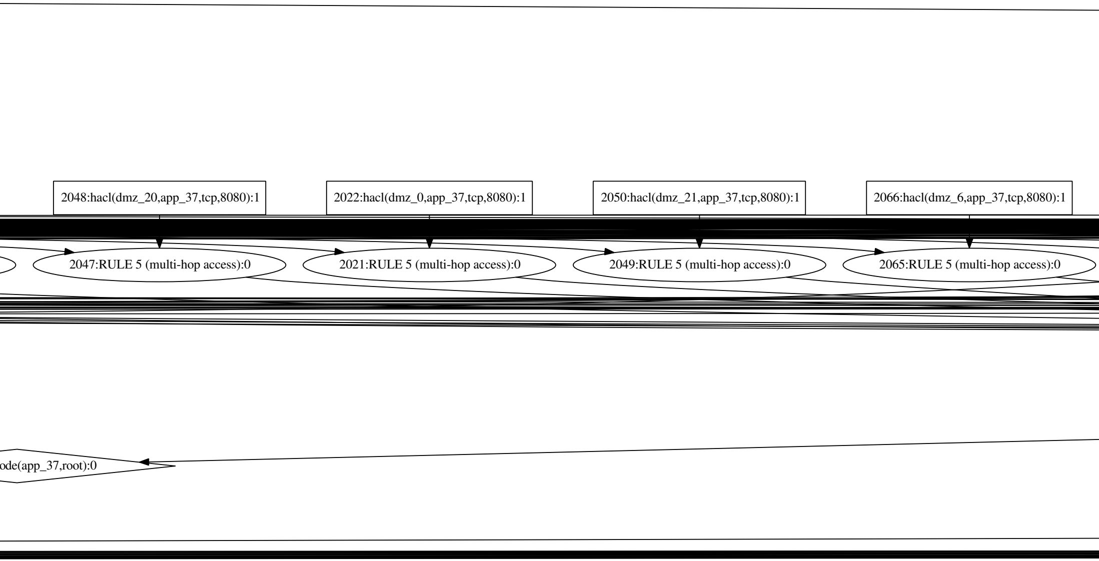

198:vulExists(dmz_4,vul_web,httpd,remoteExploit,privEscalation):1
exploit of a server program):0
,tcp,8080):1
1993:hacl(dmz_22,app_36,tcp,8080):1
1997:hacl(dmz_24,app_36,tcp,8080):1
1969:hacl(dmz_11,app_36,tcp,8080):1
1765:hacl(dmz_3,app_32,tcp,8080):1
1708:hacl(dmz_
access):0
1992:RULE 5 (multi-hop access):0
1996:RULE 5 (multi-hop access):0
1968:RULE 5 (multi-hop access):0
1764:RULE 5 (multi-hop access):0
1961:netAccess(app_36,tcp,8080):0
2014:networkServiceInfo(app_36,tomcat,tcp,8080,root):1
2015:vulExists(app_36,vul_app,tomcat,remoteExploit,privEscalation):1
1960:RULE 2 (remote exploit of a server program):0

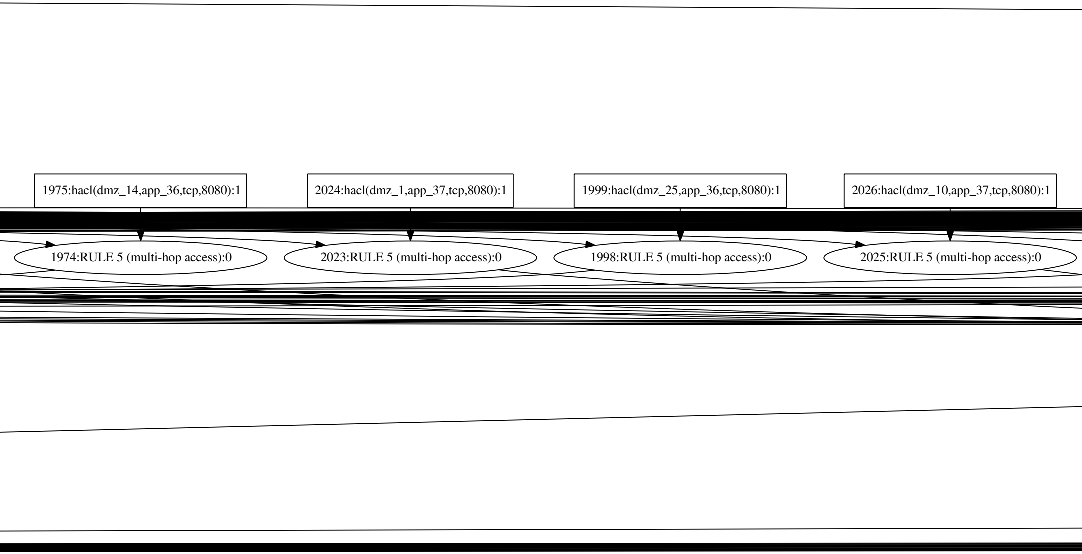

_4,app_31,tcp,8080):1
1649:hacl(dmz_4,app_30,tcp,8080):1
192:execCode(dmz_4,root):0
2038:hacl(dmz_16,app_37,tcp,8080):1
1883:hacl(dmz_3,app_34,tcp,8080):1
1707:RULE 5 (multi-hop access):0
1648:RULE 5 (multi-hop access):0
2037:RULE 5 (multi-hop access):0
1882:RULE 5 (multi-hop access):0
2017:hacl(app_37,db_51,tcp,5432):1
2016:RULE 5 (multi-hop access):0

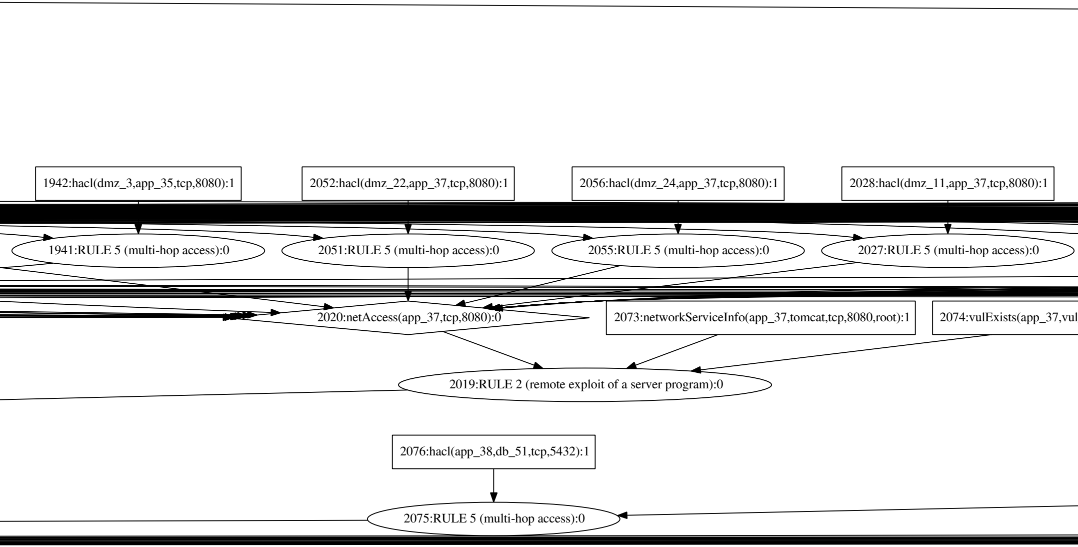

2040:hacl(dmz_17,app_37,tcp,8080):1
1826:hacl(dmz_4,app_33,tcp,8080):1
2042:hacl(dmz_18,app_37,tcp,8080):1
1918:hacl(dmz_15,app_35,tcp,8080):1
039:RULE 5 (multi-hop access):0
1825:RULE 5 (multi-hop access):0
2041:RULE 5 (multi-hop access):0
1917:RULE 5 (multi-hop access):0

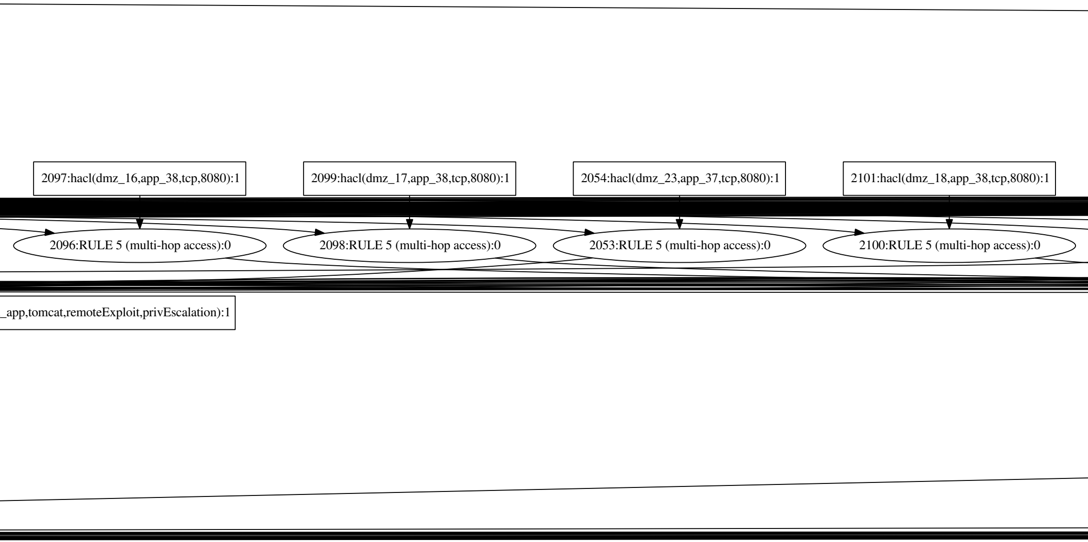

2009:hacl(dmz_7,app_36,tcp,8080):1
2064:hacl(dmz_5,app_37,tcp,8080):1
2044:hacl(dmz_19,app_37,tcp,8080):1
1995:hacl(dmz_23,app_36,tcp,8080):1
2008:RULE 5 (multi-hop access):0
2063:RULE 5 (multi-hop access):0
2043:RULE 5 (multi-hop access):0
1994:RULE 5 (multi-hop access):0

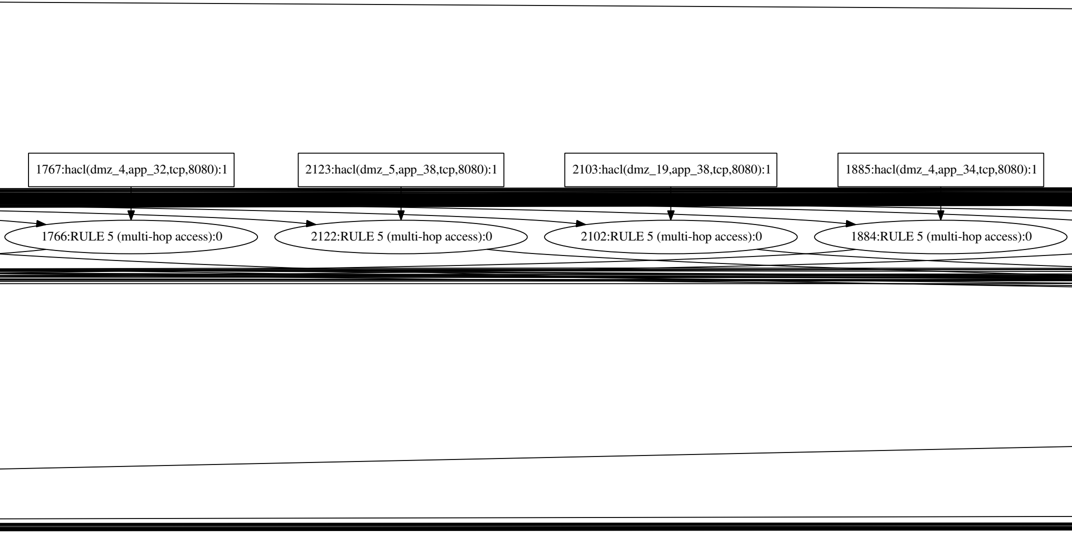

241:hacl(internet,dmz_9,tcp,80):1
240:RULE 6 (direct network access):0
2011:hacl(dmz_8,app_36,tcp,8080):1
1971:hacl(dmz_12,app_36,tcp,8080):1
2046:hacl(dmz_2,app_37,tcp,8080):1
1973:hacl(dmz_13,app_36,tcp,8080):1
2010:RULE 5 (multi-hop access):0
1970:RULE 5 (multi-hop access):0
2045:RULE 5 (multi-hop access):0
1972:RULE 5 (multi-hop access):0
2018:execC

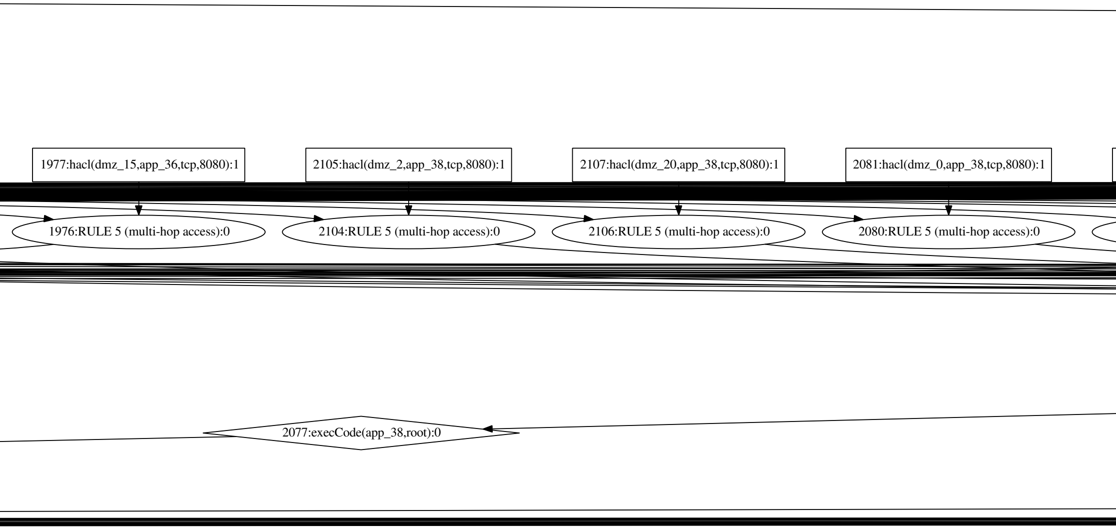

2048:hacl(dmz_20,app_37,tcp,8080):1
2022:hacl(dmz_0,app_37,tcp,8080):1
2050:hacl(dmz_21,app_37,tcp,8080):1
2066:hacl(dmz_6,app_37,tcp,8080):1
2047:RULE 5 (multi-hop access):0
2021:RULE 5 (multi-hop access):0
2049:RULE 5 (multi-hop access):0
2065:RULE 5 (multi-hop access):0
ode(app_37,root):0

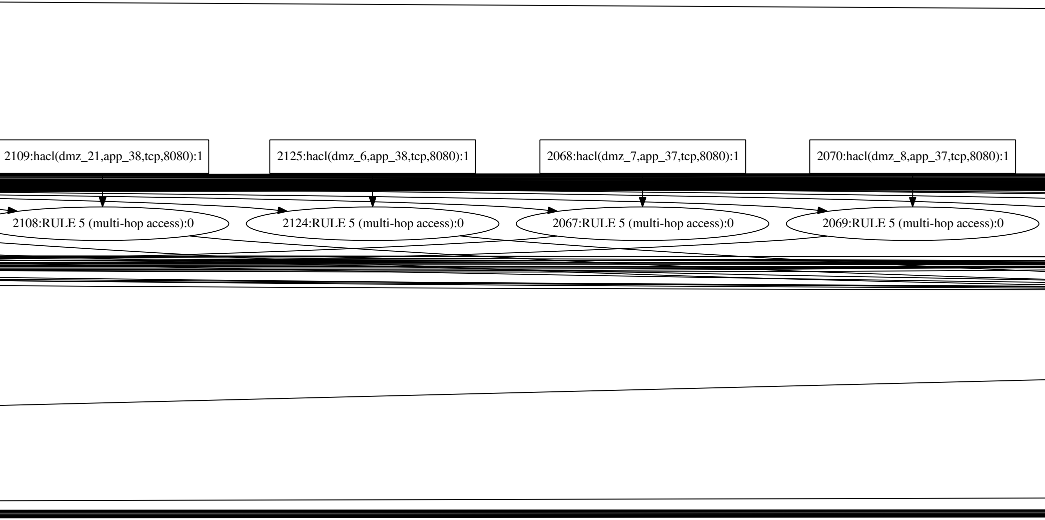

1975:hacl(dmz_14,app_36,tcp,8080):1
2024:hacl(dmz_1,app_37,tcp,8080):1
1999:hacl(dmz_25,app_36,tcp,8080):1
2026:hacl(dmz_10,app_37,tcp,8080):1
1974:RULE 5 (multi-hop access):0
2023:RULE 5 (multi-hop access):0
1998:RULE 5 (multi-hop access):0
2025:RULE 5 (multi-hop access):0

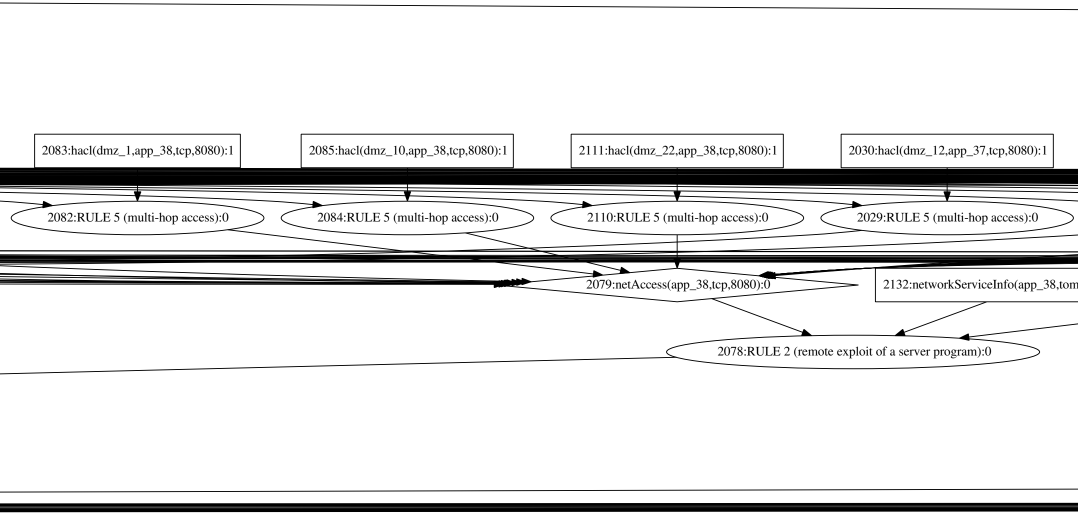

1942:hacl(dmz_3,app_35,tcp,8080):1
2052:hacl(dmz_22,app_37,tcp,8080):1
2056:hacl(dmz_24,app_37,tcp,8080):1
2028:hacl(dmz_11,app_37,tcp,8080):1
1941:RULE 5 (multi-hop access):0
2051:RULE 5 (multi-hop access):0
2055:RULE 5 (multi-hop access):0
2027:RULE 5 (multi-hop access):0
2020:netAccess(app_37,tcp,8080):0
2073:networkServiceInfo(app_37,tomcat,tcp,8080,root):1
2074:vulExists(app_37,vul
2019:RULE 2 (remote exploit of a server program):0
2076:hacl(app_38,db_51,tcp,5432):1
2075:RULE 5 (multi-hop access):0

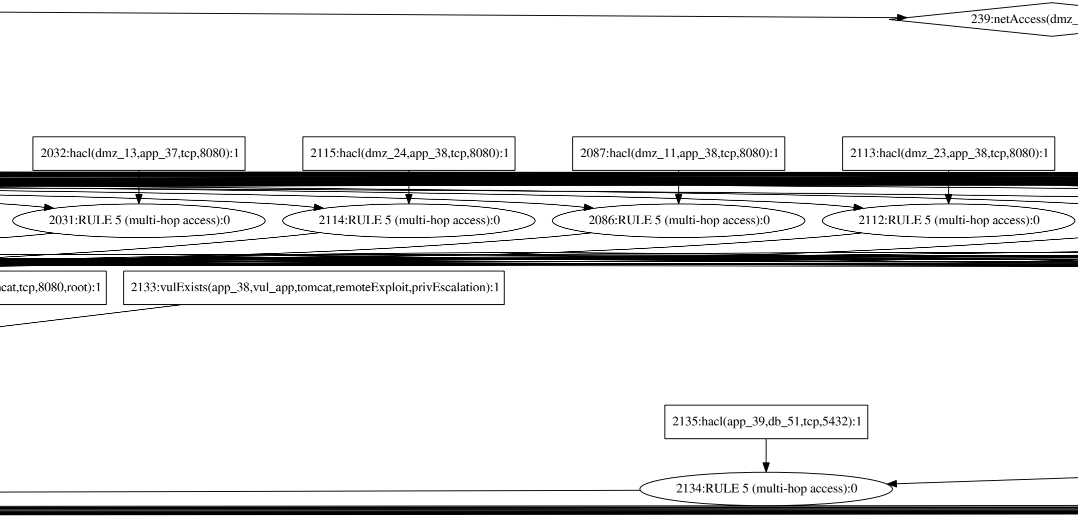

2097:hacl(dmz_16,app_38,tcp,8080):1
2099:hacl(dmz_17,app_38,tcp,8080):1
2054:hacl(dmz_23,app_37,tcp,8080):1
2101:hacl(dmz_18,app_38,tcp,8080):1
2096:RULE 5 (multi-hop access):0
2098:RULE 5 (multi-hop access):0
2053:RULE 5 (multi-hop access):0
2100:RULE 5 (multi-hop access):0
_app,tomcat,remoteExploit,privEscalation):1

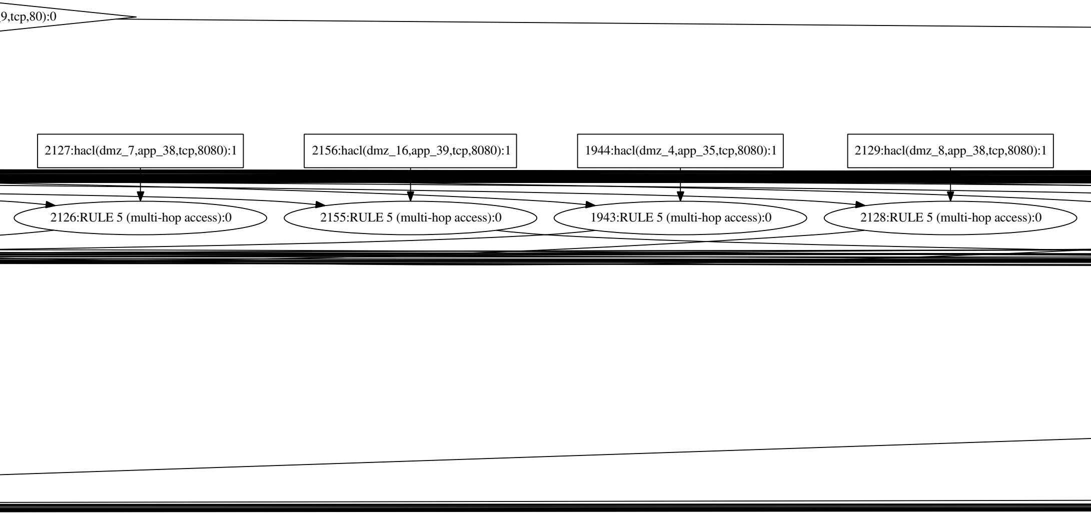

1767:hacl(dmz_4,app_32,tcp,8080):1
2123:hacl(dmz_5,app_38,tcp,8080):1
2103:hacl(dmz_19,app_38,tcp,8080):1
1885:hacl(dmz_4,app_34,tcp,8080):1
1766:RULE 5 (multi-hop access):0
2122:RULE 5 (multi-hop access):0
2102:RULE 5 (multi-hop access):0
1884:RULE 5 (multi-hop access):0

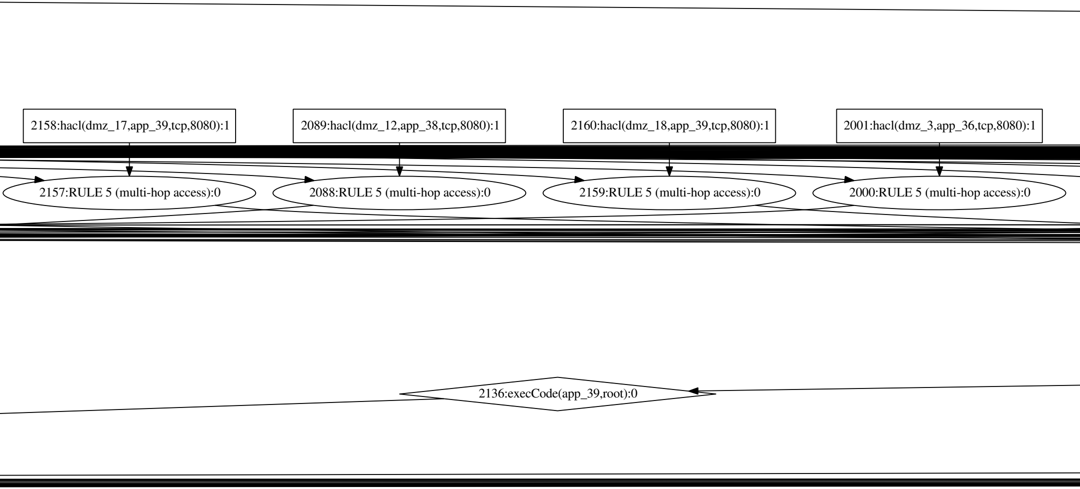

1977:hacl(dmz_15,app_36,tcp,8080):1
2105:hacl(dmz_2,app_38,tcp,8080):1
2107:hacl(dmz_20,app_38,tcp,8080):1
2081:hacl(dmz_0,app_38,tcp,8080):1
1976:RULE 5 (multi-hop access):0
2104:RULE 5 (multi-hop access):0
2106:RULE 5 (multi-hop access):0
2080:RULE 5 (multi-hop access):0
2077:execCode(app_38,root):0

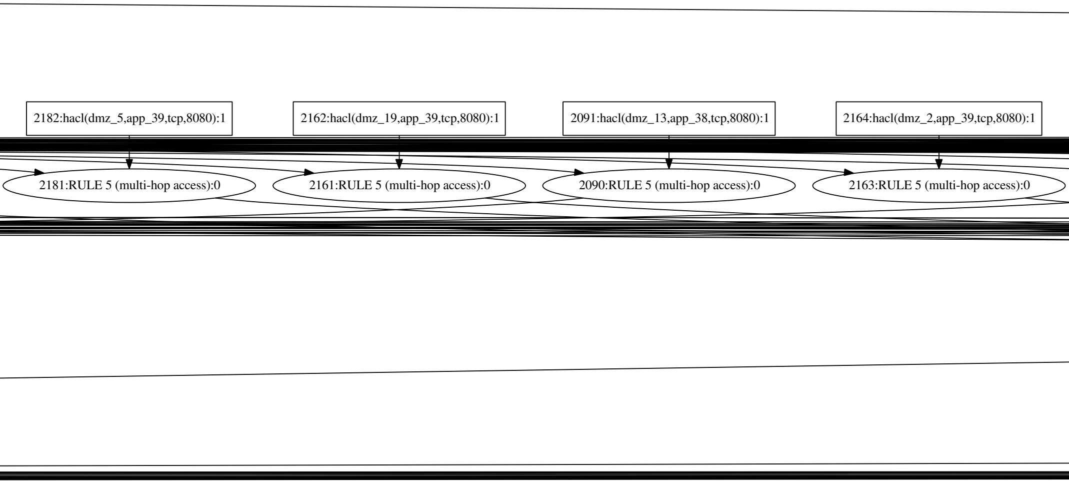

2109:hacl(dmz_21,app_38,tcp,8080):1
2125:hacl(dmz_6,app_38,tcp,8080):1
2068:hacl(dmz_7,app_37,tcp,8080):1
2070:hacl(dmz_8,app_37,tcp,8080):1
2108:RULE 5 (multi-hop access):0
2124:RULE 5 (multi-hop access):0
2067:RULE 5 (multi-hop access):0
2069:RULE 5 (multi-hop access):0

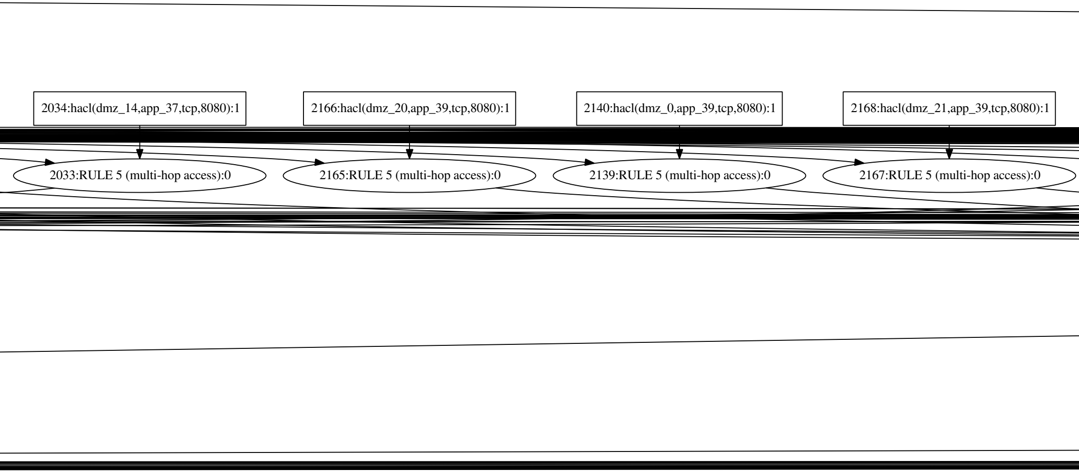

2083:hacl(dmz_1,app_38,tcp,8080):1
2085:hacl(dmz_10,app_38,tcp,8080):1
2111:hacl(dmz_22,app_38,tcp,8080):1
2030:hacl(dmz_12,app_37,tcp,8080):1
2082:RULE 5 (multi-hop access):0
2084:RULE 5 (multi-hop access):0
2110:RULE 5 (multi-hop access):0
2029:RULE 5 (multi-hop access):0
2079:netAccess(app_38,tcp,8080):0
2132:networkServiceInfo(app_38,tom
2078:RULE 2 (remote exploit of a server program):0

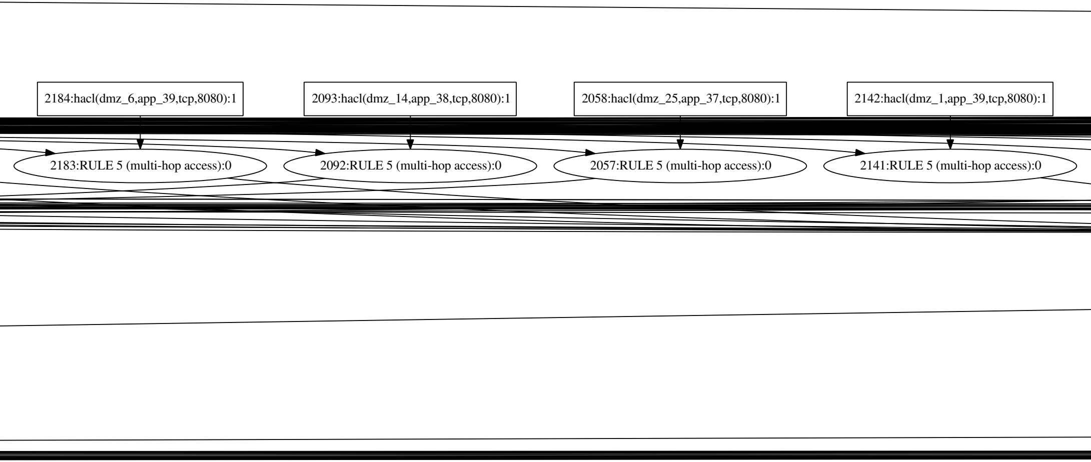

239:netAccess(dmz_
2032:hacl(dmz_13,app_37,tcp,8080):1
2115:hacl(dmz_24,app_38,tcp,8080):1
2087:hacl(dmz_11,app_38,tcp,8080):1
2113:hacl(dmz_23,app_38,tcp,8080):1
2031:RULE 5 (multi-hop access):0
2114:RULE 5 (multi-hop access):0
2086:RULE 5 (multi-hop access):0
2112:RULE 5 (multi-hop access):0
cat,tcp,8080,root):1
2133:vulExists(app_38,vul_app,tomcat,remoteExploit,privEscalation):1
2135:hacl(app_39,db_51,tcp,5432):1
2134:RULE 5 (multi-hop access):0

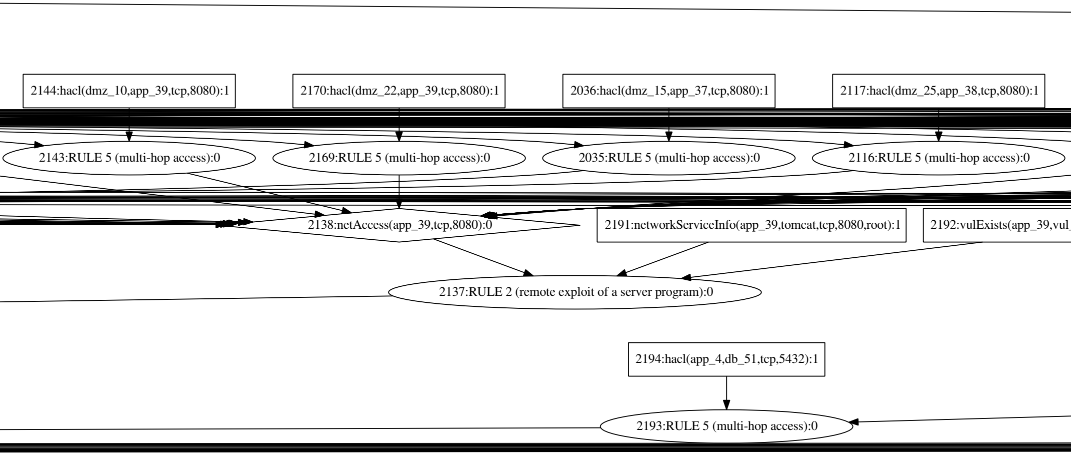

9,tcp,80):0
2127:hacl(dmz_7,app_38,tcp,8080):1
2156:hacl(dmz_16,app_39,tcp,8080):1
1944:hacl(dmz_4,app_35,tcp,8080):1
2129:hacl(dmz_8,app_38,tcp,8080):1
2126:RULE 5 (multi-hop access):0
2155:RULE 5 (multi-hop access):0
1943:RULE 5 (multi-hop access):0
2128:RULE 5 (multi-hop access):0

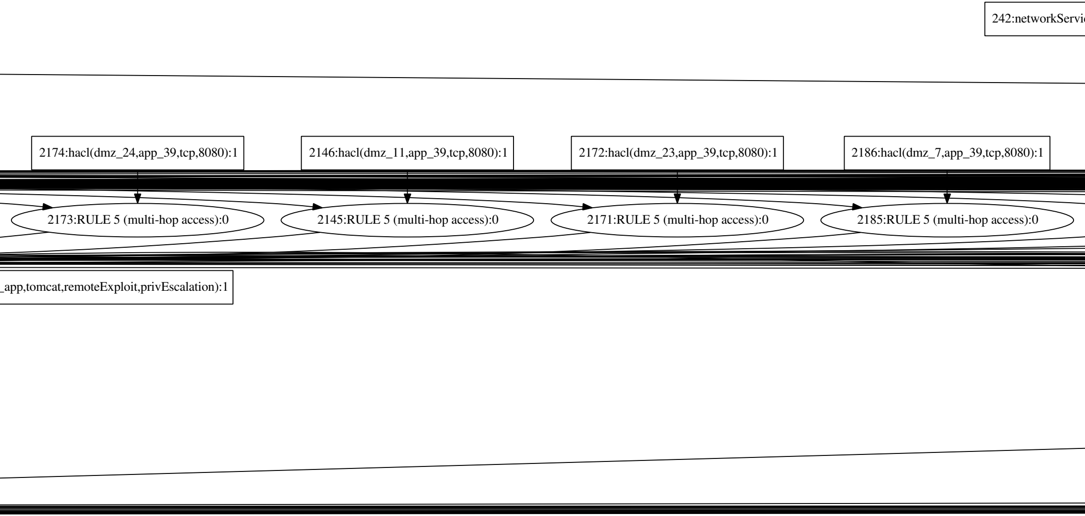

2158:hacl(dmz_17,app_39,tcp,8080):1
2089:hacl(dmz_12,app_38,tcp,8080):1
2160:hacl(dmz_18,app_39,tcp,8080):1
2001:hacl(dmz_3,app_36,tcp,8080):1
2157:RULE 5 (multi-hop access):0
2088:RULE 5 (multi-hop access):0
2159:RULE 5 (multi-hop access):0
2000:RULE 5 (multi-hop access):0
2136:execCode(app_39,root):0

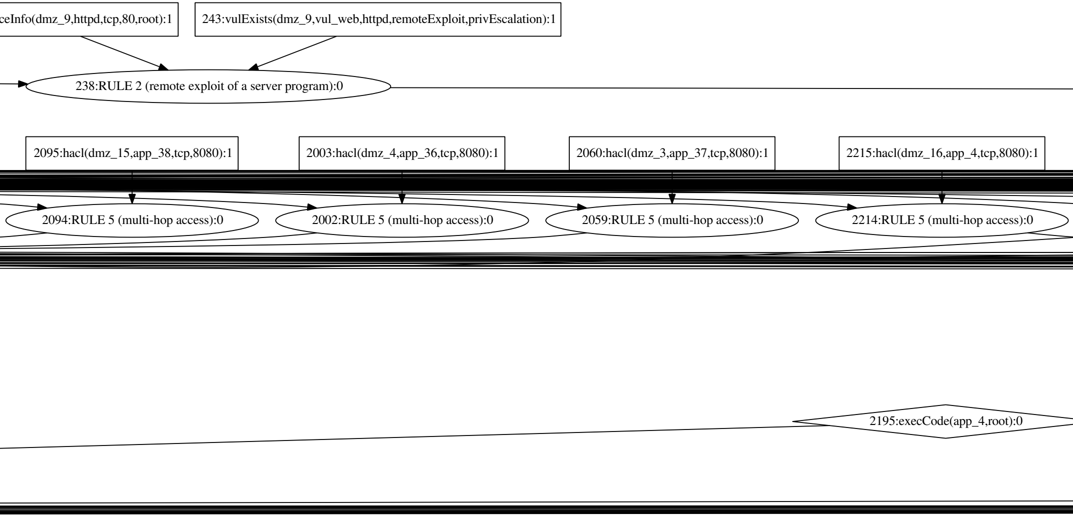

2182:hacl(dmz_5,app_39,tcp,8080):1
2162:hacl(dmz_19,app_39,tcp,8080):1
2091:hacl(dmz_13,app_38,tcp,8080):1
2164:hacl(dmz_2,app_39,tcp,8080):1
2181:RULE 5 (multi-hop access):0
2161:RULE 5 (multi-hop access):0
2090:RULE 5 (multi-hop access):0
2163:RULE 5 (multi-hop access):0

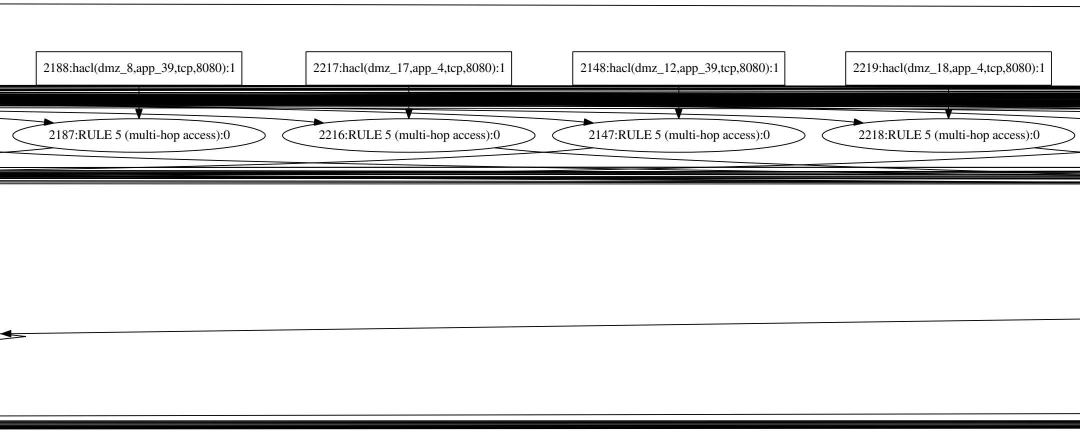

2034:hacl(dmz_14,app_37,tcp,8080):1
2166:hacl(dmz_20,app_39,tcp,8080):1
2140:hacl(dmz_0,app_39,tcp,8080):1
2168:hacl(dmz_21,app_39,tcp,8080):1
2033:RULE 5 (multi-hop access):0
2165:RULE 5 (multi-hop access):0
2139:RULE 5 (multi-hop access):0
2167:RULE 5 (multi-hop access):0

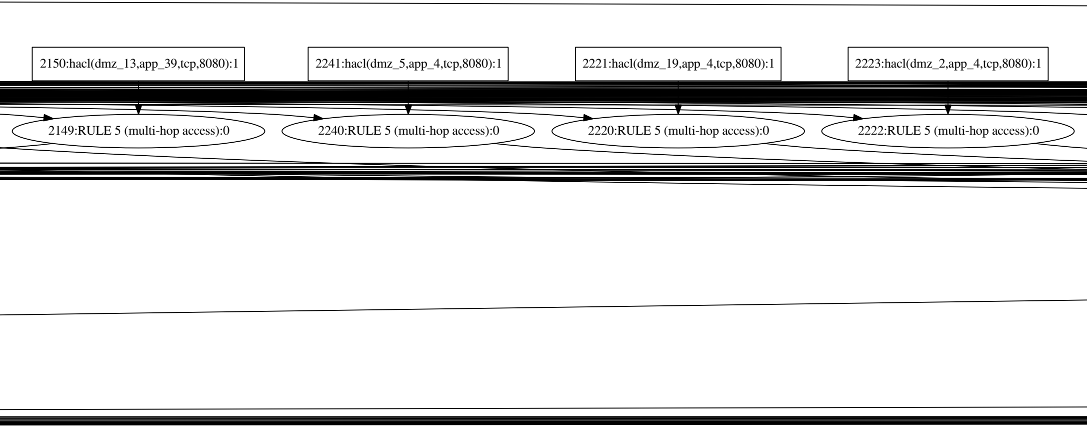

2184:hacl(dmz_6,app_39,tcp,8080):1
2093:hacl(dmz_14,app_38,tcp,8080):1
2058:hacl(dmz_25,app_37,tcp,8080):1
2142:hacl(dmz_1,app_39,tcp,8080):1
2183:RULE 5 (multi-hop access):0
2092:RULE 5 (multi-hop access):0
2057:RULE 5 (multi-hop access):0
2141:RULE 5 (multi-hop access):0

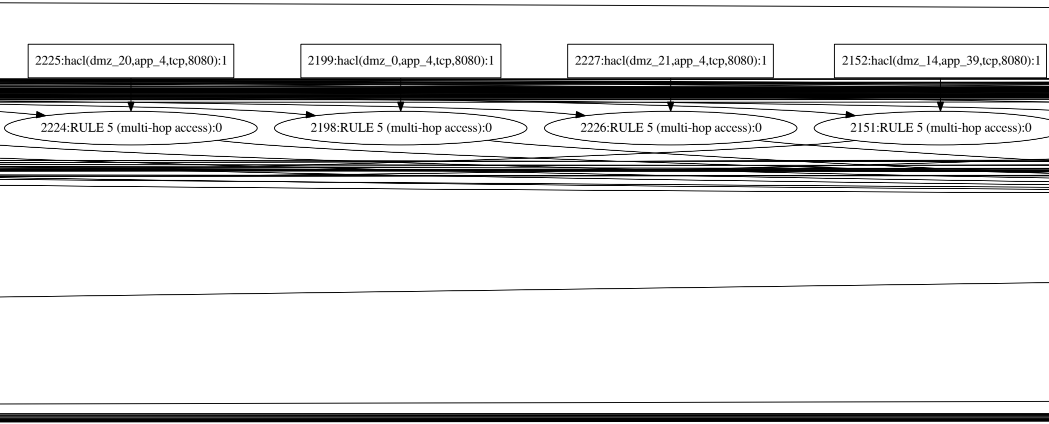

2144:hacl(dmz_10,app_39,tcp,8080):1
2170:hacl(dmz_22,app_39,tcp,8080):1
2036:hacl(dmz_15,app_37,tcp,8080):1
2117:hacl(dmz_25,app_38,tcp,8080):1
2143:RULE 5 (multi-hop access):0
2169:RULE 5 (multi-hop access):0
2035:RULE 5 (multi-hop access):0
2116:RULE 5 (multi-hop access):0
2138:netAccess(app_39,tcp,8080):0
2191:networkServiceInfo(app_39,tomcat,tcp,8080,root):1
2192:vulExists(app_39,vul_
2137:RULE 2 (remote exploit of a server program):0
2194:hacl(app_4,db_51,tcp,5432):1
2193:RULE 5 (multi-hop access):0


242:networkServi
2174:hacl(dmz_24,app_39,tcp,8080):1
2146:hacl(dmz_11,app_39,tcp,8080):1
2172:hacl(dmz_23,app_39,tcp,8080):1
2186:hacl(dmz_7,app_39,tcp,8080):1
2173:RULE 5 (multi-hop access):0
2145:RULE 5 (multi-hop access):0
2171:RULE 5 (multi-hop access):0
2185:RULE 5 (multi-hop access):0
_app,tomcat,remoteExploit,privEscalation):1

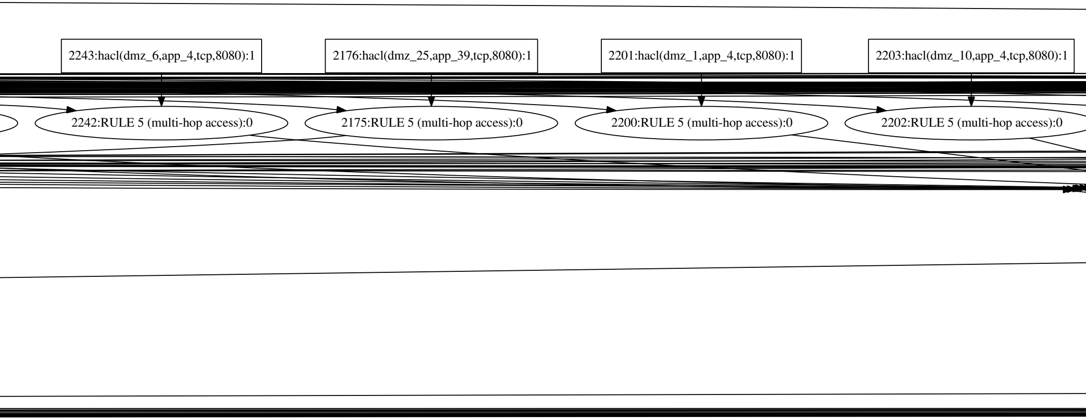

ceInfo(dmz_9,httpd,tcp,80,root):1
243:vulExists(dmz_9,vul_web,httpd,remoteExploit,privEscalation):1
238:RULE 2 (remote exploit of a server program):0
2095:hacl(dmz_15,app_38,tcp,8080):1
2003:hacl(dmz_4,app_36,tcp,8080):1
2060:hacl(dmz_3,app_37,tcp,8080):1
2215:hacl(dmz_16,app_4,tcp,8080):1
2094:RULE 5 (multi-hop access):0
2002:RULE 5 (multi-hop access):0
2059:RULE 5 (multi-hop access):0
2214:RULE 5 (multi-hop access):0
2195:execCode(app_4,root):0

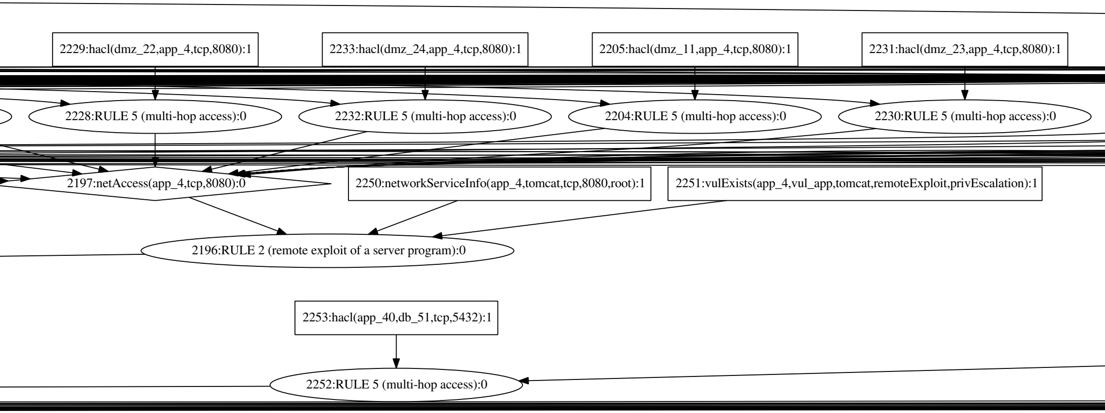

2188:hacl(dmz_8,app_39,tcp,8080):1
2217:hacl(dmz_17,app_4,tcp,8080):1
2148:hacl(dmz_12,app_39,tcp,8080):1
2219:hacl(dmz_18,app_4,tcp,8080):1
2187:RULE 5 (multi-hop access):0
2216:RULE 5 (multi-hop access):0
2147:RULE 5 (multi-hop access):0
2218:RULE 5 (multi-hop access):0

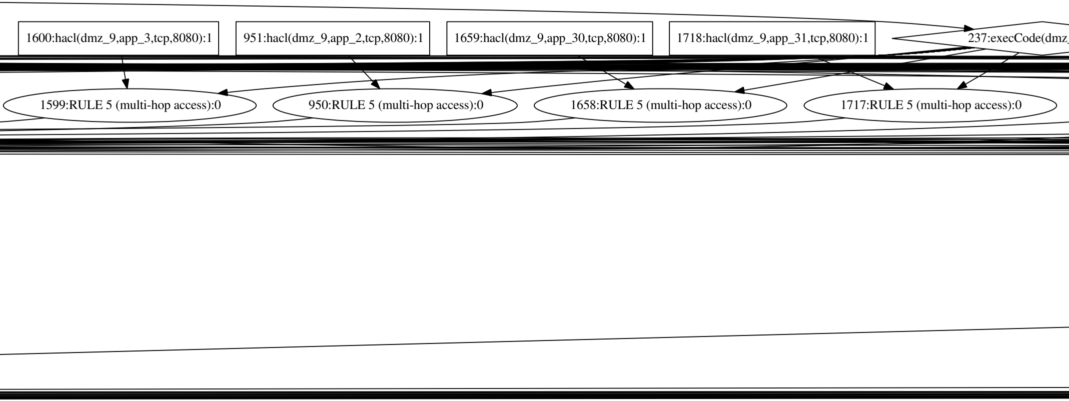

2150:hacl(dmz_13,app_39,tcp,8080):1
2241:hacl(dmz_5,app_4,tcp,8080):1
2221:hacl(dmz_19,app_4,tcp,8080):1
2223:hacl(dmz_2,app_4,tcp,8080):1
2149:RULE 5 (multi-hop access):0
2240:RULE 5 (multi-hop access):0
2220:RULE 5 (multi-hop access):0
2222:RULE 5 (multi-hop access):0

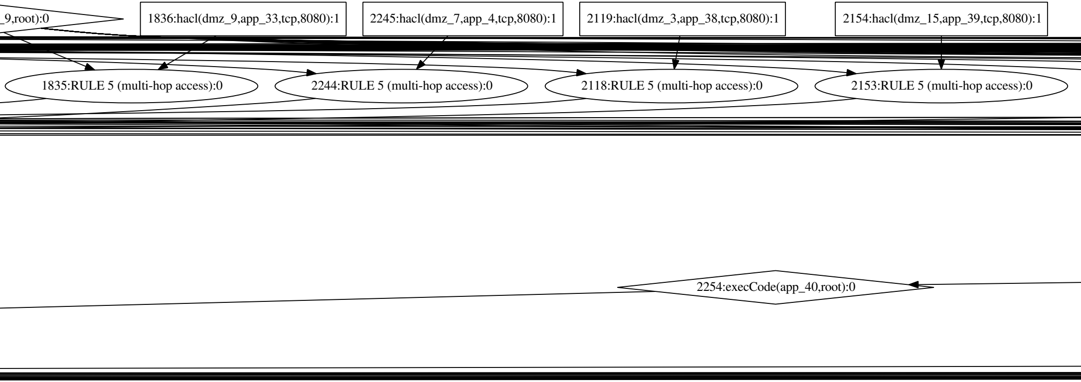


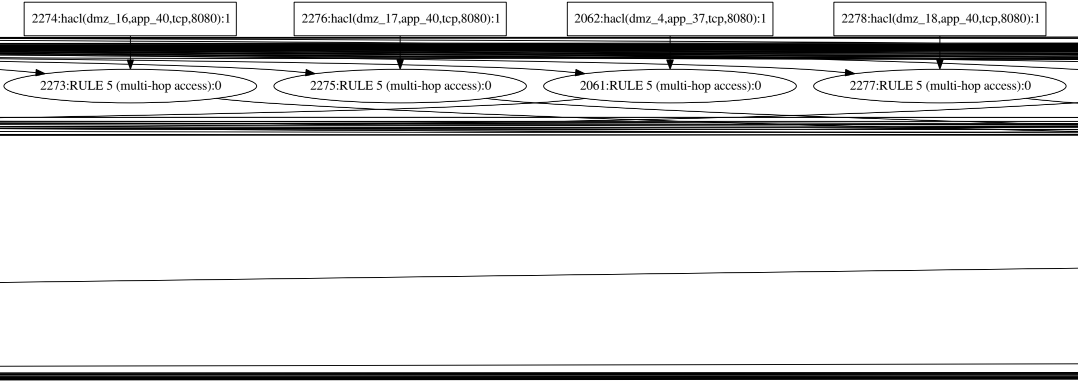
2225:hacl(dmz_20,app_4,tcp,8080):1
2199:hacl(dmz_0,app_4,tcp,8080):1
2227:hacl(dmz_21,app_4,tcp,8080):1
2152:hacl(dmz_14,app_39,tcp,8080):1
2224:RULE 5 (multi-hop access):0
2198:RULE 5 (multi-hop access):0
2226:RULE 5 (multi-hop access):0
2151:RULE 5 (multi-hop access):0

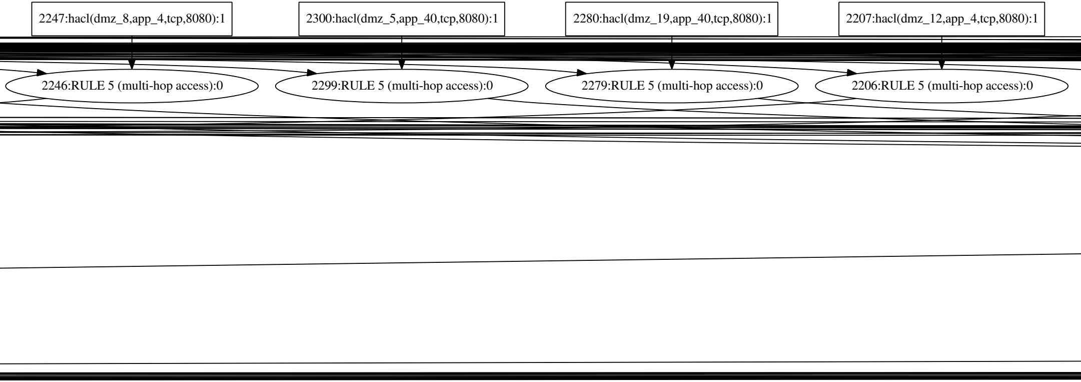

2243:hacl(dmz_6,app_4,tcp,8080):1
2176:hacl(dmz_25,app_39,tcp,8080):1
2201:hacl(dmz_1,app_4,tcp,8080):1
2203:hacl(dmz_10,app_4,tcp,8080):1
2242:RULE 5 (multi-hop access):0
2175:RULE 5 (multi-hop access):0
2200:RULE 5 (multi-hop access):0
2202:RULE 5 (multi-hop access):0

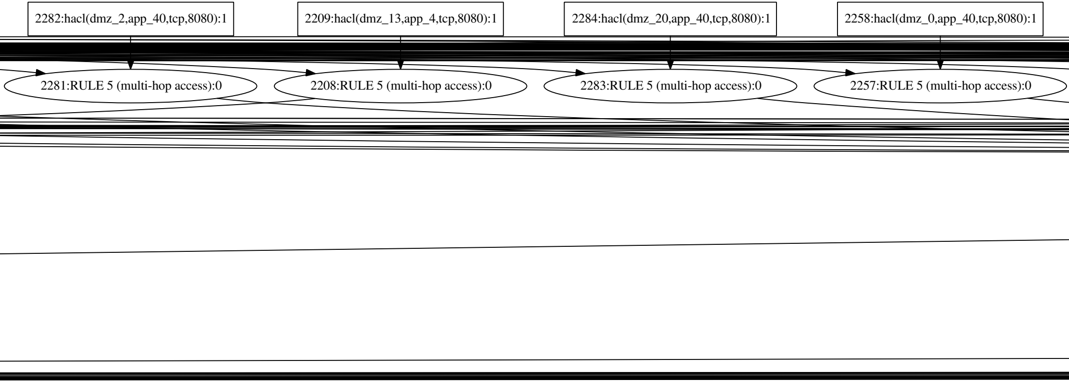

2229:hacl(dmz_22,app_4,tcp,8080):1
2233:hacl(dmz_24,app_4,tcp,8080):1
2205:hacl(dmz_11,app_4,tcp,8080):1
2231:hacl(dmz_23,app_4,tcp,8080):1
2228:RULE 5 (multi-hop access):0
2232:RULE 5 (multi-hop access):0
2204:RULE 5 (multi-hop access):0
2230:RULE 5 (multi-hop access):0
2197:netAccess(app_4,tcp,8080):0
2250:networkServiceInfo(app_4,tomcat,tcp,8080,root):1
2251:vulExists(app_4,vul_app,tomcat,remoteExploit,privEscalation):1
2196:RULE 2 (remote exploit of a server program):0
2253:hacl(app_40,db_51,tcp,5432):1
2252:RULE 5 (multi-hop access):0

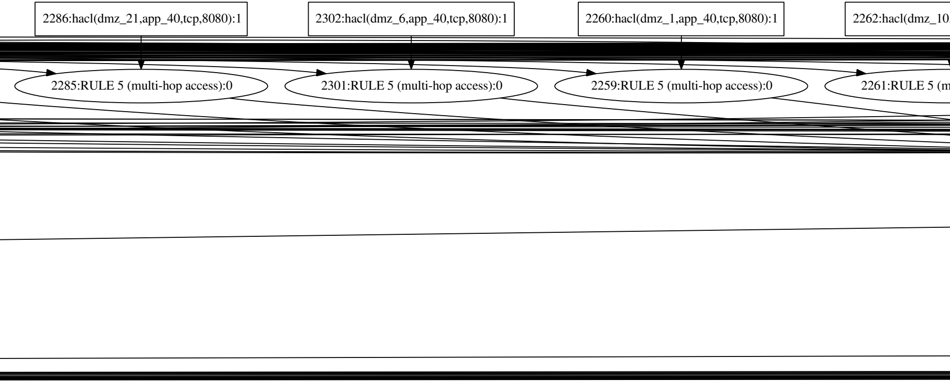

1600:hacl(dmz_9,app_3,tcp,8080):1
951:hacl(dmz_9,app_2,tcp,8080):1
1659:hacl(dmz_9,app_30,tcp,8080):1
1718:hacl(dmz_9,app_31,tcp,8080):1
237:execCode(dmz_
1599:RULE 5 (multi-hop access):0
950:RULE 5 (multi-hop access):0
1658:RULE 5 (multi-hop access):0
1717:RULE 5 (multi-hop access):0

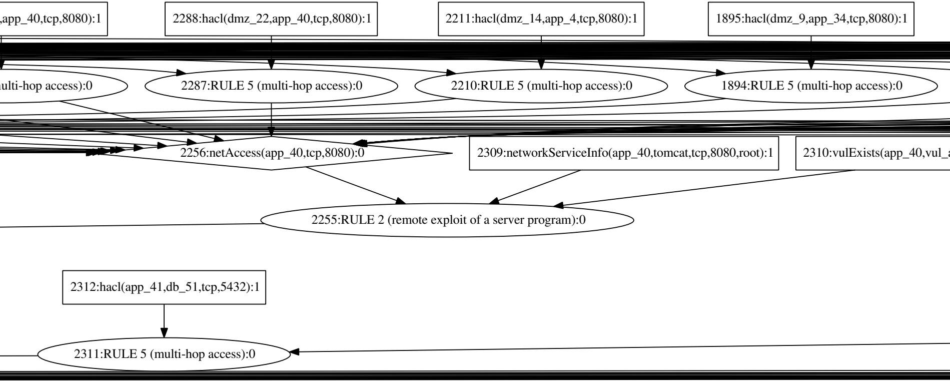

_9,root):0
1836:hacl(dmz_9,app_33,tcp,8080):1
2245:hacl(dmz_7,app_4,tcp,8080):1
2119:hacl(dmz_3,app_38,tcp,8080):1
2154:hacl(dmz_15,app_39,tcp,8080):1
1835:RULE 5 (multi-hop access):0
2244:RULE 5 (multi-hop access):0
2118:RULE 5 (multi-hop access):0
2153:RULE 5 (multi-hop access):0
2254:execCode(app_40,root):0

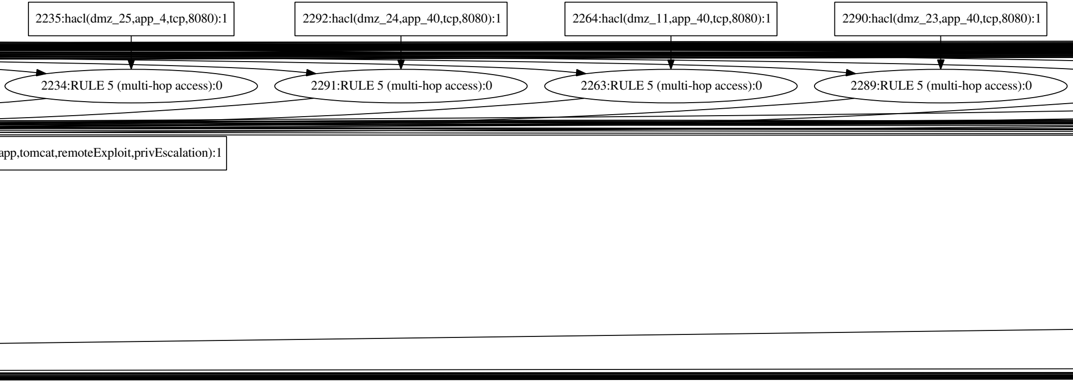

2274:hacl(dmz_16,app_40,tcp,8080):1
2276:hacl(dmz_17,app_40,tcp,8080):1
2062:hacl(dmz_4,app_37,tcp,8080):1
2278:hacl(dmz_18,app_40,tcp,8080):1
2273:RULE 5 (multi-hop access):0
2275:RULE 5 (multi-hop access):0
2061:RULE 5 (multi-hop access):0
2277:RULE 5 (multi-hop access):0

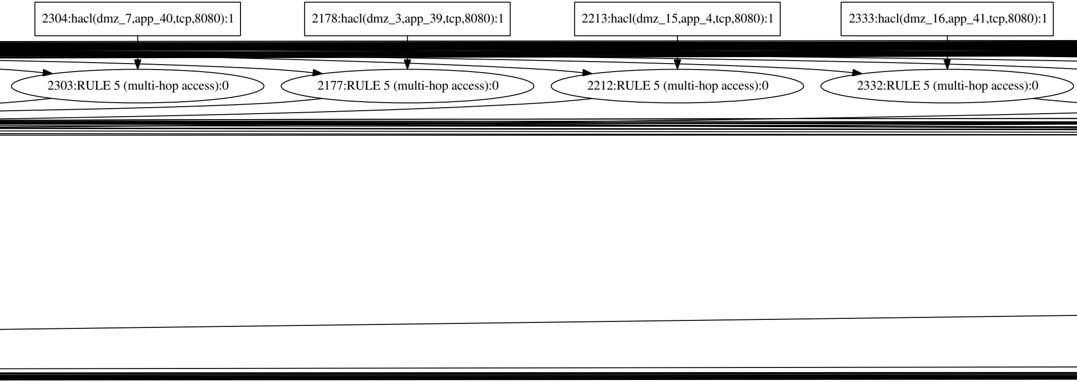

2247:hacl(dmz_8,app_4,tcp,8080):1
2300:hacl(dmz_5,app_40,tcp,8080):1
2280:hacl(dmz_19,app_40,tcp,8080):1
2207:hacl(dmz_12,app_4,tcp,8080):1
2246:RULE 5 (multi-hop access):0
2299:RULE 5 (multi-hop access):0
2279:RULE 5 (multi-hop access):0
2206:RULE 5 (multi-hop access):0

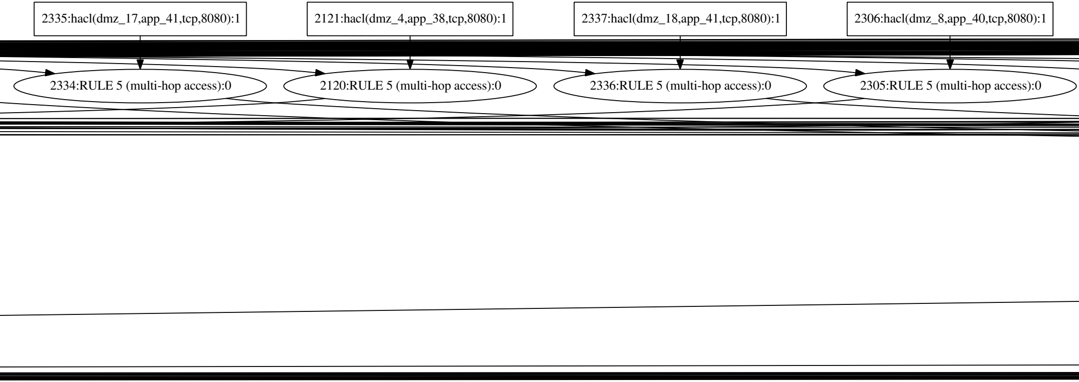

2282:hacl(dmz_2,app_40,tcp,8080):1
2209:hacl(dmz_13,app_4,tcp,8080):1
2284:hacl(dmz_20,app_40,tcp,8080):1
2258:hacl(dmz_0,app_40,tcp,8080):1
2281:RULE 5 (multi-hop access):0
2208:RULE 5 (multi-hop access):0
2283:RULE 5 (multi-hop access):0
2257:RULE 5 (multi-hop access):0

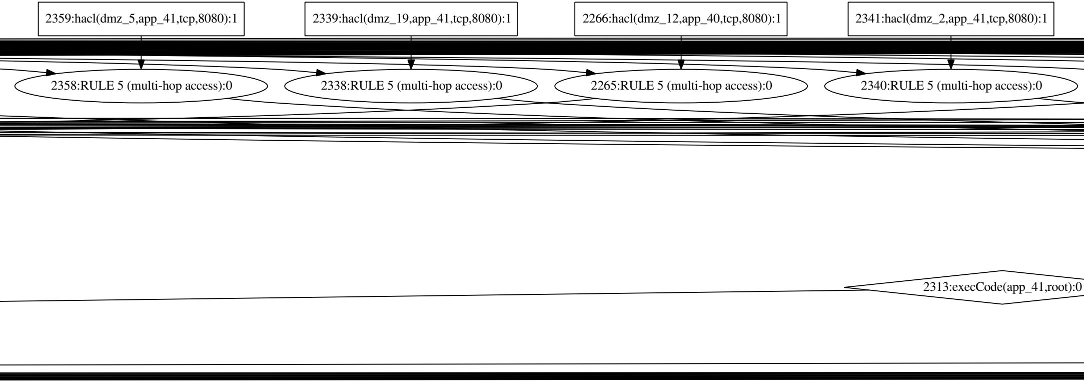

2286:hacl(dmz_21,app_40,tcp,8080):1
2302:hacl(dmz_6,app_40,tcp,8080):1
2260:hacl(dmz_1,app_40,tcp,8080):1
2262:hacl(dmz_10
2285:RULE 5 (multi-hop access):0
2301:RULE 5 (multi-hop access):0
2259:RULE 5 (multi-hop access):0
2261:RULE 5 (m

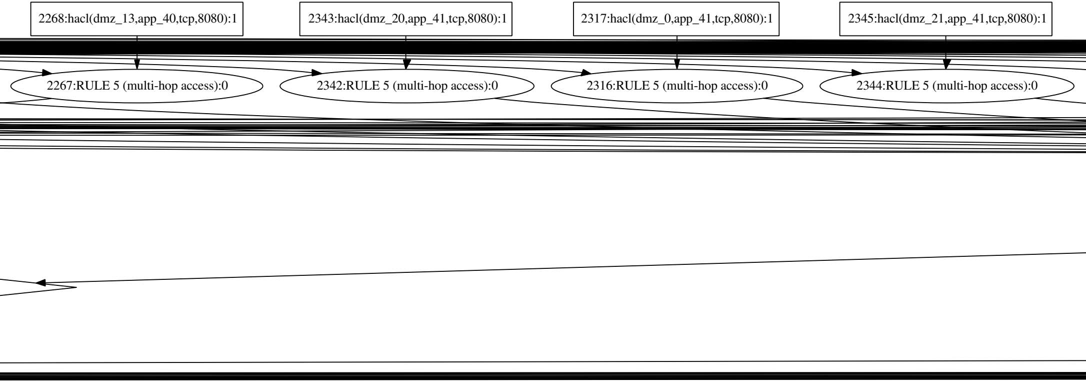

,app_40,tcp,8080):1
2288:hacl(dmz_22,app_40,tcp,8080):1
2211:hacl(dmz_14,app_4,tcp,8080):1
1895:hacl(dmz_9,app_34,tcp,8080):1
ulti-hop access):0
2287:RULE 5 (multi-hop access):0
2210:RULE 5 (multi-hop access):0
1894:RULE 5 (multi-hop access):0
2256:netAccess(app_40,tcp,8080):0
2309:networkServiceInfo(app_40,tomcat,tcp,8080,root):1
2310:vulExists(app_40,vul_a
2255:RULE 2 (remote exploit of a server program):0
2312:hacl(app_41,db_51,tcp,5432):1
2311:RULE 5 (multi-hop access):0

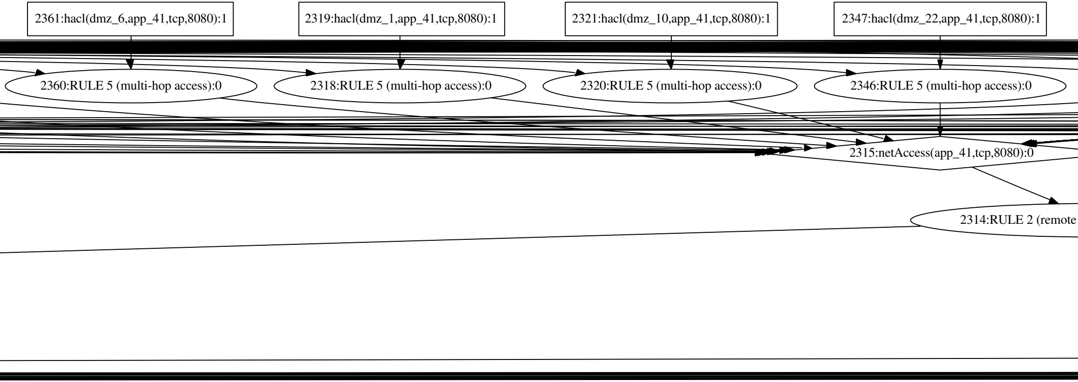

2235:hacl(dmz_25,app_4,tcp,8080):1
2292:hacl(dmz_24,app_40,tcp,8080):1
2264:hacl(dmz_11,app_40,tcp,8080):1
2290:hacl(dmz_23,app_40,tcp,8080):1
2234:RULE 5 (multi-hop access):0
2291:RULE 5 (multi-hop access):0
2263:RULE 5 (multi-hop access):0
2289:RULE 5 (multi-hop access):0
app,tomcat,remoteExploit,privEscalation):1

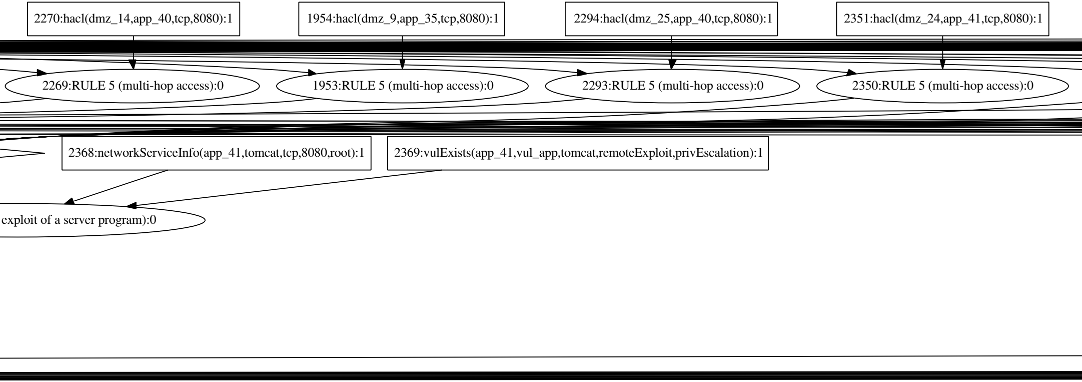

2304:hacl(dmz_7,app_40,tcp,8080):1
2178:hacl(dmz_3,app_39,tcp,8080):1
2213:hacl(dmz_15,app_4,tcp,8080):1
2333:hacl(dmz_16,app_41,tcp,8080):1
2303:RULE 5 (multi-hop access):0
2177:RULE 5 (multi-hop access):0
2212:RULE 5 (multi-hop access):0
2332:RULE 5 (multi-hop access):0

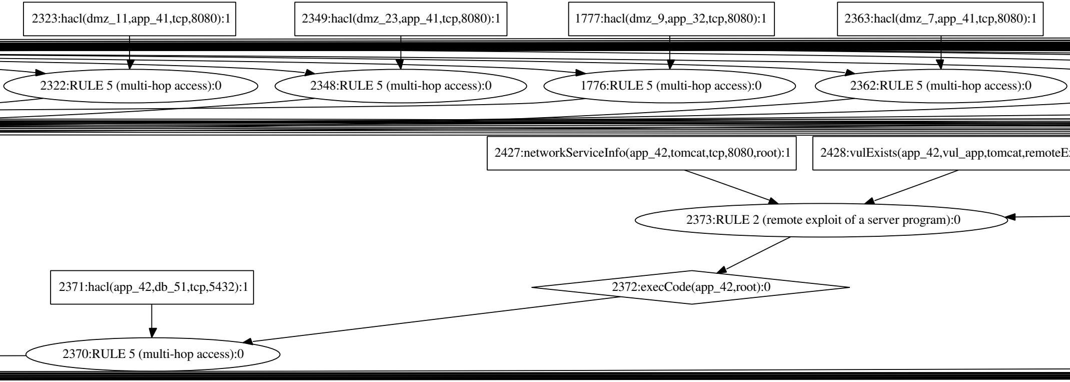

2335:hacl(dmz_17,app_41,tcp,8080):1
2121:hacl(dmz_4,app_38,tcp,8080):1
2337:hacl(dmz_18,app_41,tcp,8080):1
2306:hacl(dmz_8,app_40,tcp,8080):1
2334:RULE 5 (multi-hop access):0
2120:RULE 5 (multi-hop access):0
2336:RULE 5 (multi-hop access):0
2305:RULE 5 (multi-hop access):0

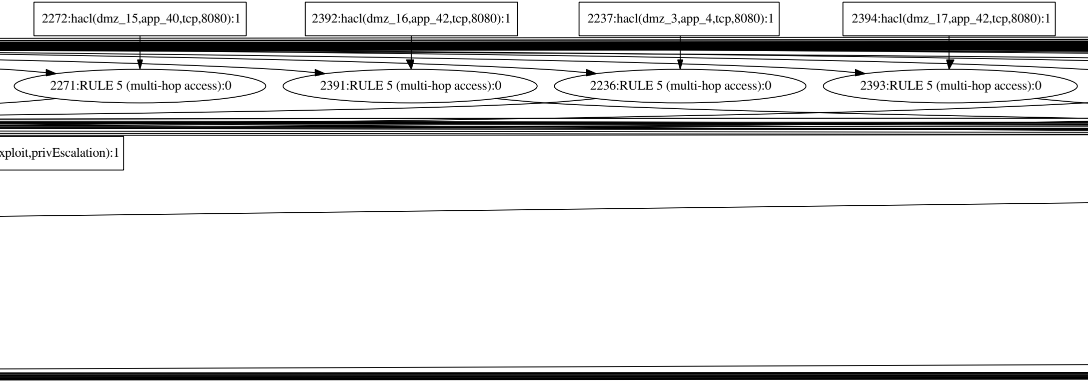

2359:hacl(dmz_5,app_41,tcp,8080):1
2339:hacl(dmz_19,app_41,tcp,8080):1
2266:hacl(dmz_12,app_40,tcp,8080):1
2341:hacl(dmz_2,app_41,tcp,8080):1
2358:RULE 5 (multi-hop access):0
2338:RULE 5 (multi-hop access):0
2265:RULE 5 (multi-hop access):0
2340:RULE 5 (multi-hop access):0
2313:execCode(app_41,root):0

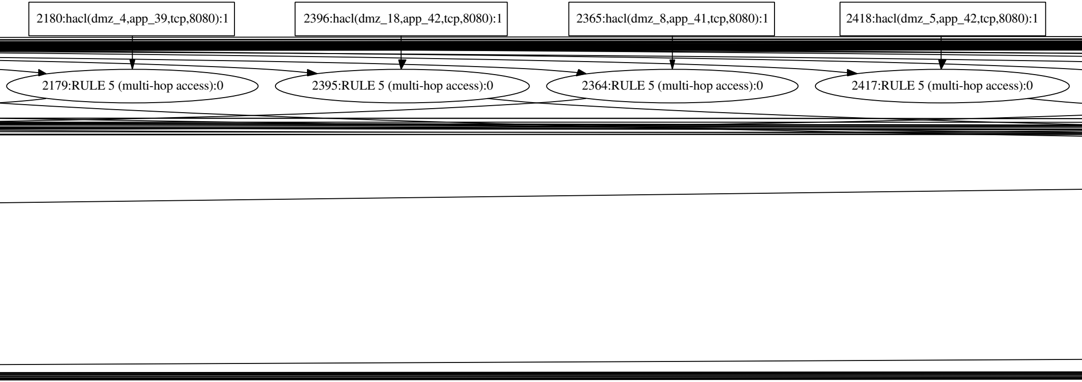

2268:hacl(dmz_13,app_40,tcp,8080):1
2343:hacl(dmz_20,app_41,tcp,8080):1
2317:hacl(dmz_0,app_41,tcp,8080):1
2345:hacl(dmz_21,app_41,tcp,8080):1
2267:RULE 5 (multi-hop access):0
2342:RULE 5 (multi-hop access):0
2316:RULE 5 (multi-hop access):0
2344:RULE 5 (multi-hop access):0

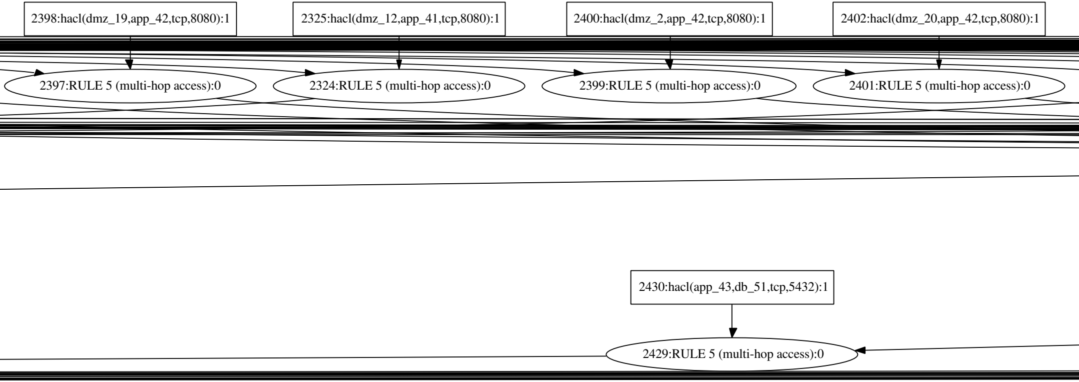

2361:hacl(dmz_6,app_41,tcp,8080):1
2319:hacl(dmz_1,app_41,tcp,8080):1
2321:hacl(dmz_10,app_41,tcp,8080):1
2347:hacl(dmz_22,app_41,tcp,8080):1
2360:RULE 5 (multi-hop access):0
2318:RULE 5 (multi-hop access):0
2320:RULE 5 (multi-hop access):0
2346:RULE 5 (multi-hop access):0
2315:netAccess(app_41,tcp,8080):0
2314:RULE 2 (remote

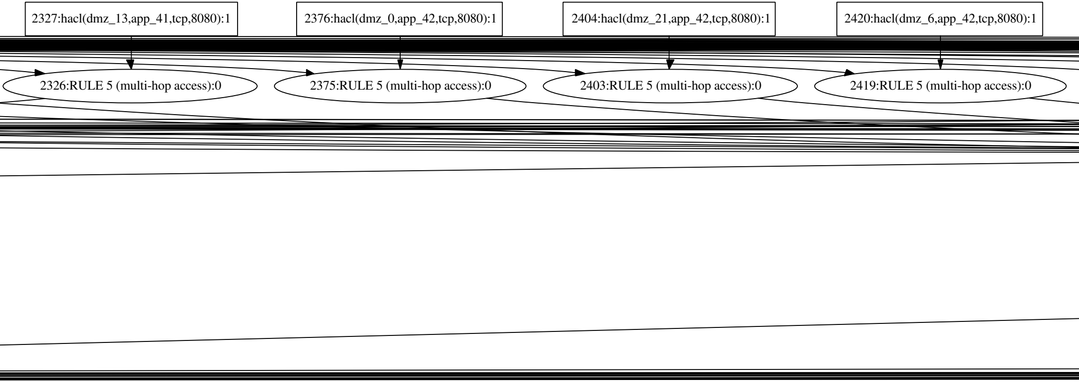

2270:hacl(dmz_14,app_40,tcp,8080):1
1954:hacl(dmz_9,app_35,tcp,8080):1
2294:hacl(dmz_25,app_40,tcp,8080):1
2351:hacl(dmz_24,app_41,tcp,8080):1
2269:RULE 5 (multi-hop access):0
1953:RULE 5 (multi-hop access):0
2293:RULE 5 (multi-hop access):0
2350:RULE 5 (multi-hop access):0
2368:networkServiceInfo(app_41,tomcat,tcp,8080,root):1
2369:vulExists(app_41,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0

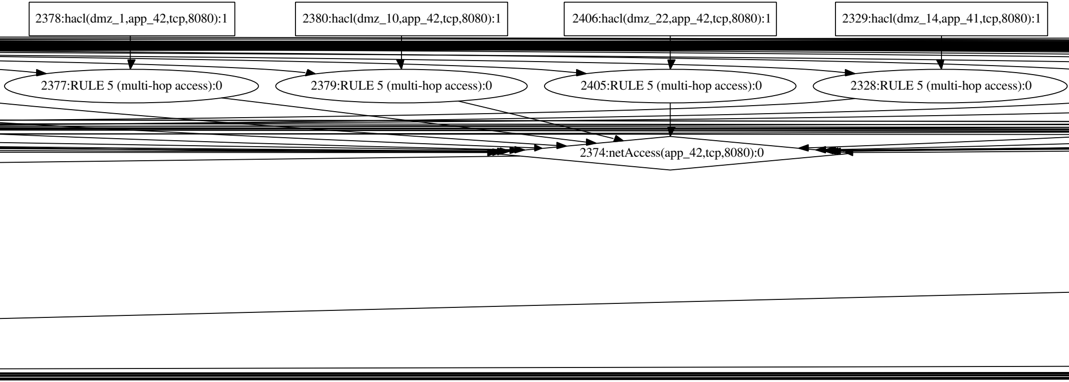

2323:hacl(dmz_11,app_41,tcp,8080):1
2349:hacl(dmz_23,app_41,tcp,8080):1
1777:hacl(dmz_9,app_32,tcp,8080):1
2363:hacl(dmz_7,app_41,tcp,8080):1
2322:RULE 5 (multi-hop access):0
2348:RULE 5 (multi-hop access):0
1776:RULE 5 (multi-hop access):0
2362:RULE 5 (multi-hop access):0
2427:networkServiceInfo(app_42,tomcat,tcp,8080,root):1
2428:vulExists(app_42,vul_app,tomcat,remoteE
2373:RULE 2 (remote exploit of a server program):0
2371:hacl(app_42,db_51,tcp,5432):1
2372:execCode(app_42,root):0
2370:RULE 5 (multi-hop access):0

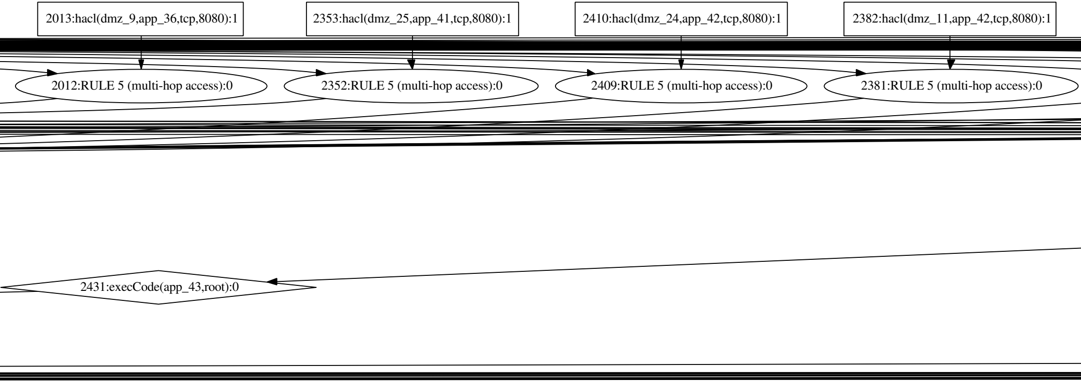

2272:hacl(dmz_15,app_40,tcp,8080):1
2392:hacl(dmz_16,app_42,tcp,8080):1
2237:hacl(dmz_3,app_4,tcp,8080):1
2394:hacl(dmz_17,app_42,tcp,8080):1
2271:RULE 5 (multi-hop access):0
2391:RULE 5 (multi-hop access):0
2236:RULE 5 (multi-hop access):0
2393:RULE 5 (multi-hop access):0
xploit,privEscalation):1

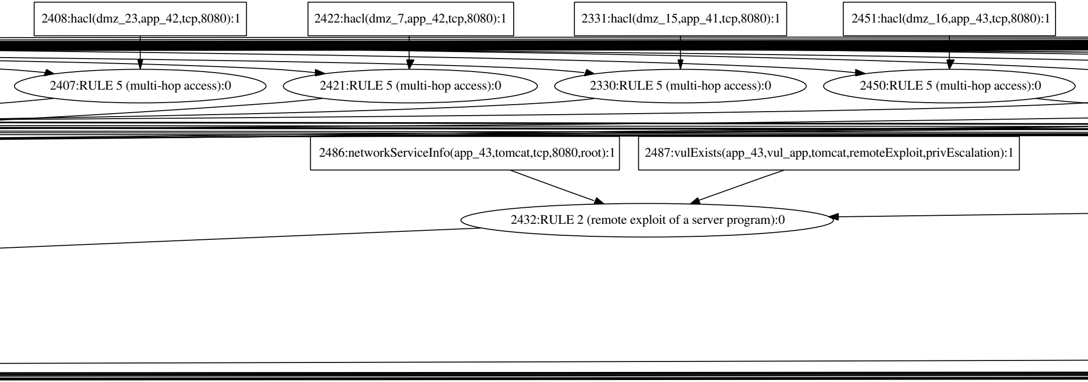

2180:hacl(dmz_4,app_39,tcp,8080):1
2396:hacl(dmz_18,app_42,tcp,8080):1
2365:hacl(dmz_8,app_41,tcp,8080):1
2418:hacl(dmz_5,app_42,tcp,8080):1
2179:RULE 5 (multi-hop access):0
2395:RULE 5 (multi-hop access):0
2364:RULE 5 (multi-hop access):0
2417:RULE 5 (multi-hop access):0

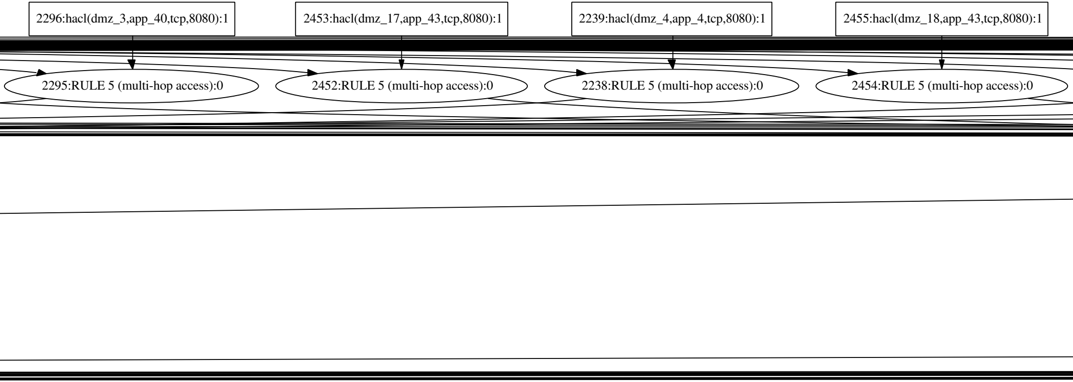

2398:hacl(dmz_19,app_42,tcp,8080):1
2325:hacl(dmz_12,app_41,tcp,8080):1
2400:hacl(dmz_2,app_42,tcp,8080):1
2402:hacl(dmz_20,app_42,tcp,8080):1
2397:RULE 5 (multi-hop access):0
2324:RULE 5 (multi-hop access):0
2399:RULE 5 (multi-hop access):0
2401:RULE 5 (multi-hop access):0
2430:hacl(app_43,db_51,tcp,5432):1
2429:RULE 5 (multi-hop access):0

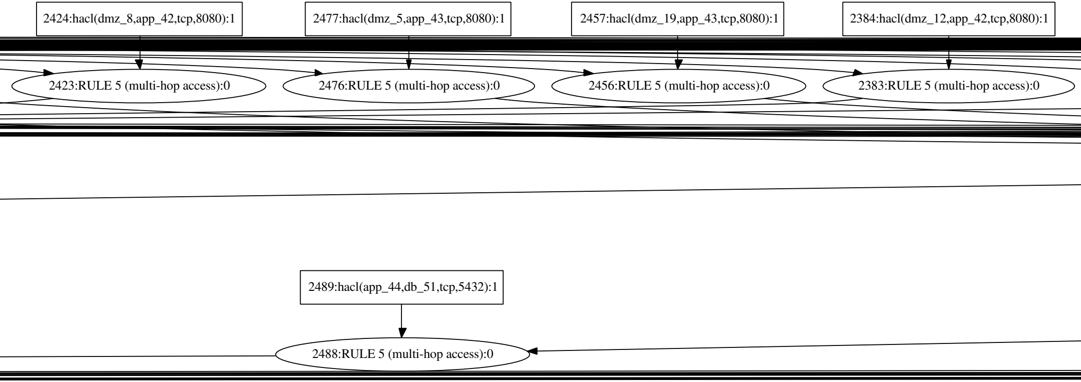

2327:hacl(dmz_13,app_41,tcp,8080):1
2376:hacl(dmz_0,app_42,tcp,8080):1
2404:hacl(dmz_21,app_42,tcp,8080):1
2420:hacl(dmz_6,app_42,tcp,8080):1
2326:RULE 5 (multi-hop access):0
2375:RULE 5 (multi-hop access):0
2403:RULE 5 (multi-hop access):0
2419:RULE 5 (multi-hop access):0

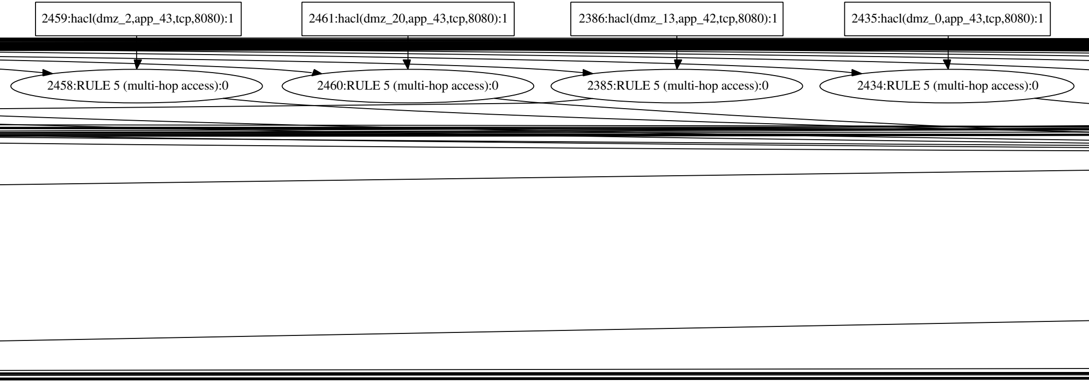

2378:hacl(dmz_1,app_42,tcp,8080):1
2380:hacl(dmz_10,app_42,tcp,8080):1
2406:hacl(dmz_22,app_42,tcp,8080):1
2329:hacl(dmz_14,app_41,tcp,8080):1
2377:RULE 5 (multi-hop access):0
2379:RULE 5 (multi-hop access):0
2405:RULE 5 (multi-hop access):0
2328:RULE 5 (multi-hop access):0
2374:netAccess(app_42,tcp,8080):0

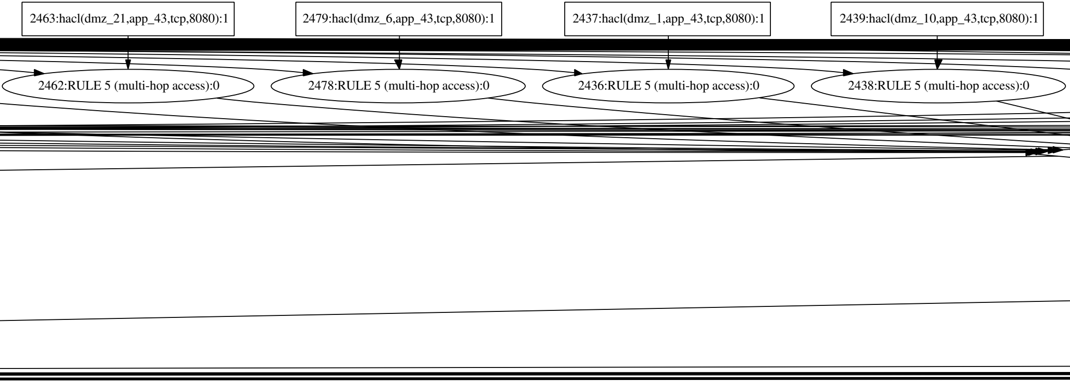

2013:hacl(dmz_9,app_36,tcp,8080):1
2353:hacl(dmz_25,app_41,tcp,8080):1
2410:hacl(dmz_24,app_42,tcp,8080):1
2382:hacl(dmz_11,app_42,tcp,8080):1
2012:RULE 5 (multi-hop access):0
2352:RULE 5 (multi-hop access):0
2409:RULE 5 (multi-hop access):0
2381:RULE 5 (multi-hop access):0
2431:execCode(app_43,root):0

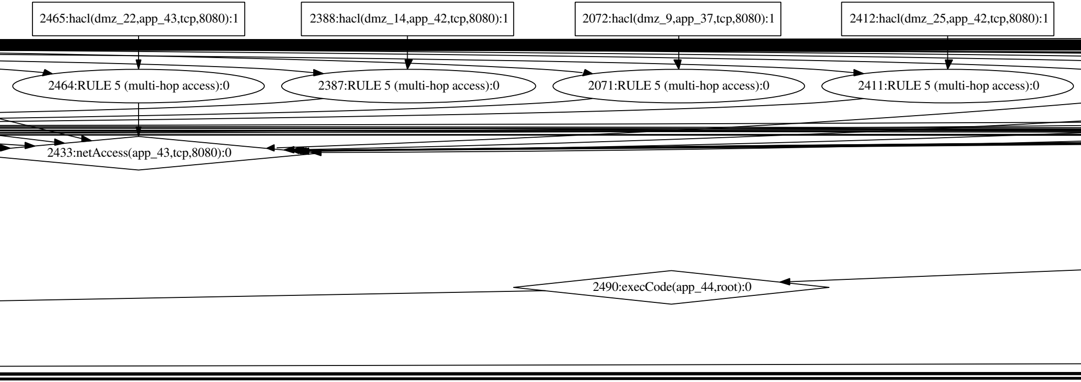

2408:hacl(dmz_23,app_42,tcp,8080):1
2422:hacl(dmz_7,app_42,tcp,8080):1
2331:hacl(dmz_15,app_41,tcp,8080):1
2451:hacl(dmz_16,app_43,tcp,8080):1
2407:RULE 5 (multi-hop access):0
2421:RULE 5 (multi-hop access):0
2330:RULE 5 (multi-hop access):0
2450:RULE 5 (multi-hop access):0
2486:networkServiceInfo(app_43,tomcat,tcp,8080,root):1
2487:vulExists(app_43,vul_app,tomcat,remoteExploit,privEscalation):1
2432:RULE 2 (remote exploit of a server program):0

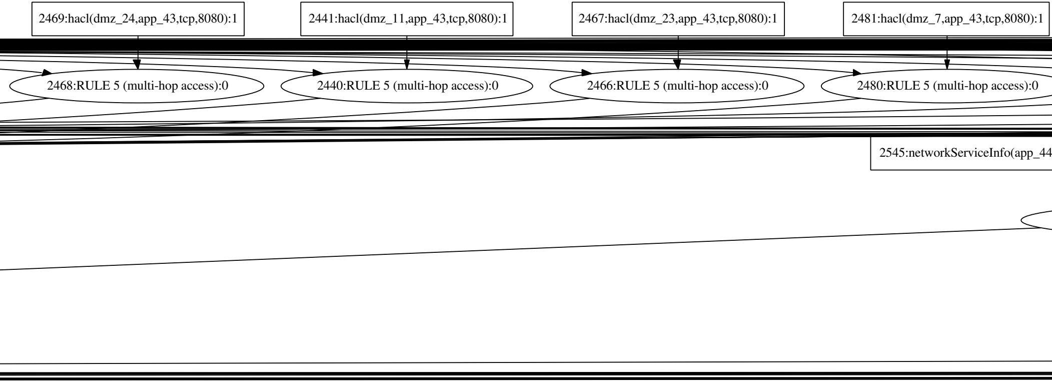

2296:hacl(dmz_3,app_40,tcp,8080):1
2453:hacl(dmz_17,app_43,tcp,8080):1
2239:hacl(dmz_4,app_4,tcp,8080):1
2455:hacl(dmz_18,app_43,tcp,8080):1
2295:RULE 5 (multi-hop access):0
2452:RULE 5 (multi-hop access):0
2238:RULE 5 (multi-hop access):0
2454:RULE 5 (multi-hop access):0

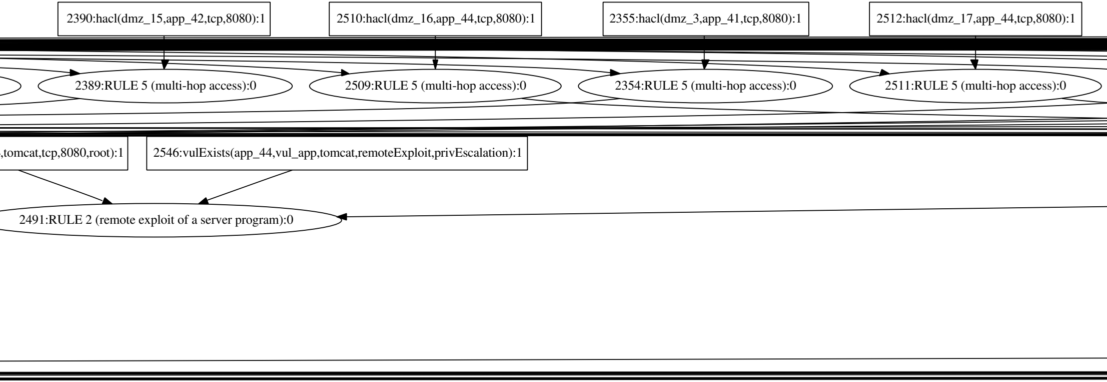

2424:hacl(dmz_8,app_42,tcp,8080):1
2477:hacl(dmz_5,app_43,tcp,8080):1
2457:hacl(dmz_19,app_43,tcp,8080):1
2384:hacl(dmz_12,app_42,tcp,8080):1
2423:RULE 5 (multi-hop access):0
2476:RULE 5 (multi-hop access):0
2456:RULE 5 (multi-hop access):0
2383:RULE 5 (multi-hop access):0
2489:hacl(app_44,db_51,tcp,5432):1
2488:RULE 5 (multi-hop access):0

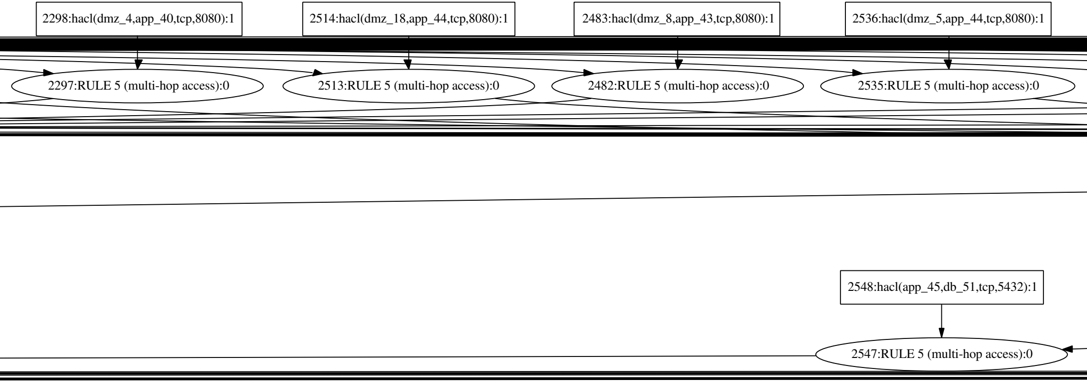

2459:hacl(dmz_2,app_43,tcp,8080):1
2461:hacl(dmz_20,app_43,tcp,8080):1
2386:hacl(dmz_13,app_42,tcp,8080):1
2435:hacl(dmz_0,app_43,tcp,8080):1
2458:RULE 5 (multi-hop access):0
2460:RULE 5 (multi-hop access):0
2385:RULE 5 (multi-hop access):0
2434:RULE 5 (multi-hop access):0

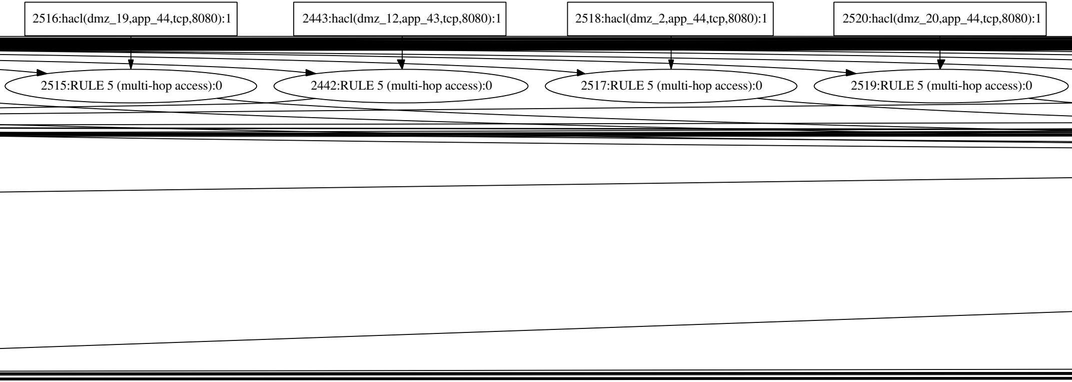

2463:hacl(dmz_21,app_43,tcp,8080):1
2479:hacl(dmz_6,app_43,tcp,8080):1
2437:hacl(dmz_1,app_43,tcp,8080):1
2439:hacl(dmz_10,app_43,tcp,8080):1
2462:RULE 5 (multi-hop access):0
2478:RULE 5 (multi-hop access):0
2436:RULE 5 (multi-hop access):0
2438:RULE 5 (multi-hop access):0

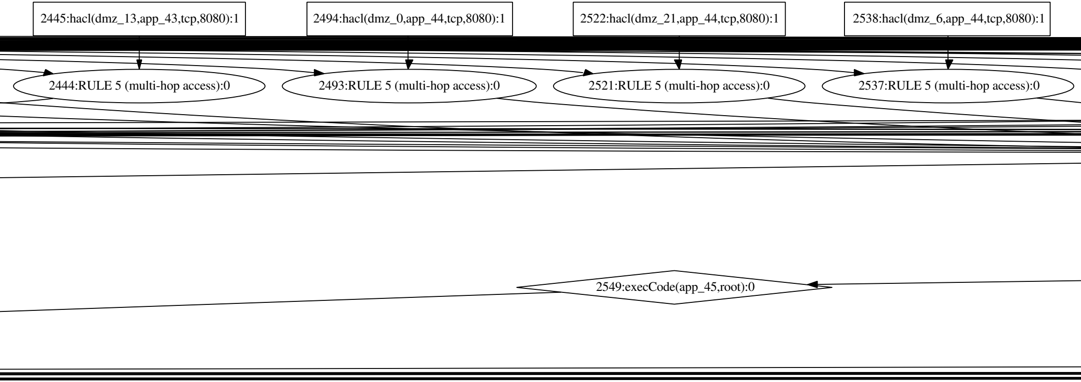

2465:hacl(dmz_22,app_43,tcp,8080):1
2388:hacl(dmz_14,app_42,tcp,8080):1
2072:hacl(dmz_9,app_37,tcp,8080):1
2412:hacl(dmz_25,app_42,tcp,8080):1
2464:RULE 5 (multi-hop access):0
2387:RULE 5 (multi-hop access):0
2071:RULE 5 (multi-hop access):0
2411:RULE 5 (multi-hop access):0
2433:netAccess(app_43,tcp,8080):0
2490:execCode(app_44,root):0

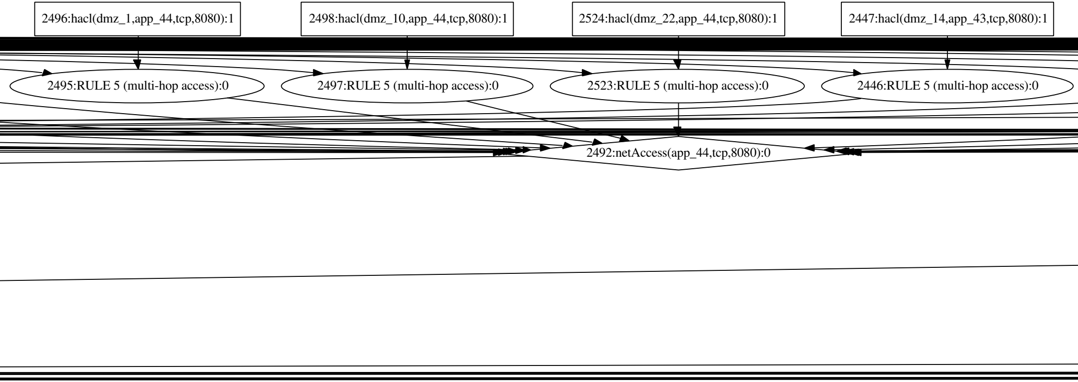

2469:hacl(dmz_24,app_43,tcp,8080):1
2441:hacl(dmz_11,app_43,tcp,8080):1
2467:hacl(dmz_23,app_43,tcp,8080):1
2481:hacl(dmz_7,app_43,tcp,8080):1
2468:RULE 5 (multi-hop access):0
2440:RULE 5 (multi-hop access):0
2466:RULE 5 (multi-hop access):0
2480:RULE 5 (multi-hop access):0
2545:networkServiceInfo(app_44

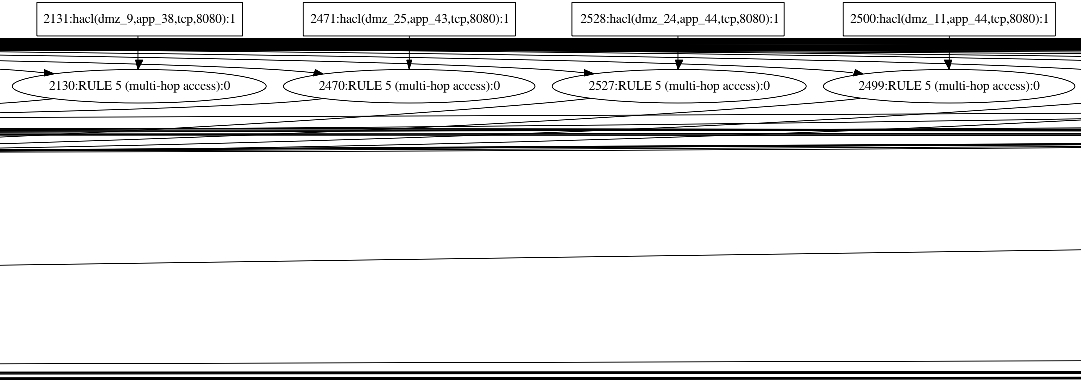

2390:hacl(dmz_15,app_42,tcp,8080):1
2510:hacl(dmz_16,app_44,tcp,8080):1
2355:hacl(dmz_3,app_41,tcp,8080):1
2512:hacl(dmz_17,app_44,tcp,8080):1
2389:RULE 5 (multi-hop access):0
2509:RULE 5 (multi-hop access):0
2354:RULE 5 (multi-hop access):0
2511:RULE 5 (multi-hop access):0
,tomcat,tcp,8080,root):1
2546:vulExists(app_44,vul_app,tomcat,remoteExploit,privEscalation):1
2491:RULE 2 (remote exploit of a server program):0

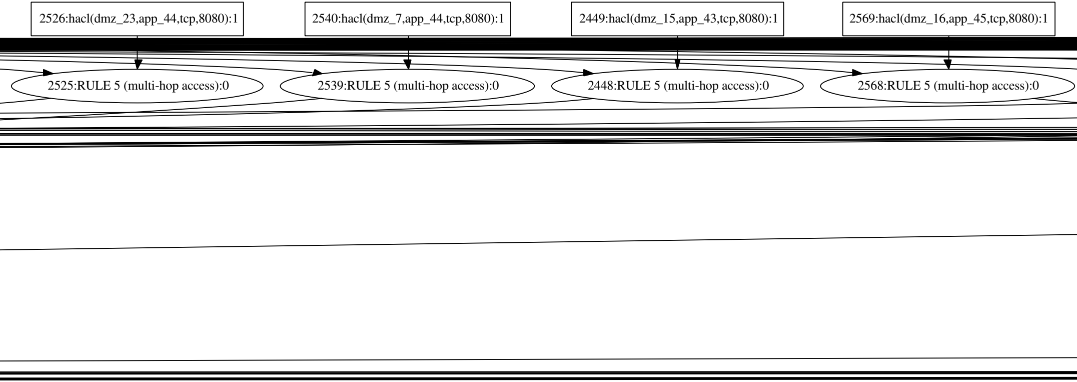

2298:hacl(dmz_4,app_40,tcp,8080):1
2514:hacl(dmz_18,app_44,tcp,8080):1
2483:hacl(dmz_8,app_43,tcp,8080):1
2536:hacl(dmz_5,app_44,tcp,8080):1
2297:RULE 5 (multi-hop access):0
2513:RULE 5 (multi-hop access):0
2482:RULE 5 (multi-hop access):0
2535:RULE 5 (multi-hop access):0
2548:hacl(app_45,db_51,tcp,5432):1
2547:RULE 5 (multi-hop access):0

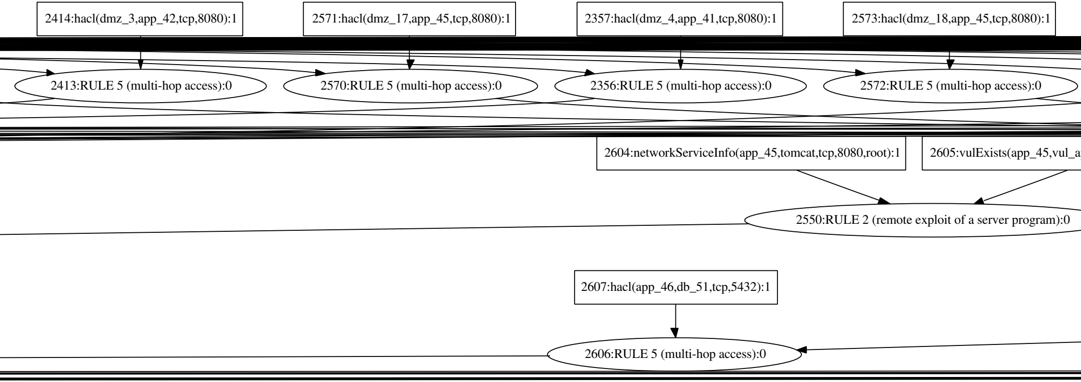

2516:hacl(dmz_19,app_44,tcp,8080):1
2443:hacl(dmz_12,app_43,tcp,8080):1
2518:hacl(dmz_2,app_44,tcp,8080):1
2520:hacl(dmz_20,app_44,tcp,8080):1
2515:RULE 5 (multi-hop access):0
2442:RULE 5 (multi-hop access):0
2517:RULE 5 (multi-hop access):0
2519:RULE 5 (multi-hop access):0

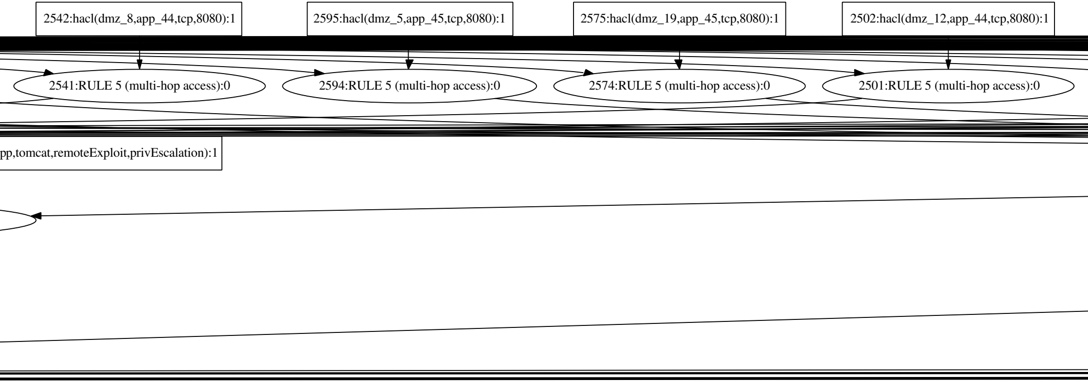

2445:hacl(dmz_13,app_43,tcp,8080):1
2494:hacl(dmz_0,app_44,tcp,8080):1
2522:hacl(dmz_21,app_44,tcp,8080):1
2538:hacl(dmz_6,app_44,tcp,8080):1
2444:RULE 5 (multi-hop access):0
2493:RULE 5 (multi-hop access):0
2521:RULE 5 (multi-hop access):0
2537:RULE 5 (multi-hop access):0
2549:execCode(app_45,root):0

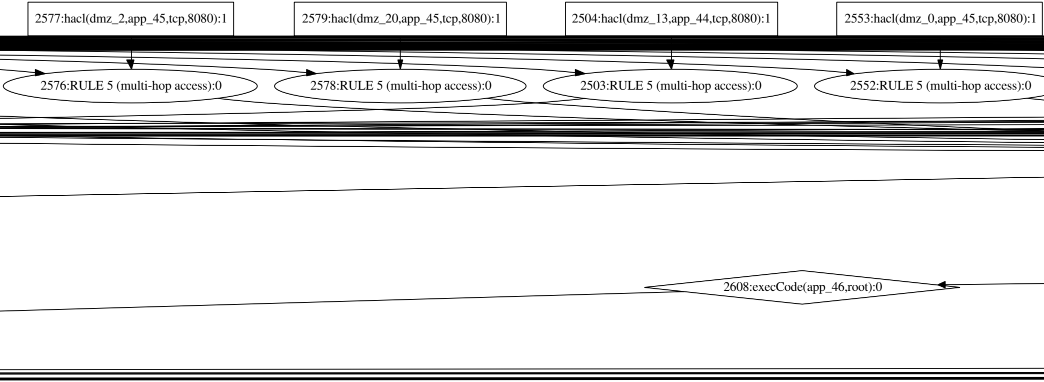

2496:hacl(dmz_1,app_44,tcp,8080):1
2498:hacl(dmz_10,app_44,tcp,8080):1
2524:hacl(dmz_22,app_44,tcp,8080):1
2447:hacl(dmz_14,app_43,tcp,8080):1
2495:RULE 5 (multi-hop access):0
2497:RULE 5 (multi-hop access):0
2523:RULE 5 (multi-hop access):0
2446:RULE 5 (multi-hop access):0
2492:netAccess(app_44,tcp,8080):0

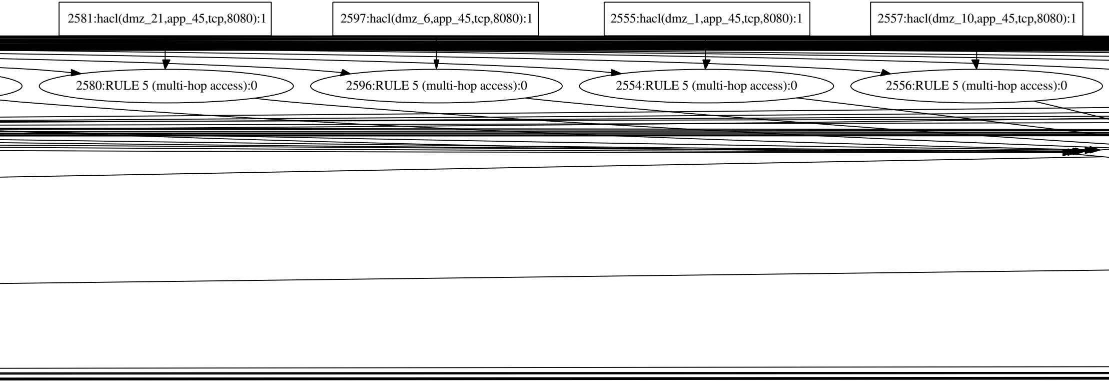

2131:hacl(dmz_9,app_38,tcp,8080):1
2471:hacl(dmz_25,app_43,tcp,8080):1
2528:hacl(dmz_24,app_44,tcp,8080):1
2500:hacl(dmz_11,app_44,tcp,8080):1
2130:RULE 5 (multi-hop access):0
2470:RULE 5 (multi-hop access):0
2527:RULE 5 (multi-hop access):0
2499:RULE 5 (multi-hop access):0

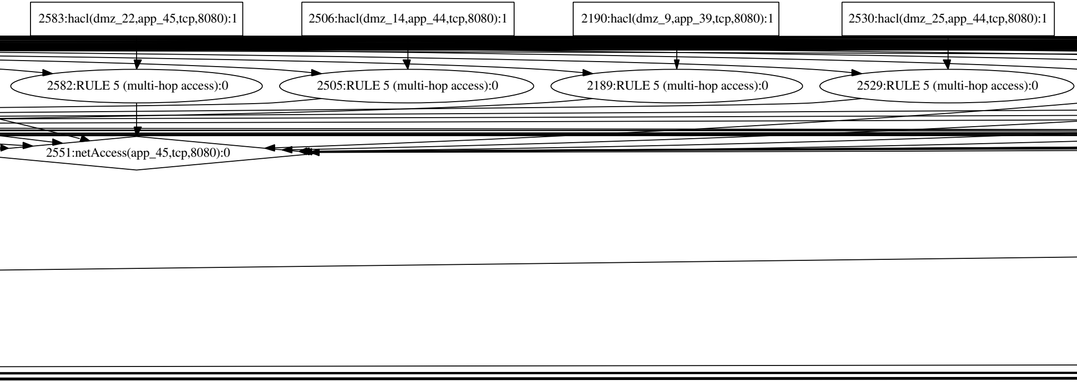

2526:hacl(dmz_23,app_44,tcp,8080):1
2540:hacl(dmz_7,app_44,tcp,8080):1
2449:hacl(dmz_15,app_43,tcp,8080):1
2569:hacl(dmz_16,app_45,tcp,8080):1
2525:RULE 5 (multi-hop access):0
2539:RULE 5 (multi-hop access):0
2448:RULE 5 (multi-hop access):0
2568:RULE 5 (multi-hop access):0

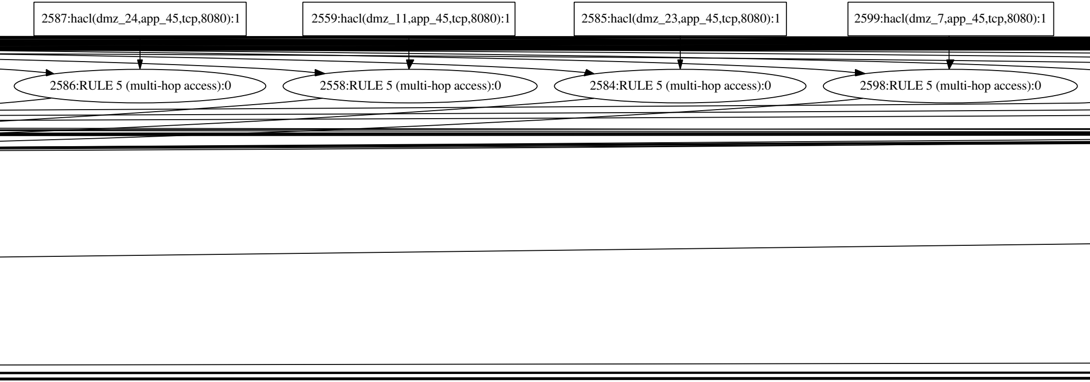

2414:hacl(dmz_3,app_42,tcp,8080):1
2571:hacl(dmz_17,app_45,tcp,8080):1
2357:hacl(dmz_4,app_41,tcp,8080):1
2573:hacl(dmz_18,app_45,tcp,8080):1
2413:RULE 5 (multi-hop access):0
2570:RULE 5 (multi-hop access):0
2356:RULE 5 (multi-hop access):0
2572:RULE 5 (multi-hop access):0
2604:networkServiceInfo(app_45,tomcat,tcp,8080,root):1
2605:vulExists(app_45,vul_a
2550:RULE 2 (remote exploit of a server program):0
2607:hacl(app_46,db_51,tcp,5432):1
2606:RULE 5 (multi-hop access):0

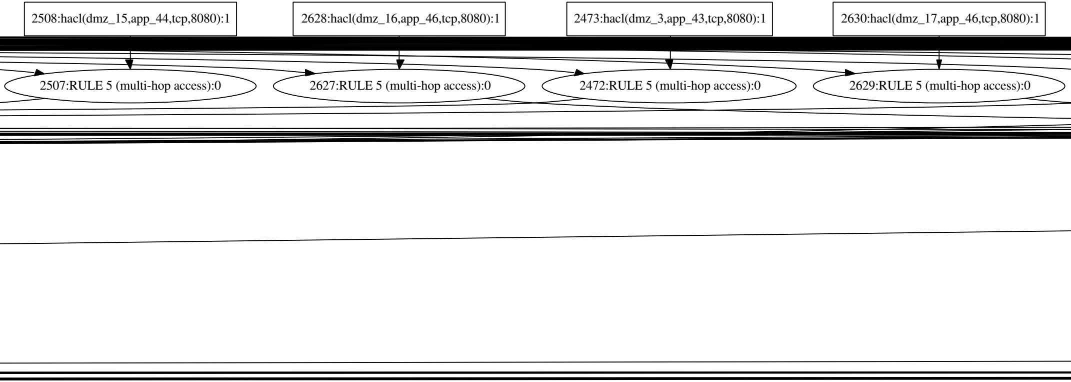

2542:hacl(dmz_8,app_44,tcp,8080):1
2595:hacl(dmz_5,app_45,tcp,8080):1
2575:hacl(dmz_19,app_45,tcp,8080):1
2502:hacl(dmz_12,app_44,tcp,8080):1
2541:RULE 5 (multi-hop access):0
2594:RULE 5 (multi-hop access):0
2574:RULE 5 (multi-hop access):0
2501:RULE 5 (multi-hop access):0
pp,tomcat,remoteExploit,privEscalation):1

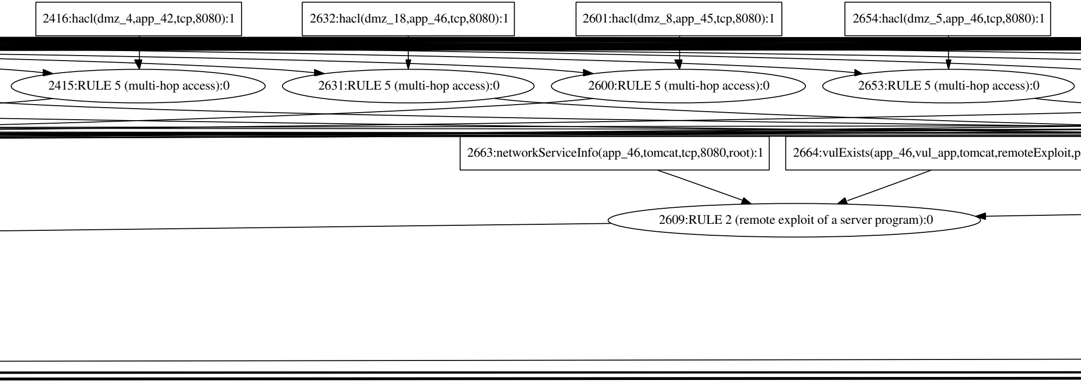

2577:hacl(dmz_2,app_45,tcp,8080):1
2579:hacl(dmz_20,app_45,tcp,8080):1
2504:hacl(dmz_13,app_44,tcp,8080):1
2553:hacl(dmz_0,app_45,tcp,8080):1
2576:RULE 5 (multi-hop access):0
2578:RULE 5 (multi-hop access):0
2503:RULE 5 (multi-hop access):0
2552:RULE 5 (multi-hop access):0
2608:execCode(app_46,root):0

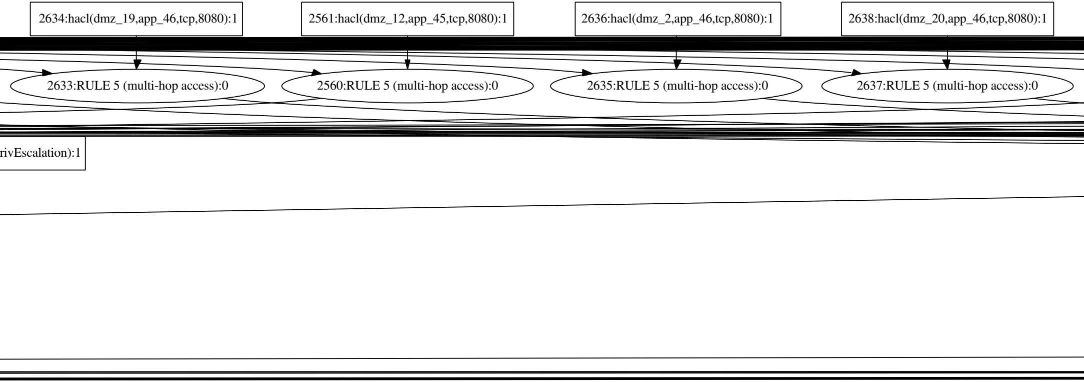

2581:hacl(dmz_21,app_45,tcp,8080):1
2597:hacl(dmz_6,app_45,tcp,8080):1
2555:hacl(dmz_1,app_45,tcp,8080):1
2557:hacl(dmz_10,app_45,tcp,8080):1
2580:RULE 5 (multi-hop access):0
2596:RULE 5 (multi-hop access):0
2554:RULE 5 (multi-hop access):0
2556:RULE 5 (multi-hop access):0

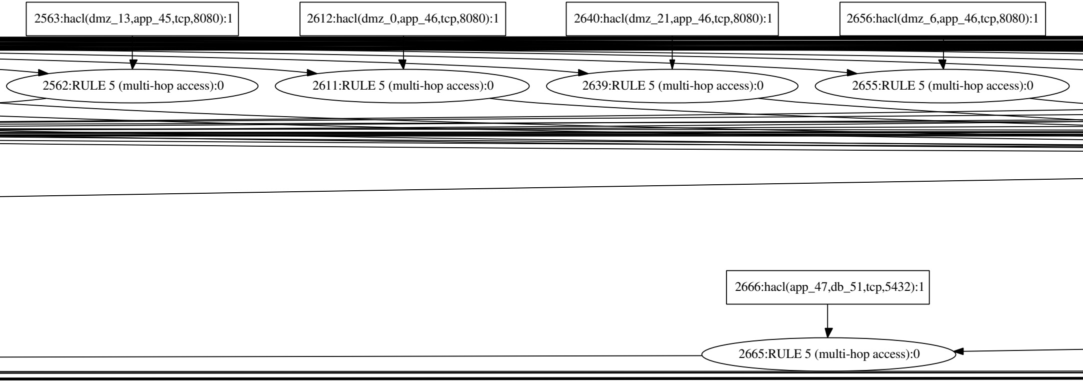

2583:hacl(dmz_22,app_45,tcp,8080):1
2506:hacl(dmz_14,app_44,tcp,8080):1
2190:hacl(dmz_9,app_39,tcp,8080):1
2530:hacl(dmz_25,app_44,tcp,8080):1
2582:RULE 5 (multi-hop access):0
2505:RULE 5 (multi-hop access):0
2189:RULE 5 (multi-hop access):0
2529:RULE 5 (multi-hop access):0
2551:netAccess(app_45,tcp,8080):0

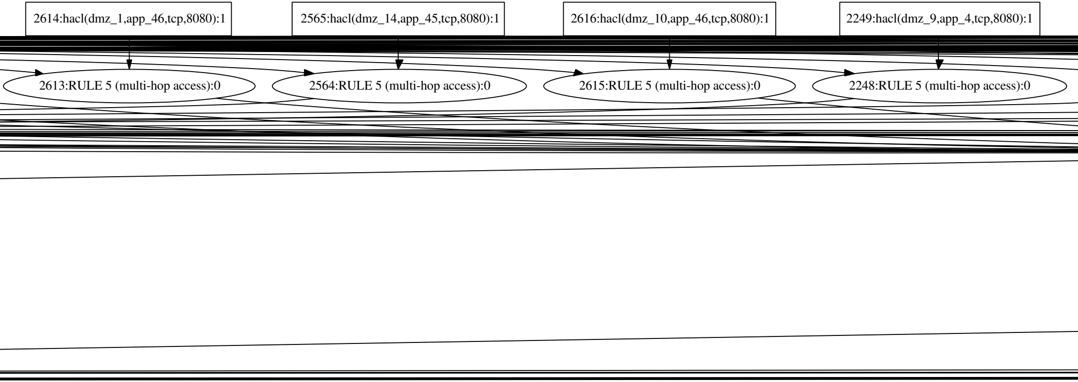

2587:hacl(dmz_24,app_45,tcp,8080):1
2559:hacl(dmz_11,app_45,tcp,8080):1
2585:hacl(dmz_23,app_45,tcp,8080):1
2599:hacl(dmz_7,app_45,tcp,8080):1
2586:RULE 5 (multi-hop access):0
2558:RULE 5 (multi-hop access):0
2584:RULE 5 (multi-hop access):0
2598:RULE 5 (multi-hop access):0

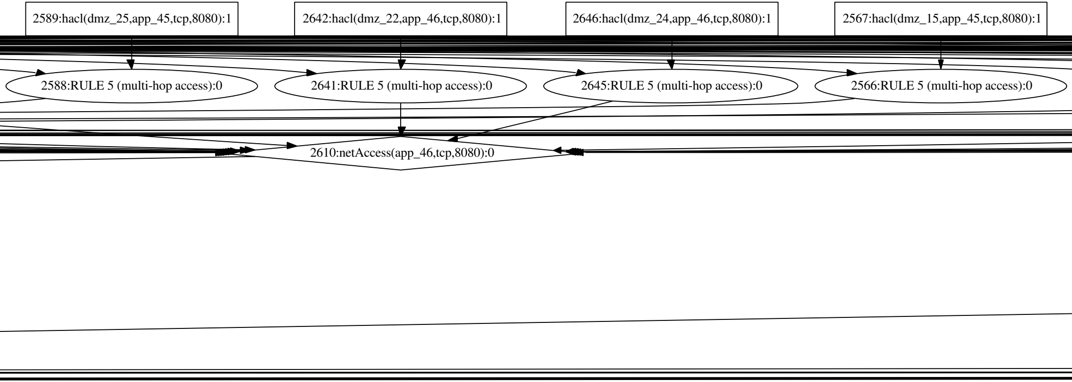

2508:hacl(dmz_15,app_44,tcp,8080):1
2628:hacl(dmz_16,app_46,tcp,8080):1
2473:hacl(dmz_3,app_43,tcp,8080):1
2630:hacl(dmz_17,app_46,tcp,8080):1
2507:RULE 5 (multi-hop access):0
2627:RULE 5 (multi-hop access):0
2472:RULE 5 (multi-hop access):0
2629:RULE 5 (multi-hop access):0

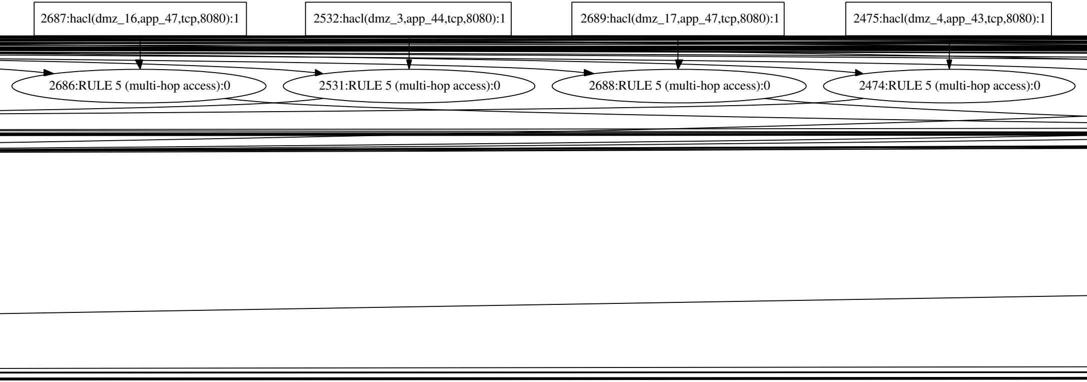

2416:hacl(dmz_4,app_42,tcp,8080):1
2632:hacl(dmz_18,app_46,tcp,8080):1
2601:hacl(dmz_8,app_45,tcp,8080):1
2654:hacl(dmz_5,app_46,tcp,8080):1
2415:RULE 5 (multi-hop access):0
2631:RULE 5 (multi-hop access):0
2600:RULE 5 (multi-hop access):0
2653:RULE 5 (multi-hop access):0
2663:networkServiceInfo(app_46,tomcat,tcp,8080,root):1
2664:vulExists(app_46,vul_app,tomcat,remoteExploit,p
2609:RULE 2 (remote exploit of a server program):0

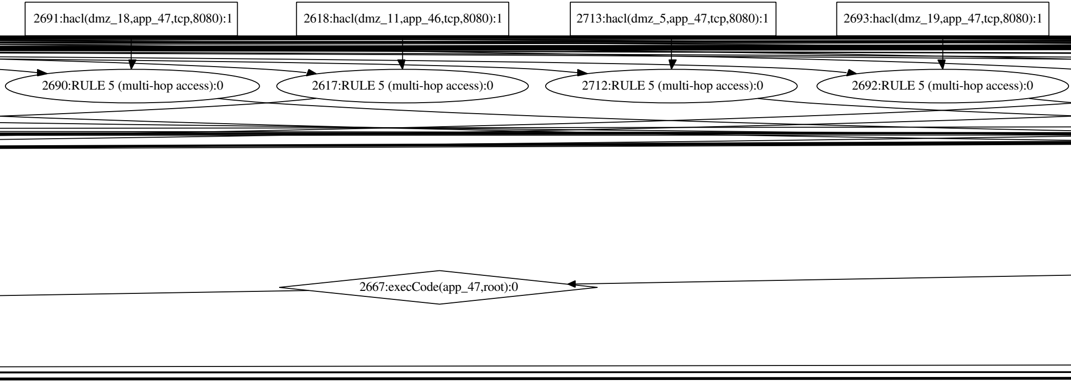

2634:hacl(dmz_19,app_46,tcp,8080):1
2561:hacl(dmz_12,app_45,tcp,8080):1
2636:hacl(dmz_2,app_46,tcp,8080):1
2638:hacl(dmz_20,app_46,tcp,8080):1
2633:RULE 5 (multi-hop access):0
2560:RULE 5 (multi-hop access):0
2635:RULE 5 (multi-hop access):0
2637:RULE 5 (multi-hop access):0
rivEscalation):1

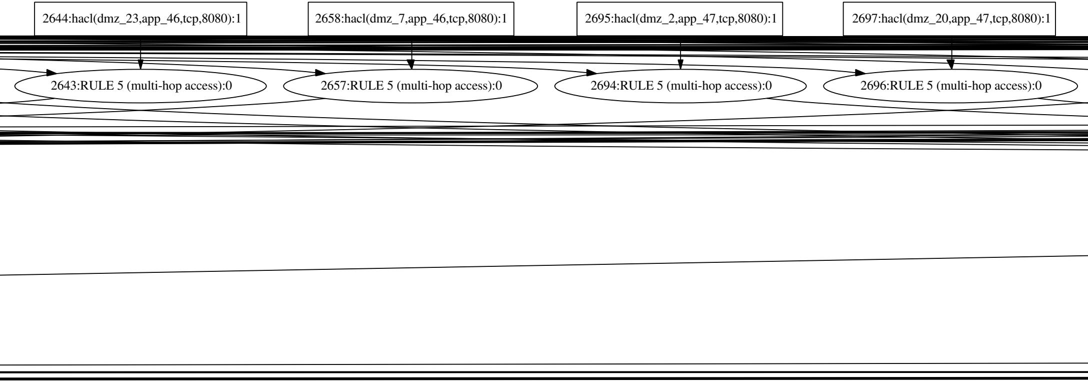

2563:hacl(dmz_13,app_45,tcp,8080):1
2612:hacl(dmz_0,app_46,tcp,8080):1
2640:hacl(dmz_21,app_46,tcp,8080):1
2656:hacl(dmz_6,app_46,tcp,8080):1
2562:RULE 5 (multi-hop access):0
2611:RULE 5 (multi-hop access):0
2639:RULE 5 (multi-hop access):0
2655:RULE 5 (multi-hop access):0
2666:hacl(app_47,db_51,tcp,5432):1
2665:RULE 5 (multi-hop access):0

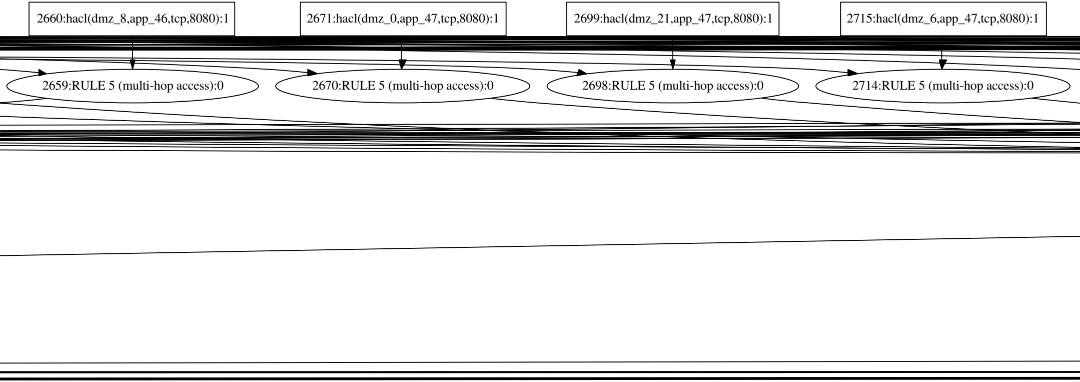

2614:hacl(dmz_1,app_46,tcp,8080):1
2565:hacl(dmz_14,app_45,tcp,8080):1
2616:hacl(dmz_10,app_46,tcp,8080):1
2249:hacl(dmz_9,app_4,tcp,8080):1
2613:RULE 5 (multi-hop access):0
2564:RULE 5 (multi-hop access):0
2615:RULE 5 (multi-hop access):0
2248:RULE 5 (multi-hop access):0

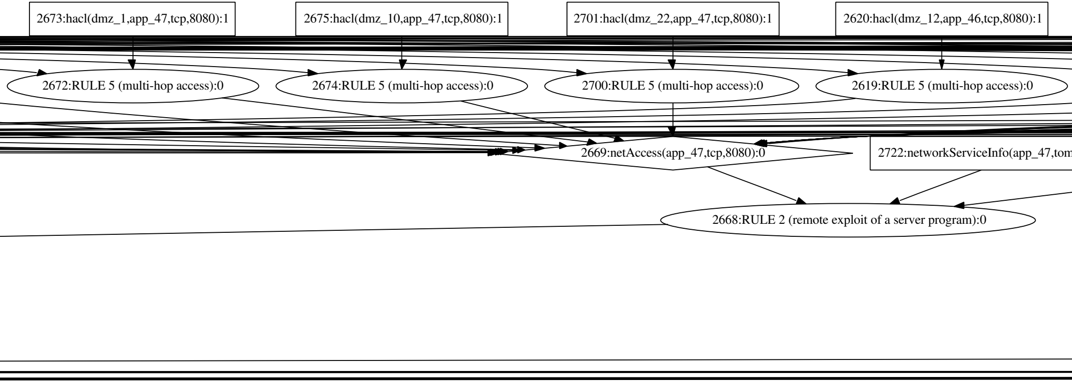

2589:hacl(dmz_25,app_45,tcp,8080):1
2642:hacl(dmz_22,app_46,tcp,8080):1
2646:hacl(dmz_24,app_46,tcp,8080):1
2567:hacl(dmz_15,app_45,tcp,8080):1
2588:RULE 5 (multi-hop access):0
2641:RULE 5 (multi-hop access):0
2645:RULE 5 (multi-hop access):0
2566:RULE 5 (multi-hop access):0
2610:netAccess(app_46,tcp,8080):0

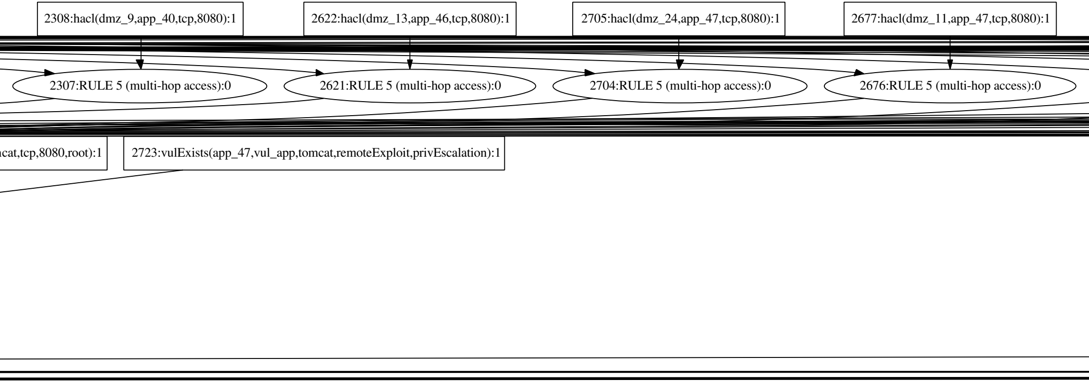

2687:hacl(dmz_16,app_47,tcp,8080):1
2532:hacl(dmz_3,app_44,tcp,8080):1
2689:hacl(dmz_17,app_47,tcp,8080):1
2475:hacl(dmz_4,app_43,tcp,8080):1
2686:RULE 5 (multi-hop access):0
2531:RULE 5 (multi-hop access):0
2688:RULE 5 (multi-hop access):0
2474:RULE 5 (multi-hop access):0

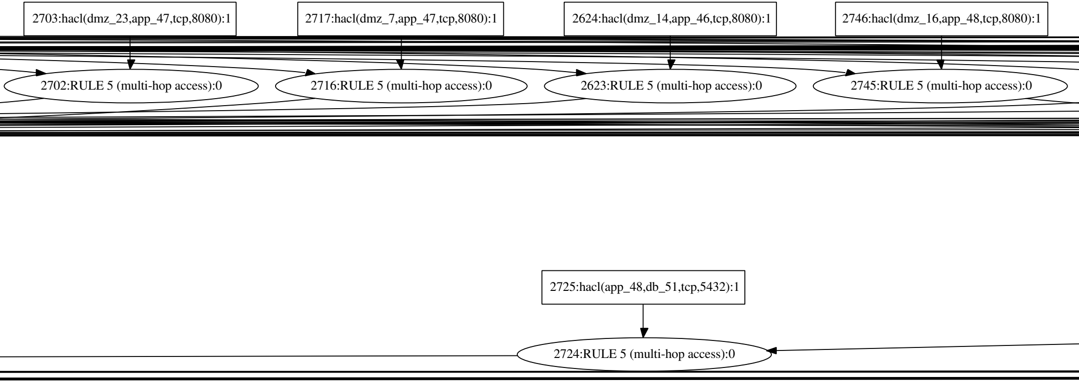

2691:hacl(dmz_18,app_47,tcp,8080):1
2618:hacl(dmz_11,app_46,tcp,8080):1
2713:hacl(dmz_5,app_47,tcp,8080):1
2693:hacl(dmz_19,app_47,tcp,8080):1
2690:RULE 5 (multi-hop access):0
2617:RULE 5 (multi-hop access):0
2712:RULE 5 (multi-hop access):0
2692:RULE 5 (multi-hop access):0
2667:execCode(app_47,root):0

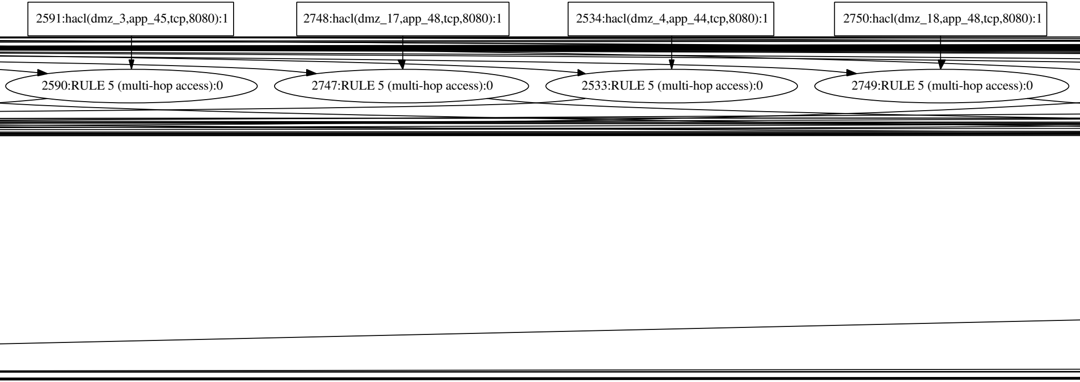

2644:hacl(dmz_23,app_46,tcp,8080):1
2658:hacl(dmz_7,app_46,tcp,8080):1
2695:hacl(dmz_2,app_47,tcp,8080):1
2697:hacl(dmz_20,app_47,tcp,8080):1
2643:RULE 5 (multi-hop access):0
2657:RULE 5 (multi-hop access):0
2694:RULE 5 (multi-hop access):0
2696:RULE 5 (multi-hop access):0

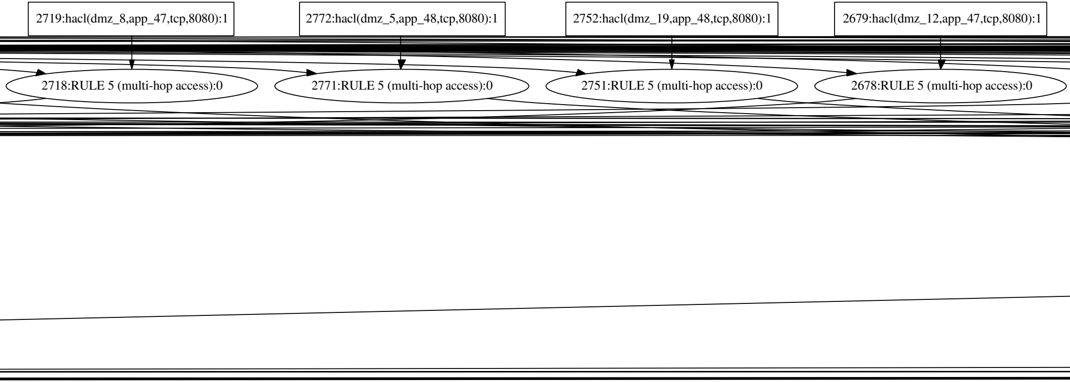

2660:hacl(dmz_8,app_46,tcp,8080):1
2671:hacl(dmz_0,app_47,tcp,8080):1
2699:hacl(dmz_21,app_47,tcp,8080):1
2715:hacl(dmz_6,app_47,tcp,8080):1
2659:RULE 5 (multi-hop access):0
2670:RULE 5 (multi-hop access):0
2698:RULE 5 (multi-hop access):0
2714:RULE 5 (multi-hop access):0

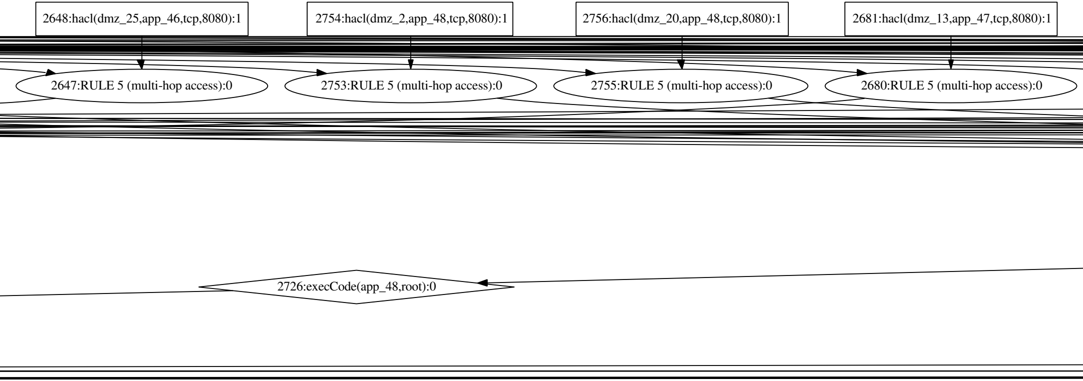

2673:hacl(dmz_1,app_47,tcp,8080):1
2675:hacl(dmz_10,app_47,tcp,8080):1
2701:hacl(dmz_22,app_47,tcp,8080):1
2620:hacl(dmz_12,app_46,tcp,8080):1
2672:RULE 5 (multi-hop access):0
2674:RULE 5 (multi-hop access):0
2700:RULE 5 (multi-hop access):0
2619:RULE 5 (multi-hop access):0
2669:netAccess(app_47,tcp,8080):0
2722:networkServiceInfo(app_47,tom
2668:RULE 2 (remote exploit of a server program):0

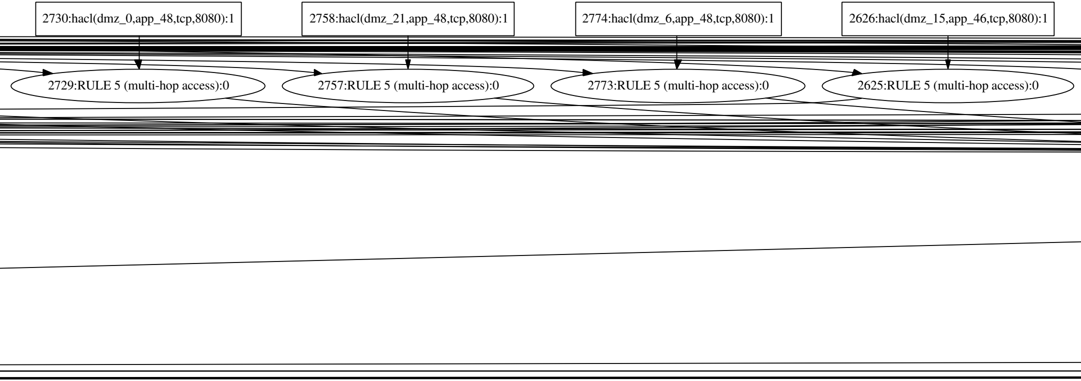

2308:hacl(dmz_9,app_40,tcp,8080):1
2622:hacl(dmz_13,app_46,tcp,8080):1
2705:hacl(dmz_24,app_47,tcp,8080):1
2677:hacl(dmz_11,app_47,tcp,8080):1
2307:RULE 5 (multi-hop access):0
2621:RULE 5 (multi-hop access):0
2704:RULE 5 (multi-hop access):0
2676:RULE 5 (multi-hop access):0
cat,tcp,8080,root):1
2723:vulExists(app_47,vul_app,tomcat,remoteExploit,privEscalation):1

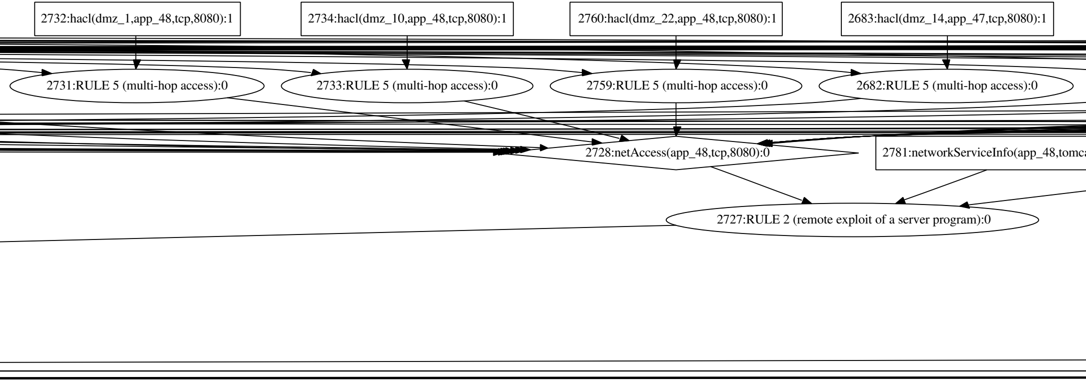

2703:hacl(dmz_23,app_47,tcp,8080):1
2717:hacl(dmz_7,app_47,tcp,8080):1
2624:hacl(dmz_14,app_46,tcp,8080):1
2746:hacl(dmz_16,app_48,tcp,8080):1
2702:RULE 5 (multi-hop access):0
2716:RULE 5 (multi-hop access):0
2623:RULE 5 (multi-hop access):0
2745:RULE 5 (multi-hop access):0
2725:hacl(app_48,db_51,tcp,5432):1
2724:RULE 5 (multi-hop access):0

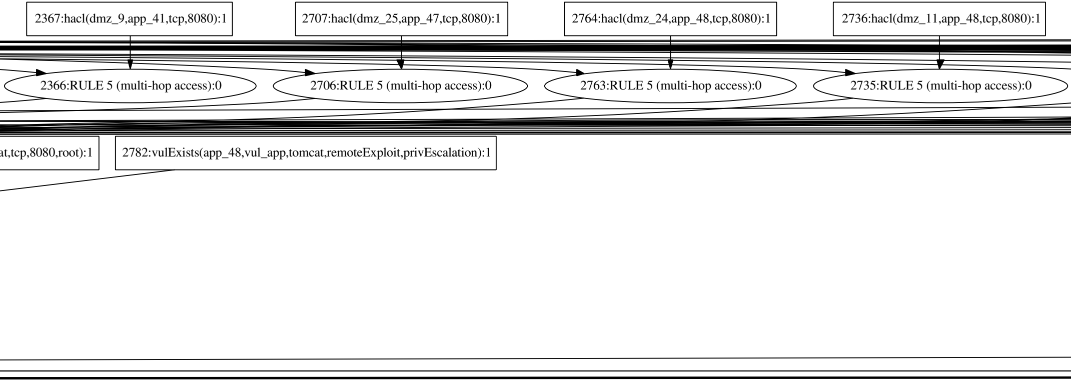

2591:hacl(dmz_3,app_45,tcp,8080):1
2748:hacl(dmz_17,app_48,tcp,8080):1
2534:hacl(dmz_4,app_44,tcp,8080):1
2750:hacl(dmz_18,app_48,tcp,8080):1
2590:RULE 5 (multi-hop access):0
2747:RULE 5 (multi-hop access):0
2533:RULE 5 (multi-hop access):0
2749:RULE 5 (multi-hop access):0

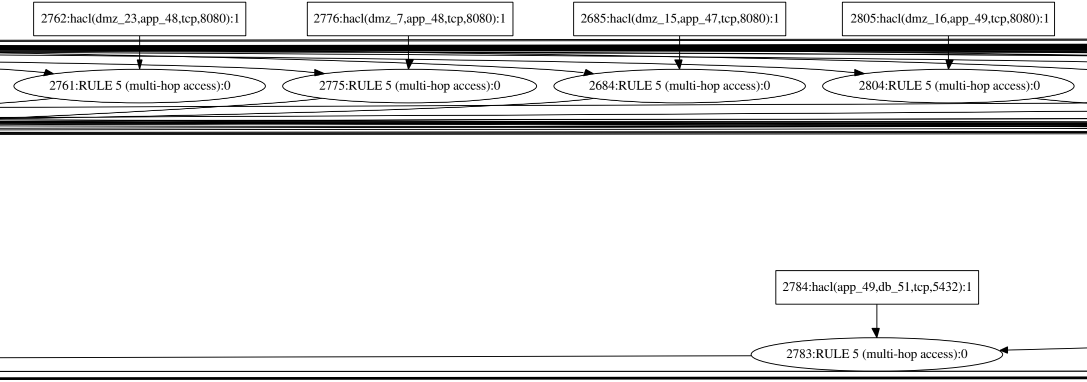

2719:hacl(dmz_8,app_47,tcp,8080):1
2772:hacl(dmz_5,app_48,tcp,8080):1
2752:hacl(dmz_19,app_48,tcp,8080):1
2679:hacl(dmz_12,app_47,tcp,8080):1
2718:RULE 5 (multi-hop access):0
2771:RULE 5 (multi-hop access):0
2751:RULE 5 (multi-hop access):0
2678:RULE 5 (multi-hop access):0

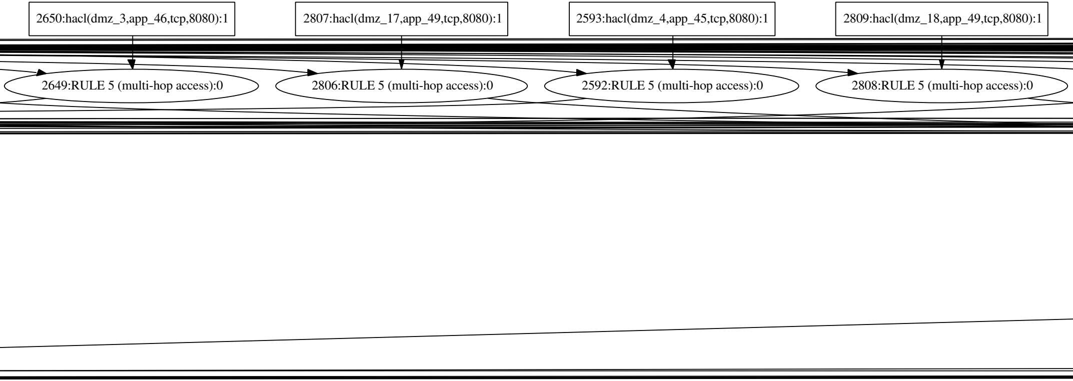

2648:hacl(dmz_25,app_46,tcp,8080):1
2754:hacl(dmz_2,app_48,tcp,8080):1
2756:hacl(dmz_20,app_48,tcp,8080):1
2681:hacl(dmz_13,app_47,tcp,8080):1
2647:RULE 5 (multi-hop access):0
2753:RULE 5 (multi-hop access):0
2755:RULE 5 (multi-hop access):0
2680:RULE 5 (multi-hop access):0
2726:execCode(app_48,root):0

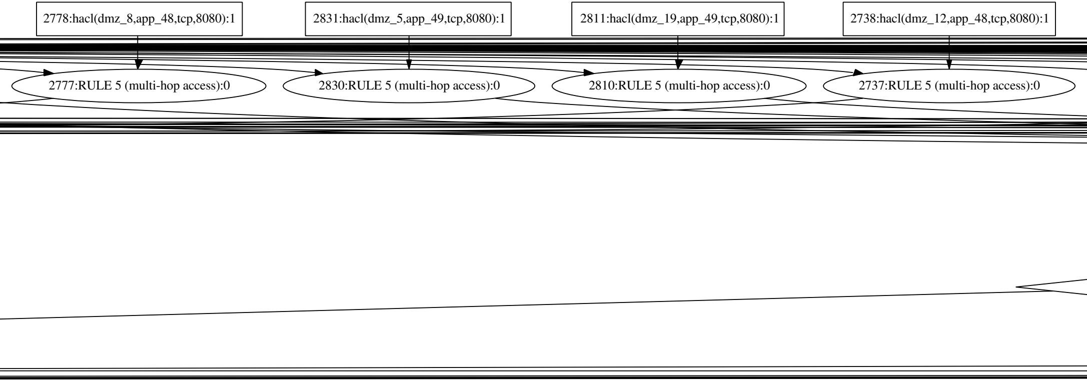

2730:hacl(dmz_0,app_48,tcp,8080):1
2758:hacl(dmz_21,app_48,tcp,8080):1
2774:hacl(dmz_6,app_48,tcp,8080):1
2626:hacl(dmz_15,app_46,tcp,8080):1
2729:RULE 5 (multi-hop access):0
2757:RULE 5 (multi-hop access):0
2773:RULE 5 (multi-hop access):0
2625:RULE 5 (multi-hop access):0

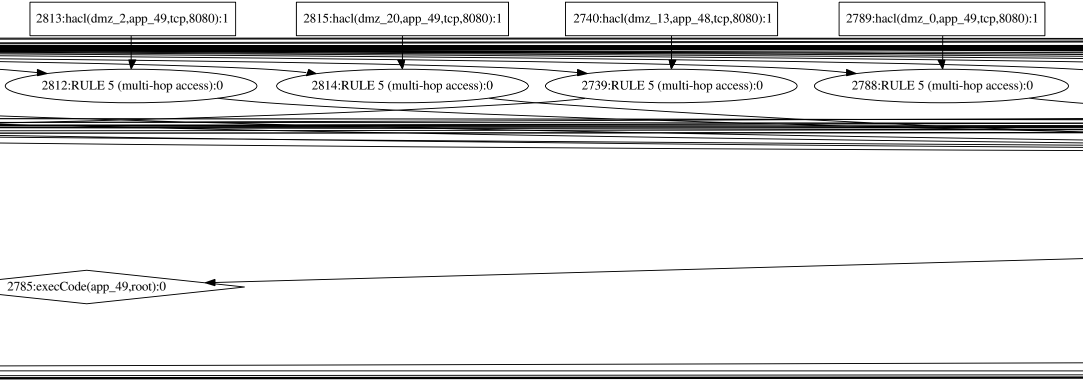

2732:hacl(dmz_1,app_48,tcp,8080):1
2734:hacl(dmz_10,app_48,tcp,8080):1
2760:hacl(dmz_22,app_48,tcp,8080):1
2683:hacl(dmz_14,app_47,tcp,8080):1
2731:RULE 5 (multi-hop access):0
2733:RULE 5 (multi-hop access):0
2759:RULE 5 (multi-hop access):0
2682:RULE 5 (multi-hop access):0
2728:netAccess(app_48,tcp,8080):0
2781:networkServiceInfo(app_48,tomca
2727:RULE 2 (remote exploit of a server program):0

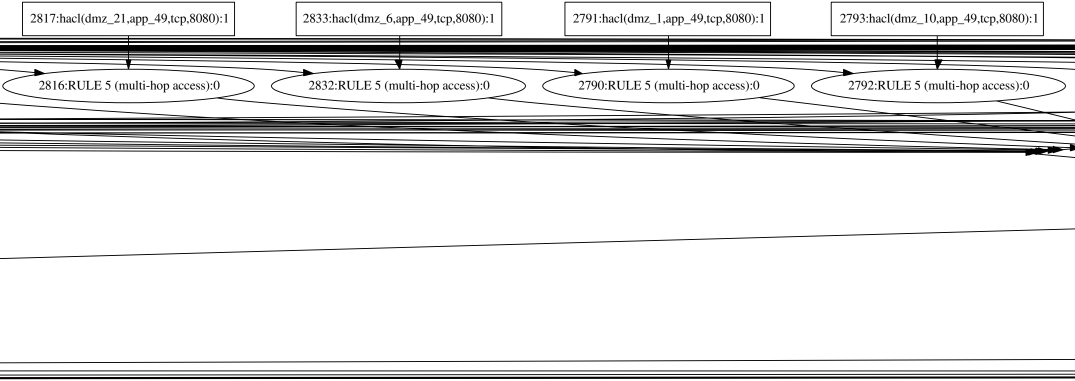

2367:hacl(dmz_9,app_41,tcp,8080):1
2707:hacl(dmz_25,app_47,tcp,8080):1
2764:hacl(dmz_24,app_48,tcp,8080):1
2736:hacl(dmz_11,app_48,tcp,8080):1
2366:RULE 5 (multi-hop access):0
2706:RULE 5 (multi-hop access):0
2763:RULE 5 (multi-hop access):0
2735:RULE 5 (multi-hop access):0
at,tcp,8080,root):1
2782:vulExists(app_48,vul_app,tomcat,remoteExploit,privEscalation):1

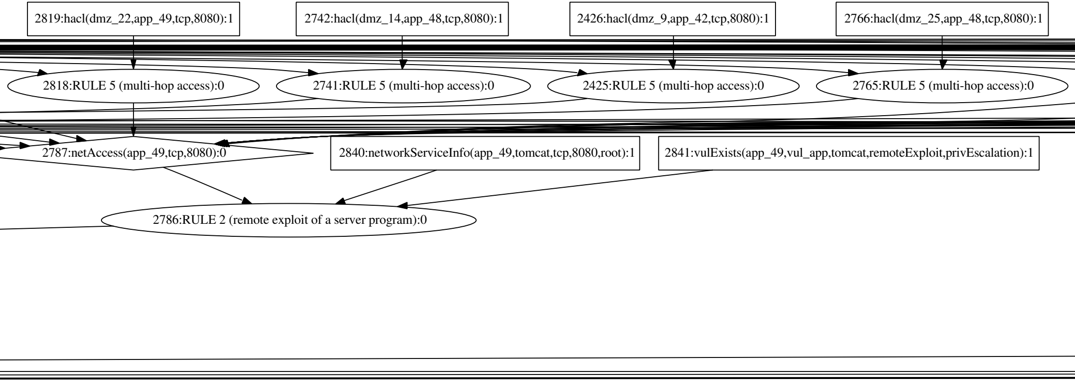

2762:hacl(dmz_23,app_48,tcp,8080):1
2776:hacl(dmz_7,app_48,tcp,8080):1
2685:hacl(dmz_15,app_47,tcp,8080):1
2805:hacl(dmz_16,app_49,tcp,8080):1
2761:RULE 5 (multi-hop access):0
2775:RULE 5 (multi-hop access):0
2684:RULE 5 (multi-hop access):0
2804:RULE 5 (multi-hop access):0
2784:hacl(app_49,db_51,tcp,5432):1
2783:RULE 5 (multi-hop access):0

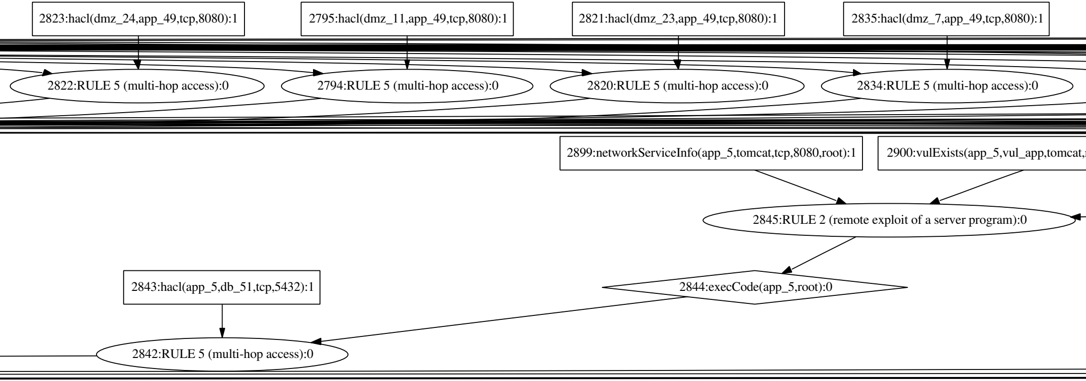

2650:hacl(dmz_3,app_46,tcp,8080):1
2807:hacl(dmz_17,app_49,tcp,8080):1
2593:hacl(dmz_4,app_45,tcp,8080):1
2809:hacl(dmz_18,app_49,tcp,8080):1
2649:RULE 5 (multi-hop access):0
2806:RULE 5 (multi-hop access):0
2592:RULE 5 (multi-hop access):0
2808:RULE 5 (multi-hop access):0

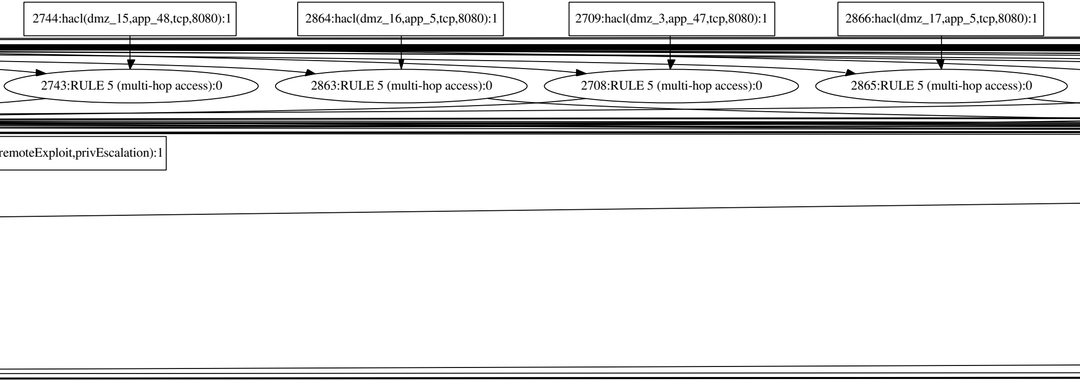

2778:hacl(dmz_8,app_48,tcp,8080):1
2831:hacl(dmz_5,app_49,tcp,8080):1
2811:hacl(dmz_19,app_49,tcp,8080):1
2738:hacl(dmz_12,app_48,tcp,8080):1
2777:RULE 5 (multi-hop access):0
2830:RULE 5 (multi-hop access):0
2810:RULE 5 (multi-hop access):0
2737:RULE 5 (multi-hop access):0

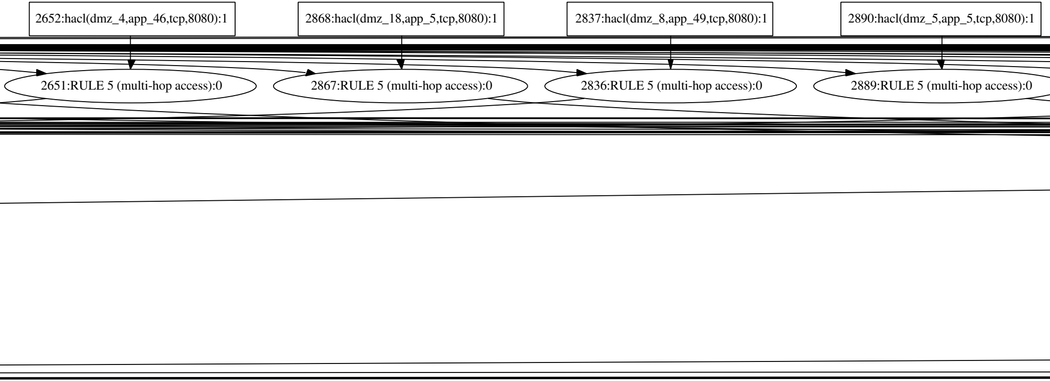

2813:hacl(dmz_2,app_49,tcp,8080):1
2815:hacl(dmz_20,app_49,tcp,8080):1
2740:hacl(dmz_13,app_48,tcp,8080):1
2789:hacl(dmz_0,app_49,tcp,8080):1
2812:RULE 5 (multi-hop access):0
2814:RULE 5 (multi-hop access):0
2739:RULE 5 (multi-hop access):0
2788:RULE 5 (multi-hop access):0
2785:execCode(app_49,root):0

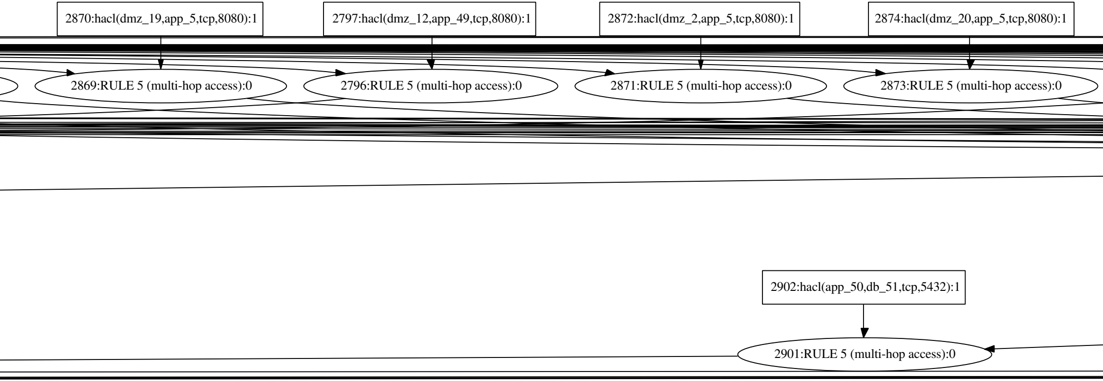

2817:hacl(dmz_21,app_49,tcp,8080):1
2833:hacl(dmz_6,app_49,tcp,8080):1
2791:hacl(dmz_1,app_49,tcp,8080):1
2793:hacl(dmz_10,app_49,tcp,8080):1
2816:RULE 5 (multi-hop access):0
2832:RULE 5 (multi-hop access):0
2790:RULE 5 (multi-hop access):0
2792:RULE 5 (multi-hop access):0

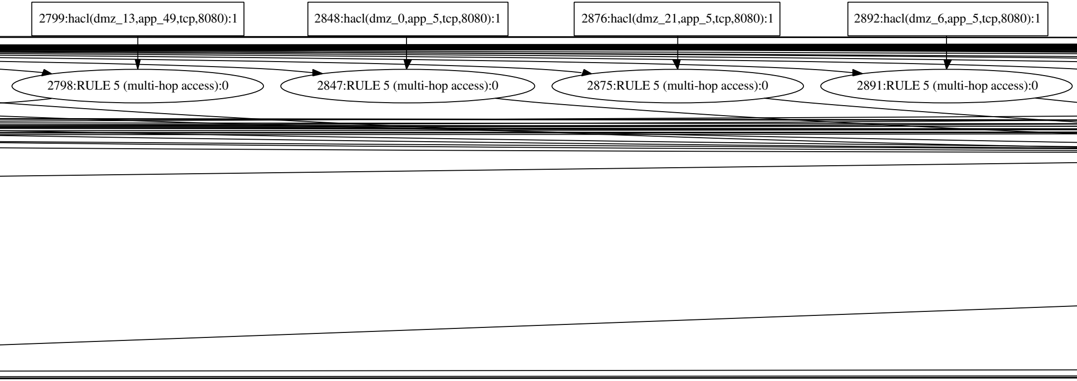

2819:hacl(dmz_22,app_49,tcp,8080):1
2742:hacl(dmz_14,app_48,tcp,8080):1
2426:hacl(dmz_9,app_42,tcp,8080):1
2766:hacl(dmz_25,app_48,tcp,8080):1
2818:RULE 5 (multi-hop access):0
2741:RULE 5 (multi-hop access):0
2425:RULE 5 (multi-hop access):0
2765:RULE 5 (multi-hop access):0
2787:netAccess(app_49,tcp,8080):0
2840:networkServiceInfo(app_49,tomcat,tcp,8080,root):1
2841:vulExists(app_49,vul_app,tomcat,remoteExploit,privEscalation):1
2786:RULE 2 (remote exploit of a server program):0

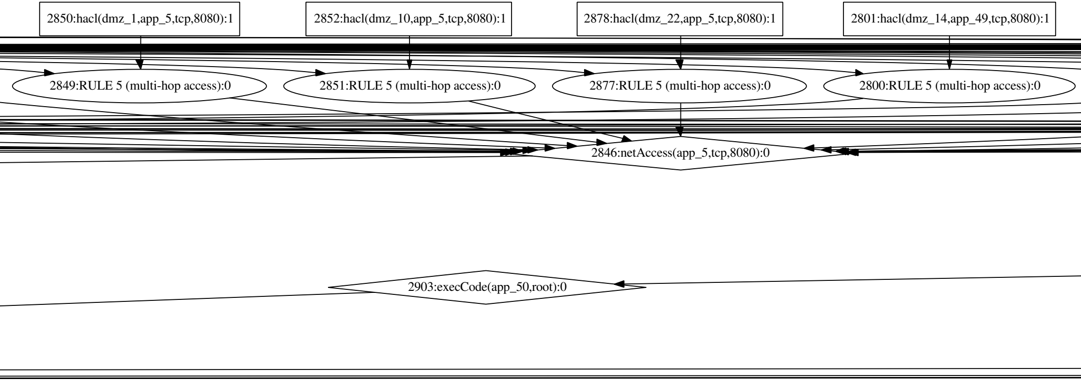

2823:hacl(dmz_24,app_49,tcp,8080):1
2795:hacl(dmz_11,app_49,tcp,8080):1
2821:hacl(dmz_23,app_49,tcp,8080):1
2835:hacl(dmz_7,app_49,tcp,8080):1
2822:RULE 5 (multi-hop access):0
2794:RULE 5 (multi-hop access):0
2820:RULE 5 (multi-hop access):0
2834:RULE 5 (multi-hop access):0
2899:networkServiceInfo(app_5,tomcat,tcp,8080,root):1
2900:vulExists(app_5,vul_app,tomcat,r
2845:RULE 2 (remote exploit of a server program):0
2843:hacl(app_5,db_51,tcp,5432):1
2844:execCode(app_5,root):0
2842:RULE 5 (multi-hop access):0

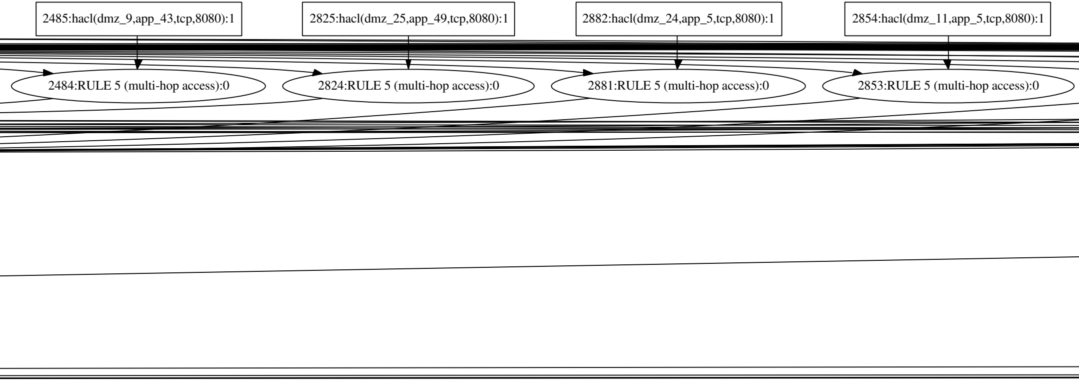

2744:hacl(dmz_15,app_48,tcp,8080):1
2864:hacl(dmz_16,app_5,tcp,8080):1
2709:hacl(dmz_3,app_47,tcp,8080):1
2866:hacl(dmz_17,app_5,tcp,8080):1
2743:RULE 5 (multi-hop access):0
2863:RULE 5 (multi-hop access):0
2708:RULE 5 (multi-hop access):0
2865:RULE 5 (multi-hop access):0
remoteExploit,privEscalation):1

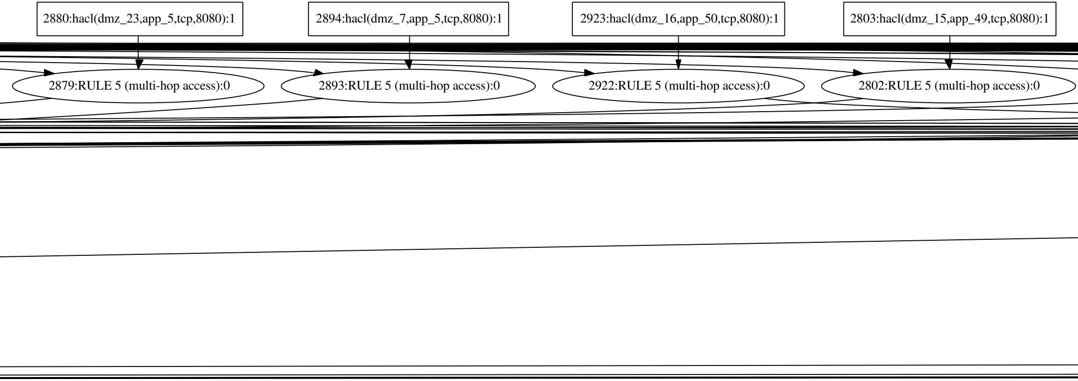

2652:hacl(dmz_4,app_46,tcp,8080):1
2868:hacl(dmz_18,app_5,tcp,8080):1
2837:hacl(dmz_8,app_49,tcp,8080):1
2890:hacl(dmz_5,app_5,tcp,8080):1
2651:RULE 5 (multi-hop access):0
2867:RULE 5 (multi-hop access):0
2836:RULE 5 (multi-hop access):0
2889:RULE 5 (multi-hop access):0

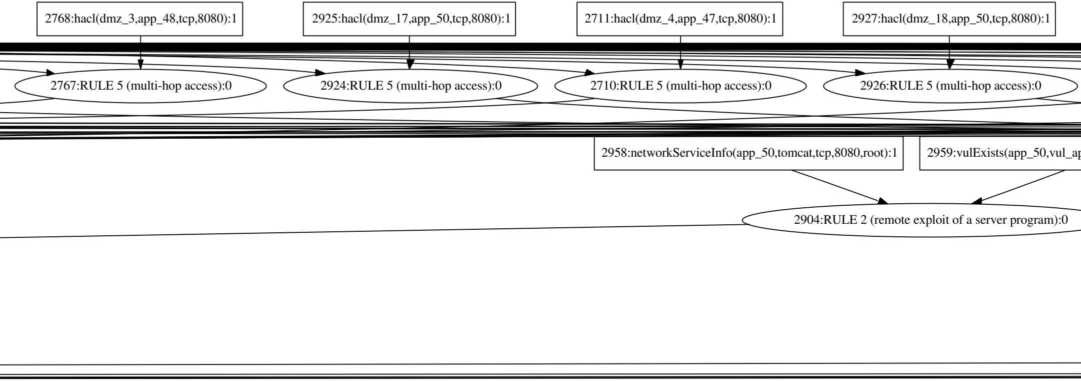

2870:hacl(dmz_19,app_5,tcp,8080):1
2797:hacl(dmz_12,app_49,tcp,8080):1
2872:hacl(dmz_2,app_5,tcp,8080):1
2874:hacl(dmz_20,app_5,tcp,8080):1
2869:RULE 5 (multi-hop access):0
2796:RULE 5 (multi-hop access):0
2871:RULE 5 (multi-hop access):0
2873:RULE 5 (multi-hop access):0
2902:hacl(app_50,db_51,tcp,5432):1
2901:RULE 5 (multi-hop access):0

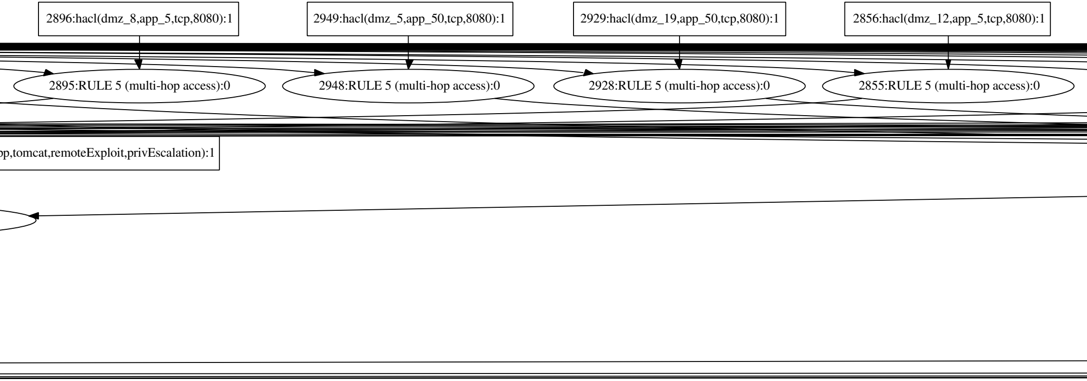

2799:hacl(dmz_13,app_49,tcp,8080):1
2848:hacl(dmz_0,app_5,tcp,8080):1
2876:hacl(dmz_21,app_5,tcp,8080):1
2892:hacl(dmz_6,app_5,tcp,8080):1
2798:RULE 5 (multi-hop access):0
2847:RULE 5 (multi-hop access):0
2875:RULE 5 (multi-hop access):0
2891:RULE 5 (multi-hop access):0

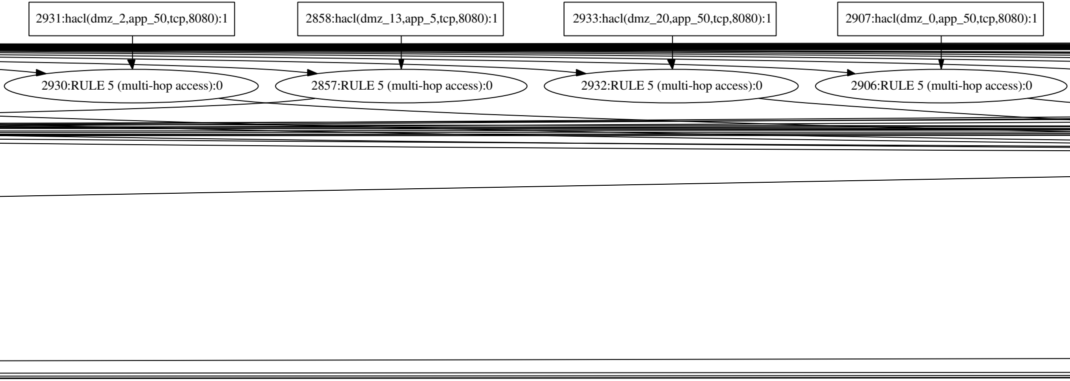

2850:hacl(dmz_1,app_5,tcp,8080):1
2852:hacl(dmz_10,app_5,tcp,8080):1
2878:hacl(dmz_22,app_5,tcp,8080):1
2801:hacl(dmz_14,app_49,tcp,8080):1
2849:RULE 5 (multi-hop access):0
2851:RULE 5 (multi-hop access):0
2877:RULE 5 (multi-hop access):0
2800:RULE 5 (multi-hop access):0
2846:netAccess(app_5,tcp,8080):0
2903:execCode(app_50,root):0

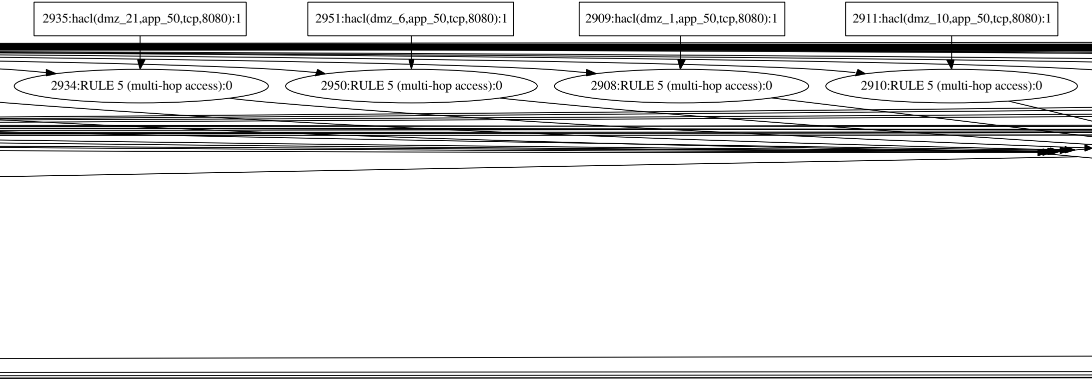

2485:hacl(dmz_9,app_43,tcp,8080):1
2825:hacl(dmz_25,app_49,tcp,8080):1
2882:hacl(dmz_24,app_5,tcp,8080):1
2854:hacl(dmz_11,app_5,tcp,8080):1
2484:RULE 5 (multi-hop access):0
2824:RULE 5 (multi-hop access):0
2881:RULE 5 (multi-hop access):0
2853:RULE 5 (multi-hop access):0

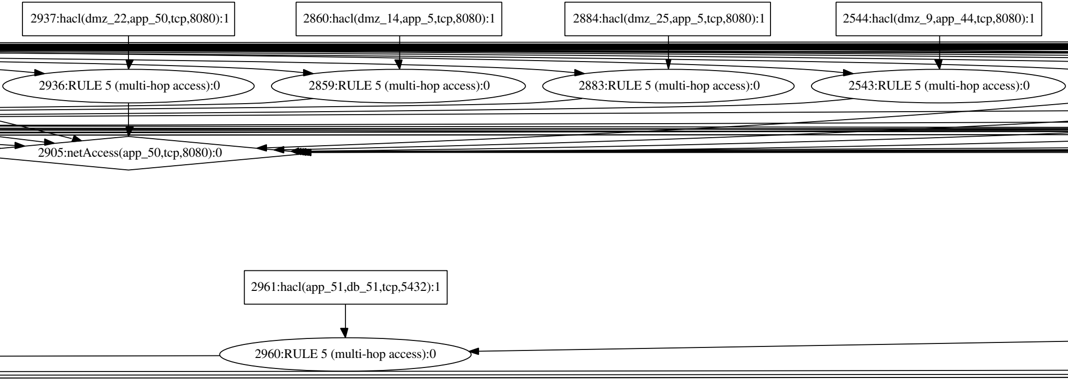

2880:hacl(dmz_23,app_5,tcp,8080):1
2894:hacl(dmz_7,app_5,tcp,8080):1
2923:hacl(dmz_16,app_50,tcp,8080):1
2803:hacl(dmz_15,app_49,tcp,8080):1
2879:RULE 5 (multi-hop access):0
2893:RULE 5 (multi-hop access):0
2922:RULE 5 (multi-hop access):0
2802:RULE 5 (multi-hop access):0

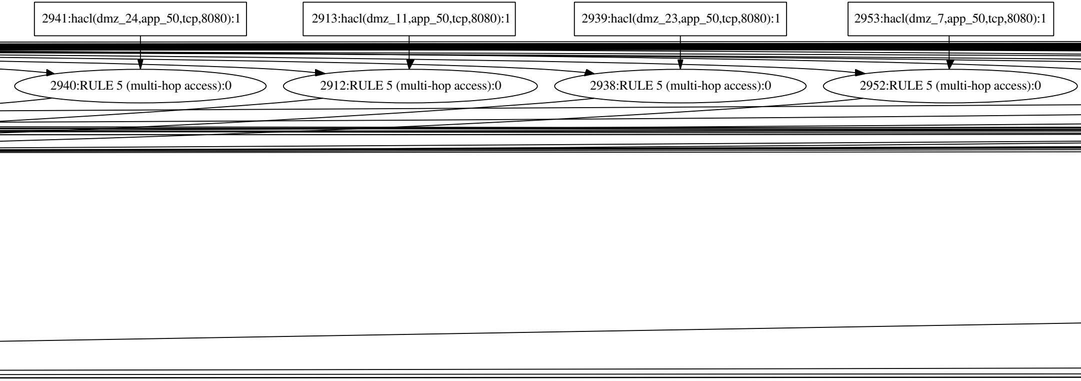

2768:hacl(dmz_3,app_48,tcp,8080):1
2925:hacl(dmz_17,app_50,tcp,8080):1
2711:hacl(dmz_4,app_47,tcp,8080):1
2927:hacl(dmz_18,app_50,tcp,8080):1
2767:RULE 5 (multi-hop access):0
2924:RULE 5 (multi-hop access):0
2710:RULE 5 (multi-hop access):0
2926:RULE 5 (multi-hop access):0
2958:networkServiceInfo(app_50,tomcat,tcp,8080,root):1
2959:vulExists(app_50,vul_ap
2904:RULE 2 (remote exploit of a server program):0

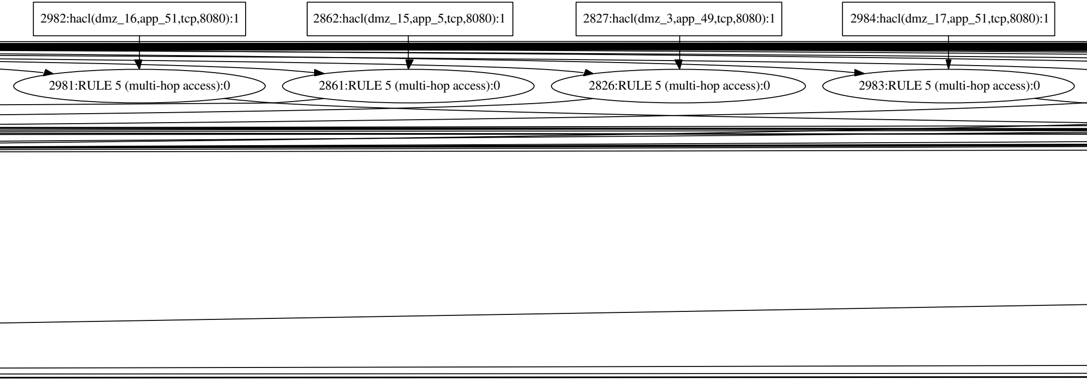

2896:hacl(dmz_8,app_5,tcp,8080):1
2949:hacl(dmz_5,app_50,tcp,8080):1
2929:hacl(dmz_19,app_50,tcp,8080):1
2856:hacl(dmz_12,app_5,tcp,8080):1
2895:RULE 5 (multi-hop access):0
2948:RULE 5 (multi-hop access):0
2928:RULE 5 (multi-hop access):0
2855:RULE 5 (multi-hop access):0
p,tomcat,remoteExploit,privEscalation):1

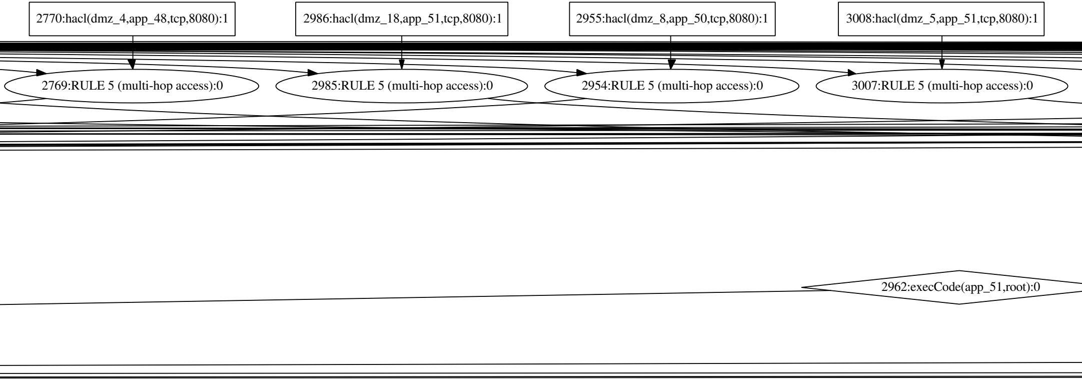

2931:hacl(dmz_2,app_50,tcp,8080):1
2858:hacl(dmz_13,app_5,tcp,8080):1
2933:hacl(dmz_20,app_50,tcp,8080):1
2907:hacl(dmz_0,app_50,tcp,8080):1
2930:RULE 5 (multi-hop access):0
2857:RULE 5 (multi-hop access):0
2932:RULE 5 (multi-hop access):0
2906:RULE 5 (multi-hop access):0

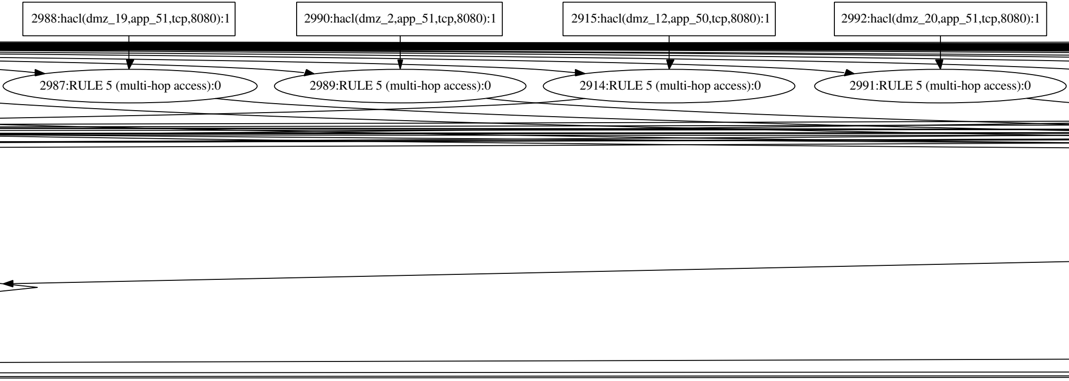

2935:hacl(dmz_21,app_50,tcp,8080):1
2951:hacl(dmz_6,app_50,tcp,8080):1
2909:hacl(dmz_1,app_50,tcp,8080):1
2911:hacl(dmz_10,app_50,tcp,8080):1
2934:RULE 5 (multi-hop access):0
2950:RULE 5 (multi-hop access):0
2908:RULE 5 (multi-hop access):0
2910:RULE 5 (multi-hop access):0

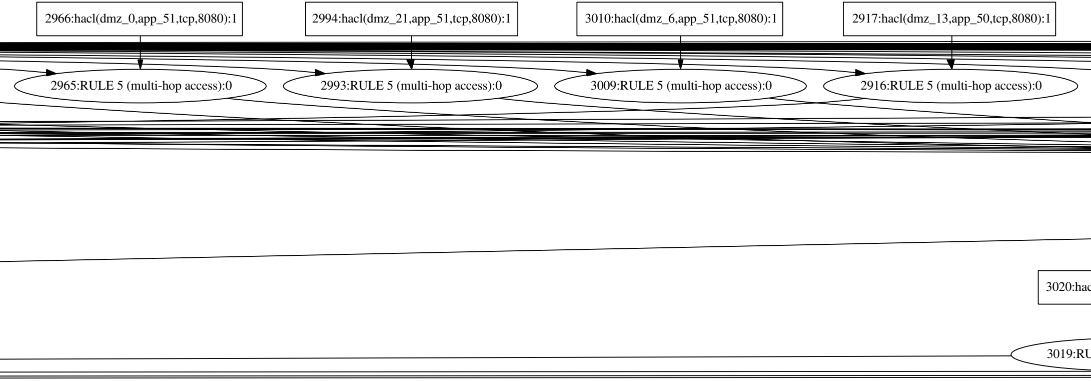

2937:hacl(dmz_22,app_50,tcp,8080):1
2860:hacl(dmz_14,app_5,tcp,8080):1
2884:hacl(dmz_25,app_5,tcp,8080):1
2544:hacl(dmz_9,app_44,tcp,8080):1
2936:RULE 5 (multi-hop access):0
2859:RULE 5 (multi-hop access):0
2883:RULE 5 (multi-hop access):0
2543:RULE 5 (multi-hop access):0
2905:netAccess(app_50,tcp,8080):0
2961:hacl(app_51,db_51,tcp,5432):1
2960:RULE 5 (multi-hop access):0

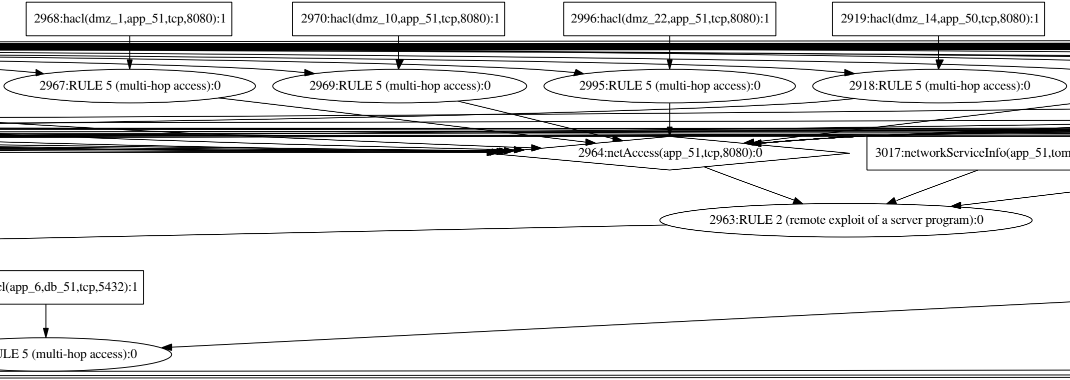

2941:hacl(dmz_24,app_50,tcp,8080):1
2913:hacl(dmz_11,app_50,tcp,8080):1
2939:hacl(dmz_23,app_50,tcp,8080):1
2953:hacl(dmz_7,app_50,tcp,8080):1
2940:RULE 5 (multi-hop access):0
2912:RULE 5 (multi-hop access):0
2938:RULE 5 (multi-hop access):0
2952:RULE 5 (multi-hop access):0

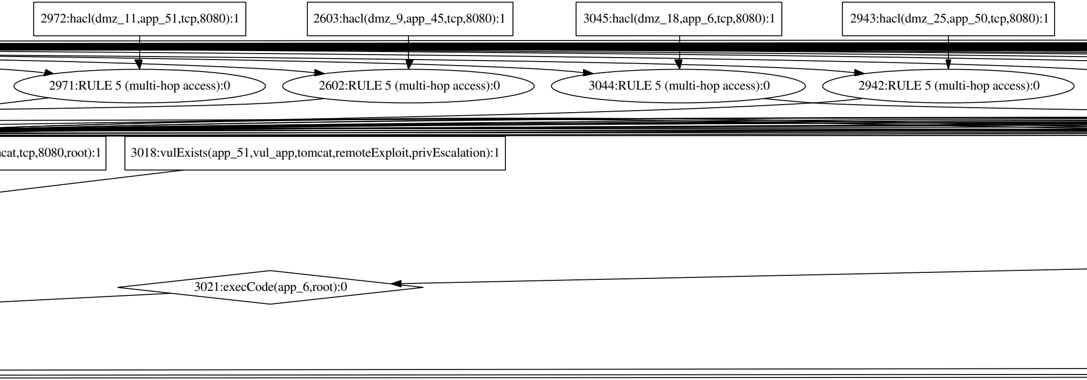

2982:hacl(dmz_16,app_51,tcp,8080):1
2862:hacl(dmz_15,app_5,tcp,8080):1
2827:hacl(dmz_3,app_49,tcp,8080):1
2984:hacl(dmz_17,app_51,tcp,8080):1
2981:RULE 5 (multi-hop access):0
2861:RULE 5 (multi-hop access):0
2826:RULE 5 (multi-hop access):0
2983:RULE 5 (multi-hop access):0

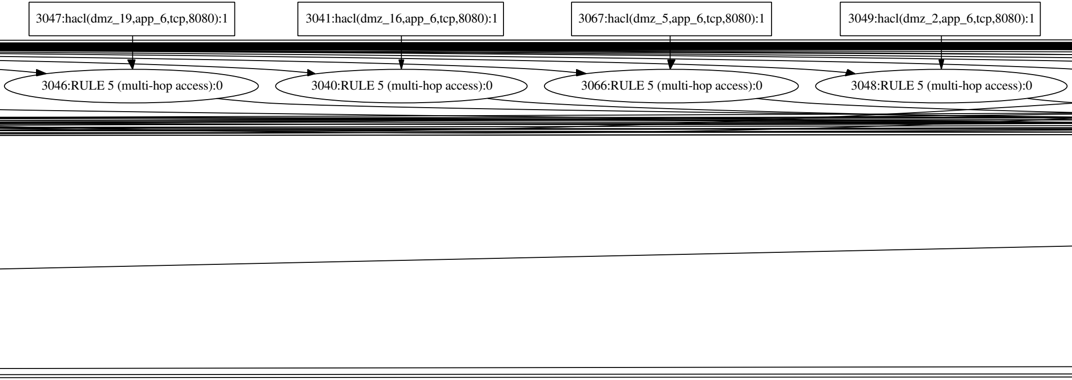

2770:hacl(dmz_4,app_48,tcp,8080):1
2986:hacl(dmz_18,app_51,tcp,8080):1
2955:hacl(dmz_8,app_50,tcp,8080):1
3008:hacl(dmz_5,app_51,tcp,8080):1
2769:RULE 5 (multi-hop access):0
2985:RULE 5 (multi-hop access):0
2954:RULE 5 (multi-hop access):0
3007:RULE 5 (multi-hop access):0
2962:execCode(app_51,root):0

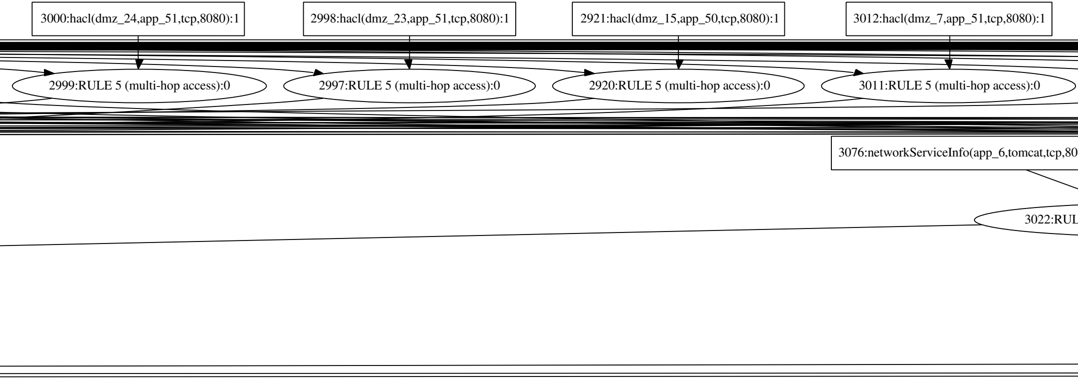

2988:hacl(dmz_19,app_51,tcp,8080):1
2990:hacl(dmz_2,app_51,tcp,8080):1
2915:hacl(dmz_12,app_50,tcp,8080):1
2992:hacl(dmz_20,app_51,tcp,8080):1
2987:RULE 5 (multi-hop access):0
2989:RULE 5 (multi-hop access):0
2914:RULE 5 (multi-hop access):0
2991:RULE 5 (multi-hop access):0

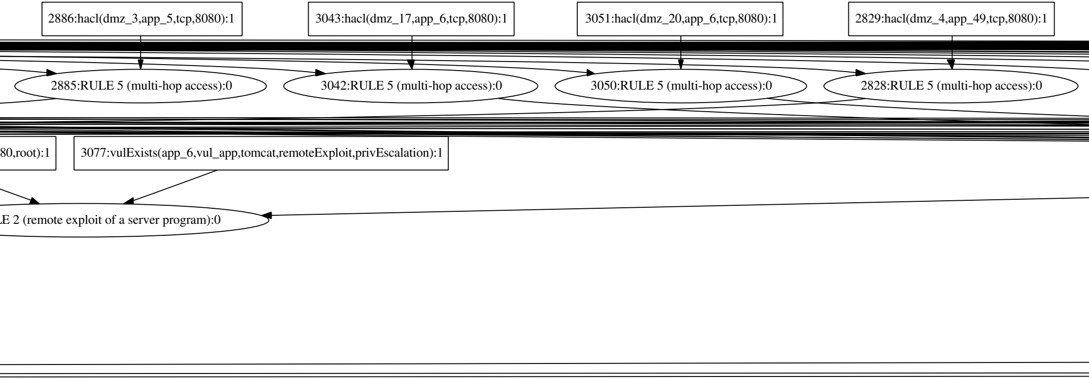

2966:hacl(dmz_0,app_51,tcp,8080):1
2994:hacl(dmz_21,app_51,tcp,8080):1
3010:hacl(dmz_6,app_51,tcp,8080):1
2917:hacl(dmz_13,app_50,tcp,8080):1
2965:RULE 5 (multi-hop access):0
2993:RULE 5 (multi-hop access):0
3009:RULE 5 (multi-hop access):0
2916:RULE 5 (multi-hop access):0
3020:hac
3019:RU

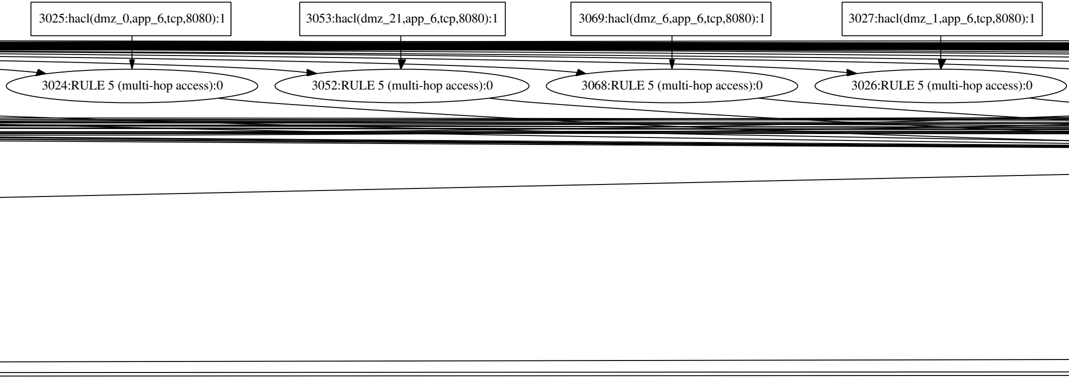

2968:hacl(dmz_1,app_51,tcp,8080):1
2970:hacl(dmz_10,app_51,tcp,8080):1
2996:hacl(dmz_22,app_51,tcp,8080):1
2919:hacl(dmz_14,app_50,tcp,8080):1
2967:RULE 5 (multi-hop access):0
2969:RULE 5 (multi-hop access):0
2995:RULE 5 (multi-hop access):0
2918:RULE 5 (multi-hop access):0
2964:netAccess(app_51,tcp,8080):0
3017:networkServiceInfo(app_51,tom
2963:RULE 2 (remote exploit of a server program):0
cl(app_6,db_51,tcp,5432):1
ULE 5 (multi-hop access):0

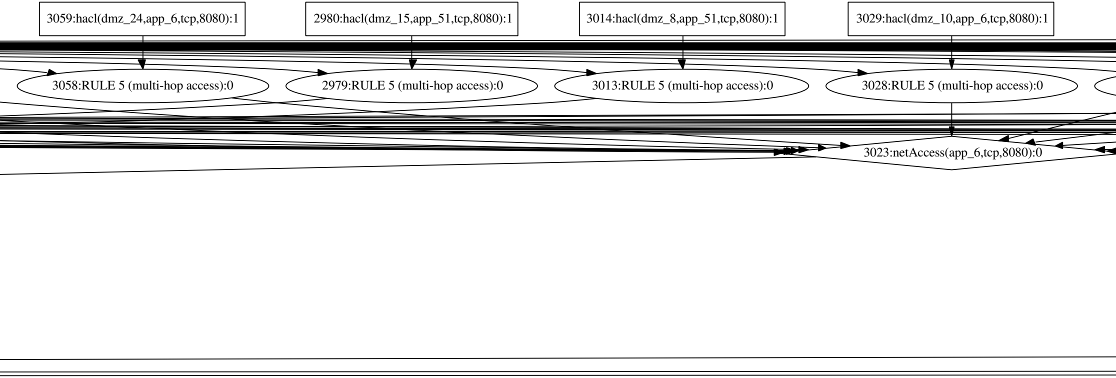

2972:hacl(dmz_11,app_51,tcp,8080):1
2603:hacl(dmz_9,app_45,tcp,8080):1
3045:hacl(dmz_18,app_6,tcp,8080):1
2943:hacl(dmz_25,app_50,tcp,8080):1
2971:RULE 5 (multi-hop access):0
2602:RULE 5 (multi-hop access):0
3044:RULE 5 (multi-hop access):0
2942:RULE 5 (multi-hop access):0
cat,tcp,8080,root):1
3018:vulExists(app_51,vul_app,tomcat,remoteExploit,privEscalation):1
3021:execCode(app_6,root):0

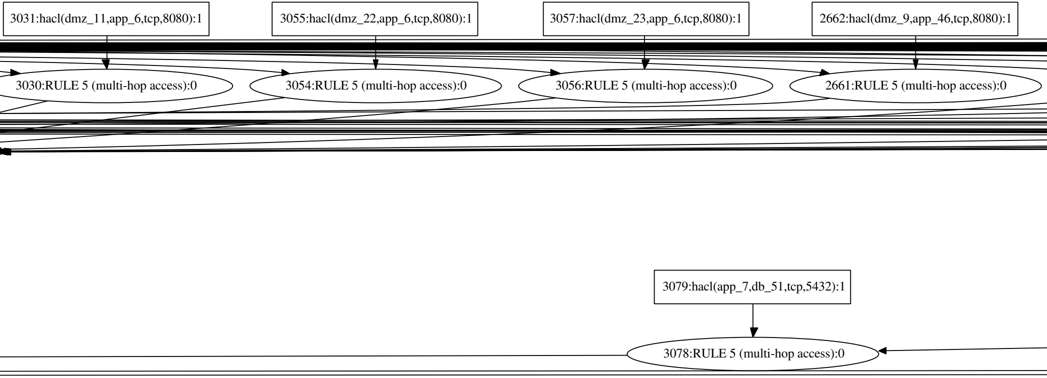

3047:hacl(dmz_19,app_6,tcp,8080):1
3041:hacl(dmz_16,app_6,tcp,8080):1
3067:hacl(dmz_5,app_6,tcp,8080):1
3049:hacl(dmz_2,app_6,tcp,8080):1
3046:RULE 5 (multi-hop access):0
3040:RULE 5 (multi-hop access):0
3066:RULE 5 (multi-hop access):0
3048:RULE 5 (multi-hop access):0

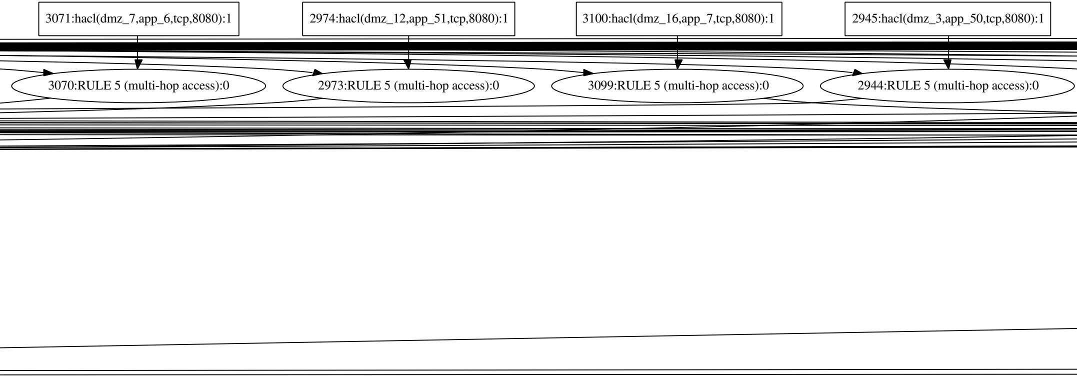

3000:hacl(dmz_24,app_51,tcp,8080):1
2998:hacl(dmz_23,app_51,tcp,8080):1
2921:hacl(dmz_15,app_50,tcp,8080):1
3012:hacl(dmz_7,app_51,tcp,8080):1
2999:RULE 5 (multi-hop access):0
2997:RULE 5 (multi-hop access):0
2920:RULE 5 (multi-hop access):0
3011:RULE 5 (multi-hop access):0
3076:networkServiceInfo(app_6,tomcat,tcp,80
3022:RUL

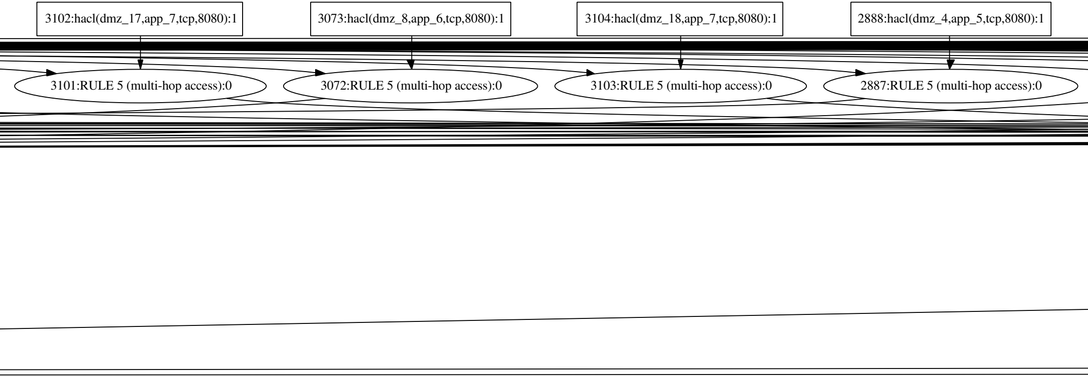

2886:hacl(dmz_3,app_5,tcp,8080):1
3043:hacl(dmz_17,app_6,tcp,8080):1
3051:hacl(dmz_20,app_6,tcp,8080):1
2829:hacl(dmz_4,app_49,tcp,8080):1
2885:RULE 5 (multi-hop access):0
3042:RULE 5 (multi-hop access):0
3050:RULE 5 (multi-hop access):0
2828:RULE 5 (multi-hop access):0
80,root):1
3077:vulExists(app_6,vul_app,tomcat,remoteExploit,privEscalation):1
LE 2 (remote exploit of a server program):0

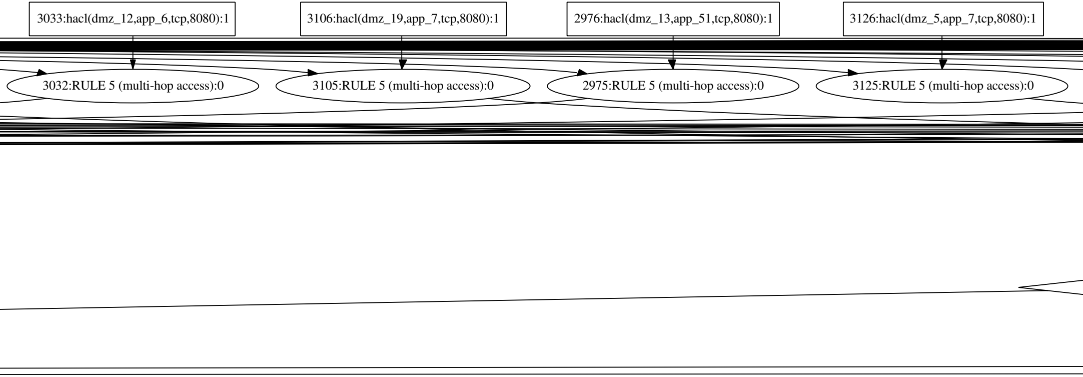

3025:hacl(dmz_0,app_6,tcp,8080):1
3053:hacl(dmz_21,app_6,tcp,8080):1
3069:hacl(dmz_6,app_6,tcp,8080):1
3027:hacl(dmz_1,app_6,tcp,8080):1
3024:RULE 5 (multi-hop access):0
3052:RULE 5 (multi-hop access):0
3068:RULE 5 (multi-hop access):0
3026:RULE 5 (multi-hop access):0

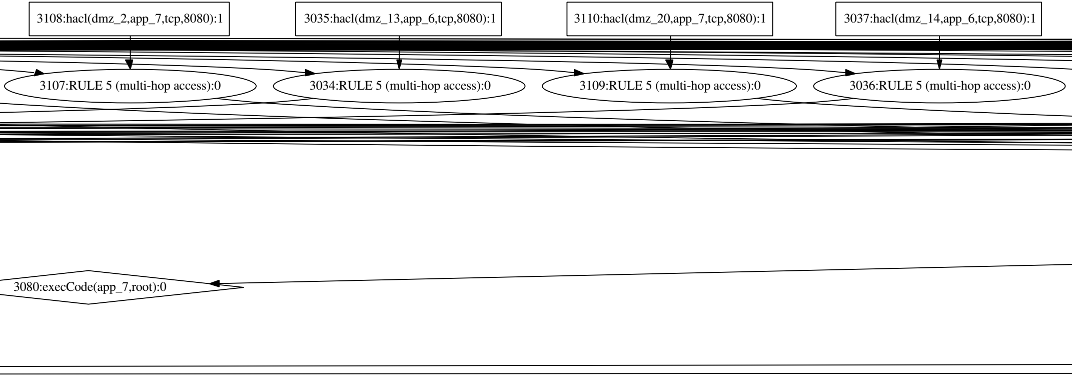

3059:hacl(dmz_24,app_6,tcp,8080):1
2980:hacl(dmz_15,app_51,tcp,8080):1
3014:hacl(dmz_8,app_51,tcp,8080):1
3029:hacl(dmz_10,app_6,tcp,8080):1
3058:RULE 5 (multi-hop access):0
2979:RULE 5 (multi-hop access):0
3013:RULE 5 (multi-hop access):0
3028:RULE 5 (multi-hop access):0
3023:netAccess(app_6,tcp,8080):0

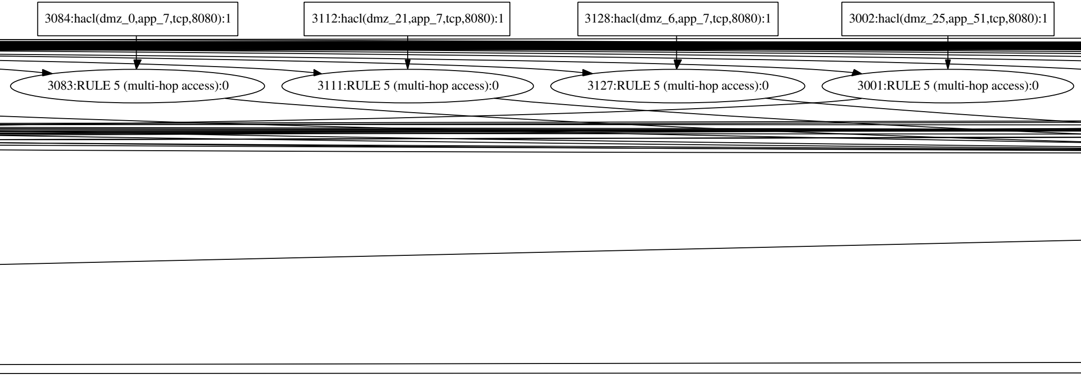

3031:hacl(dmz_11,app_6,tcp,8080):1
3055:hacl(dmz_22,app_6,tcp,8080):1
3057:hacl(dmz_23,app_6,tcp,8080):1
2662:hacl(dmz_9,app_46,tcp,8080):1
3030:RULE 5 (multi-hop access):0
3054:RULE 5 (multi-hop access):0
3056:RULE 5 (multi-hop access):0
2661:RULE 5 (multi-hop access):0
3079:hacl(app_7,db_51,tcp,5432):1
3078:RULE 5 (multi-hop access):0

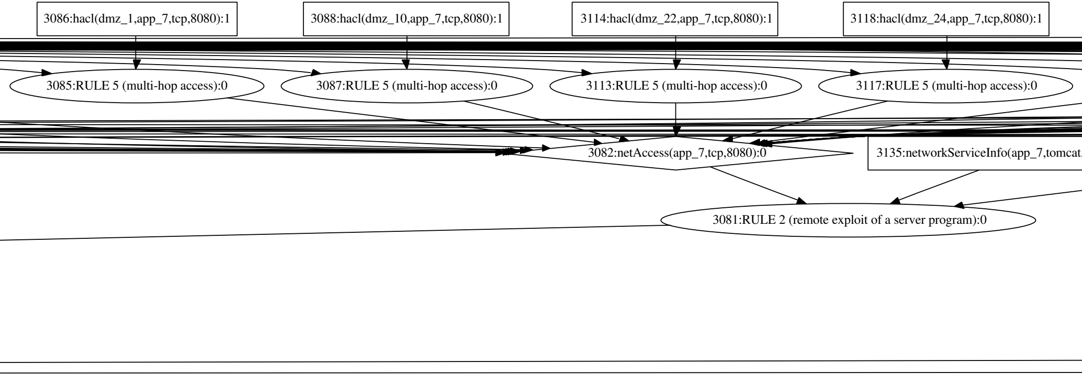

3071:hacl(dmz_7,app_6,tcp,8080):1
2974:hacl(dmz_12,app_51,tcp,8080):1
3100:hacl(dmz_16,app_7,tcp,8080):1
2945:hacl(dmz_3,app_50,tcp,8080):1
3070:RULE 5 (multi-hop access):0
2973:RULE 5 (multi-hop access):0
3099:RULE 5 (multi-hop access):0
2944:RULE 5 (multi-hop access):0

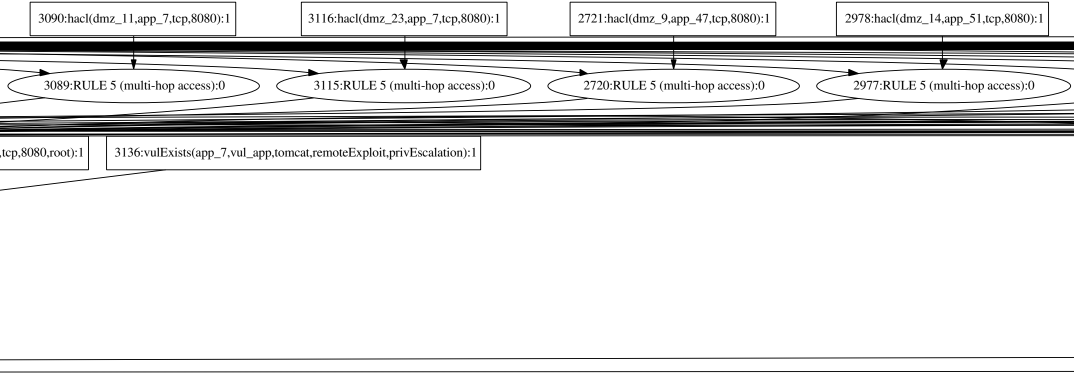

3102:hacl(dmz_17,app_7,tcp,8080):1
3073:hacl(dmz_8,app_6,tcp,8080):1
3104:hacl(dmz_18,app_7,tcp,8080):1
2888:hacl(dmz_4,app_5,tcp,8080):1
3101:RULE 5 (multi-hop access):0
3072:RULE 5 (multi-hop access):0
3103:RULE 5 (multi-hop access):0
2887:RULE 5 (multi-hop access):0

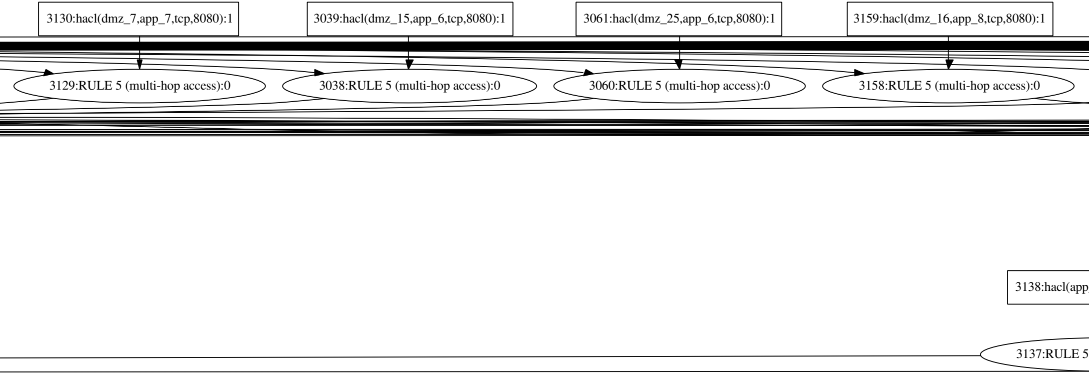

3033:hacl(dmz_12,app_6,tcp,8080):1
3106:hacl(dmz_19,app_7,tcp,8080):1
2976:hacl(dmz_13,app_51,tcp,8080):1
3126:hacl(dmz_5,app_7,tcp,8080):1
3032:RULE 5 (multi-hop access):0
3105:RULE 5 (multi-hop access):0
2975:RULE 5 (multi-hop access):0
3125:RULE 5 (multi-hop access):0

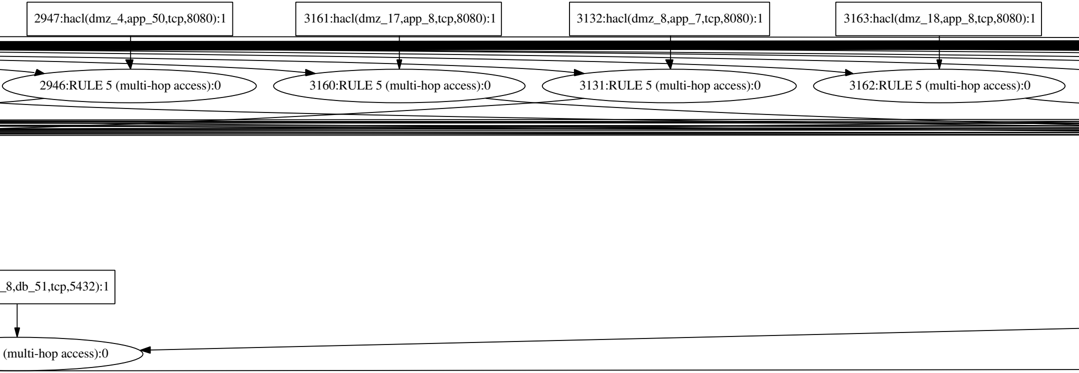

3108:hacl(dmz_2,app_7,tcp,8080):1
3035:hacl(dmz_13,app_6,tcp,8080):1
3110:hacl(dmz_20,app_7,tcp,8080):1
3037:hacl(dmz_14,app_6,tcp,8080):1
3107:RULE 5 (multi-hop access):0
3034:RULE 5 (multi-hop access):0
3109:RULE 5 (multi-hop access):0
3036:RULE 5 (multi-hop access):0
3080:execCode(app_7,root):0

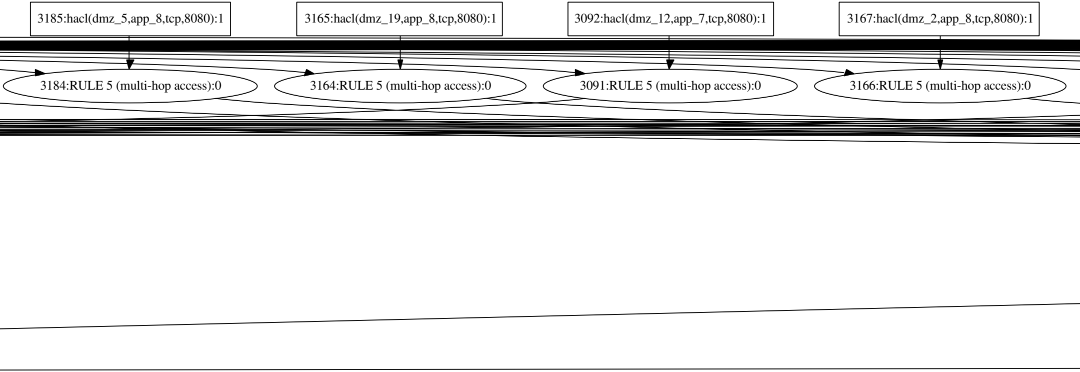

3084:hacl(dmz_0,app_7,tcp,8080):1
3112:hacl(dmz_21,app_7,tcp,8080):1
3128:hacl(dmz_6,app_7,tcp,8080):1
3002:hacl(dmz_25,app_51,tcp,8080):1
3083:RULE 5 (multi-hop access):0
3111:RULE 5 (multi-hop access):0
3127:RULE 5 (multi-hop access):0
3001:RULE 5 (multi-hop access):0

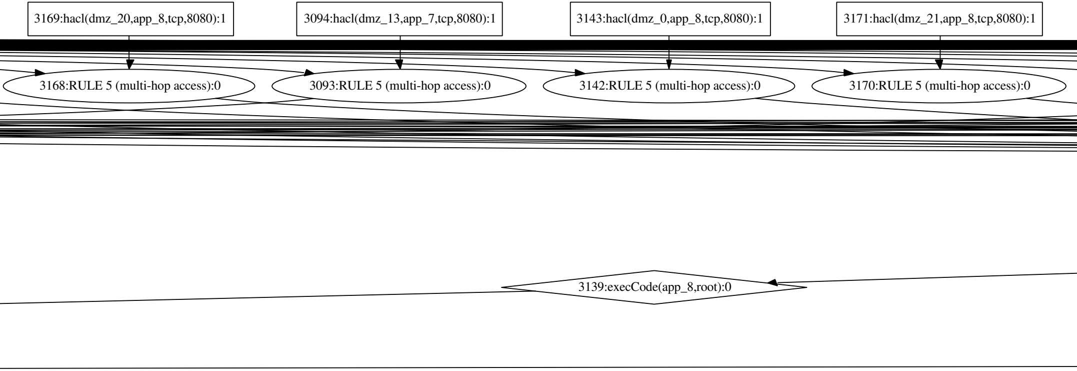

3086:hacl(dmz_1,app_7,tcp,8080):1
3088:hacl(dmz_10,app_7,tcp,8080):1
3114:hacl(dmz_22,app_7,tcp,8080):1
3118:hacl(dmz_24,app_7,tcp,8080):1
3085:RULE 5 (multi-hop access):0
3087:RULE 5 (multi-hop access):0
3113:RULE 5 (multi-hop access):0
3117:RULE 5 (multi-hop access):0
3082:netAccess(app_7,tcp,8080):0
3135:networkServiceInfo(app_7,tomcat
3081:RULE 2 (remote exploit of a server program):0

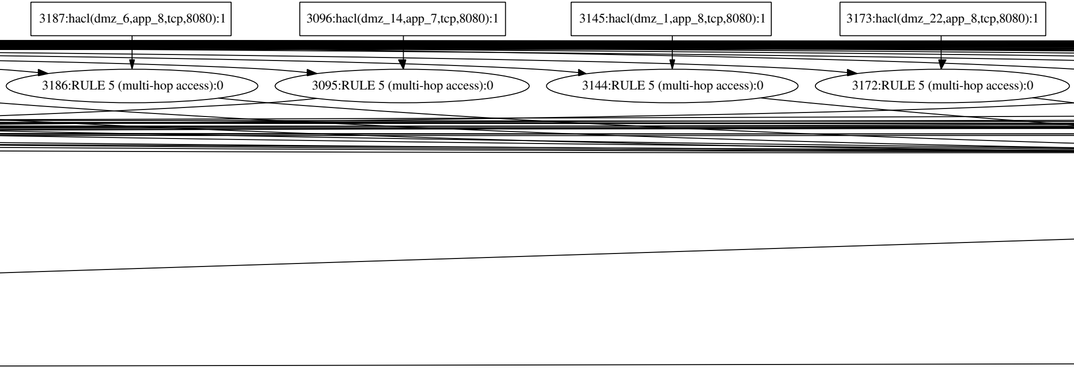

3090:hacl(dmz_11,app_7,tcp,8080):1
3116:hacl(dmz_23,app_7,tcp,8080):1
2721:hacl(dmz_9,app_47,tcp,8080):1
2978:hacl(dmz_14,app_51,tcp,8080):1
3089:RULE 5 (multi-hop access):0
3115:RULE 5 (multi-hop access):0
2720:RULE 5 (multi-hop access):0
2977:RULE 5 (multi-hop access):0
,tcp,8080,root):1
3136:vulExists(app_7,vul_app,tomcat,remoteExploit,privEscalation):1

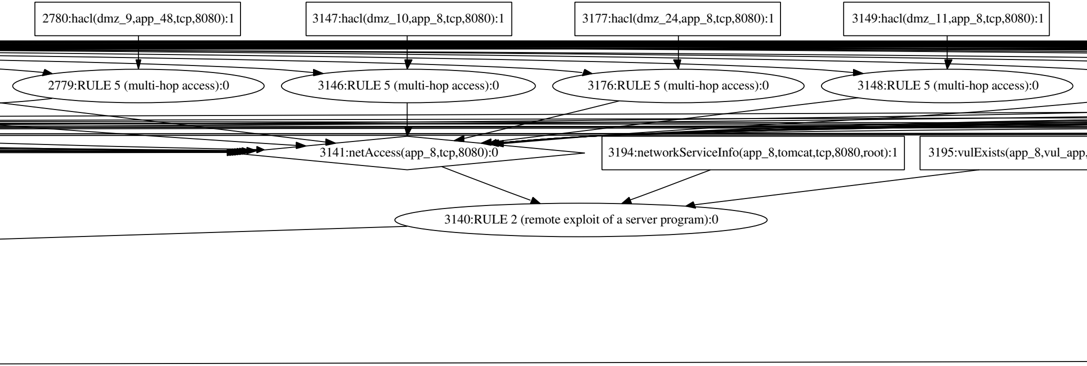

3130:hacl(dmz_7,app_7,tcp,8080):1
3039:hacl(dmz_15,app_6,tcp,8080):1
3061:hacl(dmz_25,app_6,tcp,8080):1
3159:hacl(dmz_16,app_8,tcp,8080):1
3129:RULE 5 (multi-hop access):0
3038:RULE 5 (multi-hop access):0
3060:RULE 5 (multi-hop access):0
3158:RULE 5 (multi-hop access):0
3138:hacl(app
3137:RULE 5

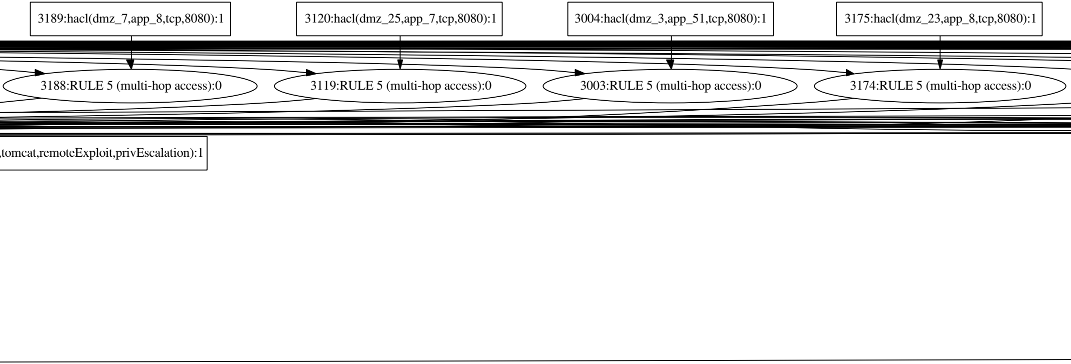

2947:hacl(dmz_4,app_50,tcp,8080):1
3161:hacl(dmz_17,app_8,tcp,8080):1
3132:hacl(dmz_8,app_7,tcp,8080):1
3163:hacl(dmz_18,app_8,tcp,8080):1
2946:RULE 5 (multi-hop access):0
3160:RULE 5 (multi-hop access):0
3131:RULE 5 (multi-hop access):0
3162:RULE 5 (multi-hop access):0
_8,db_51,tcp,5432):1
(multi-hop access):0

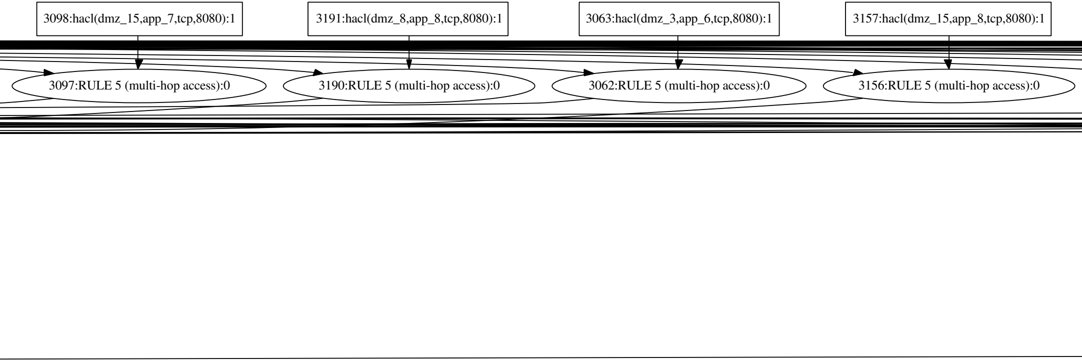

3185:hacl(dmz_5,app_8,tcp,8080):1
3165:hacl(dmz_19,app_8,tcp,8080):1
3092:hacl(dmz_12,app_7,tcp,8080):1
3167:hacl(dmz_2,app_8,tcp,8080):1
3184:RULE 5 (multi-hop access):0
3164:RULE 5 (multi-hop access):0
3091:RULE 5 (multi-hop access):0
3166:RULE 5 (multi-hop access):0

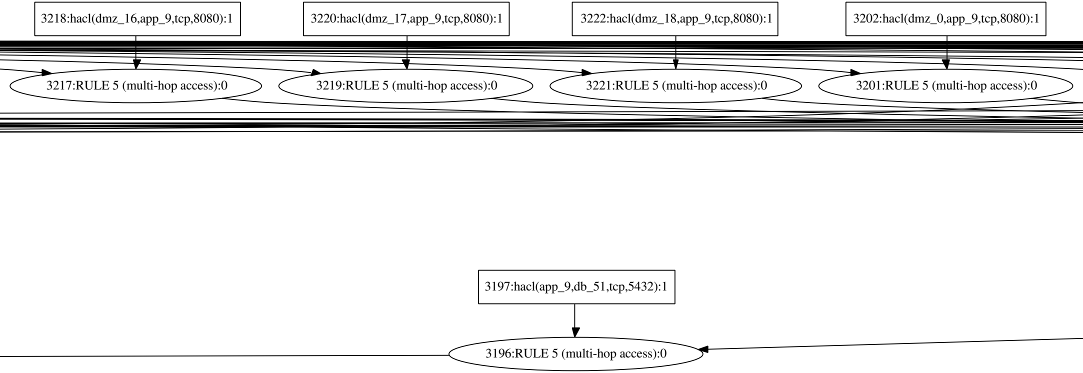

3169:hacl(dmz_20,app_8,tcp,8080):1
3094:hacl(dmz_13,app_7,tcp,8080):1
3143:hacl(dmz_0,app_8,tcp,8080):1
3171:hacl(dmz_21,app_8,tcp,8080):1
3168:RULE 5 (multi-hop access):0
3093:RULE 5 (multi-hop access):0
3142:RULE 5 (multi-hop access):0
3170:RULE 5 (multi-hop access):0
3139:execCode(app_8,root):0

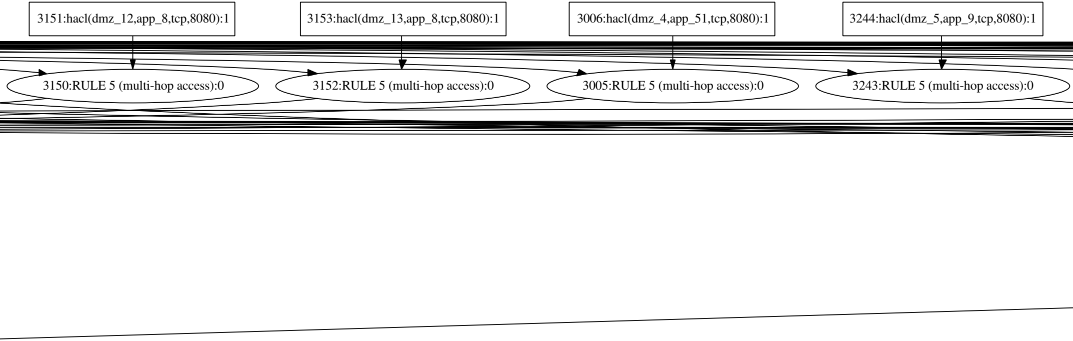

3187:hacl(dmz_6,app_8,tcp,8080):1
3096:hacl(dmz_14,app_7,tcp,8080):1
3145:hacl(dmz_1,app_8,tcp,8080):1
3173:hacl(dmz_22,app_8,tcp,8080):1
3186:RULE 5 (multi-hop access):0
3095:RULE 5 (multi-hop access):0
3144:RULE 5 (multi-hop access):0
3172:RULE 5 (multi-hop access):0

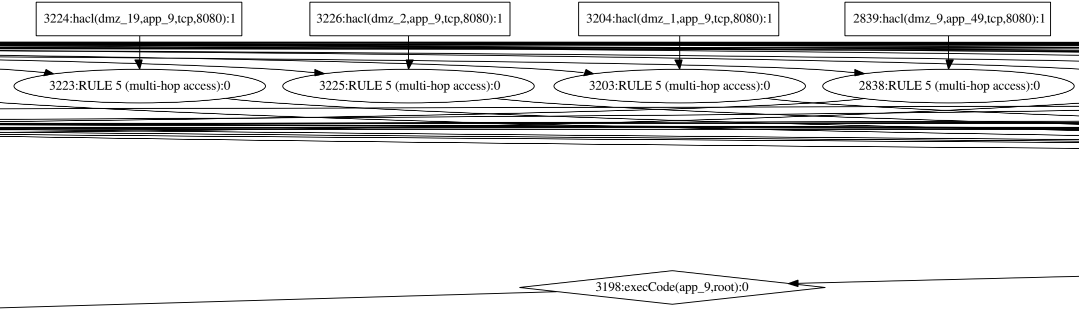

2780:hacl(dmz_9,app_48,tcp,8080):1
3147:hacl(dmz_10,app_8,tcp,8080):1
3177:hacl(dmz_24,app_8,tcp,8080):1
3149:hacl(dmz_11,app_8,tcp,8080):1
2779:RULE 5 (multi-hop access):0
3146:RULE 5 (multi-hop access):0
3176:RULE 5 (multi-hop access):0
3148:RULE 5 (multi-hop access):0
3141:netAccess(app_8,tcp,8080):0
3194:networkServiceInfo(app_8,tomcat,tcp,8080,root):1
3195:vulExists(app_8,vul_app,
3140:RULE 2 (remote exploit of a server program):0

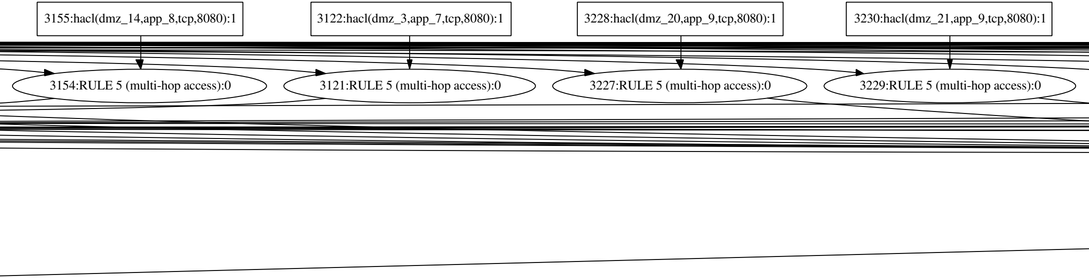

3189:hacl(dmz_7,app_8,tcp,8080):1
3120:hacl(dmz_25,app_7,tcp,8080):1
3004:hacl(dmz_3,app_51,tcp,8080):1
3175:hacl(dmz_23,app_8,tcp,8080):1
3188:RULE 5 (multi-hop access):0
3119:RULE 5 (multi-hop access):0
3003:RULE 5 (multi-hop access):0
3174:RULE 5 (multi-hop access):0
,tomcat,remoteExploit,privEscalation):1

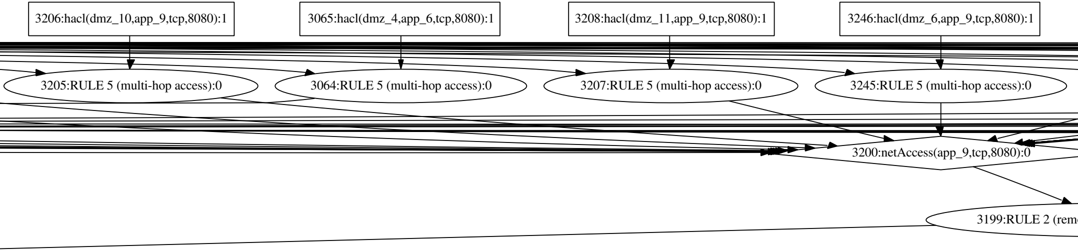

3098:hacl(dmz_15,app_7,tcp,8080):1
3191:hacl(dmz_8,app_8,tcp,8080):1
3063:hacl(dmz_3,app_6,tcp,8080):1
3157:hacl(dmz_15,app_8,tcp,8080):1
3097:RULE 5 (multi-hop access):0
3190:RULE 5 (multi-hop access):0
3062:RULE 5 (multi-hop access):0
3156:RULE 5 (multi-hop access):0

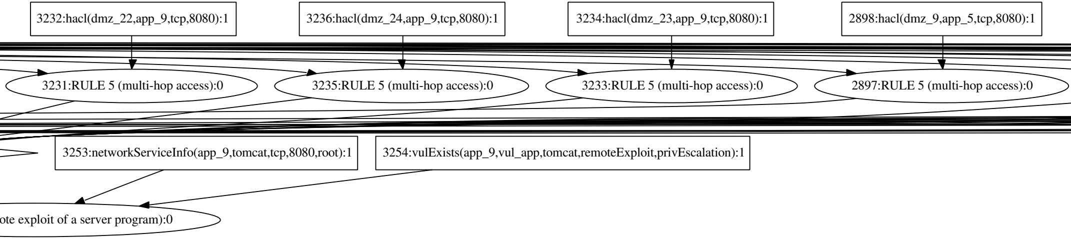

3218:hacl(dmz_16,app_9,tcp,8080):1
3220:hacl(dmz_17,app_9,tcp,8080):1
3222:hacl(dmz_18,app_9,tcp,8080):1
3202:hacl(dmz_0,app_9,tcp,8080):1
3217:RULE 5 (multi-hop access):0
3219:RULE 5 (multi-hop access):0
3221:RULE 5 (multi-hop access):0
3201:RULE 5 (multi-hop access):0
3197:hacl(app_9,db_51,tcp,5432):1
3196:RULE 5 (multi-hop access):0

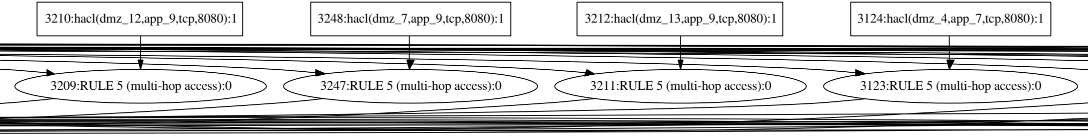

3151:hacl(dmz_12,app_8,tcp,8080):1
3153:hacl(dmz_13,app_8,tcp,8080):1
3006:hacl(dmz_4,app_51,tcp,8080):1
3244:hacl(dmz_5,app_9,tcp,8080):1
3150:RULE 5 (multi-hop access):0
3152:RULE 5 (multi-hop access):0
3005:RULE 5 (multi-hop access):0
3243:RULE 5 (multi-hop access):0

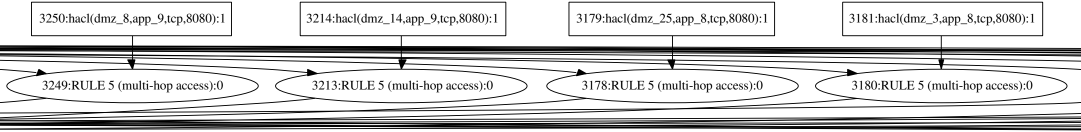

3224:hacl(dmz_19,app_9,tcp,8080):1
3226:hacl(dmz_2,app_9,tcp,8080):1
3204:hacl(dmz_1,app_9,tcp,8080):1
2839:hacl(dmz_9,app_49,tcp,8080):1
3223:RULE 5 (multi-hop access):0
3225:RULE 5 (multi-hop access):0
3203:RULE 5 (multi-hop access):0
2838:RULE 5 (multi-hop access):0
3198:execCode(app_9,root):0

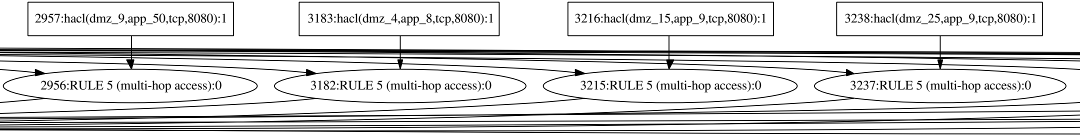

3155:hacl(dmz_14,app_8,tcp,8080):1
3122:hacl(dmz_3,app_7,tcp,8080):1
3228:hacl(dmz_20,app_9,tcp,8080):1
3230:hacl(dmz_21,app_9,tcp,8080):1
3154:RULE 5 (multi-hop access):0
3121:RULE 5 (multi-hop access):0
3227:RULE 5 (multi-hop access):0
3229:RULE 5 (multi-hop access):0

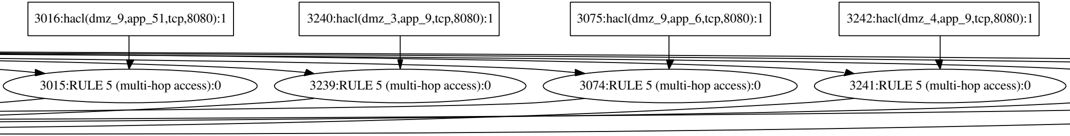

3206:hacl(dmz_10,app_9,tcp,8080):1
3065:hacl(dmz_4,app_6,tcp,8080):1
3208:hacl(dmz_11,app_9,tcp,8080):1
3246:hacl(dmz_6,app_9,tcp,8080):1
3205:RULE 5 (multi-hop access):0
3064:RULE 5 (multi-hop access):0
3207:RULE 5 (multi-hop access):0
3245:RULE 5 (multi-hop access):0
3200:netAccess(app_9,tcp,8080):0
3199:RULE 2 (rem

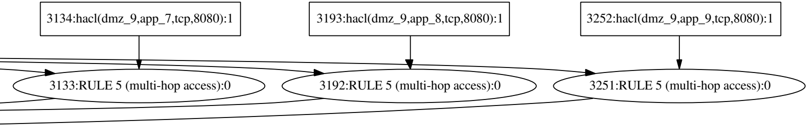

3232:hacl(dmz_22,app_9,tcp,8080):1
3236:hacl(dmz_24,app_9,tcp,8080):1
3234:hacl(dmz_23,app_9,tcp,8080):1
2898:hacl(dmz_9,app_5,tcp,8080):1
3231:RULE 5 (multi-hop access):0
3235:RULE 5 (multi-hop access):0
3233:RULE 5 (multi-hop access):0
2897:RULE 5 (multi-hop access):0
3253:networkServiceInfo(app_9,tomcat,tcp,8080,root):1
3254:vulExists(app_9,vul_app,tomcat,remoteExploit,privEscalation):1
ote exploit of a server program):0


3210:hacl(dmz_12,app_9,tcp,8080):1
3248:hacl(dmz_7,app_9,tcp,8080):1
3212:hacl(dmz_13,app_9,tcp,8080):1
3124:hacl(dmz_4,app_7,tcp,8080):1
3209:RULE 5 (multi-hop access):0
3247:RULE 5 (multi-hop access):0
3211:RULE 5 (multi-hop access):0
3123:RULE 5 (multi-hop access):0


3250:hacl(dmz_8,app_9,tcp,8080):1
3214:hacl(dmz_14,app_9,tcp,8080):1
3179:hacl(dmz_25,app_8,tcp,8080):1
3181:hacl(dmz_3,app_8,tcp,8080):1
3249:RULE 5 (multi-hop access):0
3213:RULE 5 (multi-hop access):0
3178:RULE 5 (multi-hop access):0
3180:RULE 5 (multi-hop access):0


2957:hacl(dmz_9,app_50,tcp,8080):1
3183:hacl(dmz_4,app_8,tcp,8080):1
3216:hacl(dmz_15,app_9,tcp,8080):1
3238:hacl(dmz_25,app_9,tcp,8080):1
2956:RULE 5 (multi-hop access):0
3182:RULE 5 (multi-hop access):0
3215:RULE 5 (multi-hop access):0
3237:RULE 5 (multi-hop access):0


3016:hacl(dmz_9,app_51,tcp,8080):1
3240:hacl(dmz_3,app_9,tcp,8080):1
3075:hacl(dmz_9,app_6,tcp,8080):1
3242:hacl(dmz_4,app_9,tcp,8080):1
3015:RULE 5 (multi-hop access):0
3239:RULE 5 (multi-hop access):0
3074:RULE 5 (multi-hop access):0
3241:RULE 5 (multi-hop access):0


3134:hacl(dmz_9,app_7,tcp,8080):1
3193:hacl(dmz_9,app_8,tcp,8080):1
3252:hacl(dmz_9,app_9,tcp,8080):1
3133:RULE 5 (multi-hop access):0
3192:RULE 5 (multi-hop access):0
3251:RULE 5 (multi-hop access):0